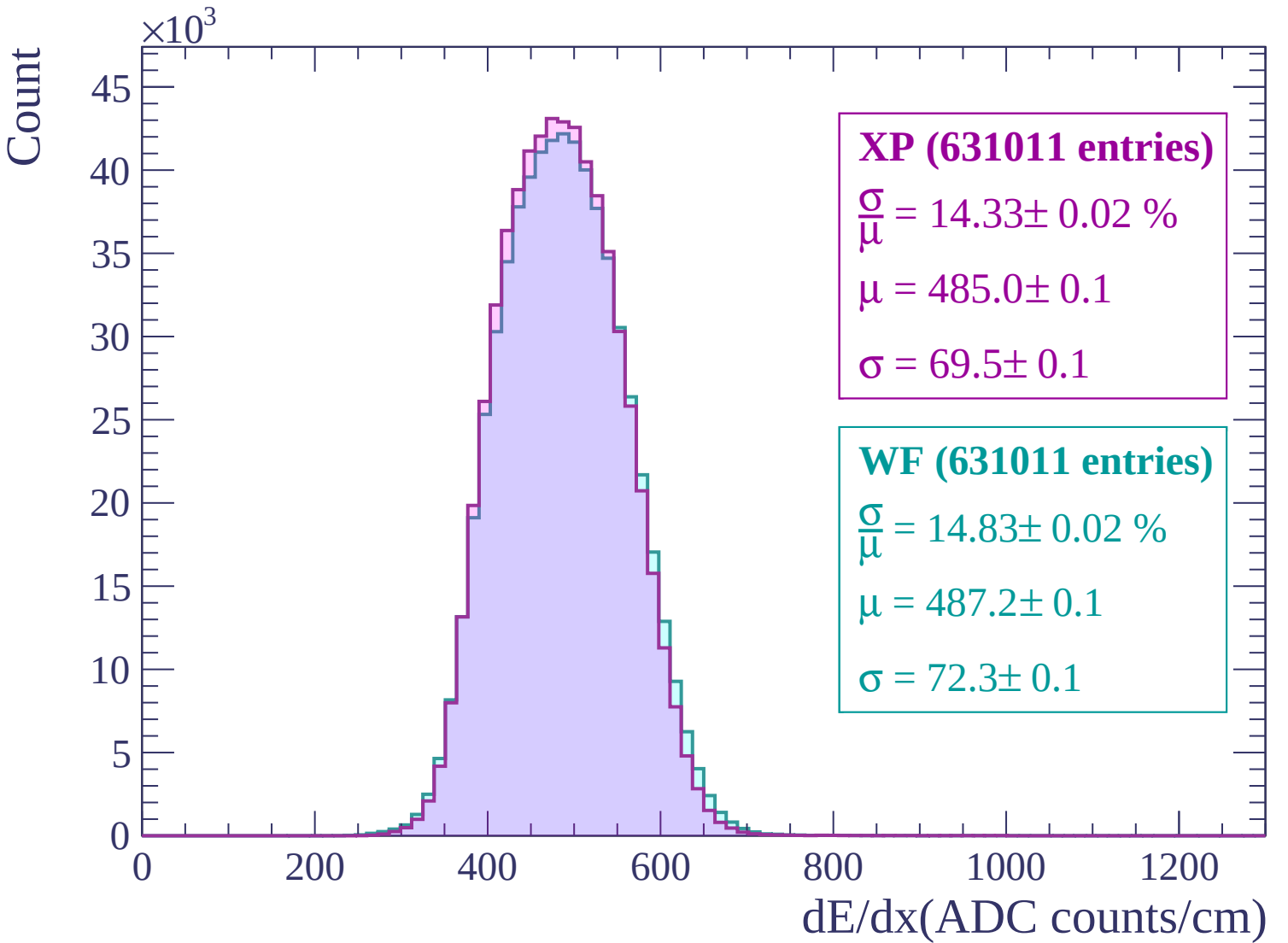
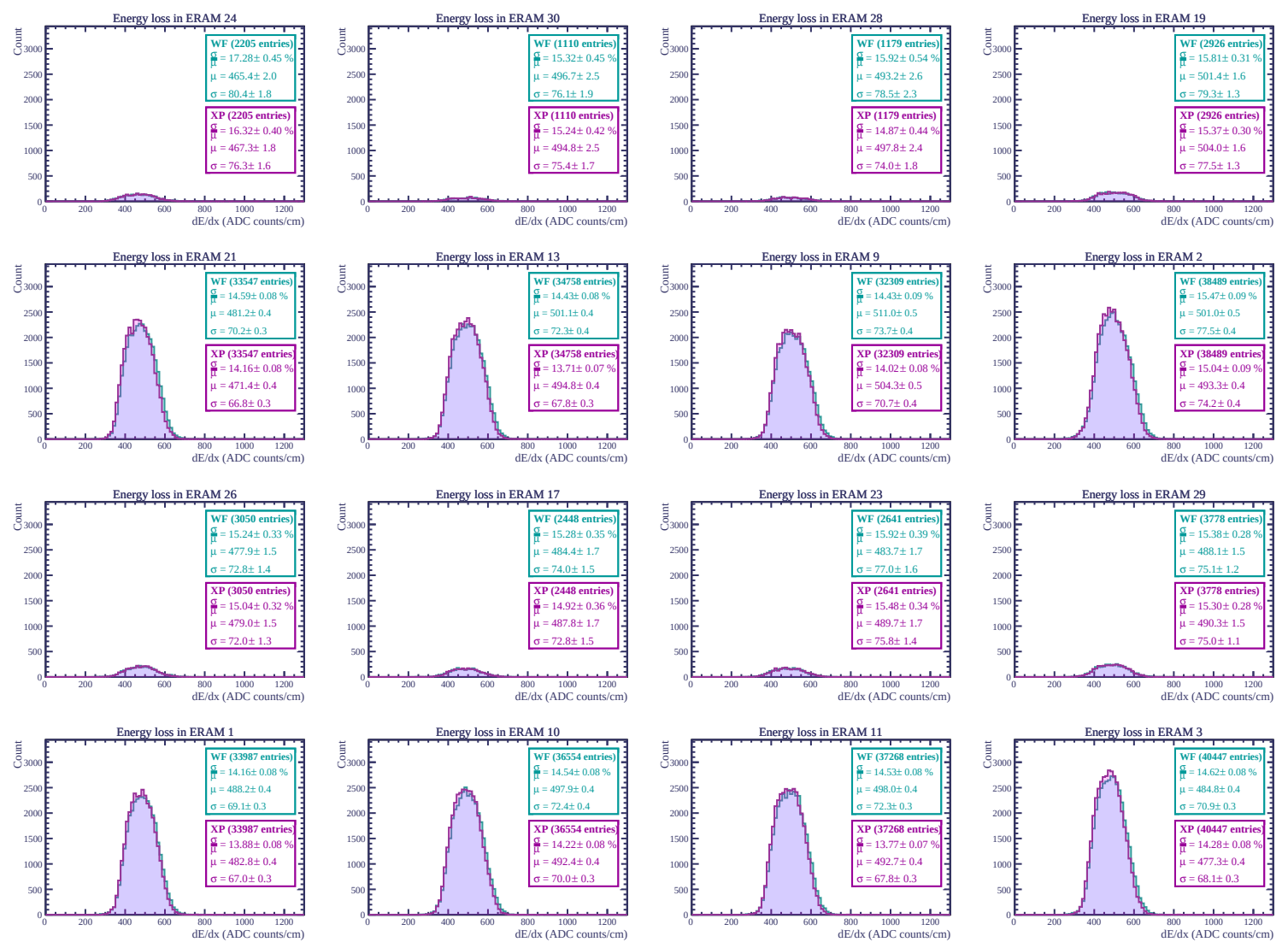
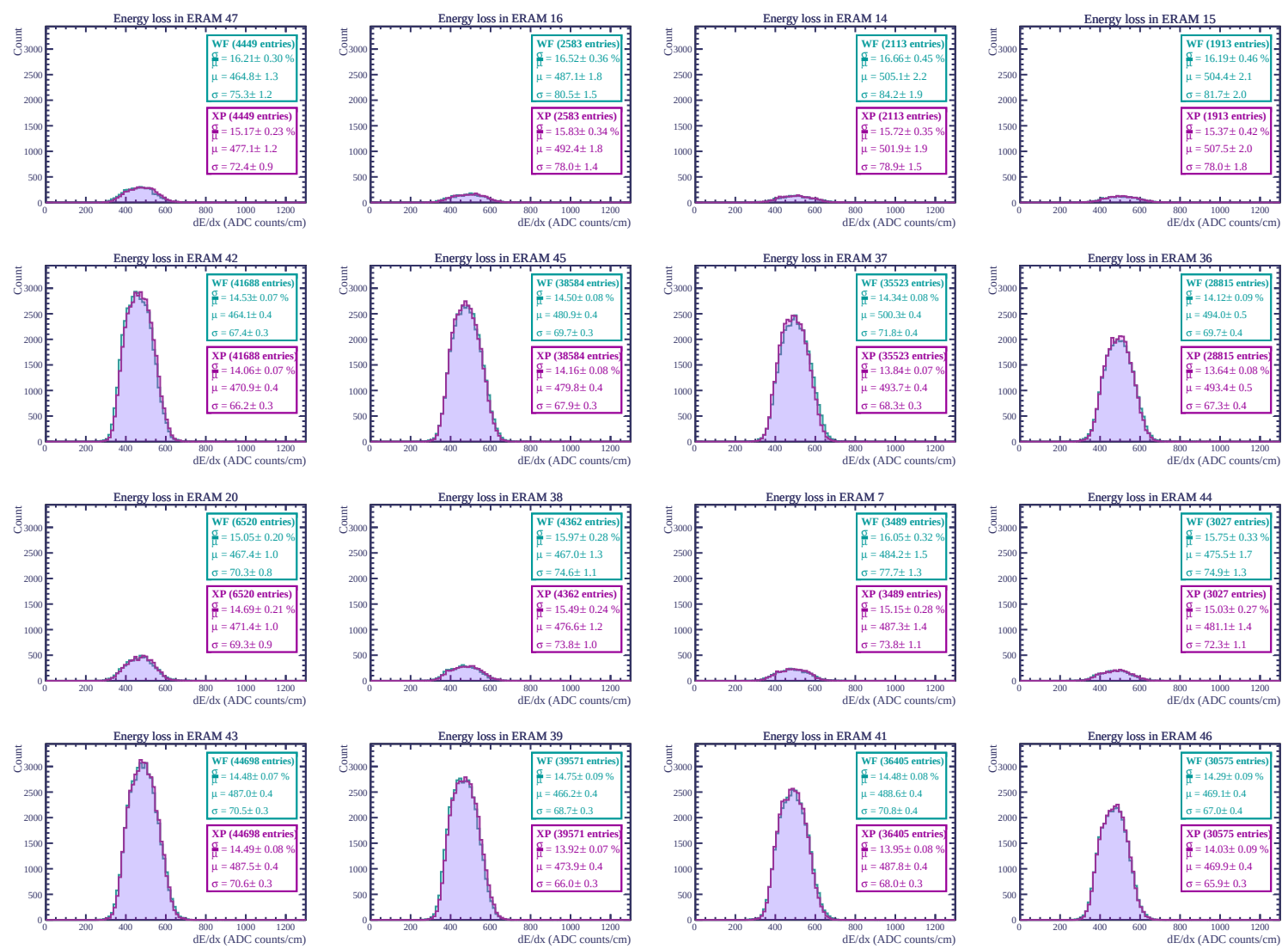
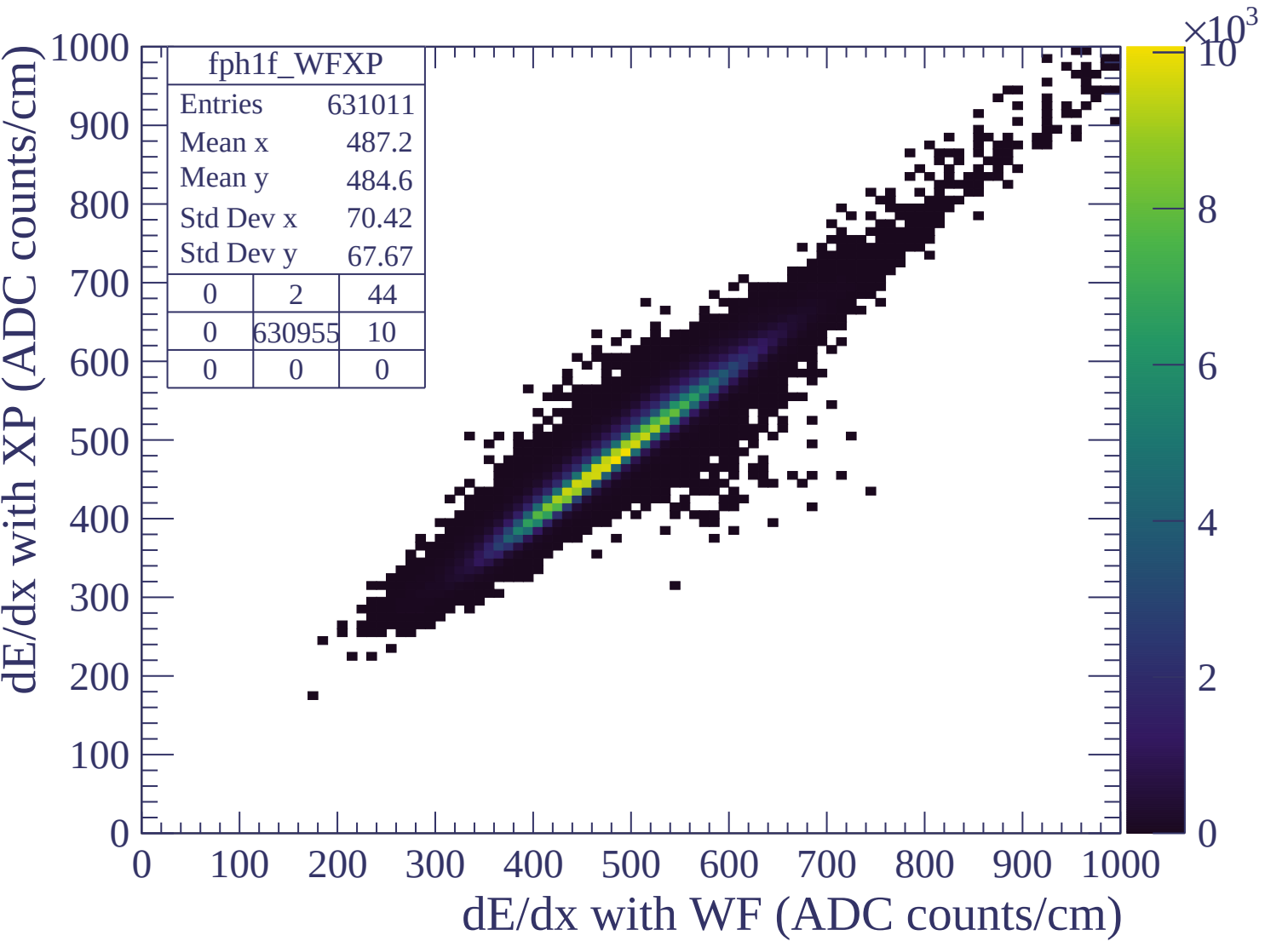
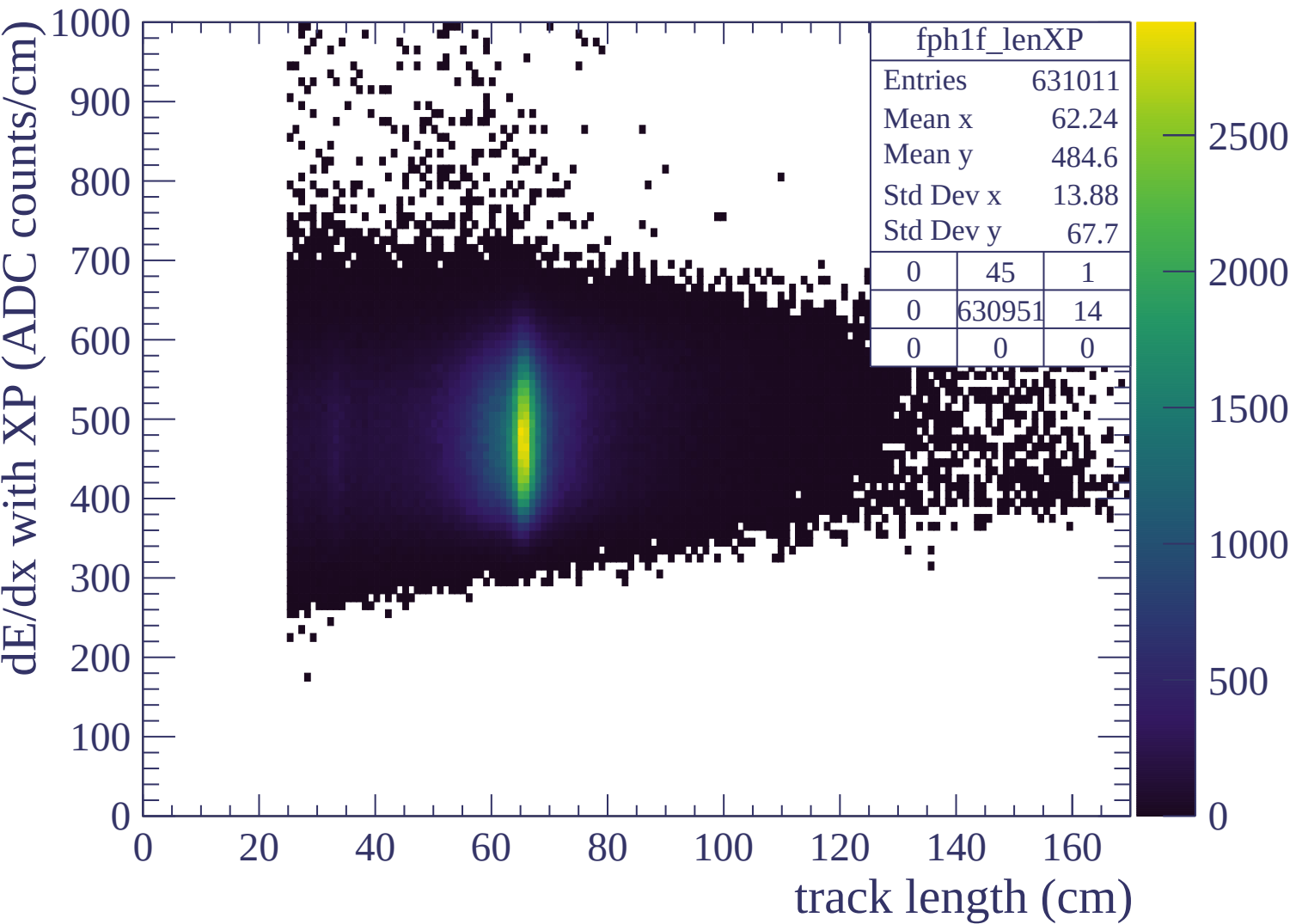


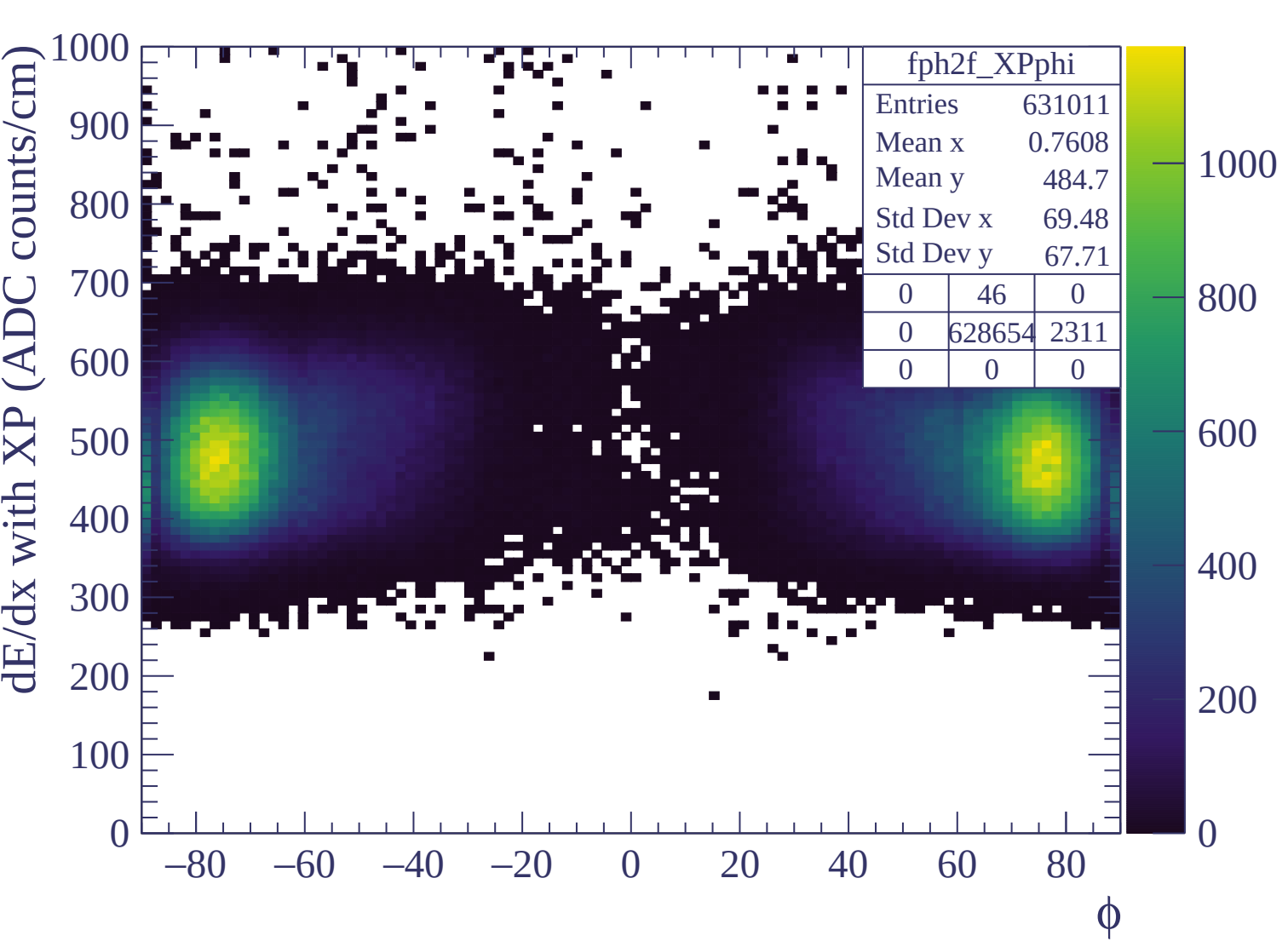


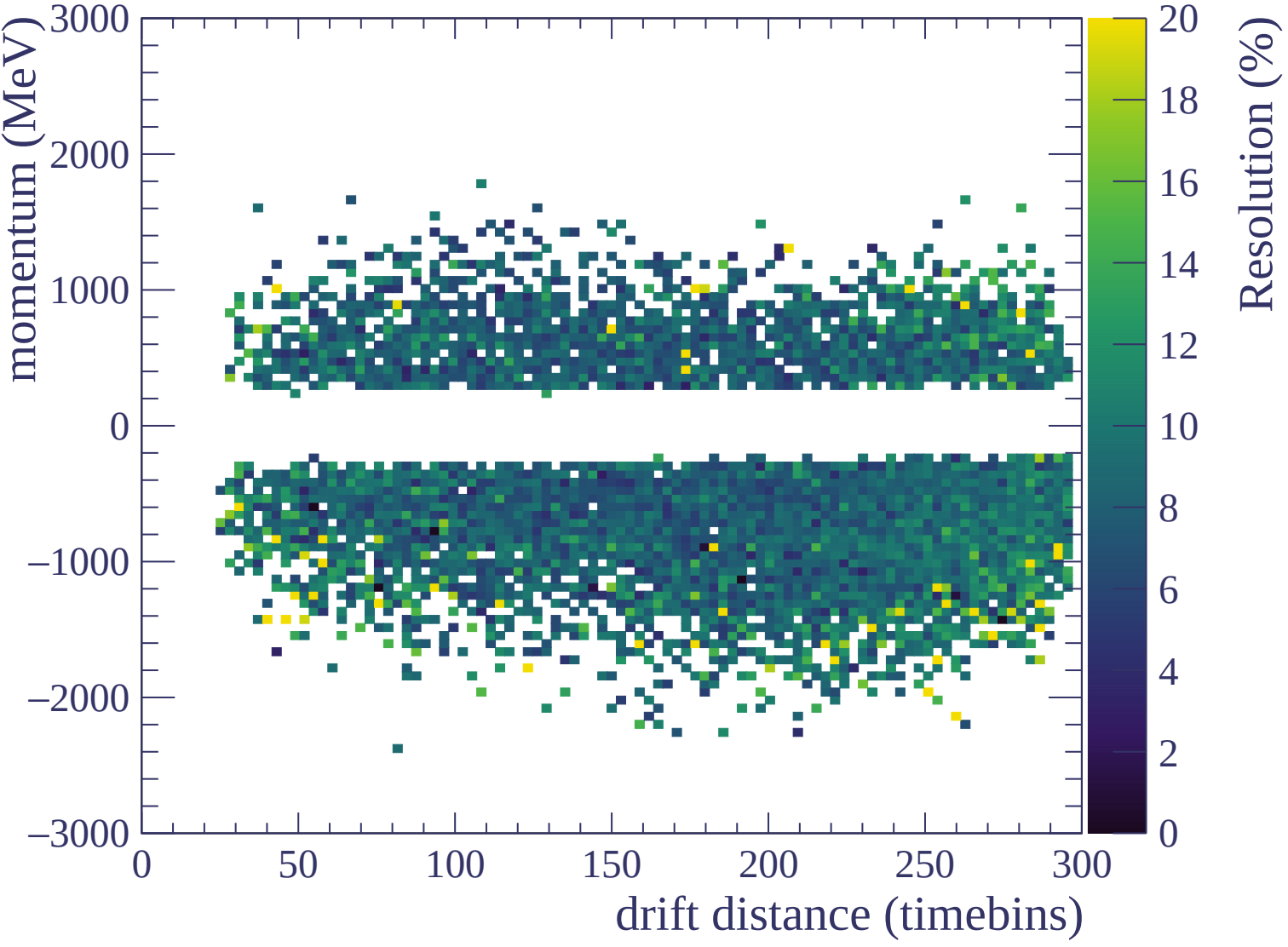


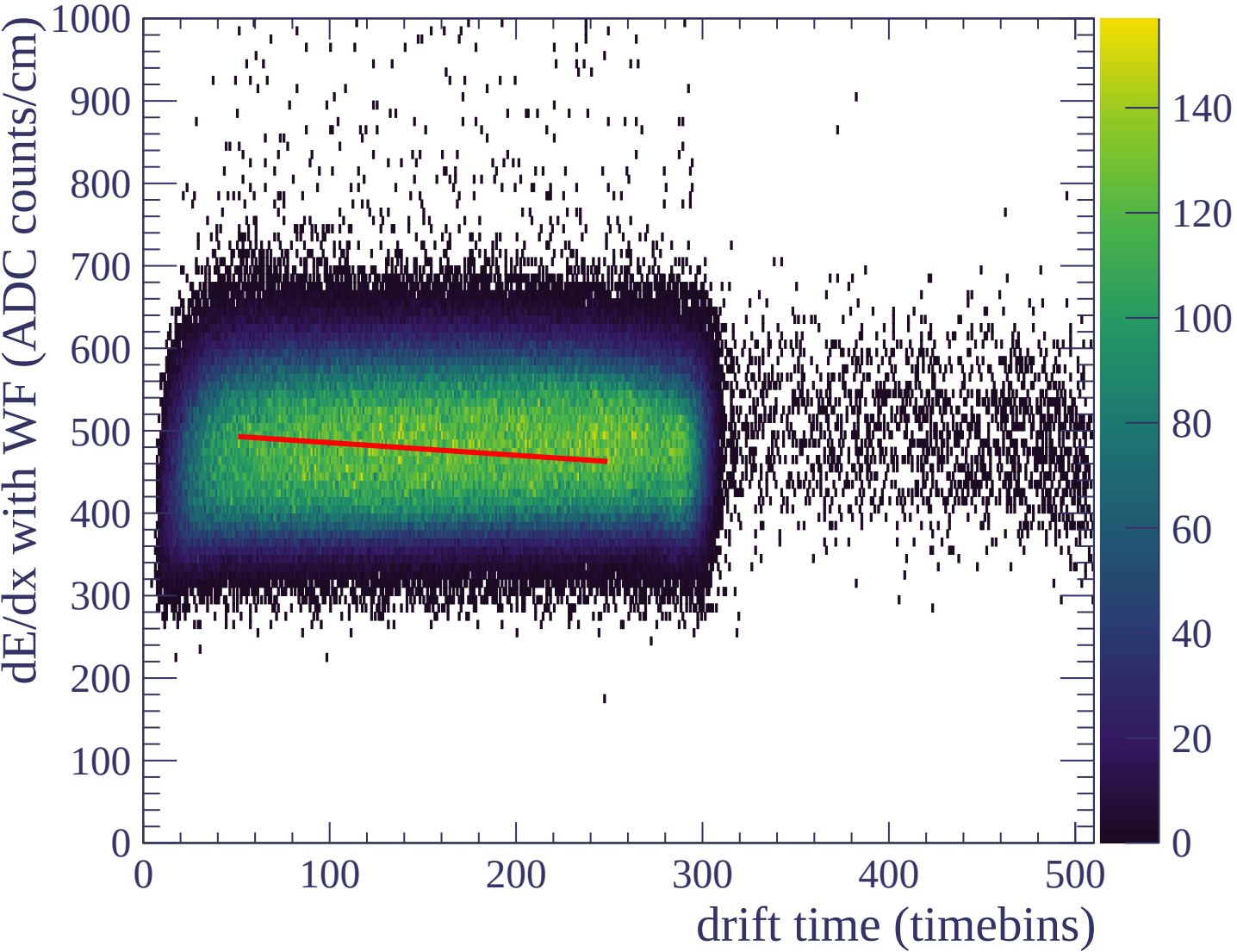


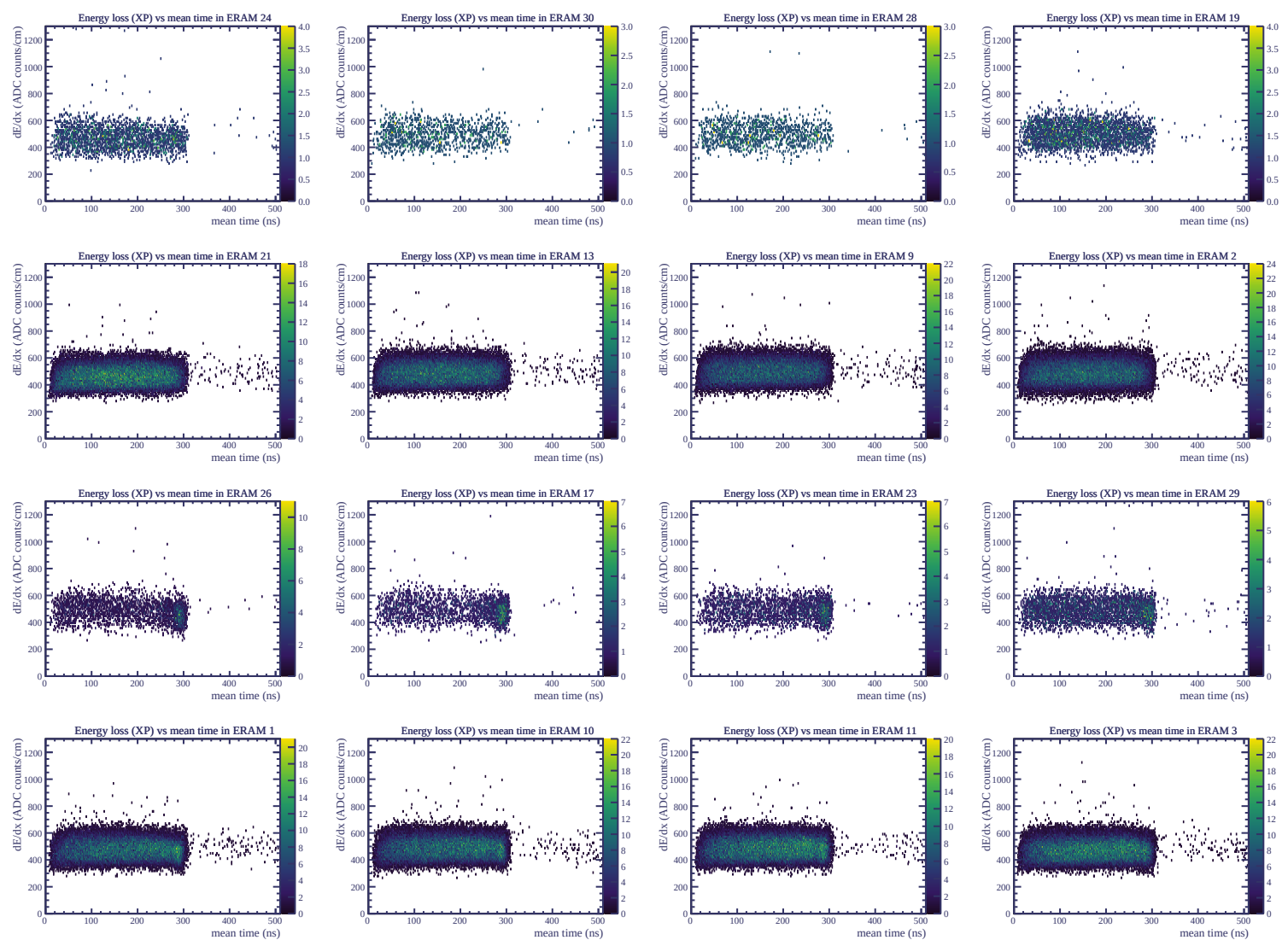


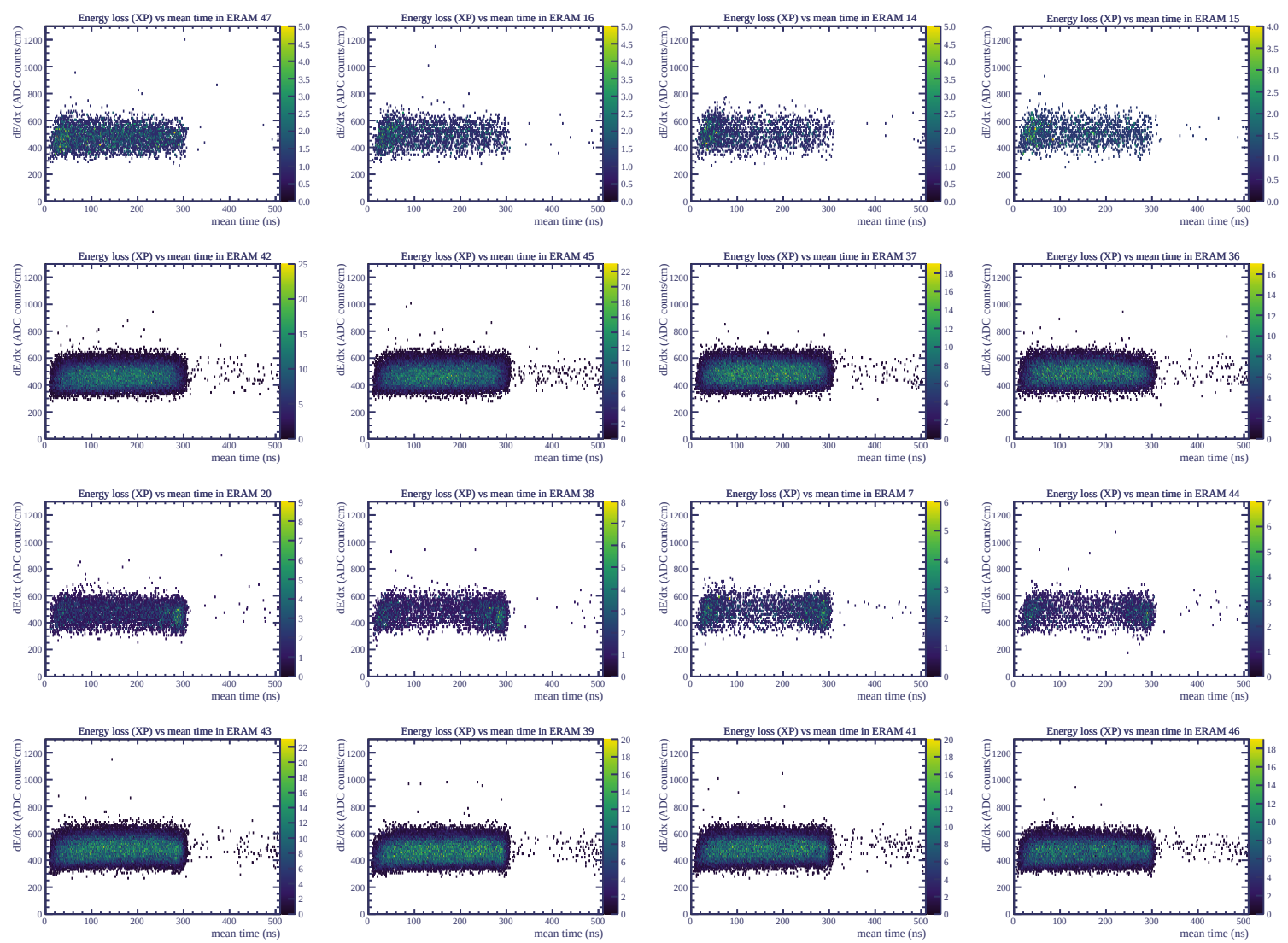


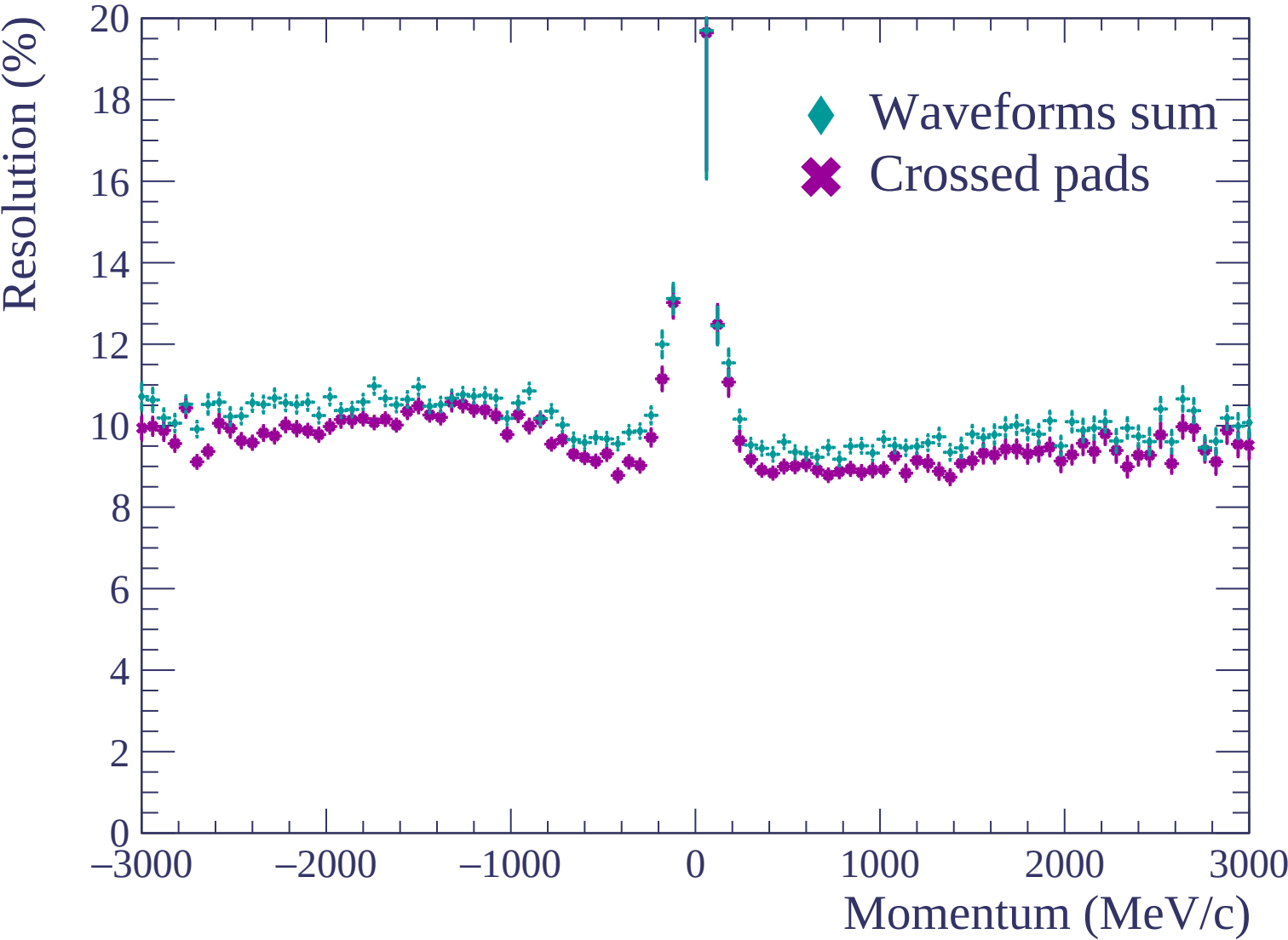


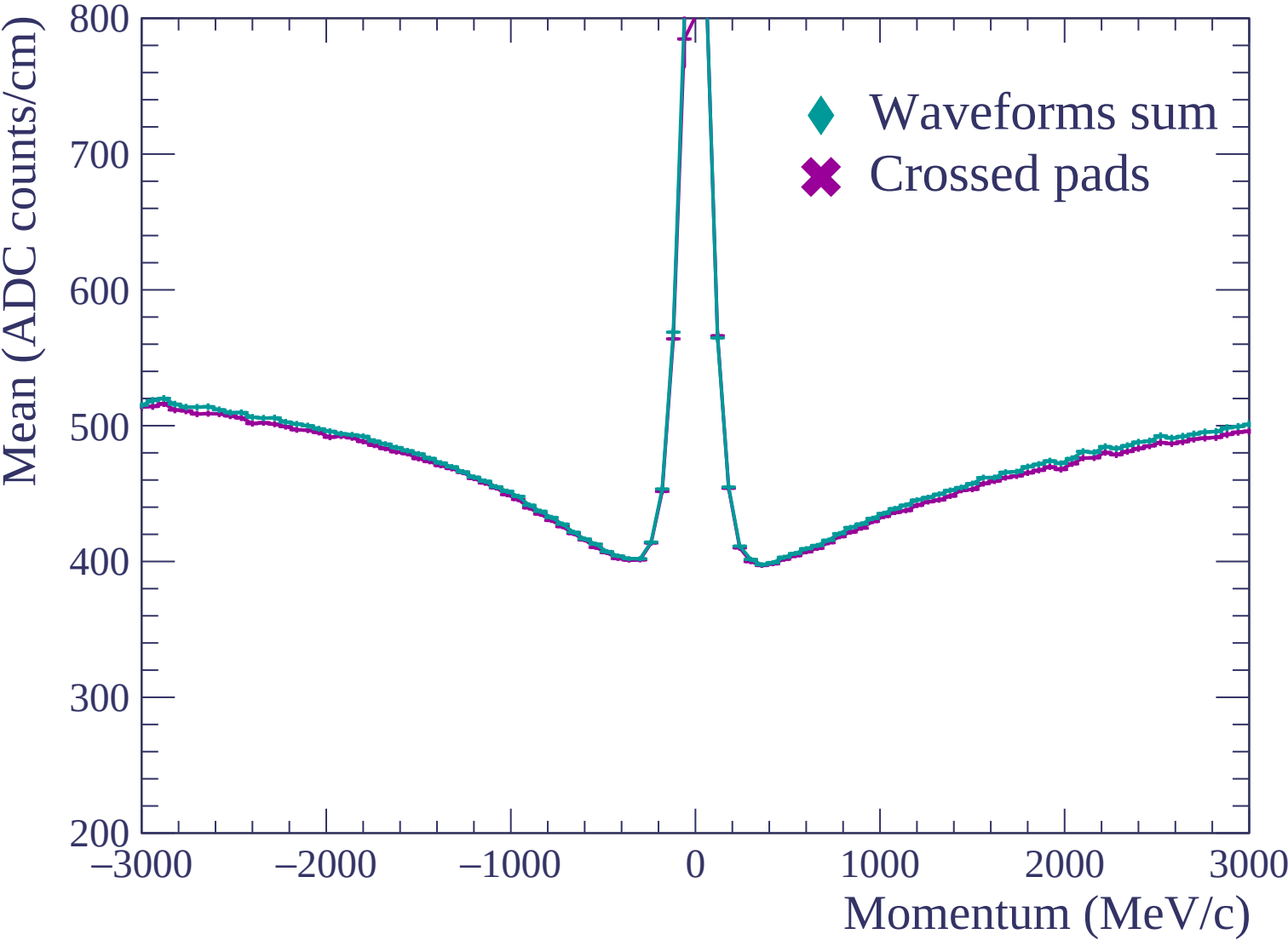


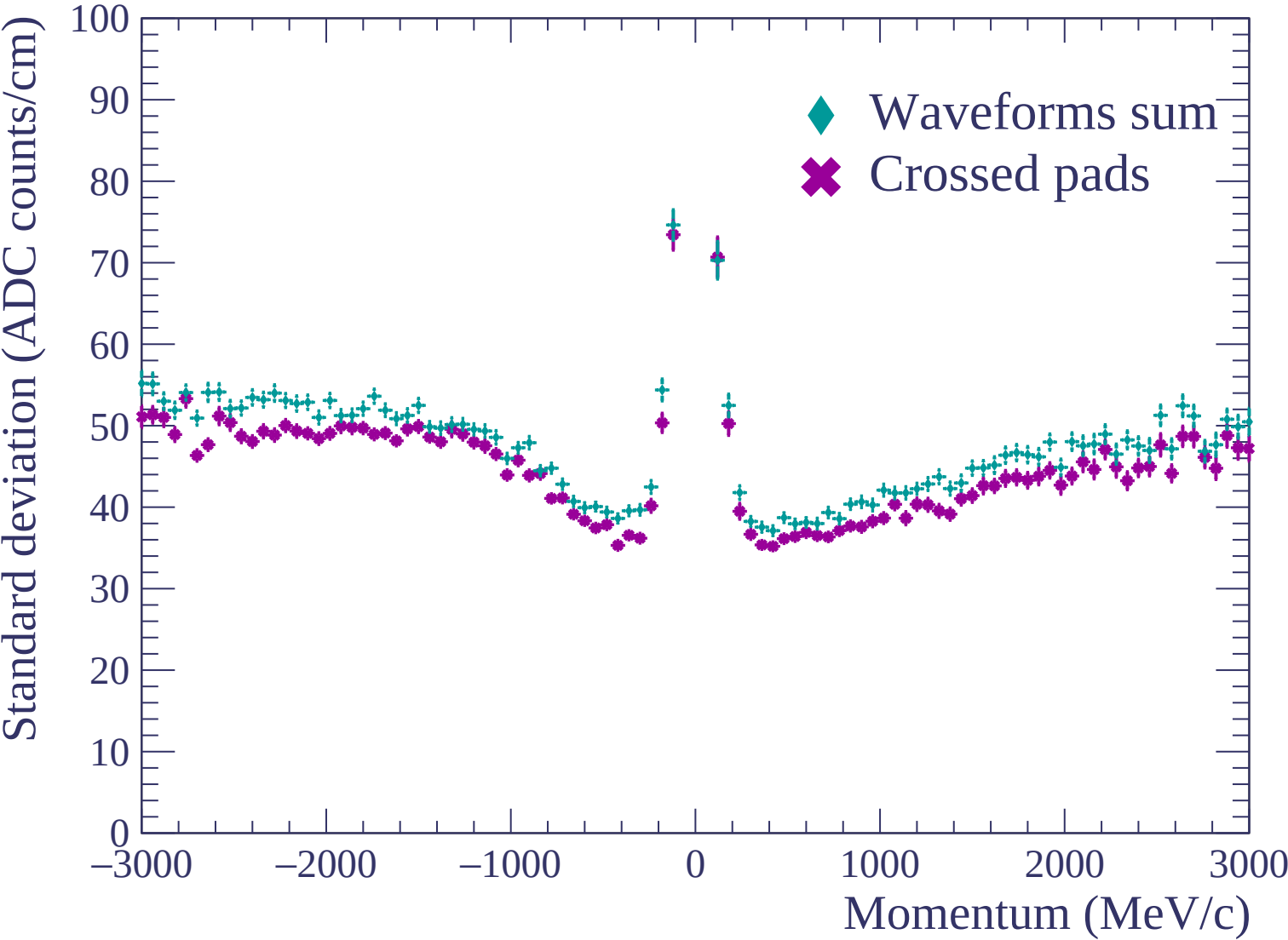


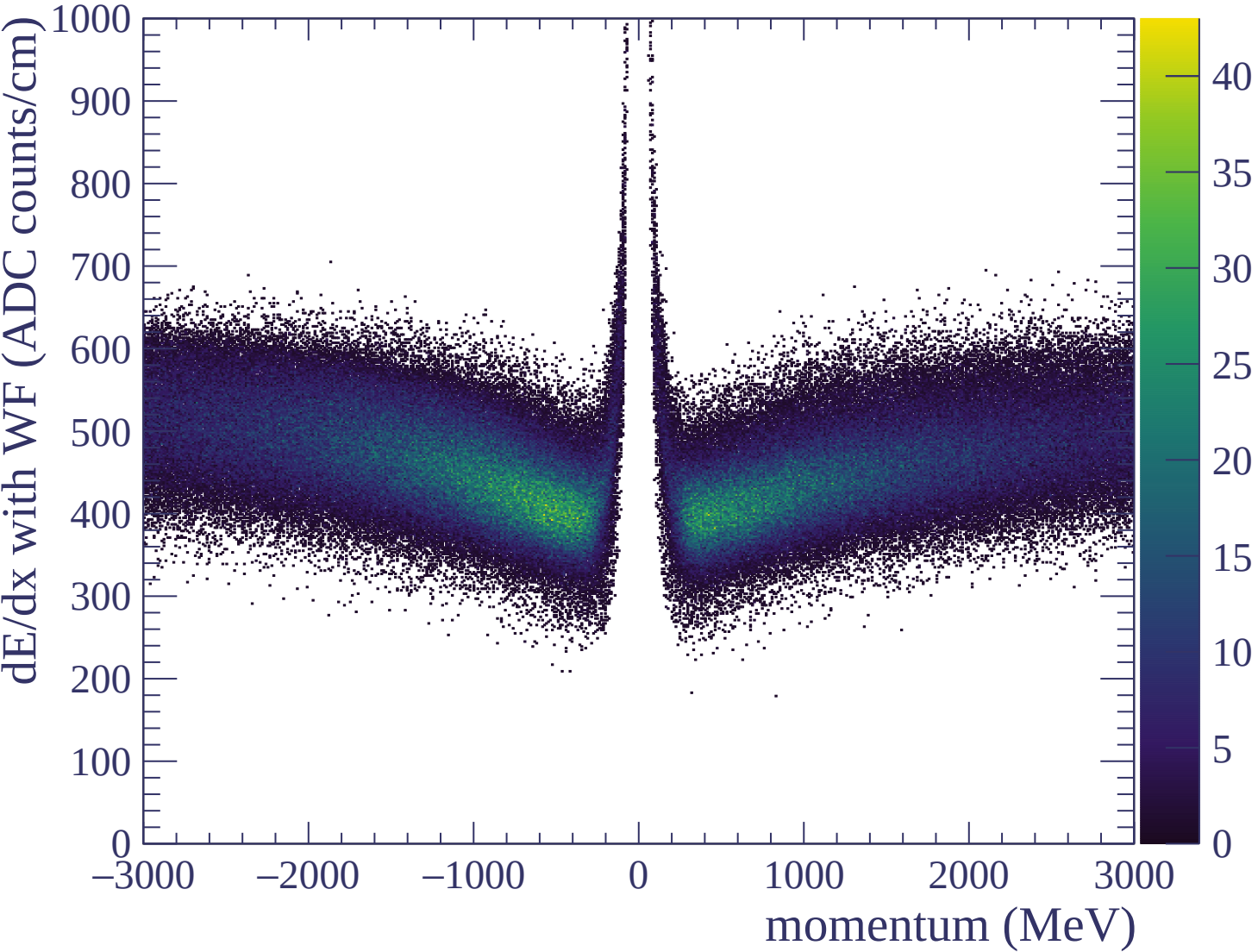


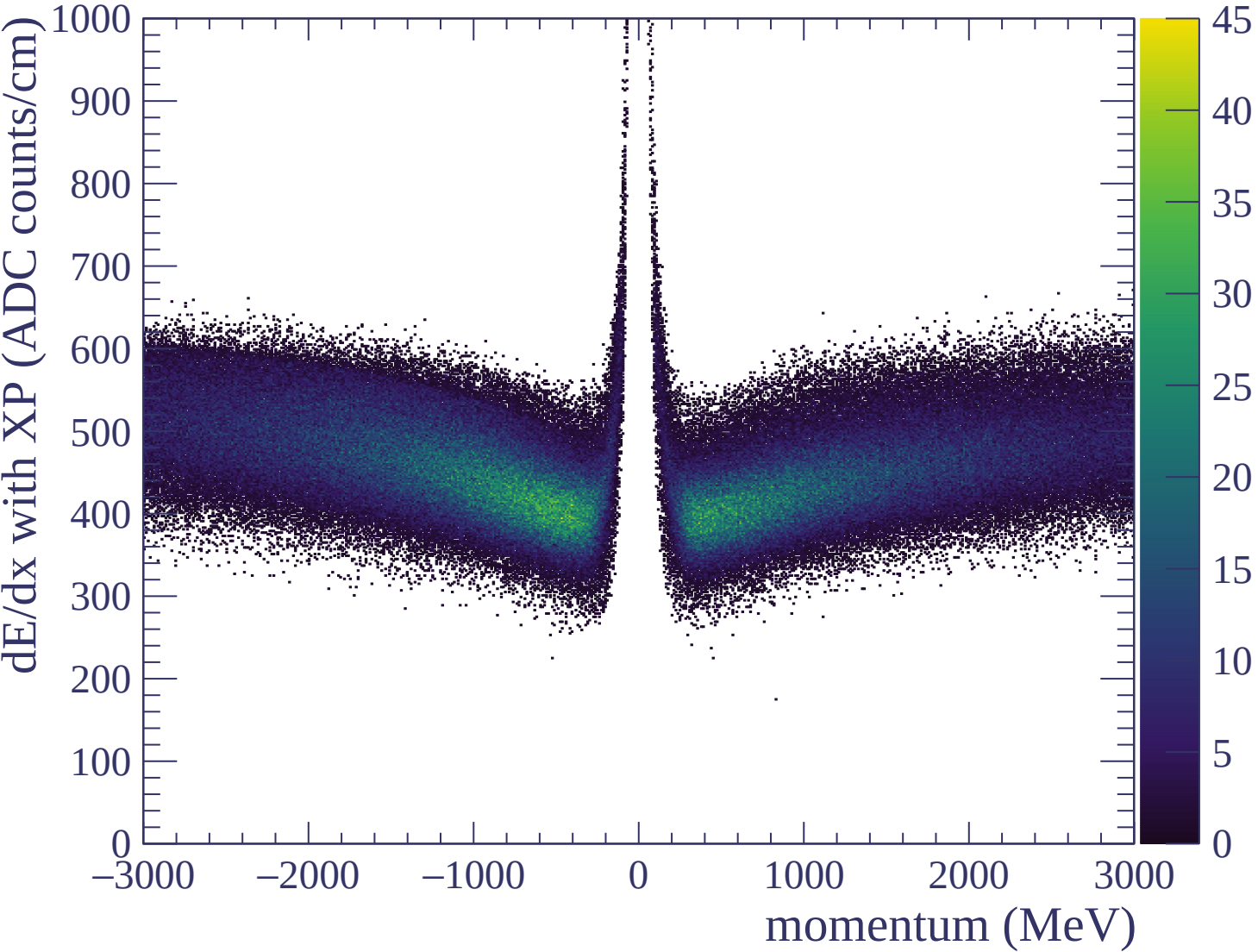


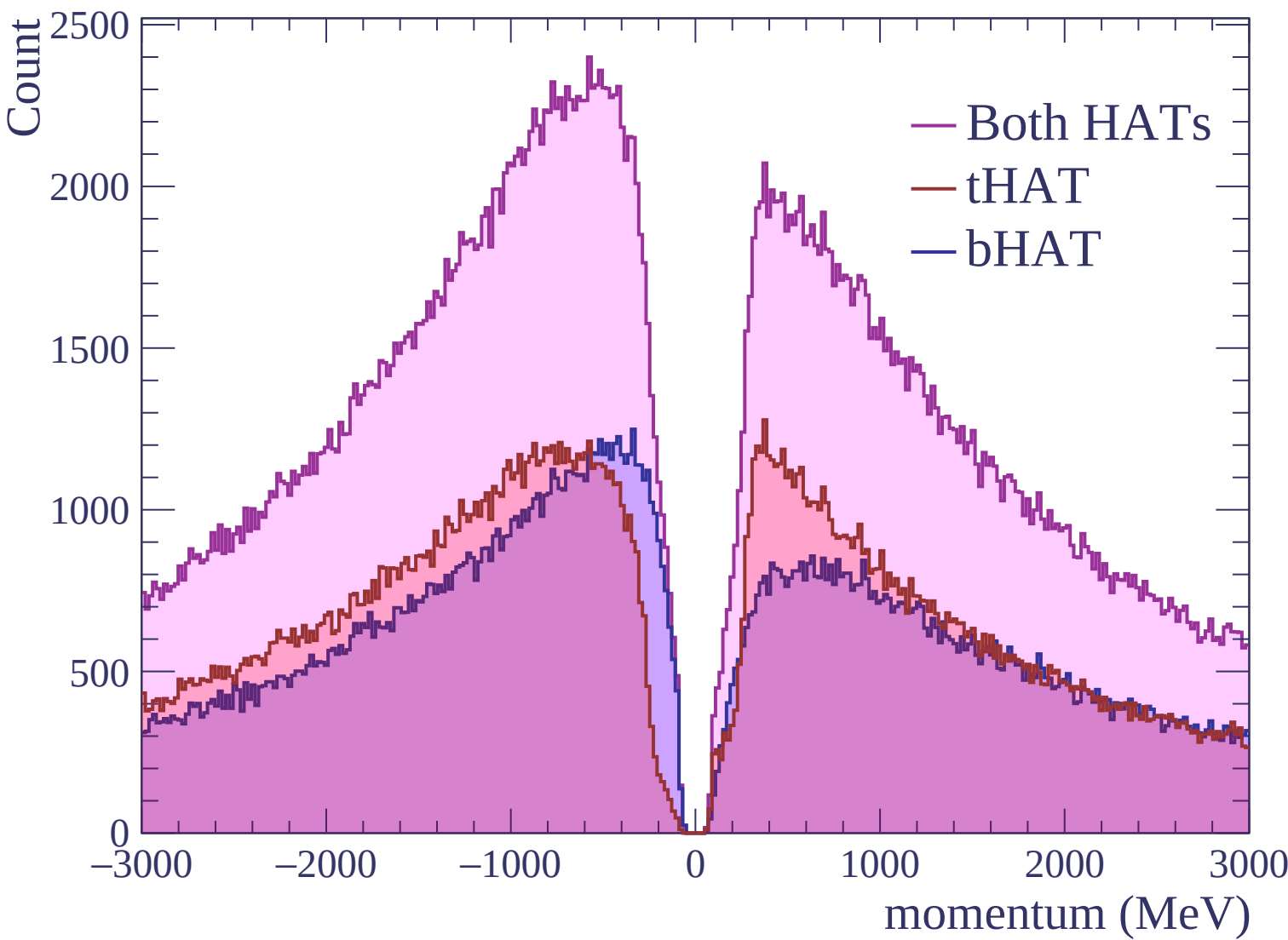


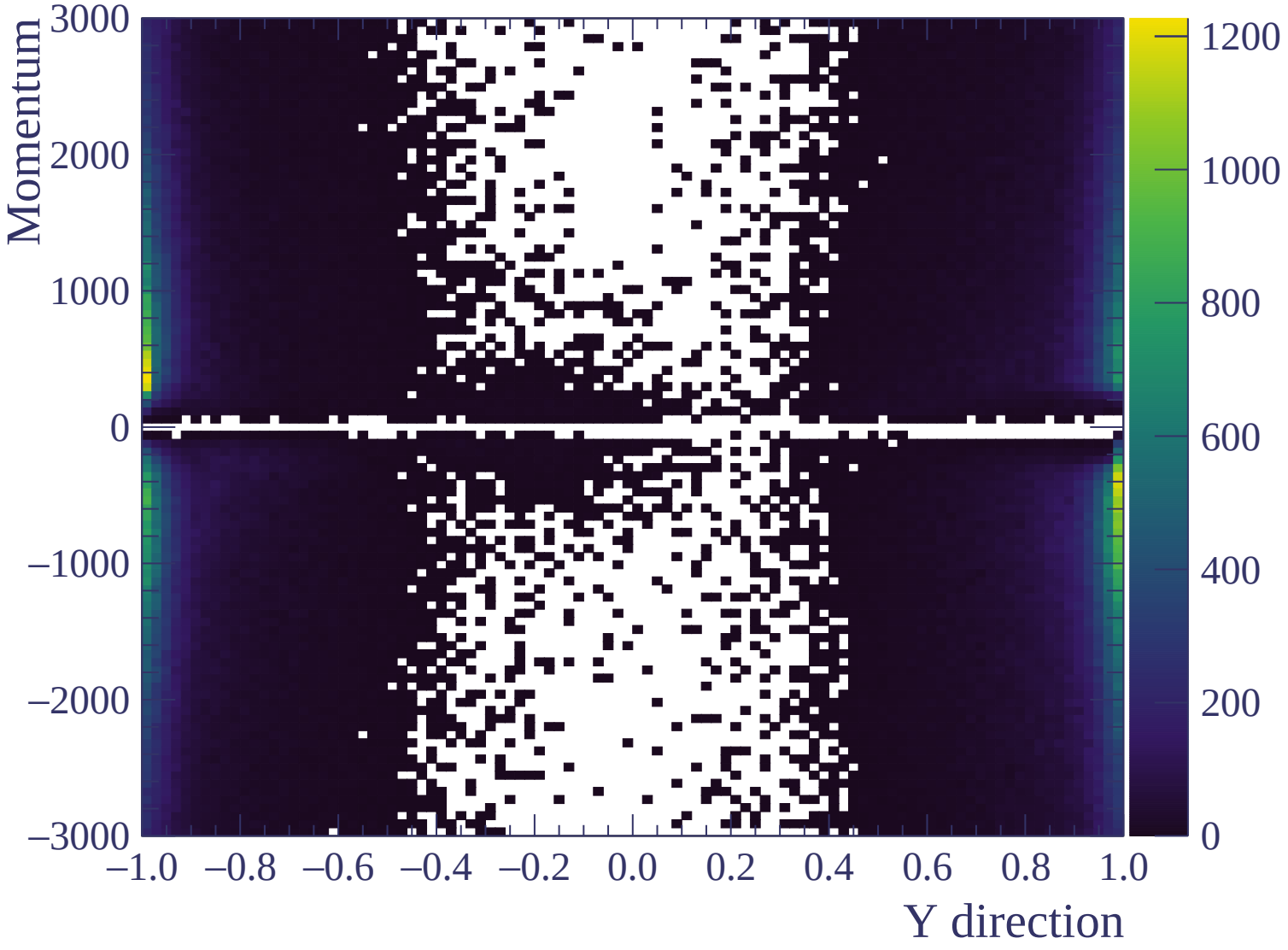


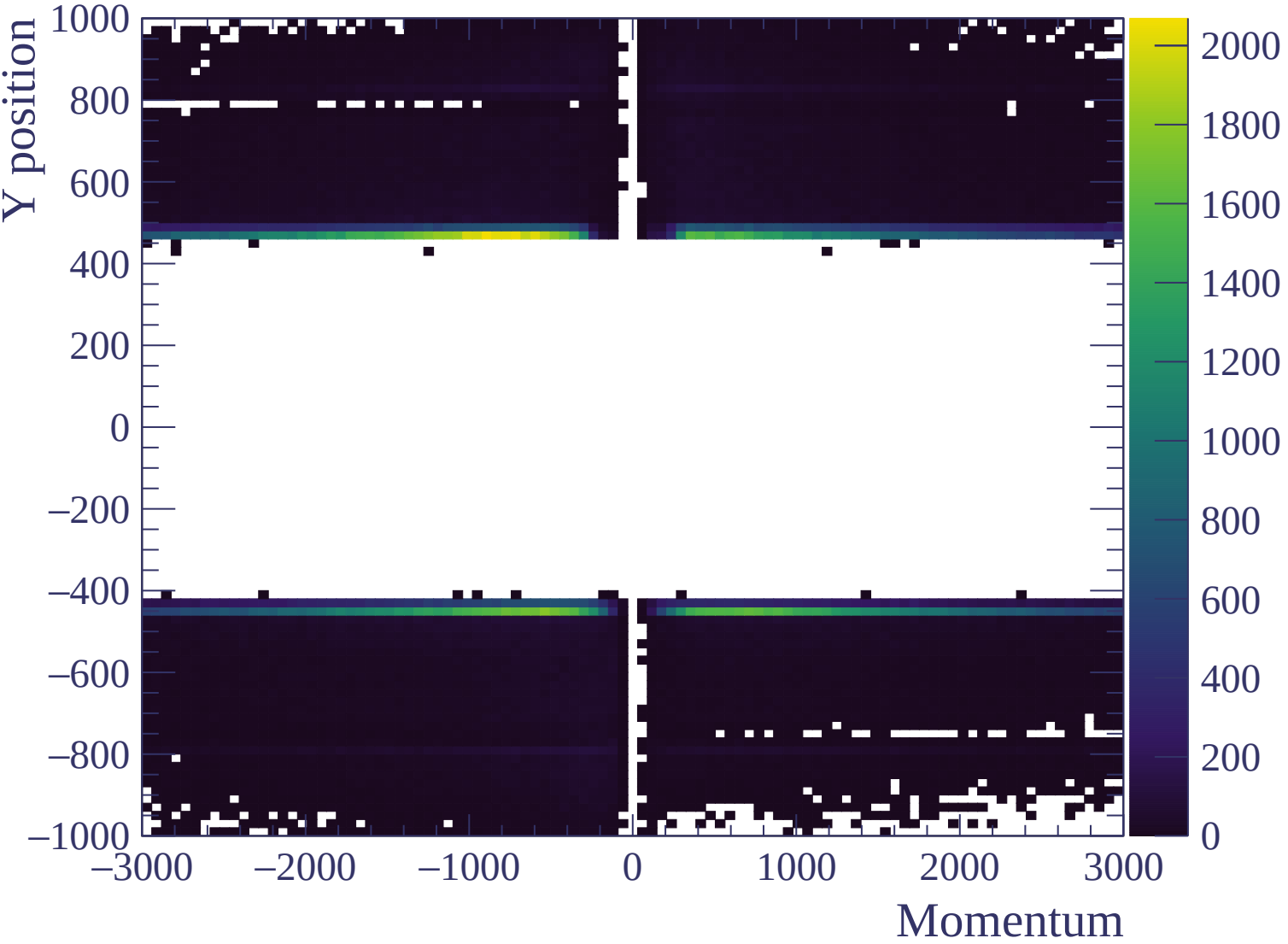


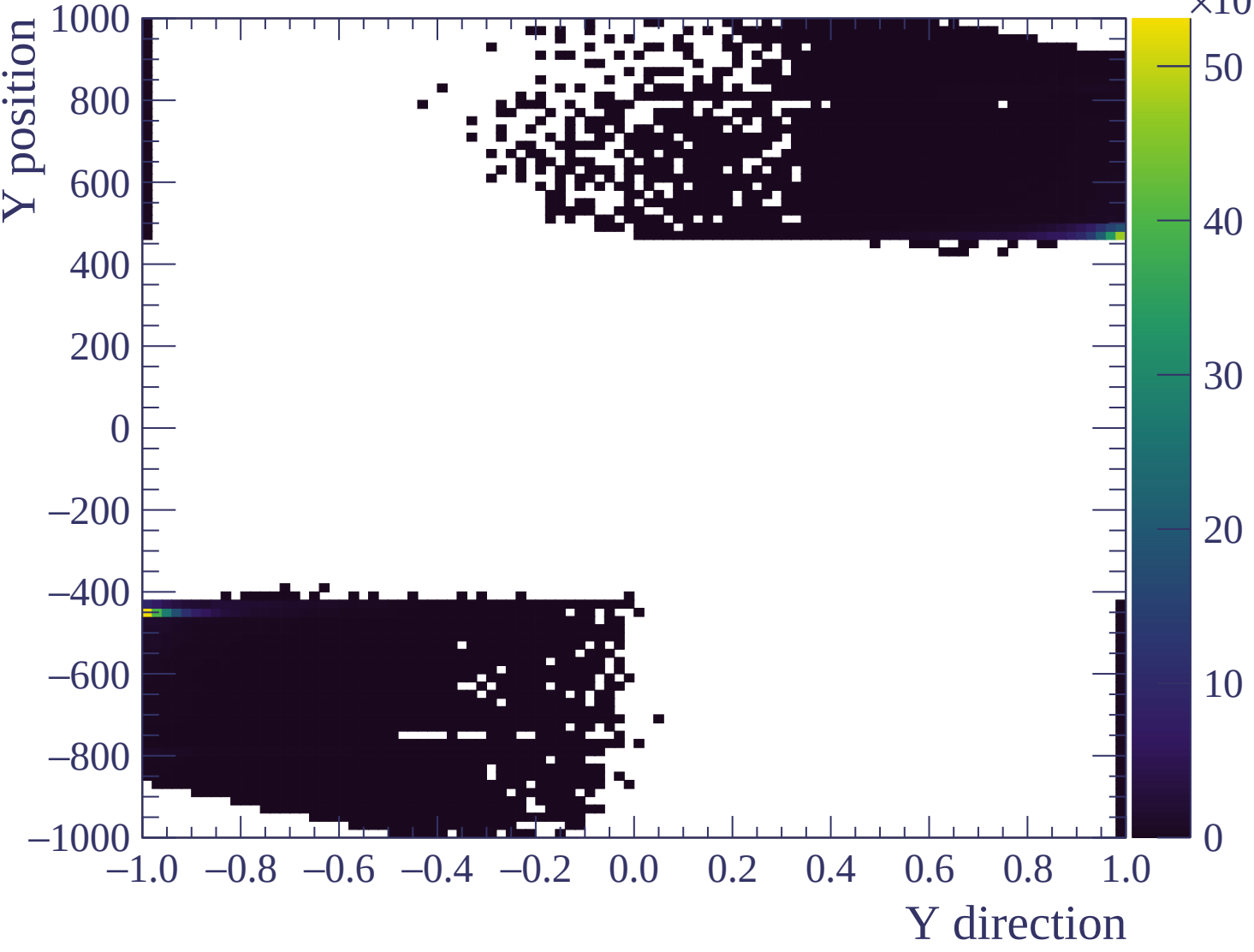


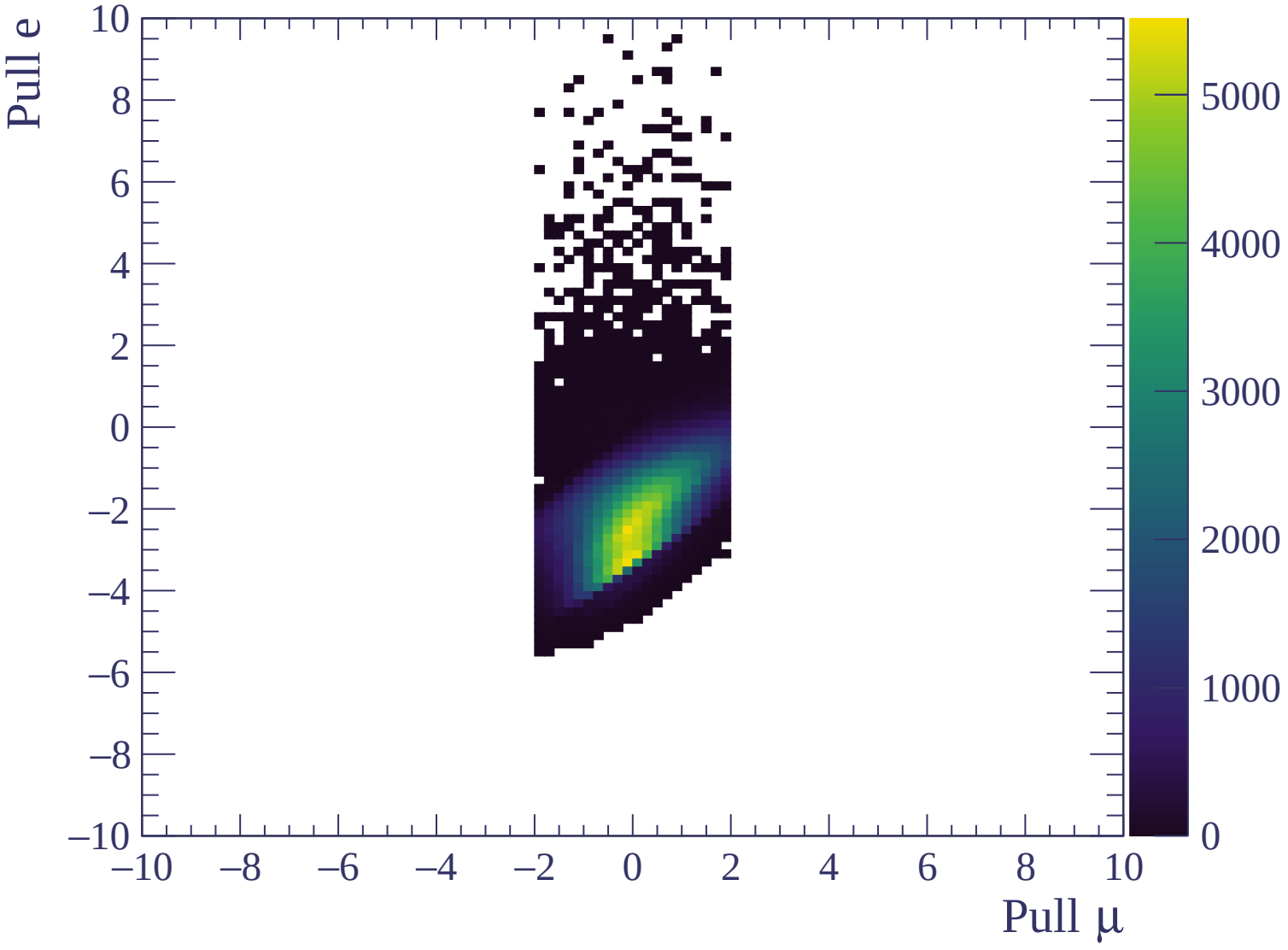






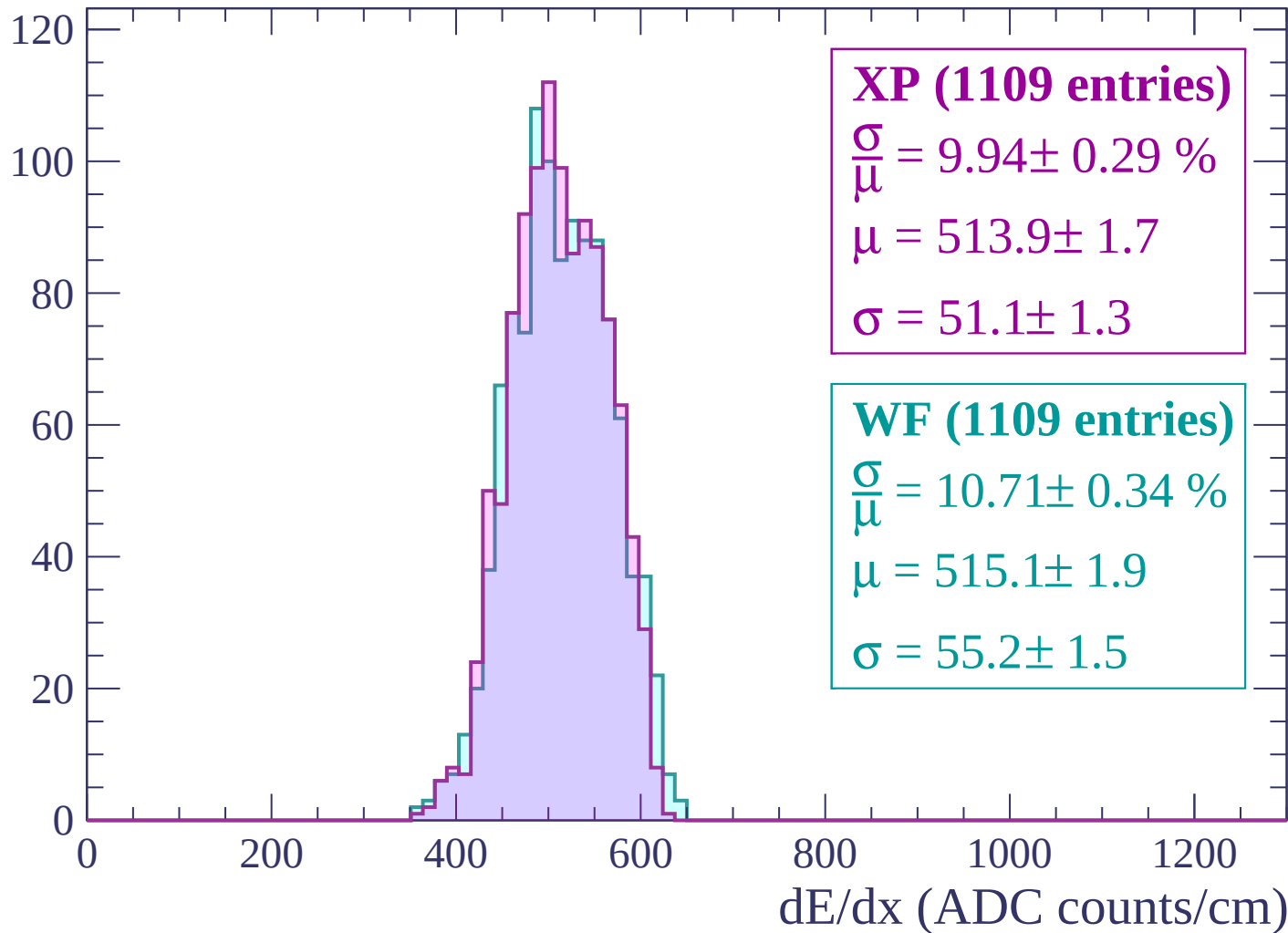






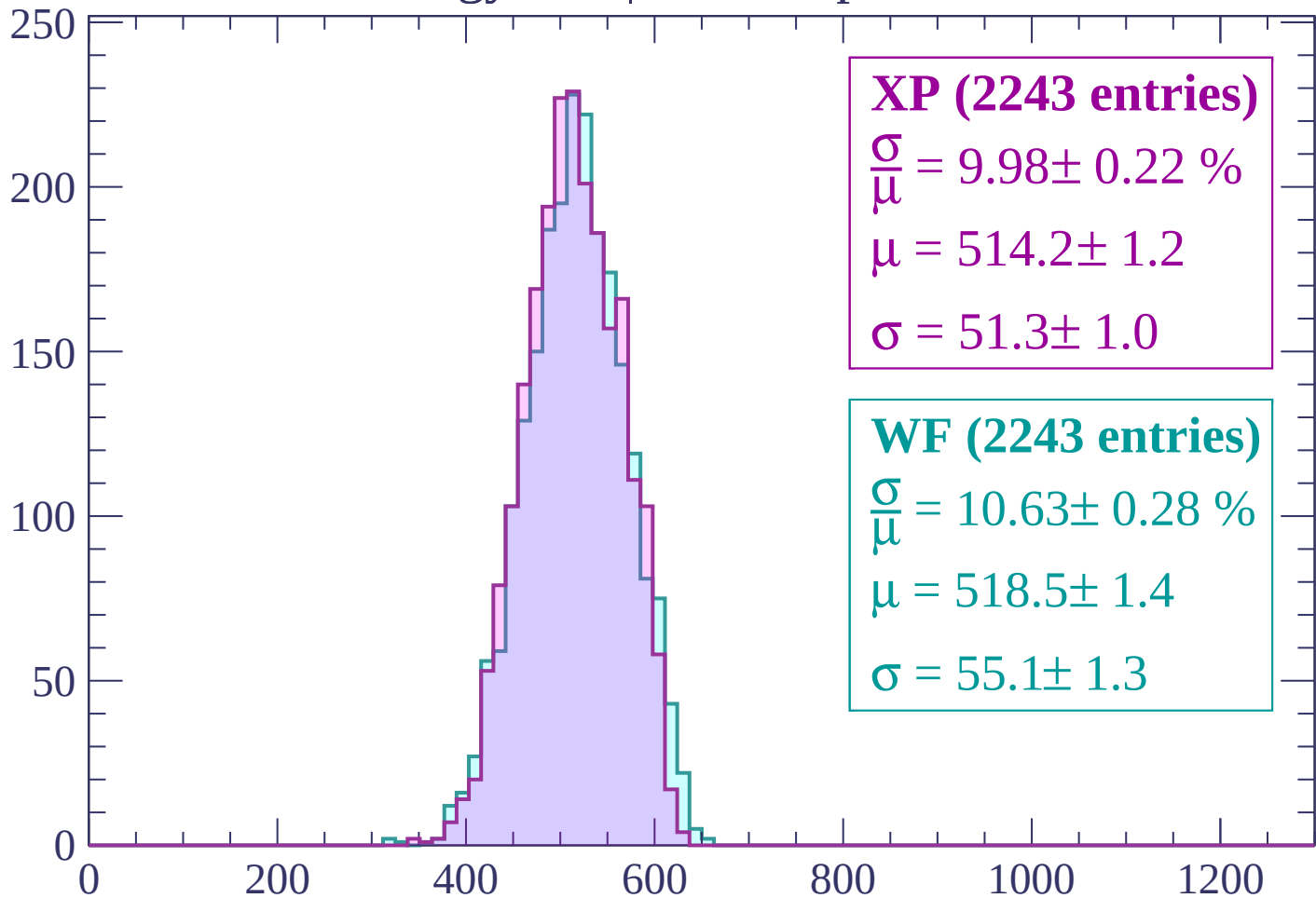
Energy loss | $-3000 < p < -2940$

Count



Energy loss | $-2940 < p < -2880$

Count



XP (2243 entries)

$$\frac{\sigma}{\mu} = 9.98 \pm 0.22 \%$$

$$\mu = 514.2 \pm 1.2$$

$$\sigma = 51.3 \pm 1.0$$

WF (2243 entries)

$$\frac{\sigma}{\mu} = 10.63 \pm 0.28 \%$$

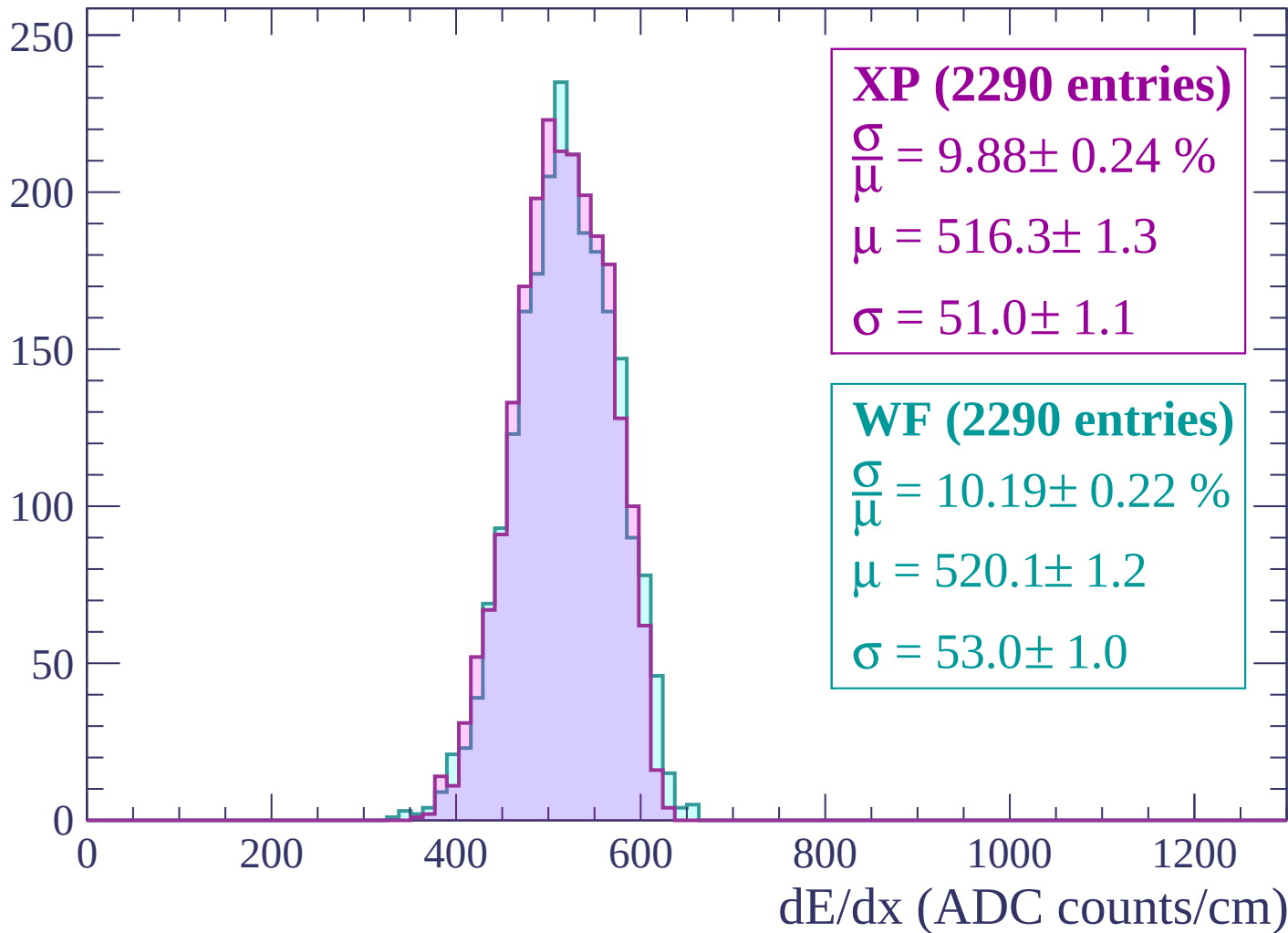
$$\mu = 518.5 \pm 1.4$$

$$\sigma = 55.1 \pm 1.3$$

dE/dx (ADC counts/cm)

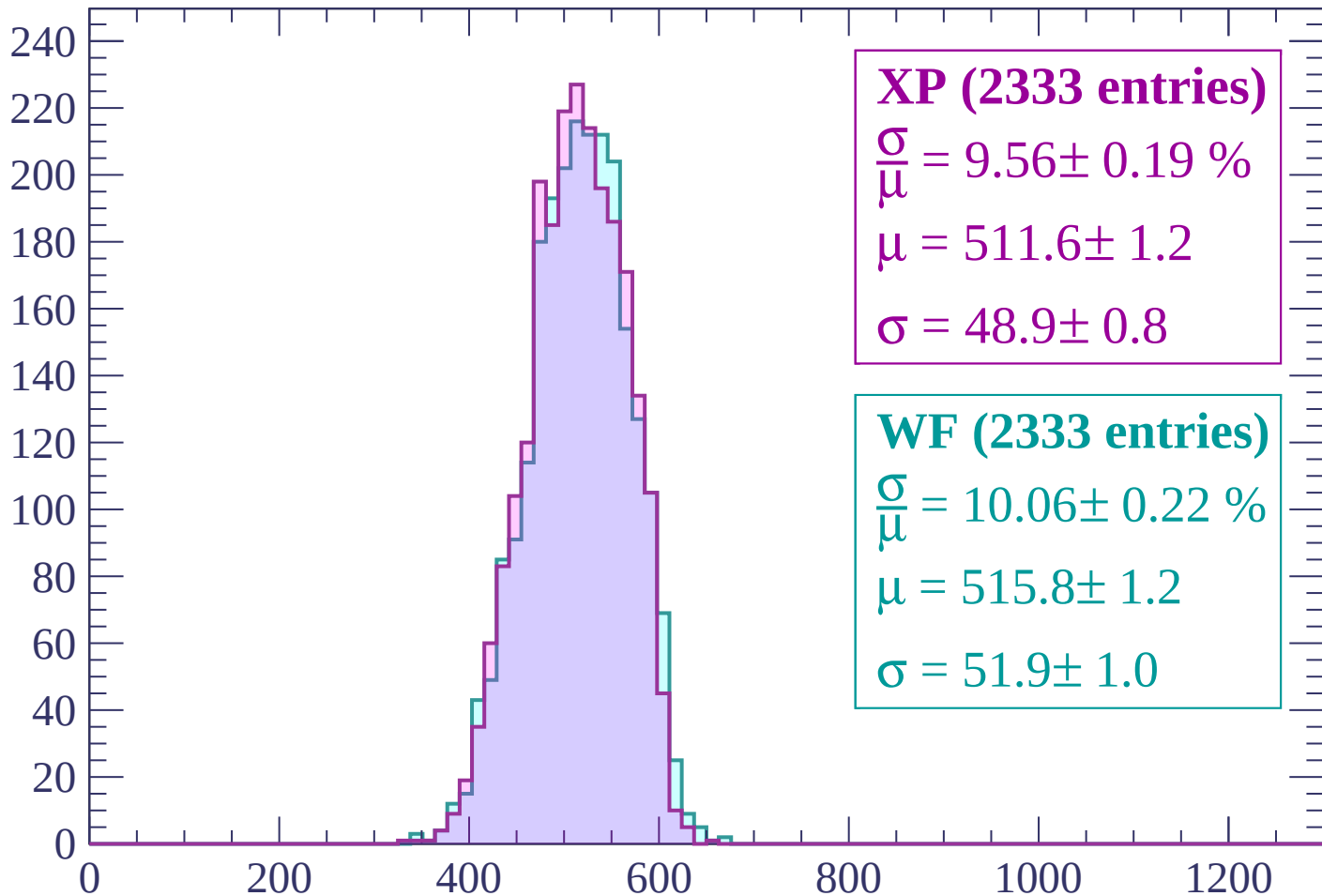
Energy loss | $-2880 < p < -2820$

Count



Energy loss | $-2820 < p < -2760$

Count



XP (2333 entries)

$$\frac{\sigma}{\mu} = 9.56 \pm 0.19 \%$$

$$\mu = 511.6 \pm 1.2$$

$$\sigma = 48.9 \pm 0.8$$

WF (2333 entries)

$$\frac{\sigma}{\mu} = 10.06 \pm 0.22 \%$$

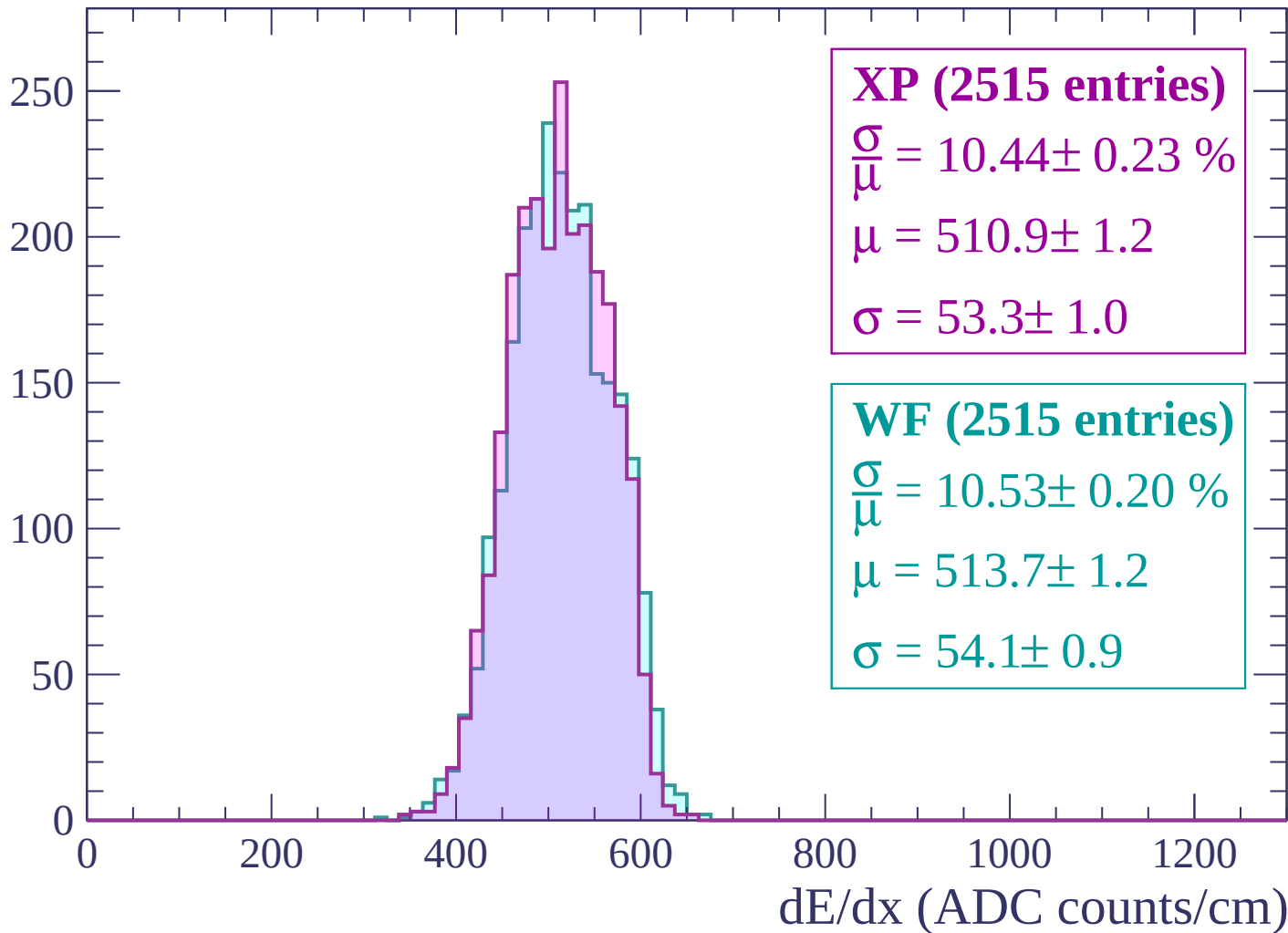
$$\mu = 515.8 \pm 1.2$$

$$\sigma = 51.9 \pm 1.0$$

dE/dx (ADC counts/cm)

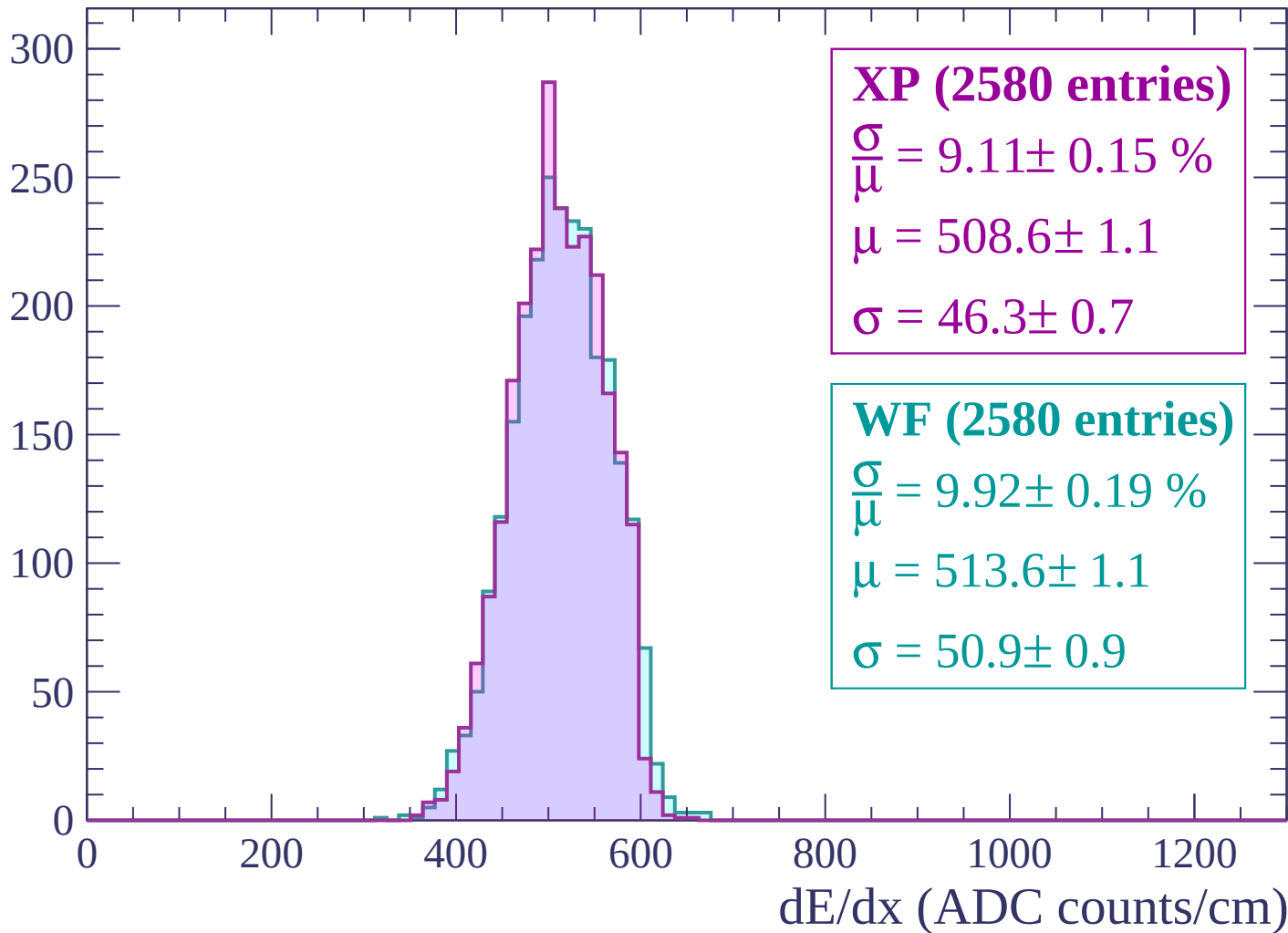
Energy loss | -2760 < p < -2700

Count



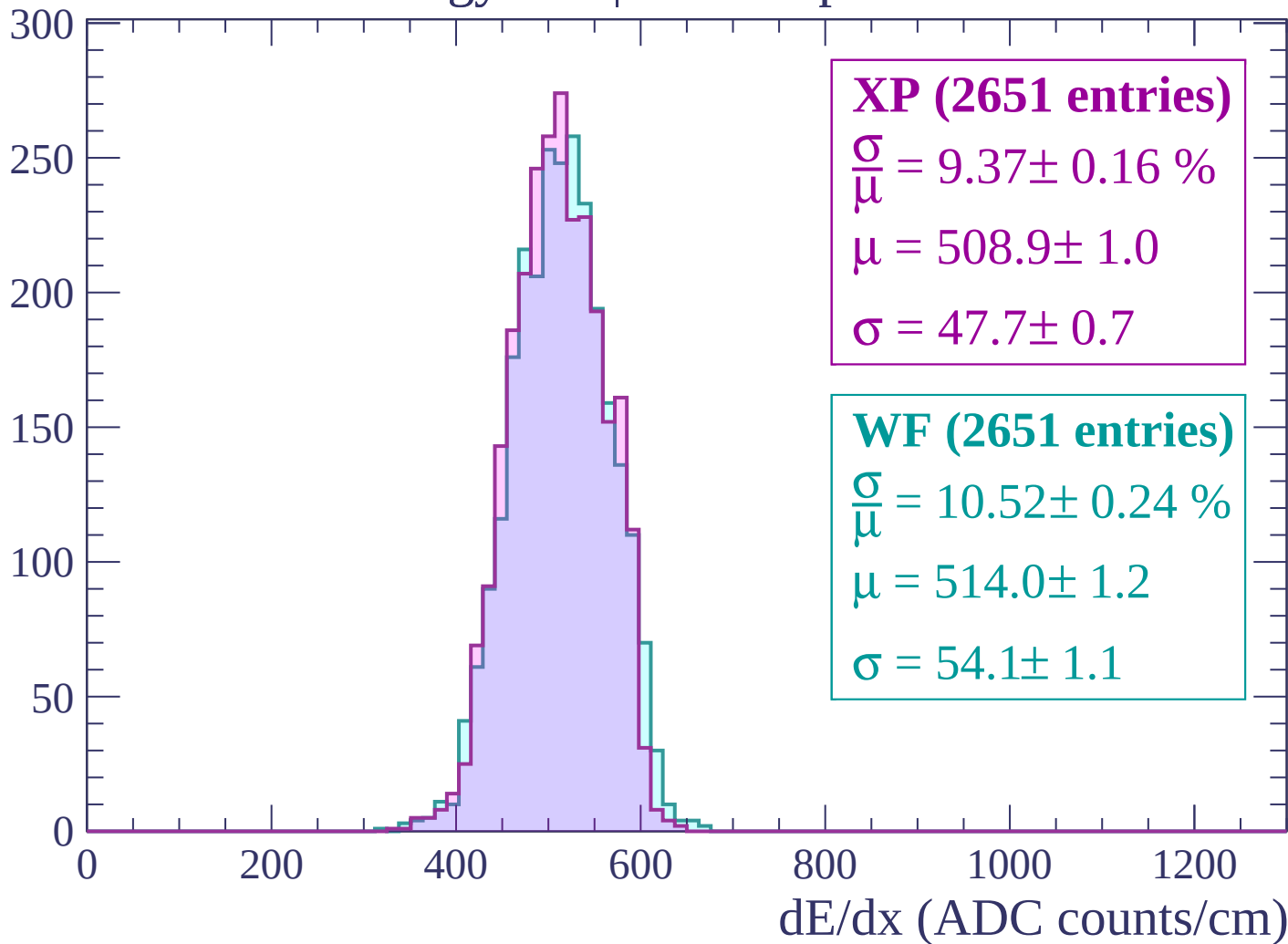
Energy loss | $-2700 < p < -2640$

Count



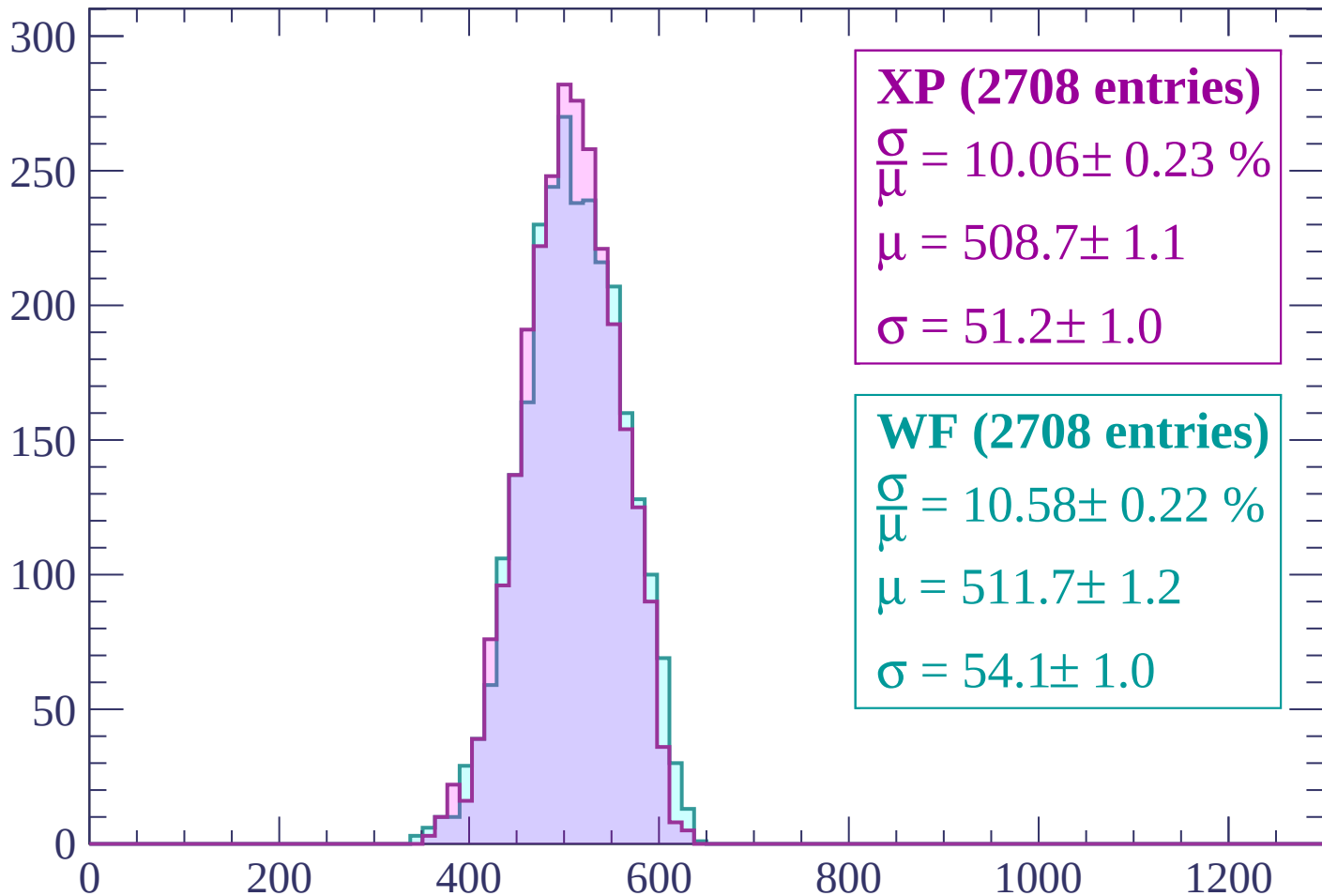
Energy loss | $-2640 < p < -2580$

Count



Energy loss | $-2580 < p < -2520$

Count



XP (2708 entries)

$$\frac{\sigma}{\mu} = 10.06 \pm 0.23 \%$$

$$\mu = 508.7 \pm 1.1$$

$$\sigma = 51.2 \pm 1.0$$

WF (2708 entries)

$$\frac{\sigma}{\mu} = 10.58 \pm 0.22 \%$$

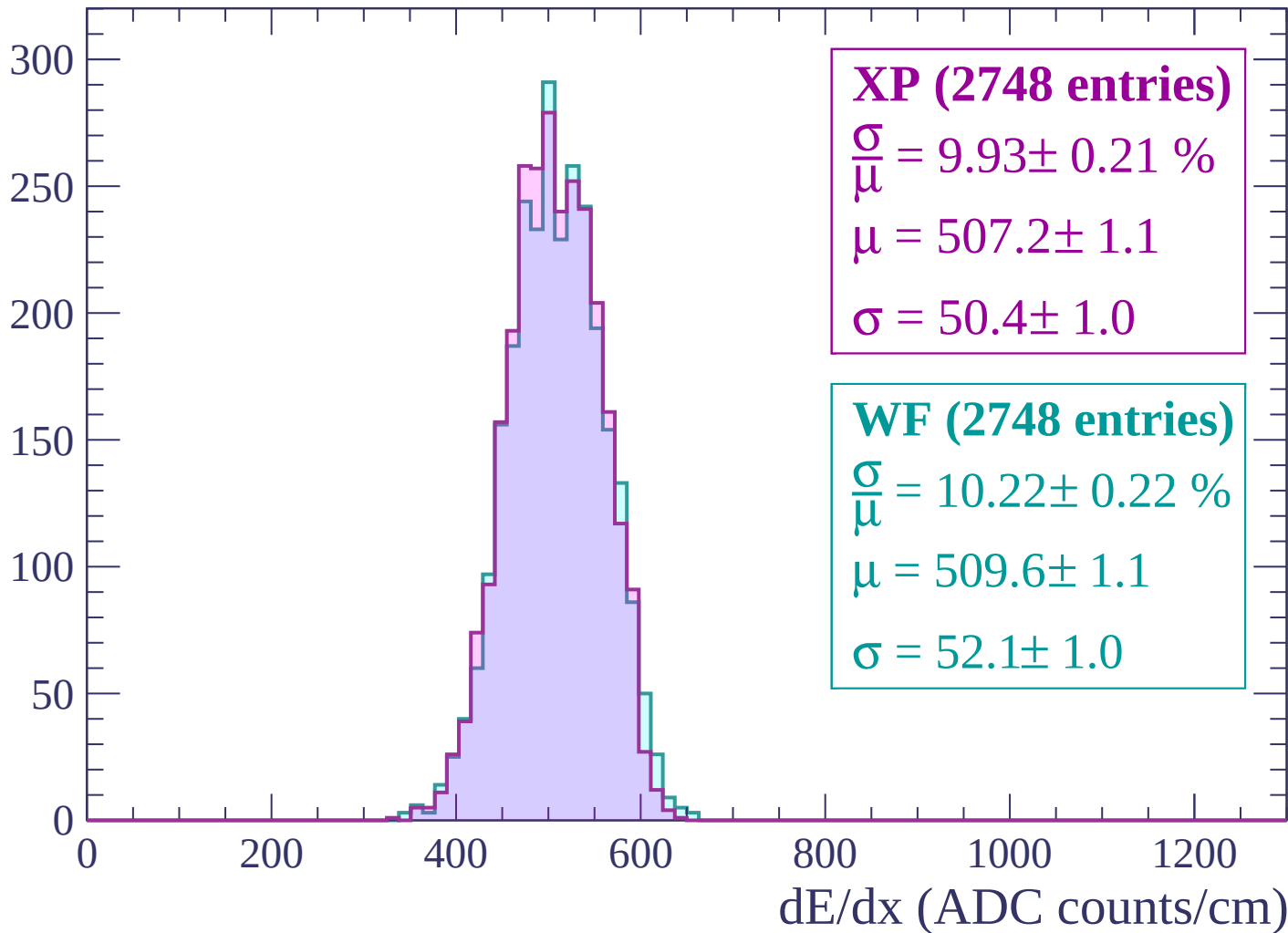
$$\mu = 511.7 \pm 1.2$$

$$\sigma = 54.1 \pm 1.0$$

dE/dx (ADC counts/cm)

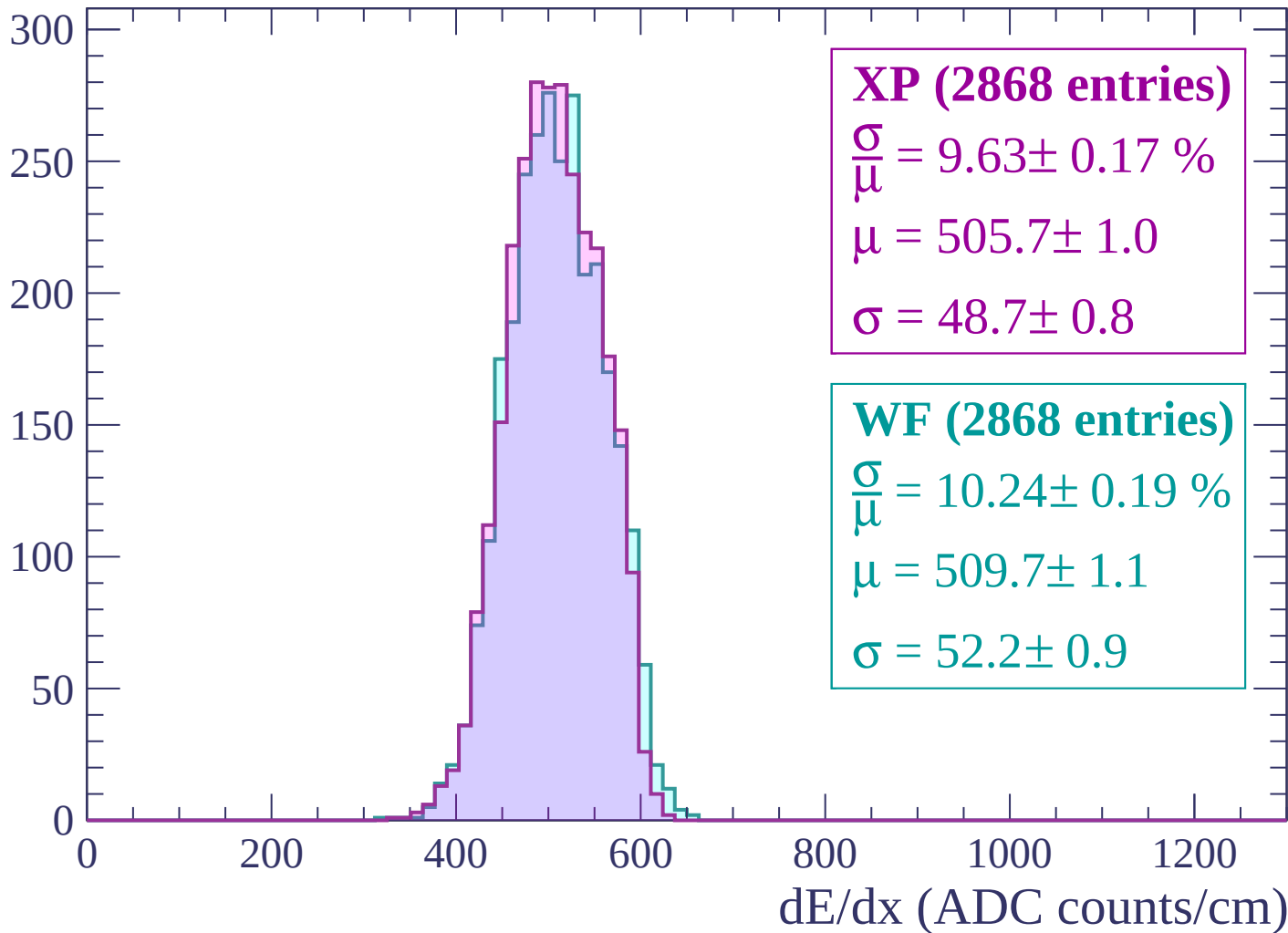
Energy loss | -2520 < p < -2460

Count



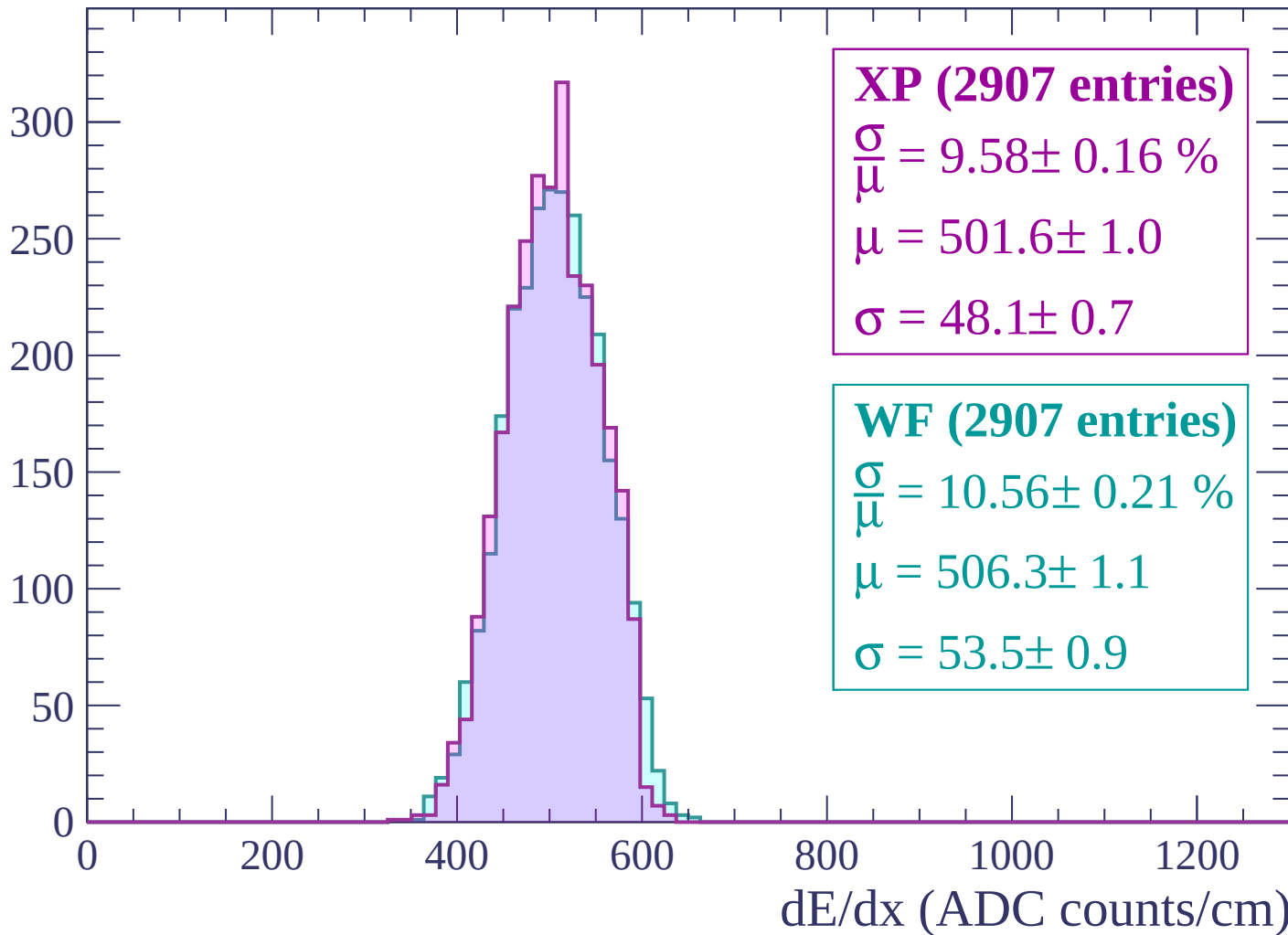
Energy loss | $-2460 < p < -2400$

Count



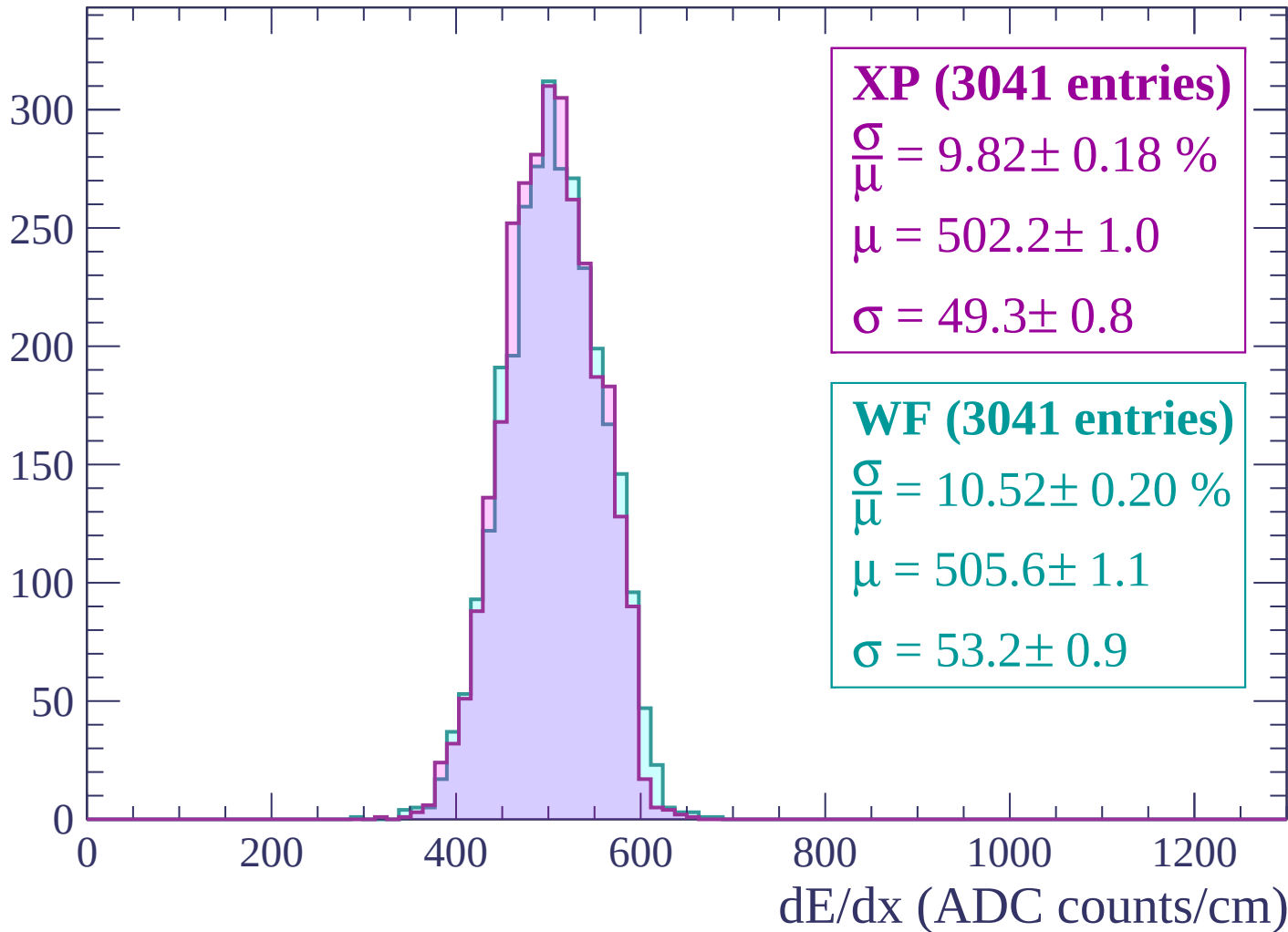
Energy loss | $-2400 < p < -2340$

Count



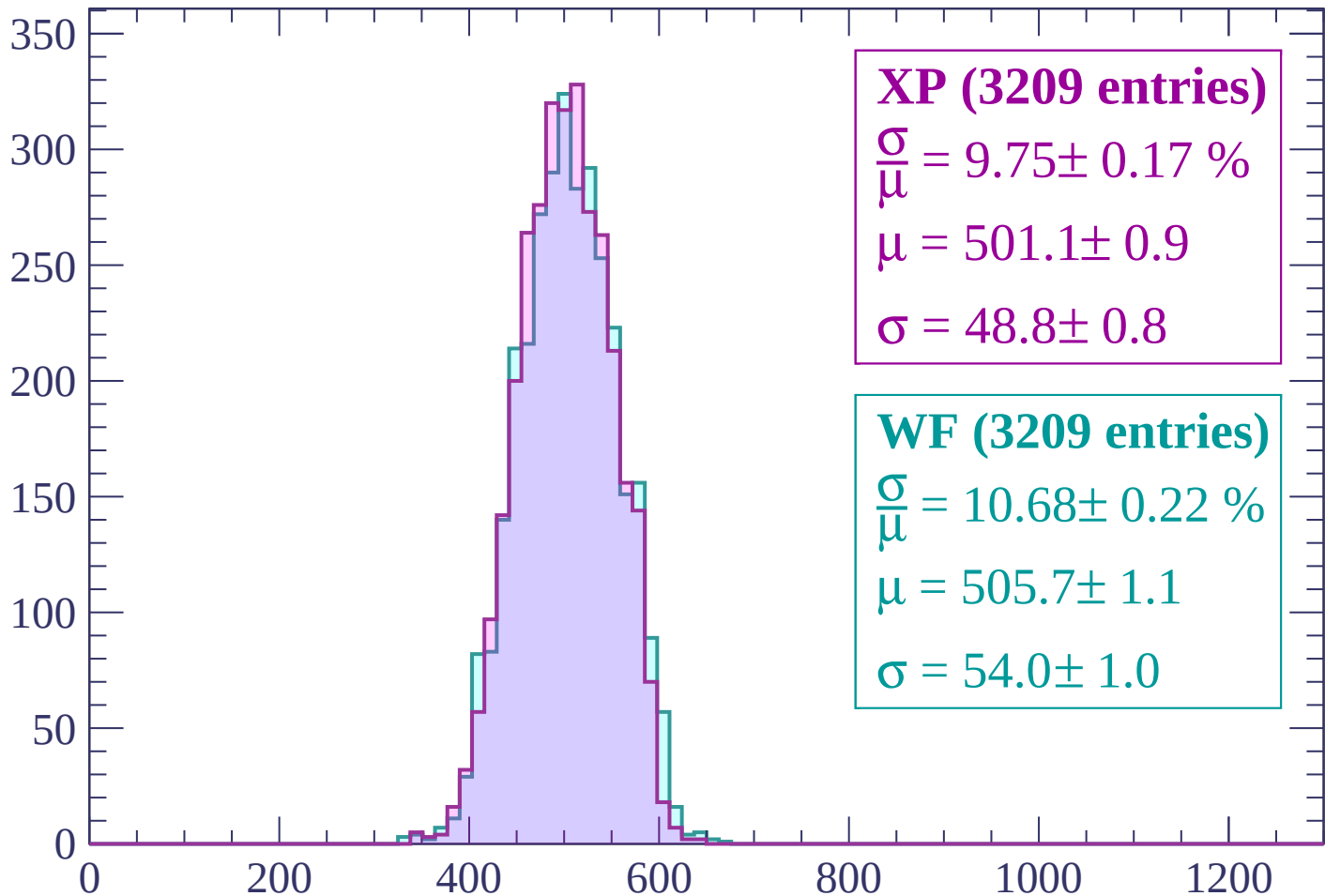
Energy loss | $-2340 < p < -2280$

Count



Energy loss | $-2280 < p < -2220$

Count



XP (3209 entries)

$$\frac{\sigma}{\mu} = 9.75 \pm 0.17 \%$$

$$\mu = 501.1 \pm 0.9$$

$$\sigma = 48.8 \pm 0.8$$

WF (3209 entries)

$$\frac{\sigma}{\mu} = 10.68 \pm 0.22 \%$$

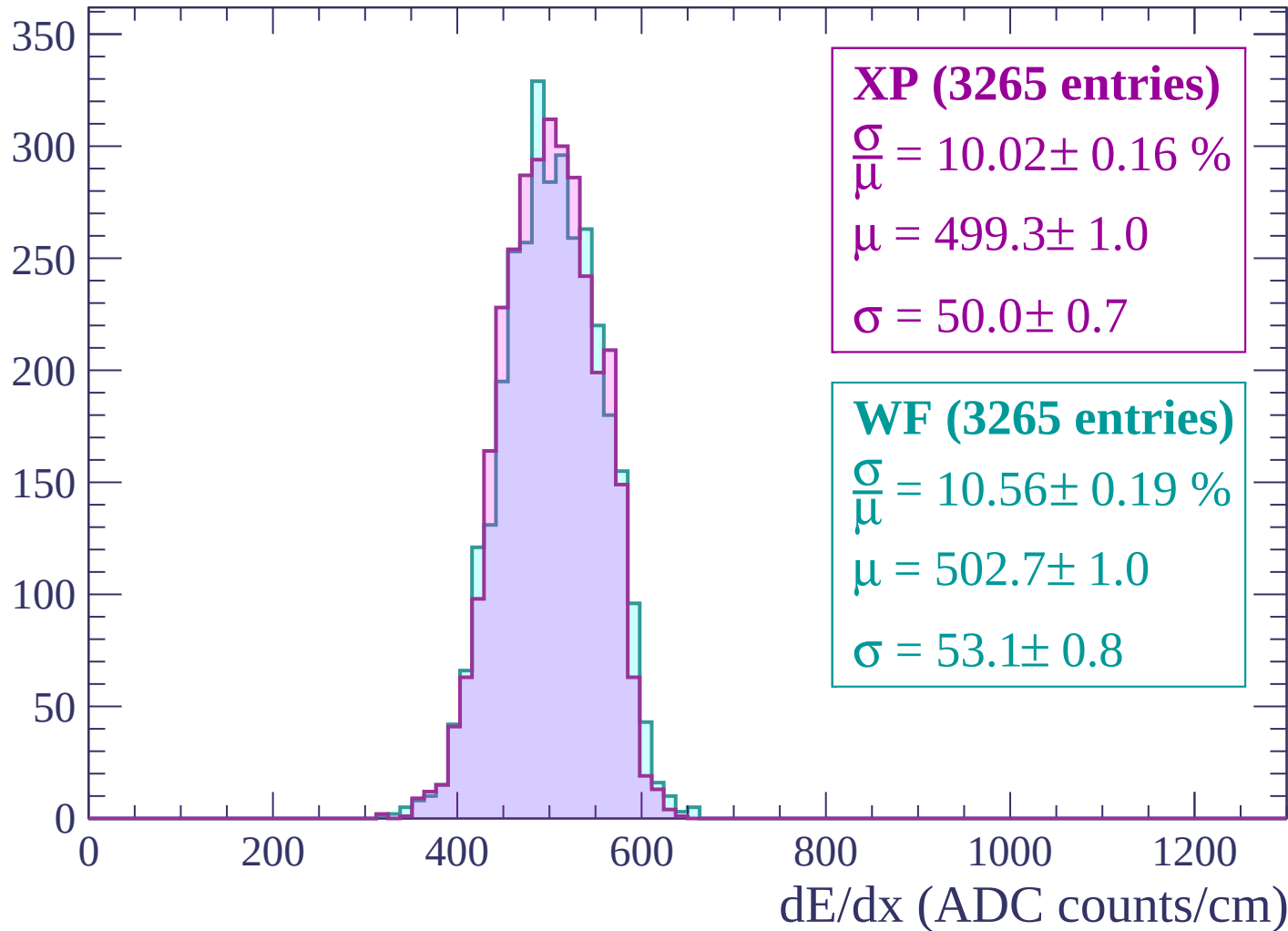
$$\mu = 505.7 \pm 1.1$$

$$\sigma = 54.0 \pm 1.0$$

dE/dx (ADC counts/cm)

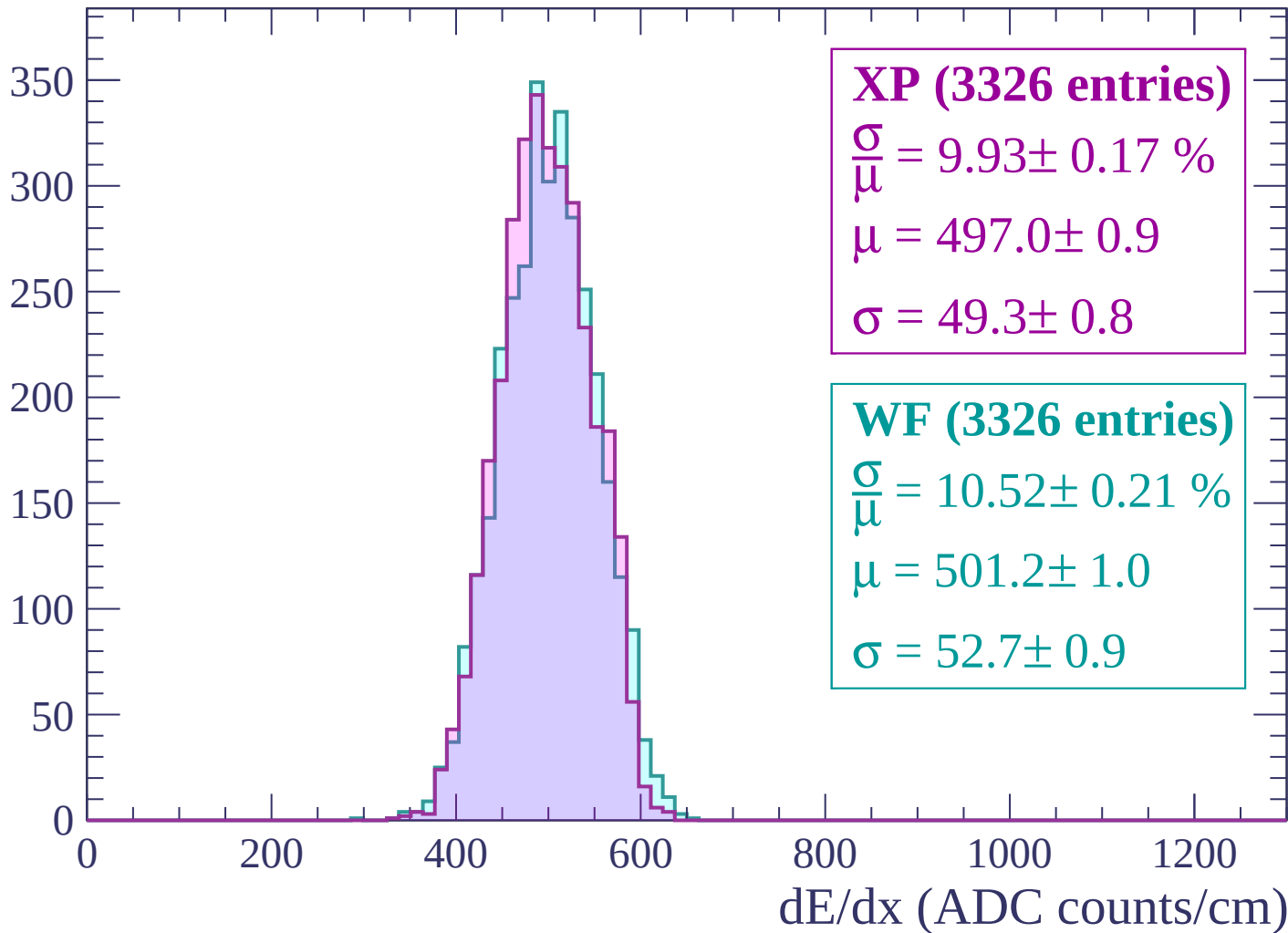
Energy loss | -2220 < p < -2160

Count



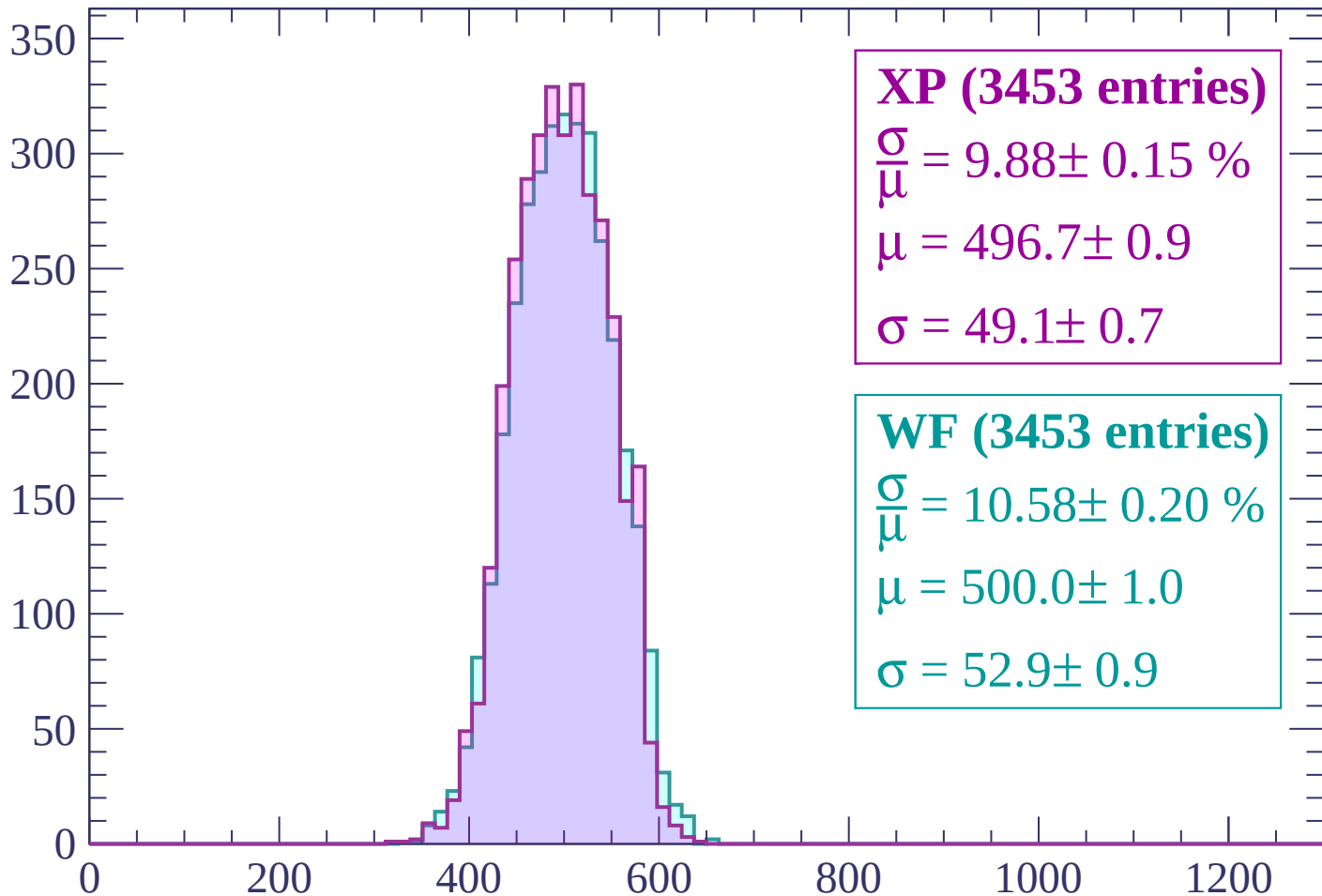
Energy loss | $-2160 < p < -2100$

Count



Energy loss | -2100 < p < -2040

Count



XP (3453 entries)

$$\frac{\sigma}{\mu} = 9.88 \pm 0.15 \%$$

$$\mu = 496.7 \pm 0.9$$

$$\sigma = 49.1 \pm 0.7$$

WF (3453 entries)

$$\frac{\sigma}{\mu} = 10.58 \pm 0.20 \%$$

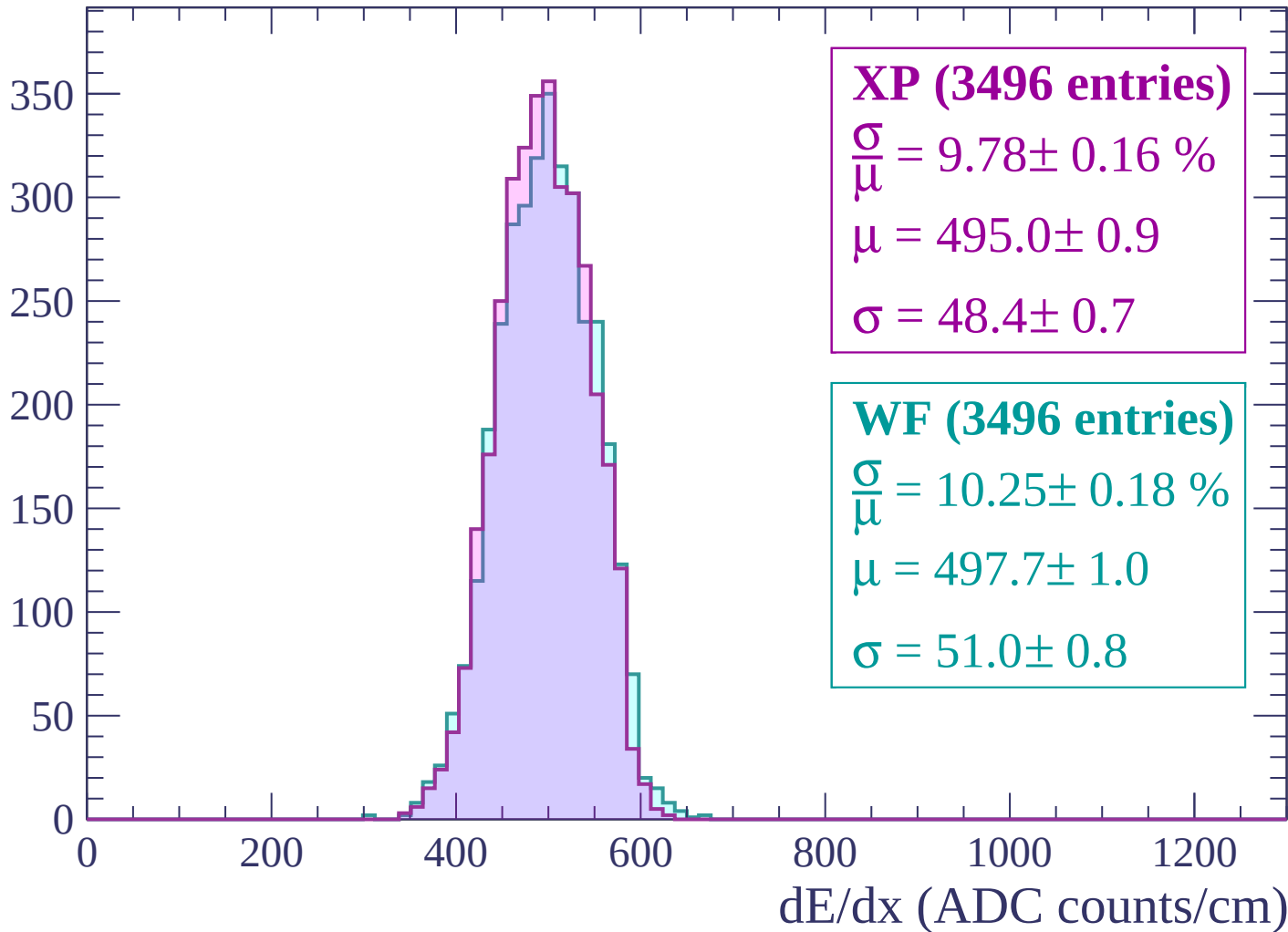
$$\mu = 500.0 \pm 1.0$$

$$\sigma = 52.9 \pm 0.9$$

dE/dx (ADC counts/cm)

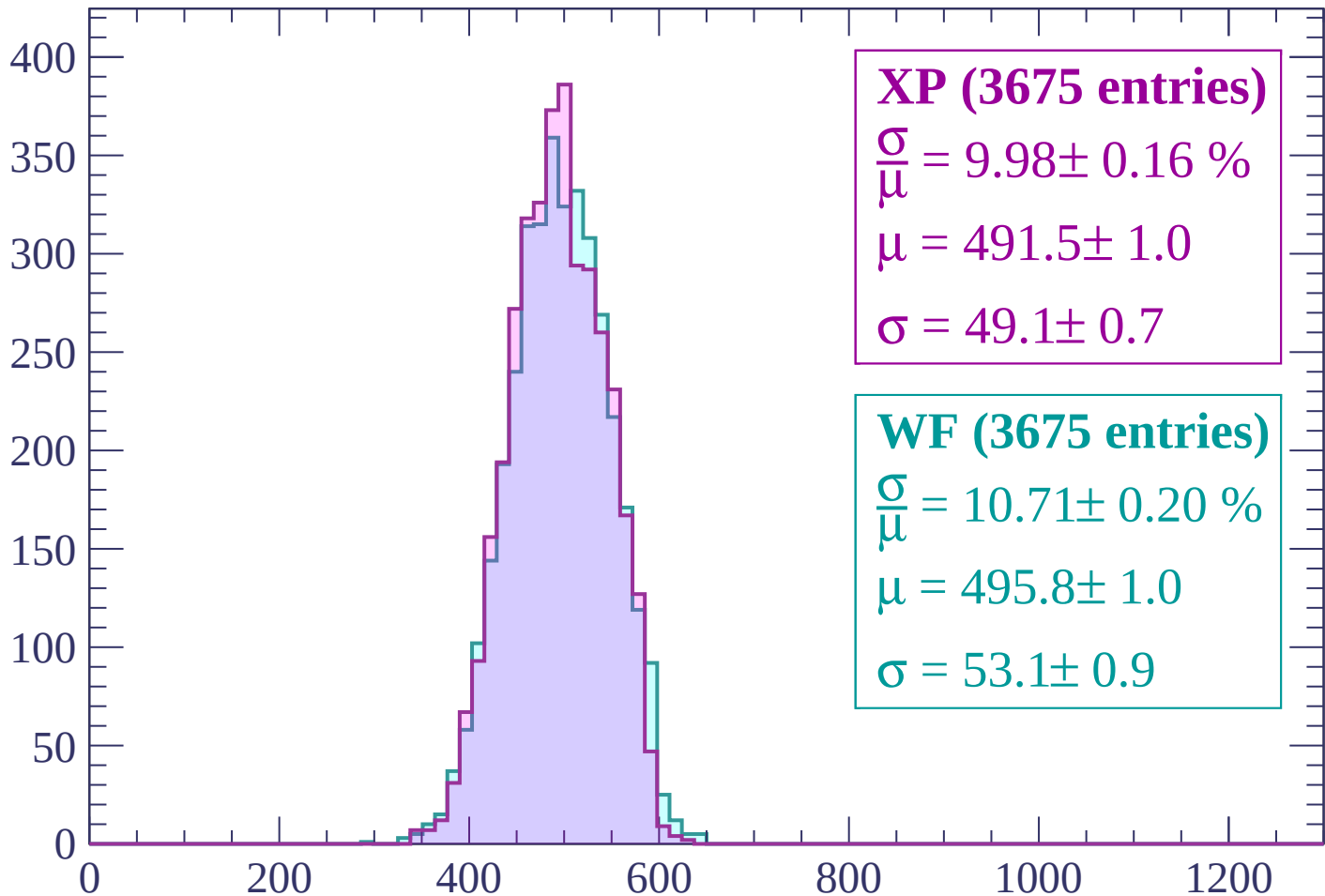
Energy loss | -2040 < p < -1980

Count



Energy loss | -1980 < p < -1920

Count



XP (3675 entries)

$$\frac{\sigma}{\mu} = 9.98 \pm 0.16 \%$$

$$\mu = 491.5 \pm 1.0$$

$$\sigma = 49.1 \pm 0.7$$

WF (3675 entries)

$$\frac{\sigma}{\mu} = 10.71 \pm 0.20 \%$$

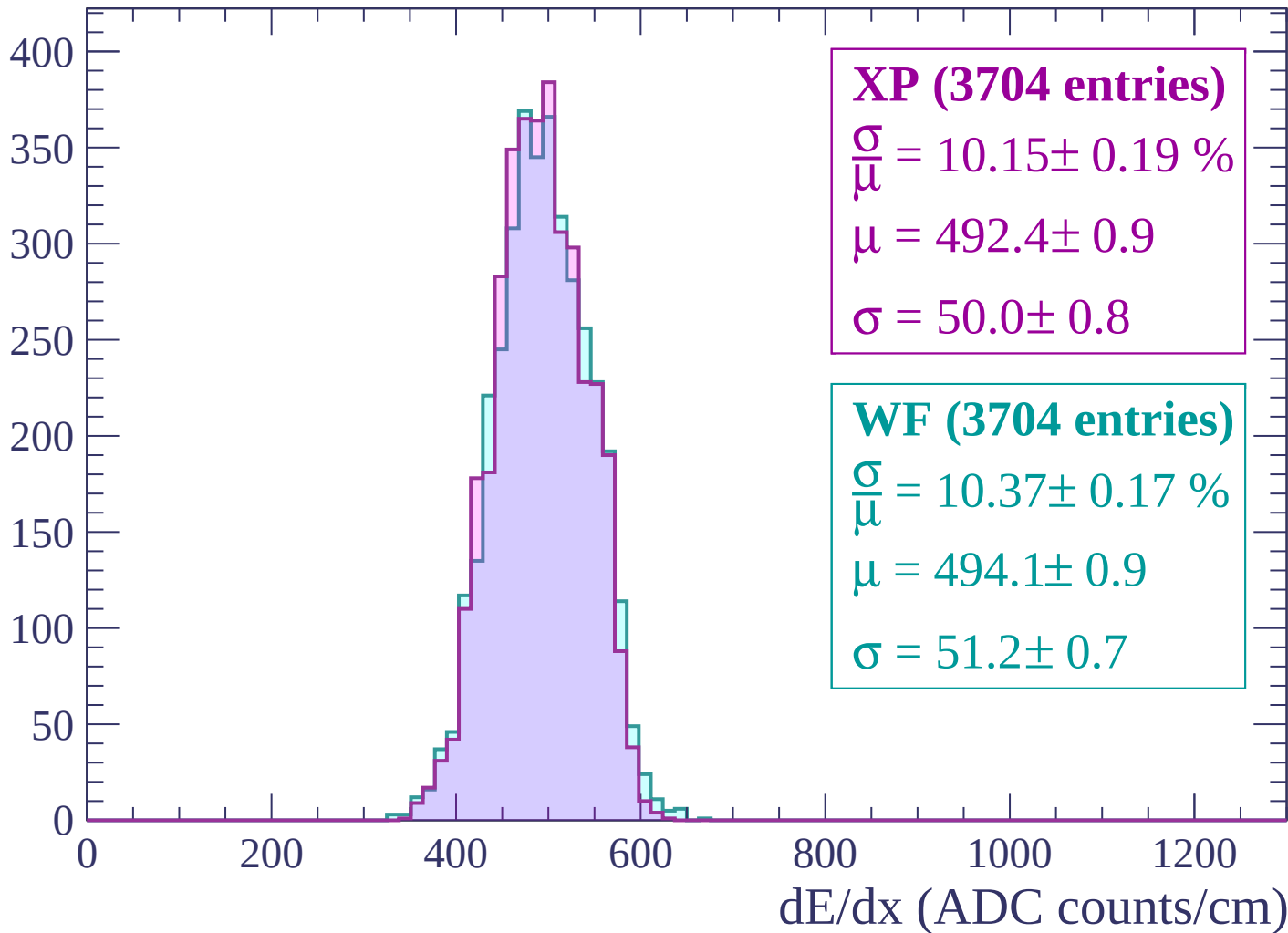
$$\mu = 495.8 \pm 1.0$$

$$\sigma = 53.1 \pm 0.9$$

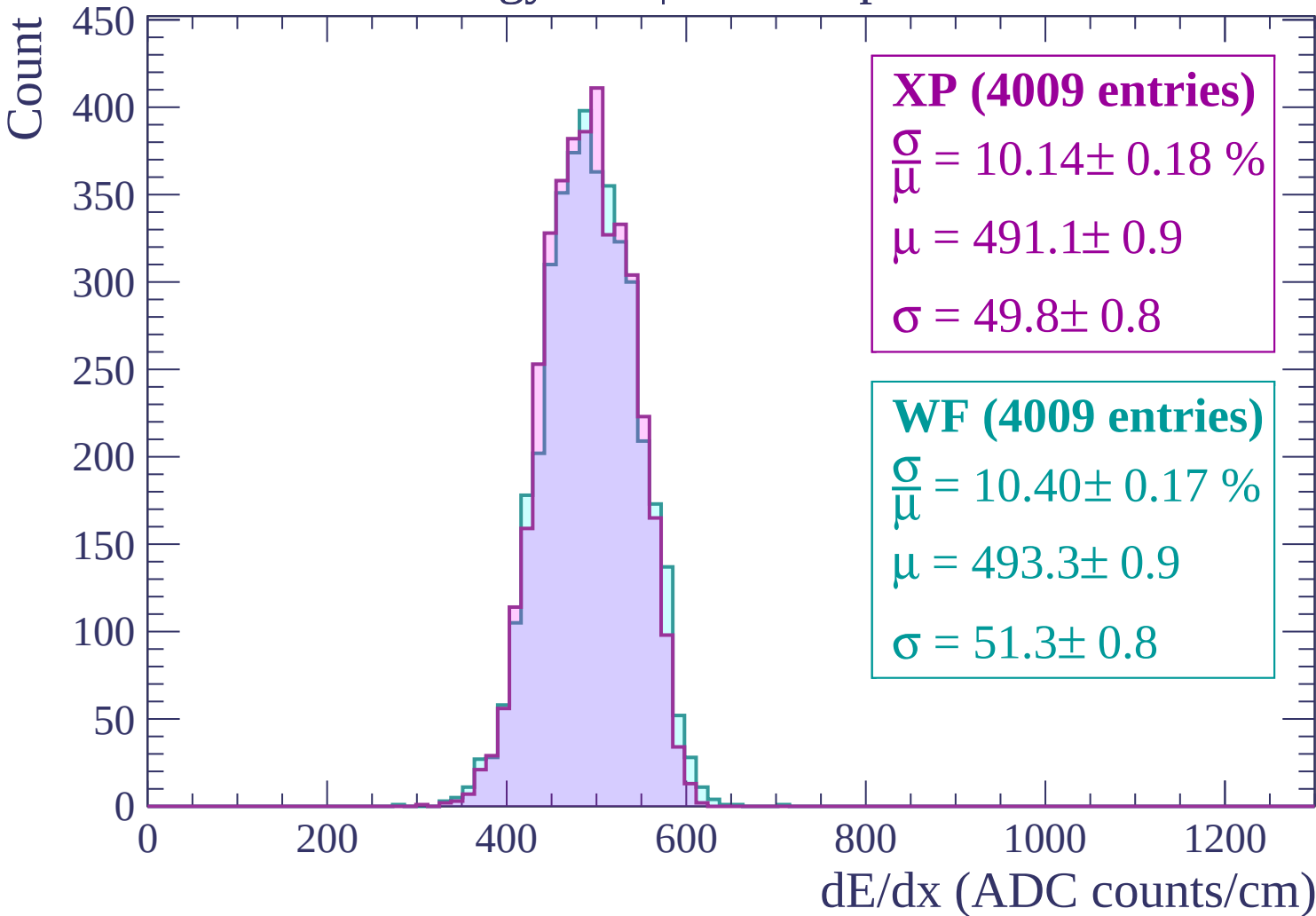
dE/dx (ADC counts/cm)

Energy loss | -1920 < p < -1860

Count

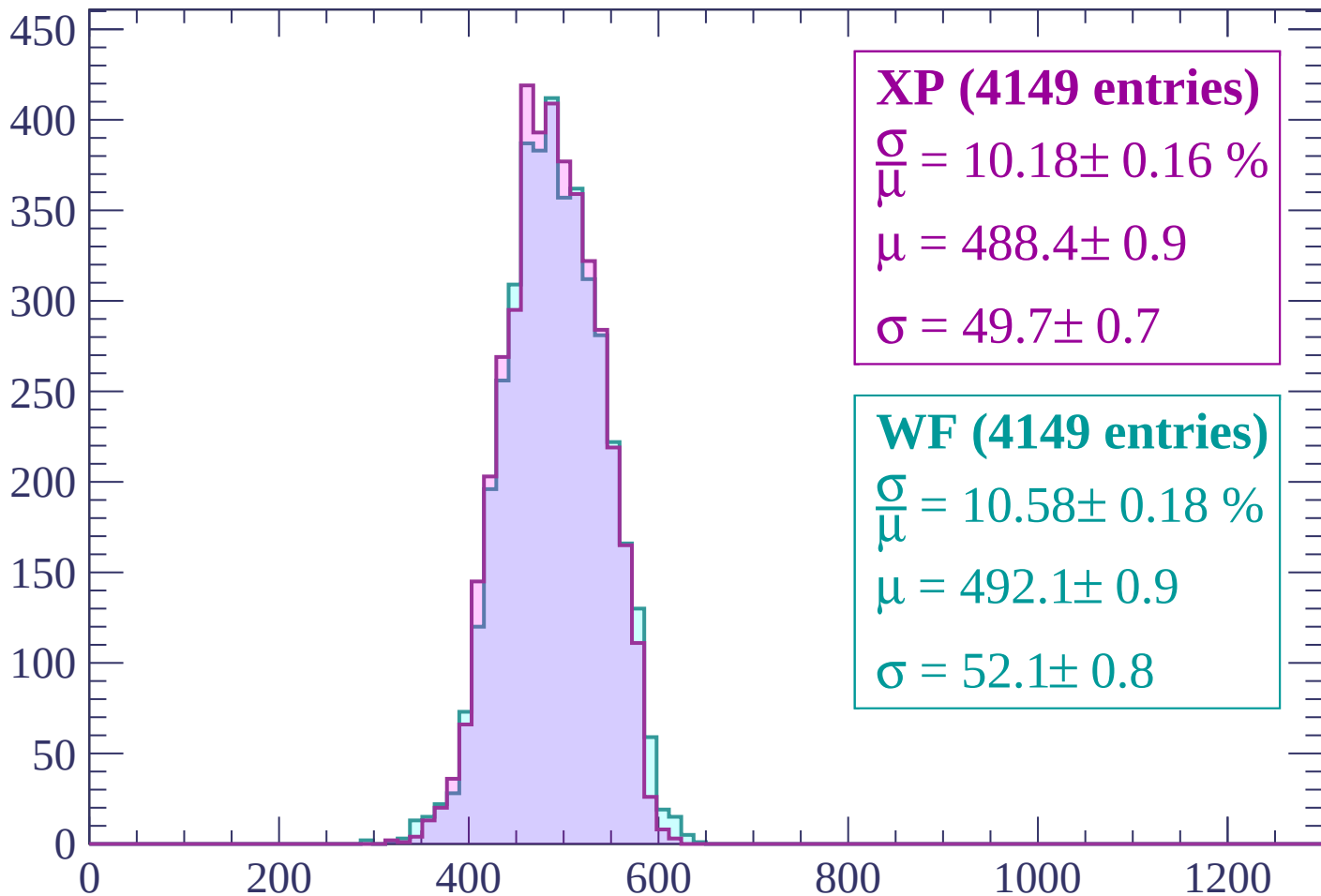


Energy loss | -1860 < p < -1800



Energy loss | $-1800 < p < -1740$

Count



XP (4149 entries)

$\frac{\sigma}{\mu} = 10.18 \pm 0.16 \%$

$\mu = 488.4 \pm 0.9$

$\sigma = 49.7 \pm 0.7$

WF (4149 entries)

$\frac{\sigma}{\mu} = 10.58 \pm 0.18 \%$

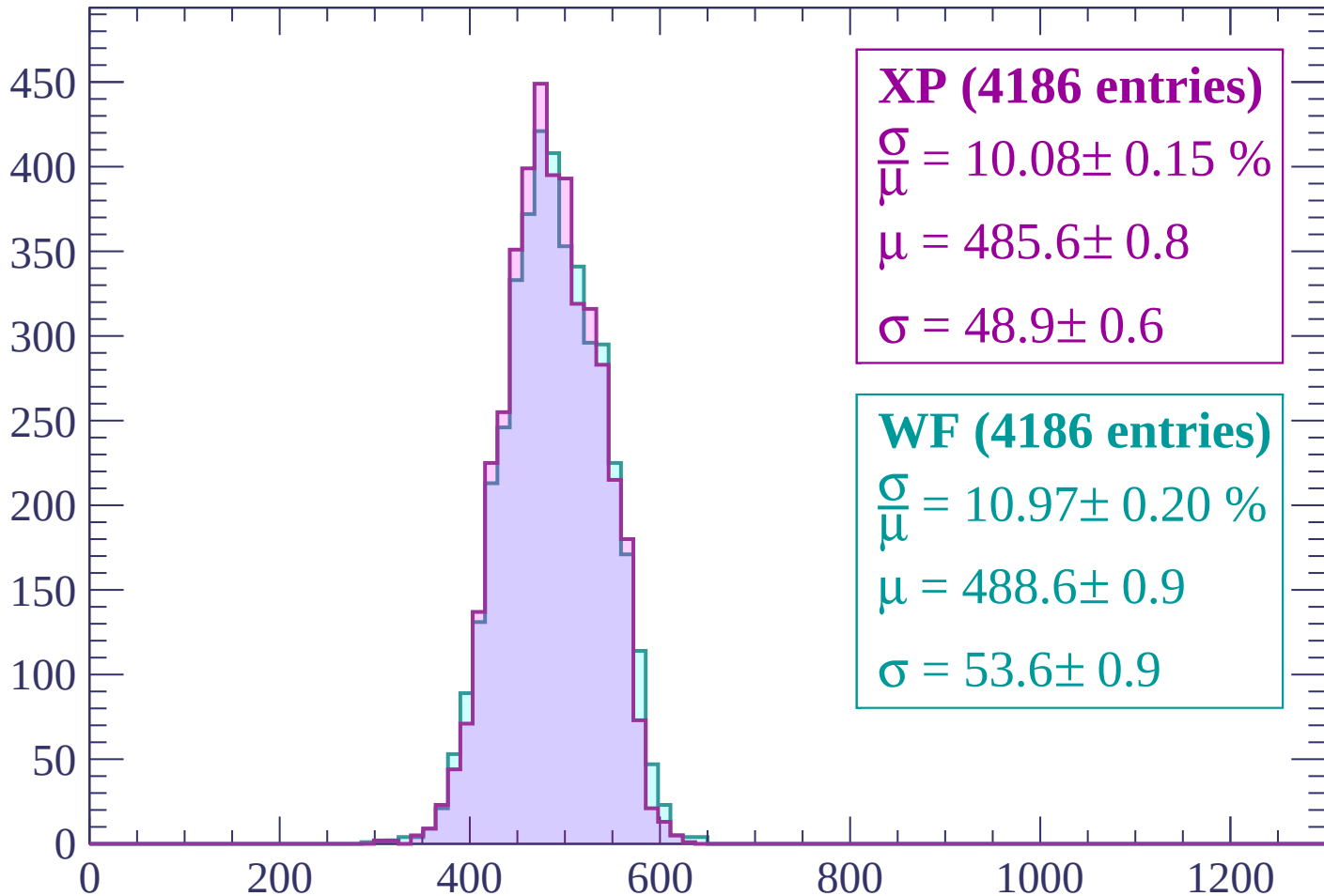
$\mu = 492.1 \pm 0.9$

$\sigma = 52.1 \pm 0.8$

dE/dx (ADC counts/cm)

Energy loss | $-1740 < p < -1680$

Count



XP (4186 entries)

$$\frac{\sigma}{\mu} = 10.08 \pm 0.15 \%$$

$$\mu = 485.6 \pm 0.8$$

$$\sigma = 48.9 \pm 0.6$$

WF (4186 entries)

$$\frac{\sigma}{\mu} = 10.97 \pm 0.20 \%$$

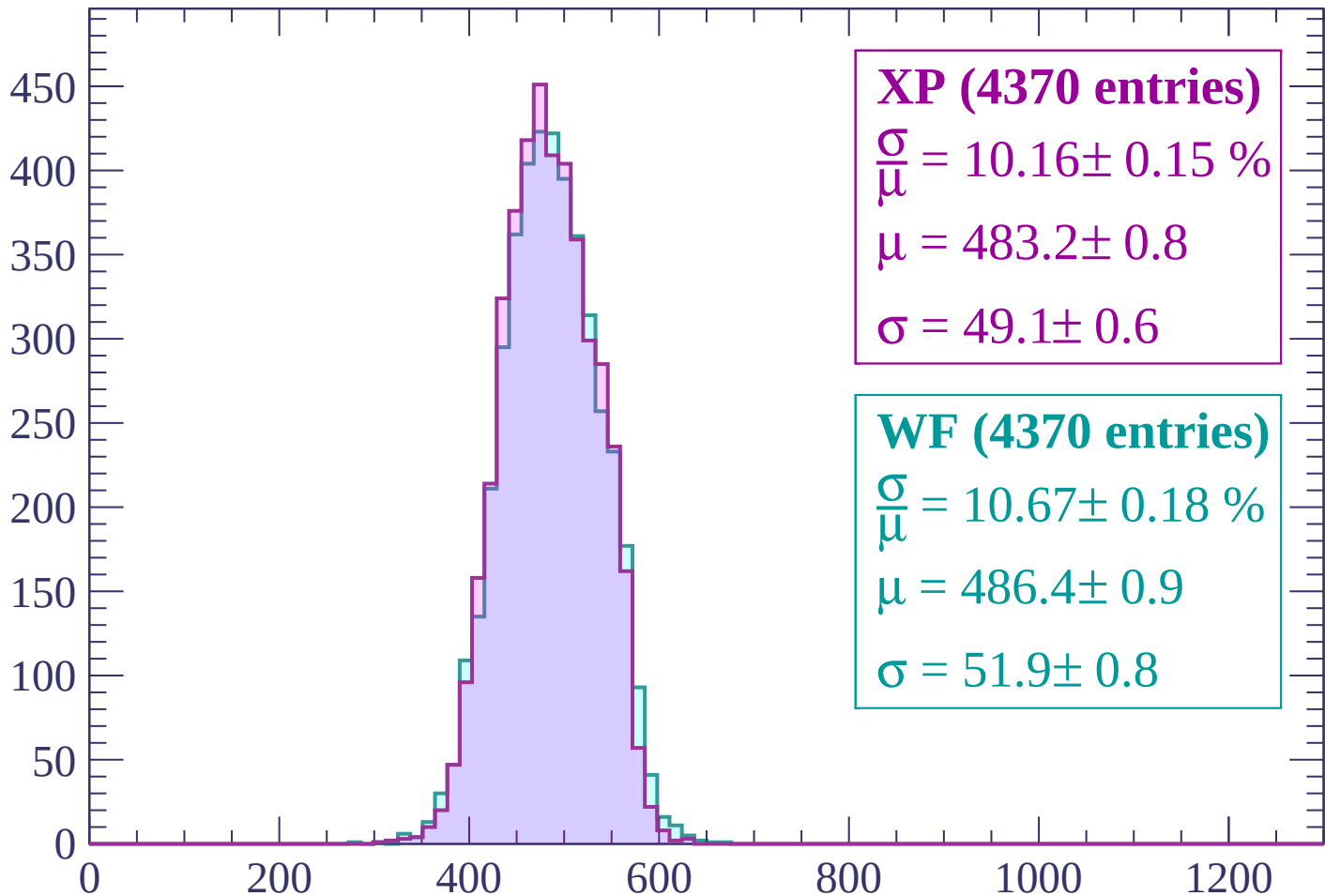
$$\mu = 488.6 \pm 0.9$$

$$\sigma = 53.6 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | -1680 < p < -1620

Count



XP (4370 entries)

$$\frac{\sigma}{\mu} = 10.16 \pm 0.15 \%$$

$$\mu = 483.2 \pm 0.8$$

$$\sigma = 49.1 \pm 0.6$$

WF (4370 entries)

$$\frac{\sigma}{\mu} = 10.67 \pm 0.18 \%$$

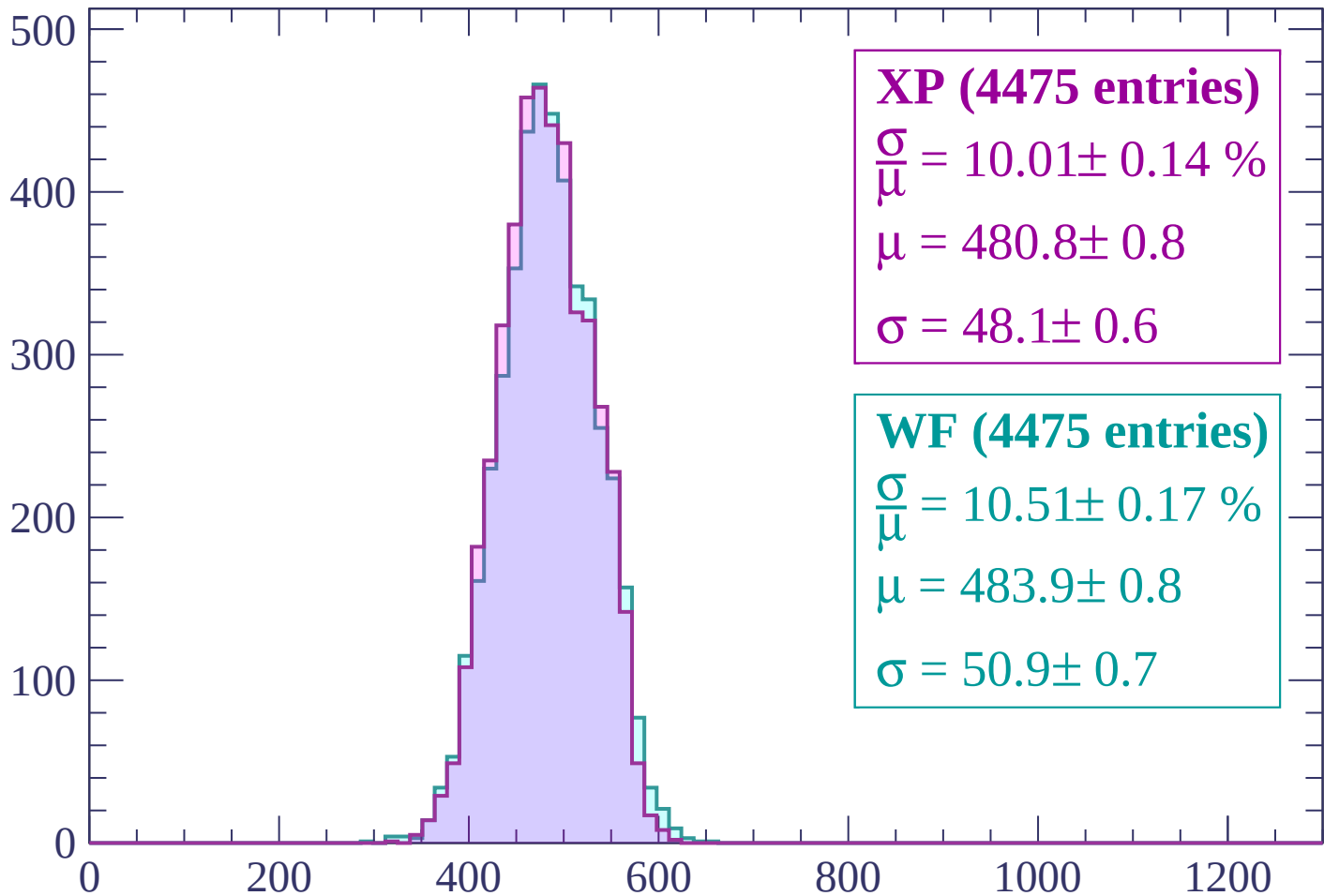
$$\mu = 486.4 \pm 0.9$$

$$\sigma = 51.9 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | -1620 < p < -1560

Count



XP (4475 entries)

$$\frac{\sigma}{\mu} = 10.01 \pm 0.14 \%$$

$$\mu = 480.8 \pm 0.8$$

$$\sigma = 48.1 \pm 0.6$$

WF (4475 entries)

$$\frac{\sigma}{\mu} = 10.51 \pm 0.17 \%$$

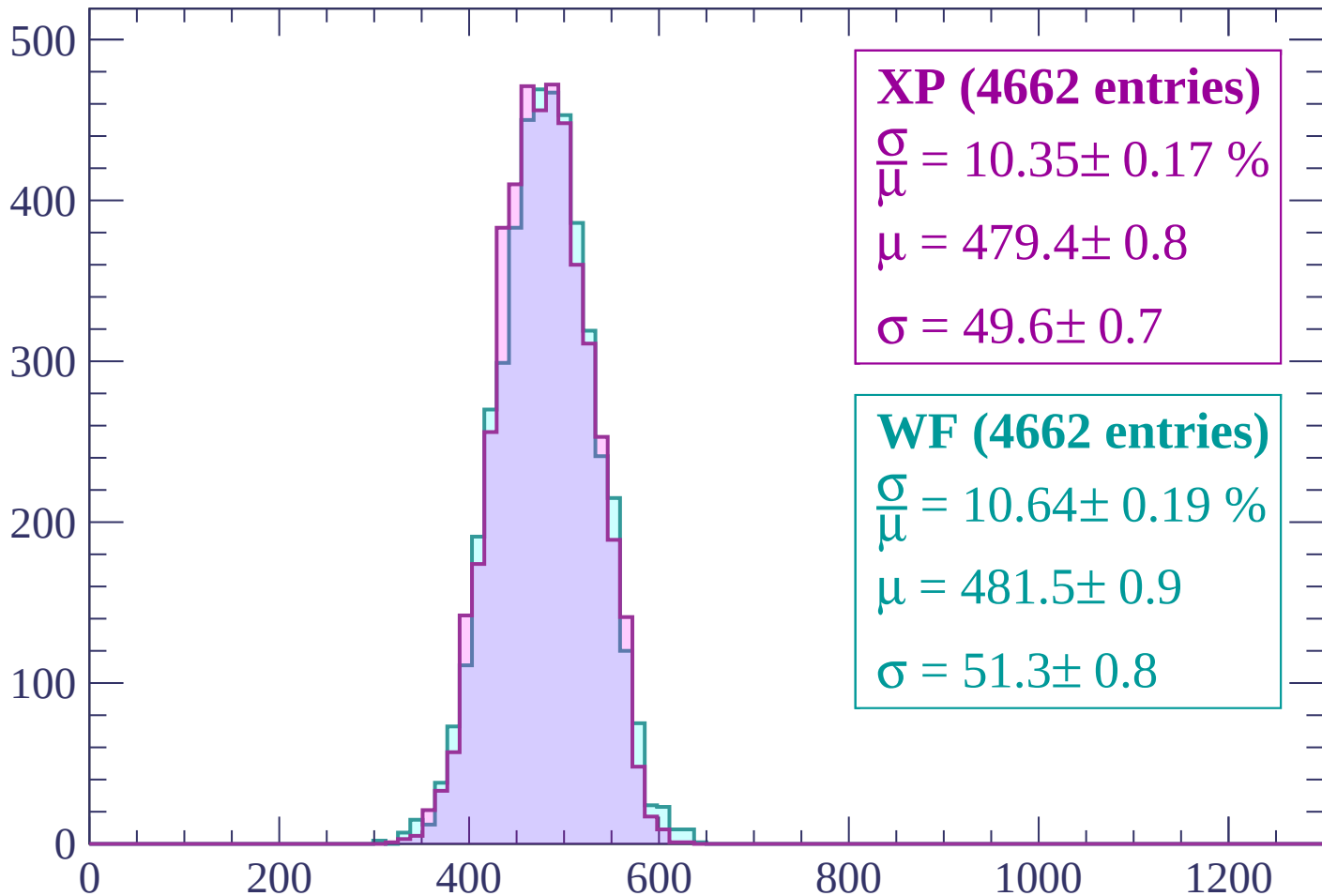
$$\mu = 483.9 \pm 0.8$$

$$\sigma = 50.9 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | -1560 < p < -1500

Count



XP (4662 entries)

$$\frac{\sigma}{\mu} = 10.35 \pm 0.17 \%$$

$$\mu = 479.4 \pm 0.8$$

$$\sigma = 49.6 \pm 0.7$$

WF (4662 entries)

$$\frac{\sigma}{\mu} = 10.64 \pm 0.19 \%$$

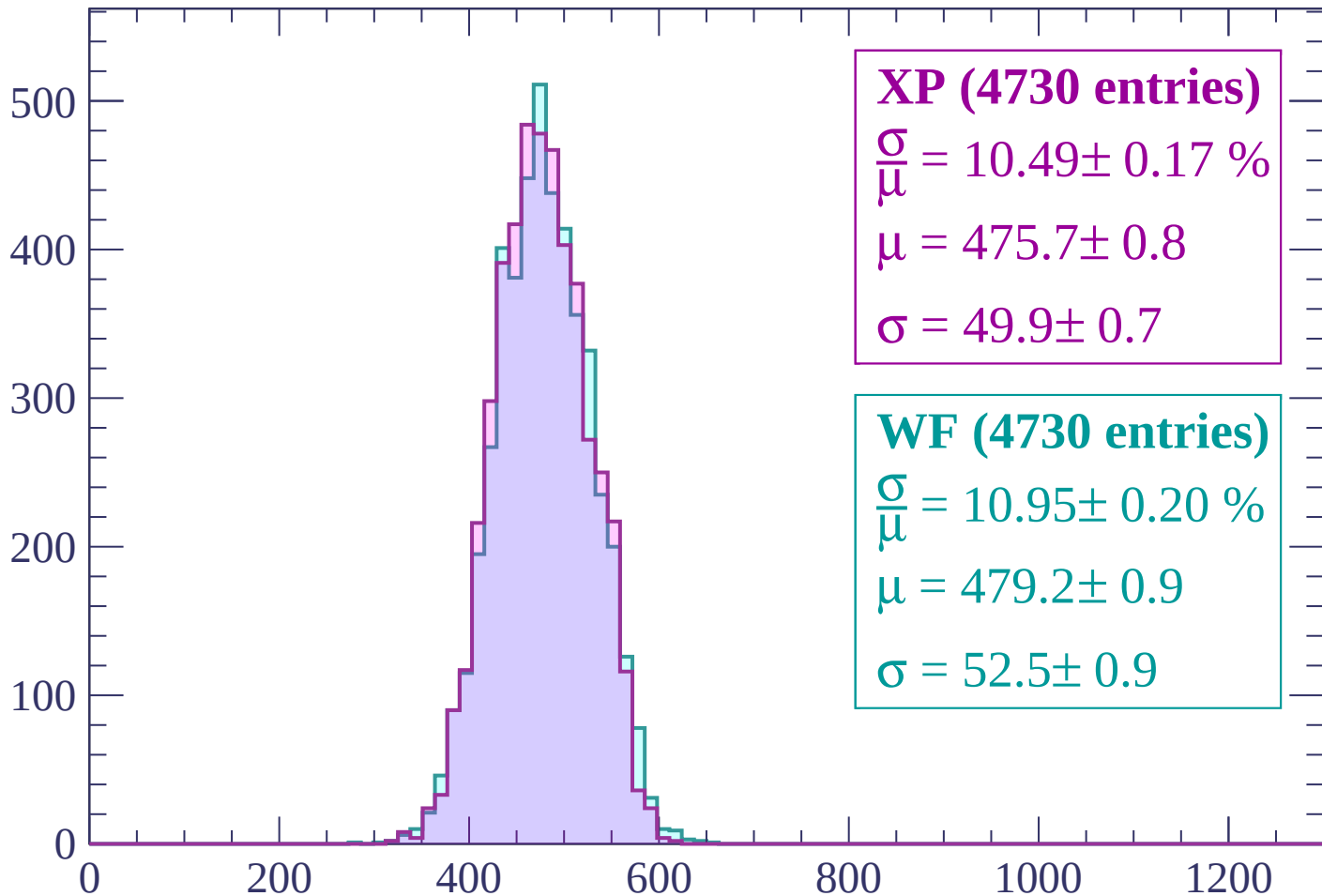
$$\mu = 481.5 \pm 0.9$$

$$\sigma = 51.3 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | -1500 < p < -1440

Count



XP (4730 entries)

$$\frac{\sigma}{\mu} = 10.49 \pm 0.17 \%$$

$$\mu = 475.7 \pm 0.8$$

$$\sigma = 49.9 \pm 0.7$$

WF (4730 entries)

$$\frac{\sigma}{\mu} = 10.95 \pm 0.20 \%$$

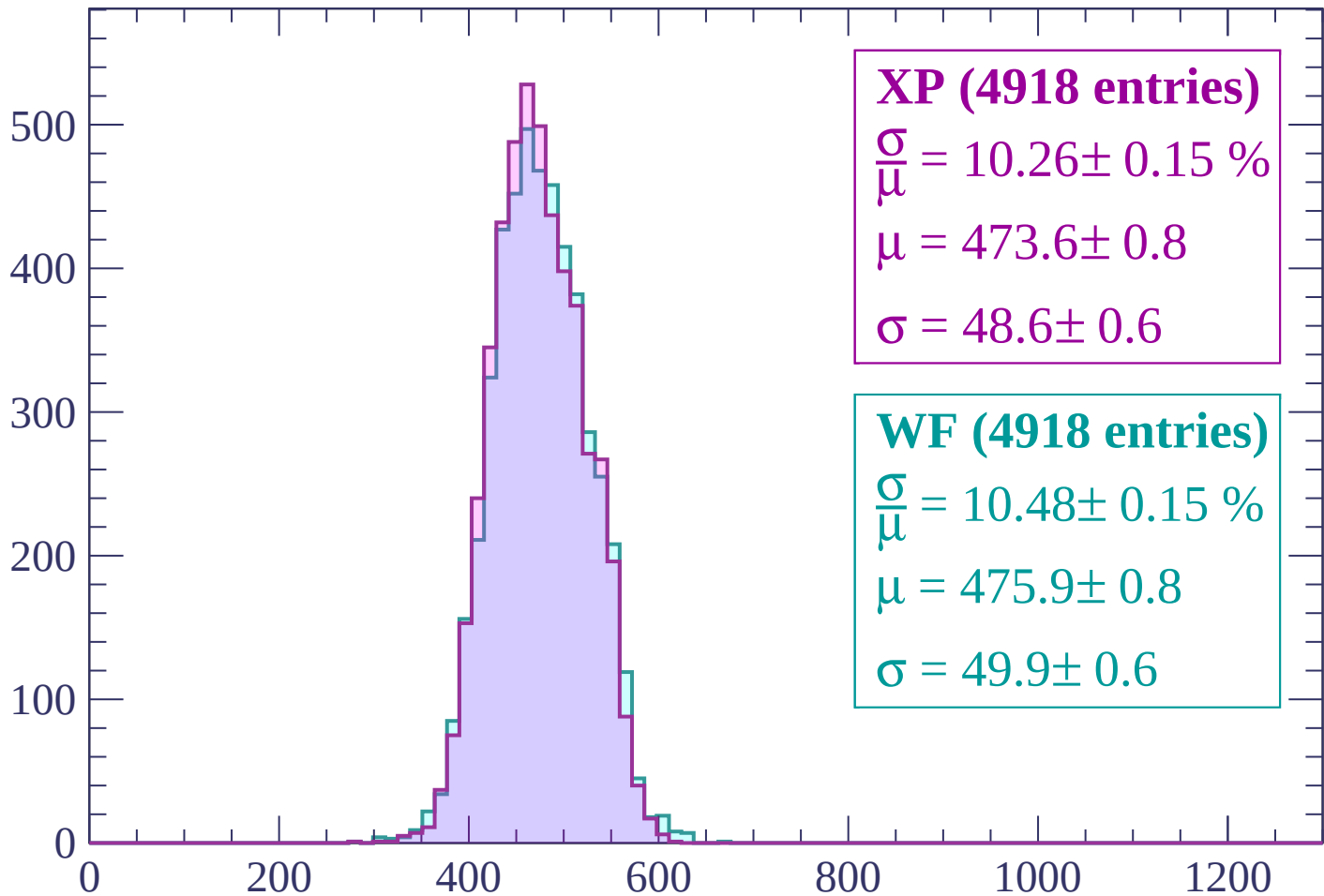
$$\mu = 479.2 \pm 0.9$$

$$\sigma = 52.5 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | $-1440 < p < -1380$

Count



XP (4918 entries)

$\frac{\sigma}{\mu} = 10.26 \pm 0.15 \%$

$\mu = 473.6 \pm 0.8$

$\sigma = 48.6 \pm 0.6$

WF (4918 entries)

$\frac{\sigma}{\mu} = 10.48 \pm 0.15 \%$

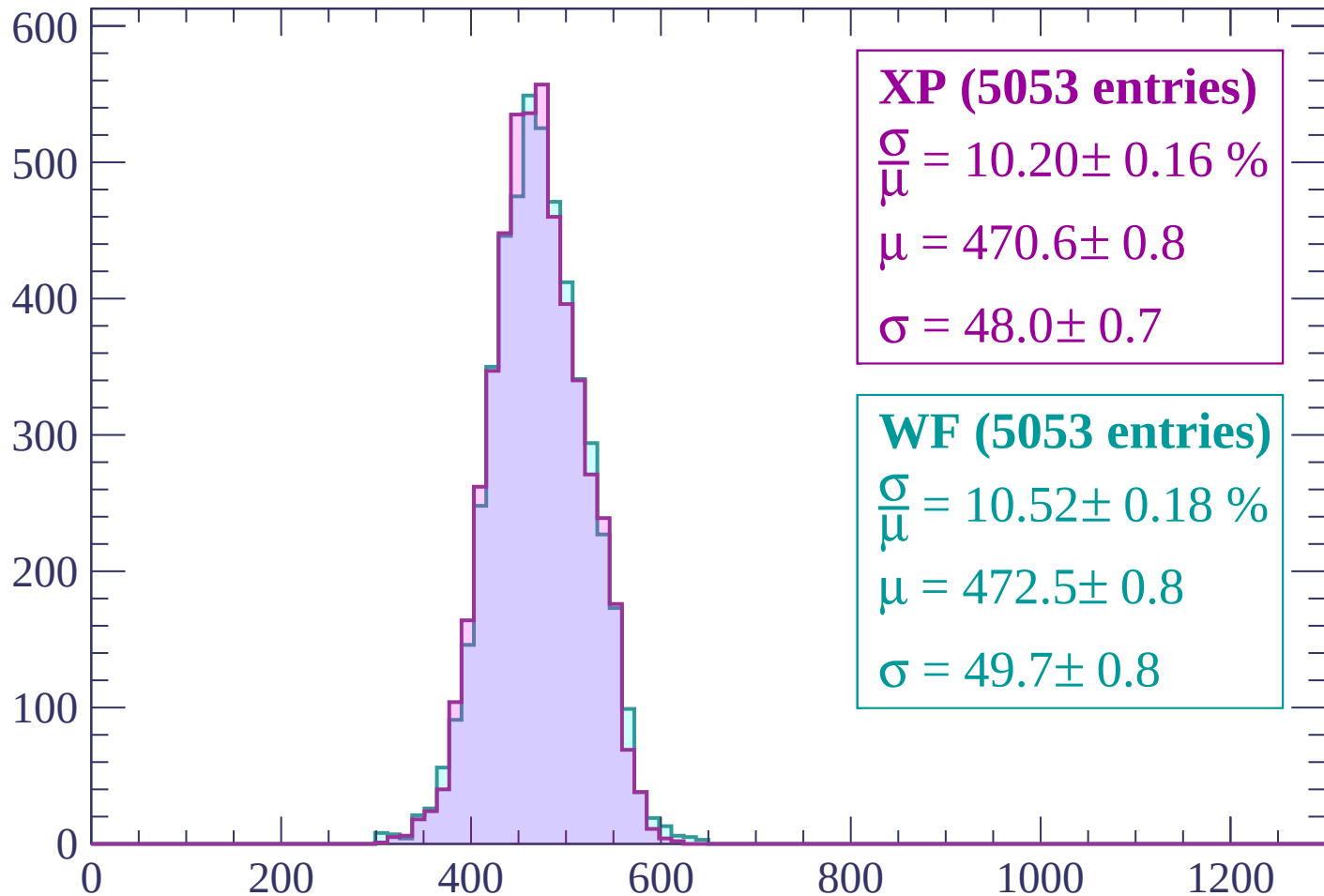
$\mu = 475.9 \pm 0.8$

$\sigma = 49.9 \pm 0.6$

dE/dx (ADC counts/cm)

Energy loss | -1380 < p < -1320

Count



XP (5053 entries)

$$\frac{\sigma}{\mu} = 10.20 \pm 0.16 \%$$

$$\mu = 470.6 \pm 0.8$$

$$\sigma = 48.0 \pm 0.7$$

WF (5053 entries)

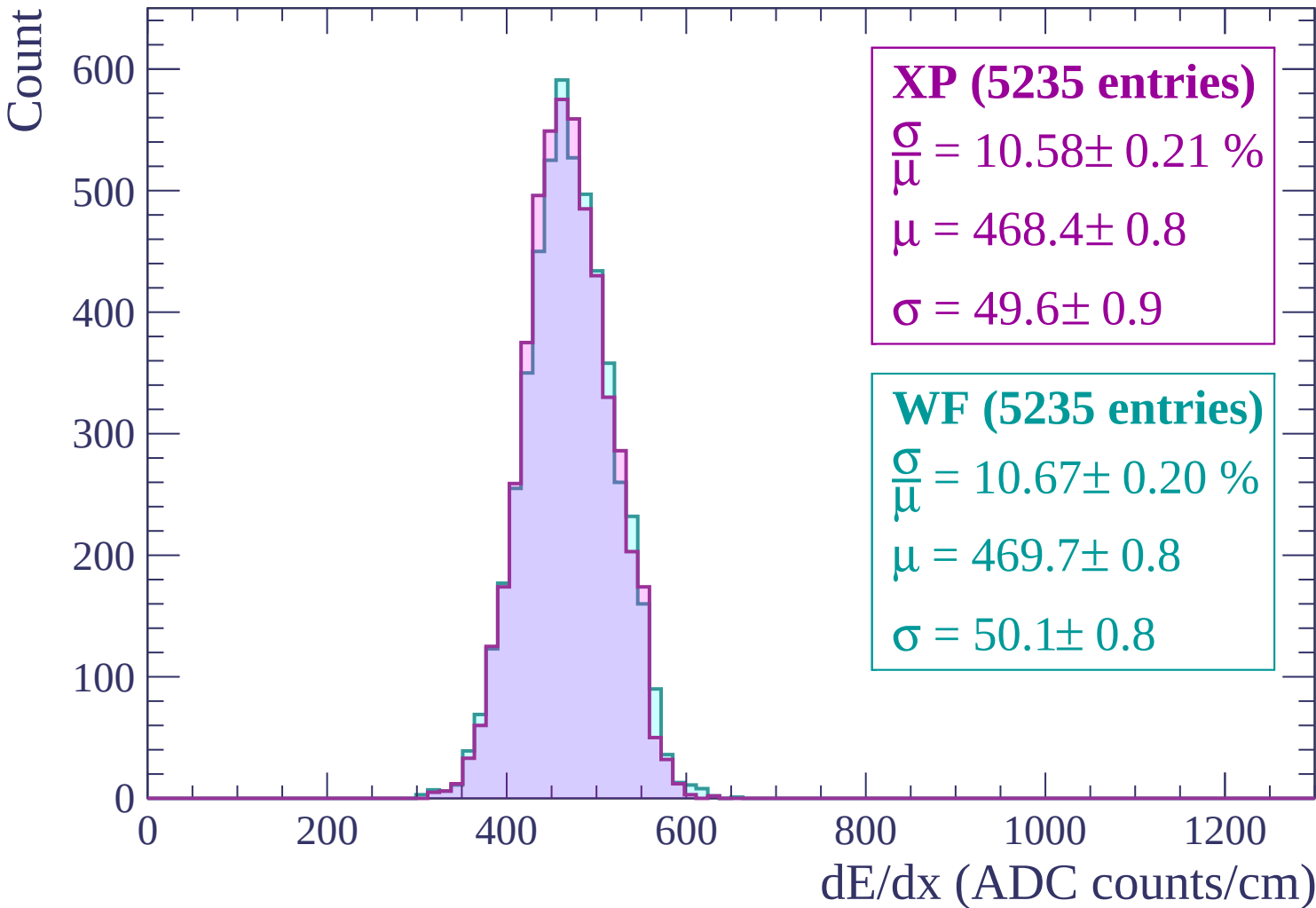
$$\frac{\sigma}{\mu} = 10.52 \pm 0.18 \%$$

$$\mu = 472.5 \pm 0.8$$

$$\sigma = 49.7 \pm 0.8$$

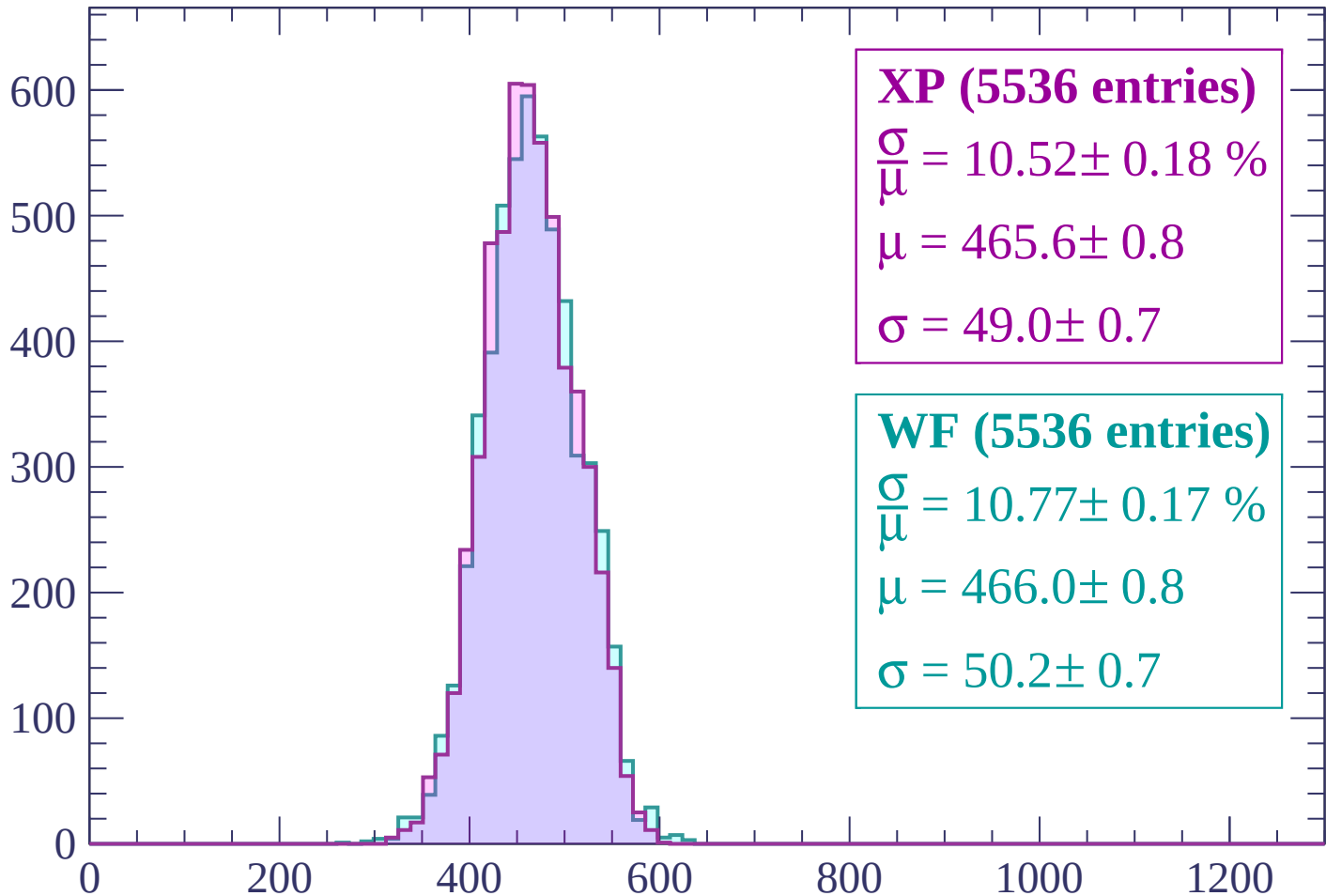
dE/dx (ADC counts/cm)

Energy loss | -1320 < p < -1260



Energy loss | -1260 < p < -1200

Count



XP (5536 entries)

$\frac{\sigma}{\mu} = 10.52 \pm 0.18 \%$

$\mu = 465.6 \pm 0.8$

$\sigma = 49.0 \pm 0.7$

WF (5536 entries)

$\frac{\sigma}{\mu} = 10.77 \pm 0.17 \%$

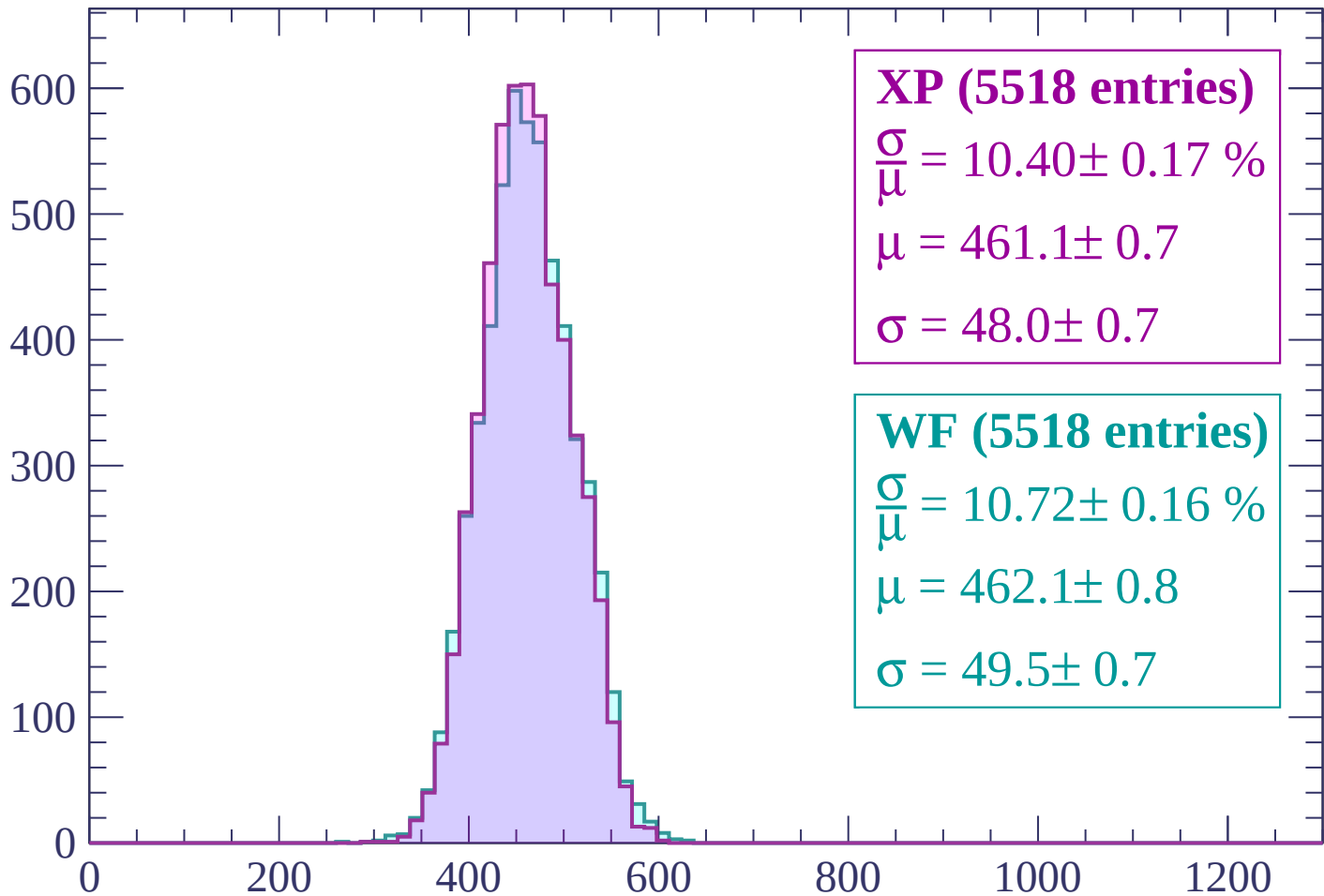
$\mu = 466.0 \pm 0.8$

$\sigma = 50.2 \pm 0.7$

dE/dx (ADC counts/cm)

Energy loss | -1200 < p < -1140

Count



XP (5518 entries)

$$\frac{\sigma}{\mu} = 10.40 \pm 0.17 \%$$

$$\mu = 461.1 \pm 0.7$$

$$\sigma = 48.0 \pm 0.7$$

WF (5518 entries)

$$\frac{\sigma}{\mu} = 10.72 \pm 0.16 \%$$

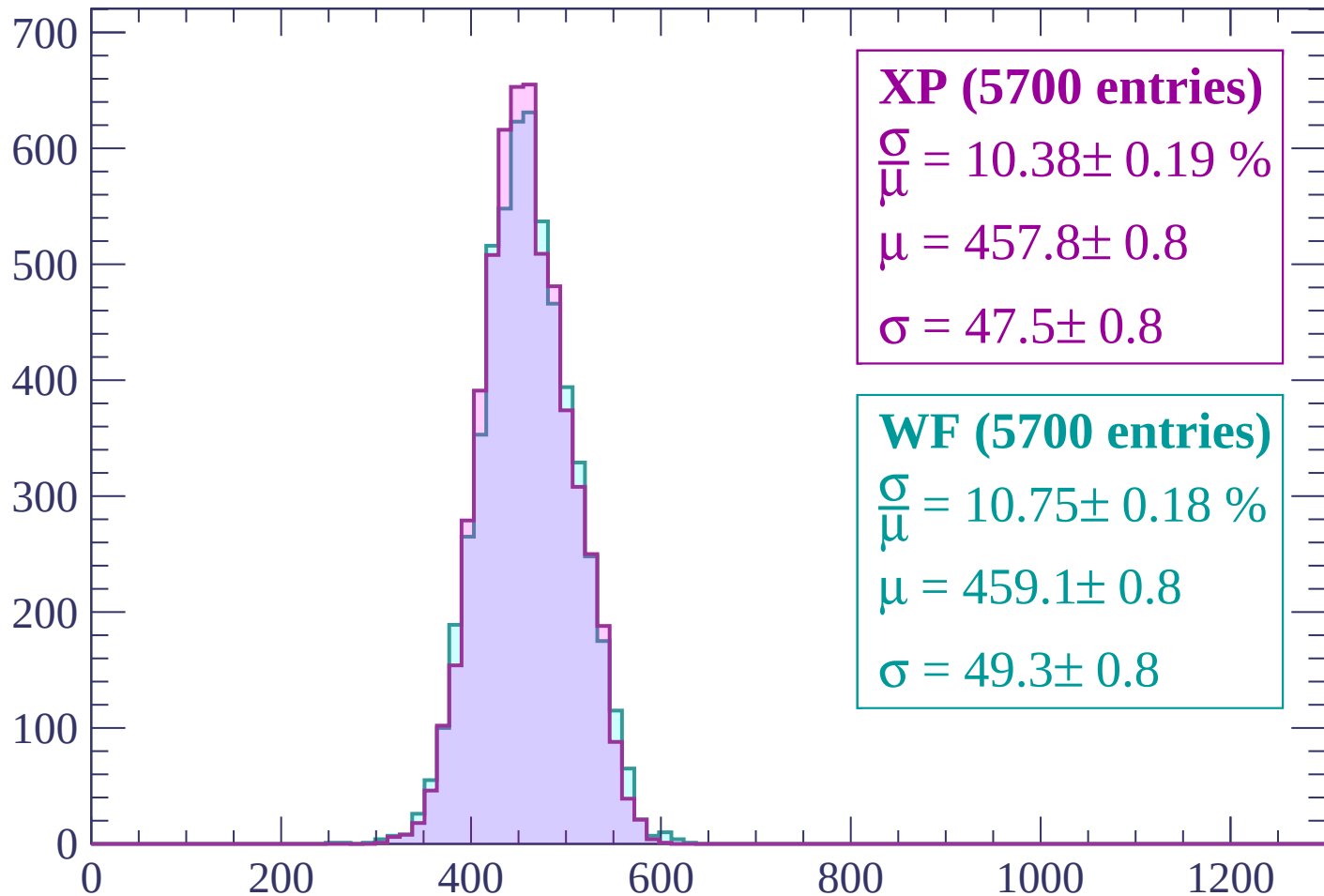
$$\mu = 462.1 \pm 0.8$$

$$\sigma = 49.5 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $-1140 < p < -1080$

Count



XP (5700 entries)

$$\frac{\sigma}{\mu} = 10.38 \pm 0.19 \%$$

$$\mu = 457.8 \pm 0.8$$

$$\sigma = 47.5 \pm 0.8$$

WF (5700 entries)

$$\frac{\sigma}{\mu} = 10.75 \pm 0.18 \%$$

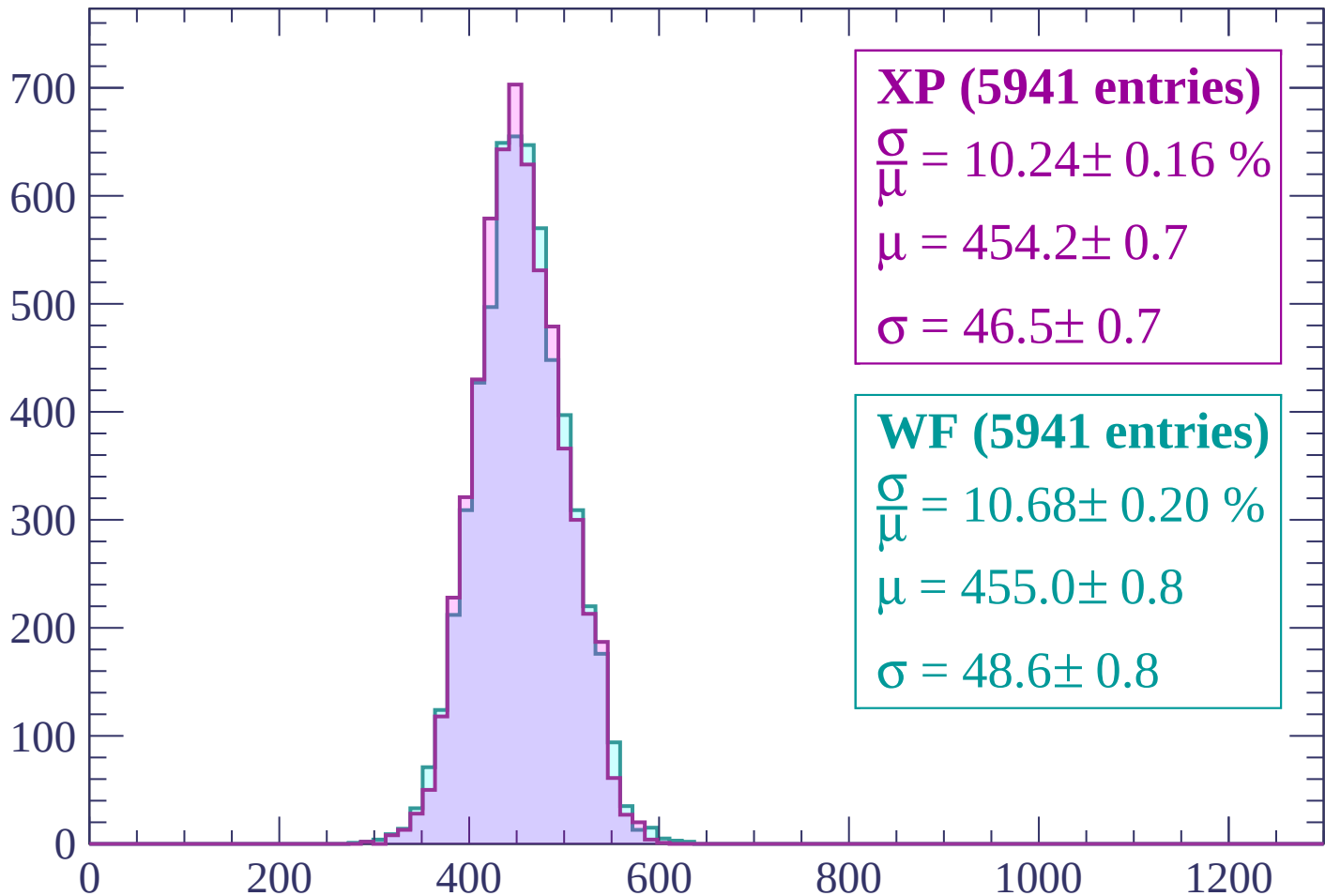
$$\mu = 459.1 \pm 0.8$$

$$\sigma = 49.3 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $-1080 < p < -1020$

Count



XP (5941 entries)

$$\frac{\sigma}{\mu} = 10.24 \pm 0.16 \%$$

$$\mu = 454.2 \pm 0.7$$

$$\sigma = 46.5 \pm 0.7$$

WF (5941 entries)

$$\frac{\sigma}{\mu} = 10.68 \pm 0.20 \%$$

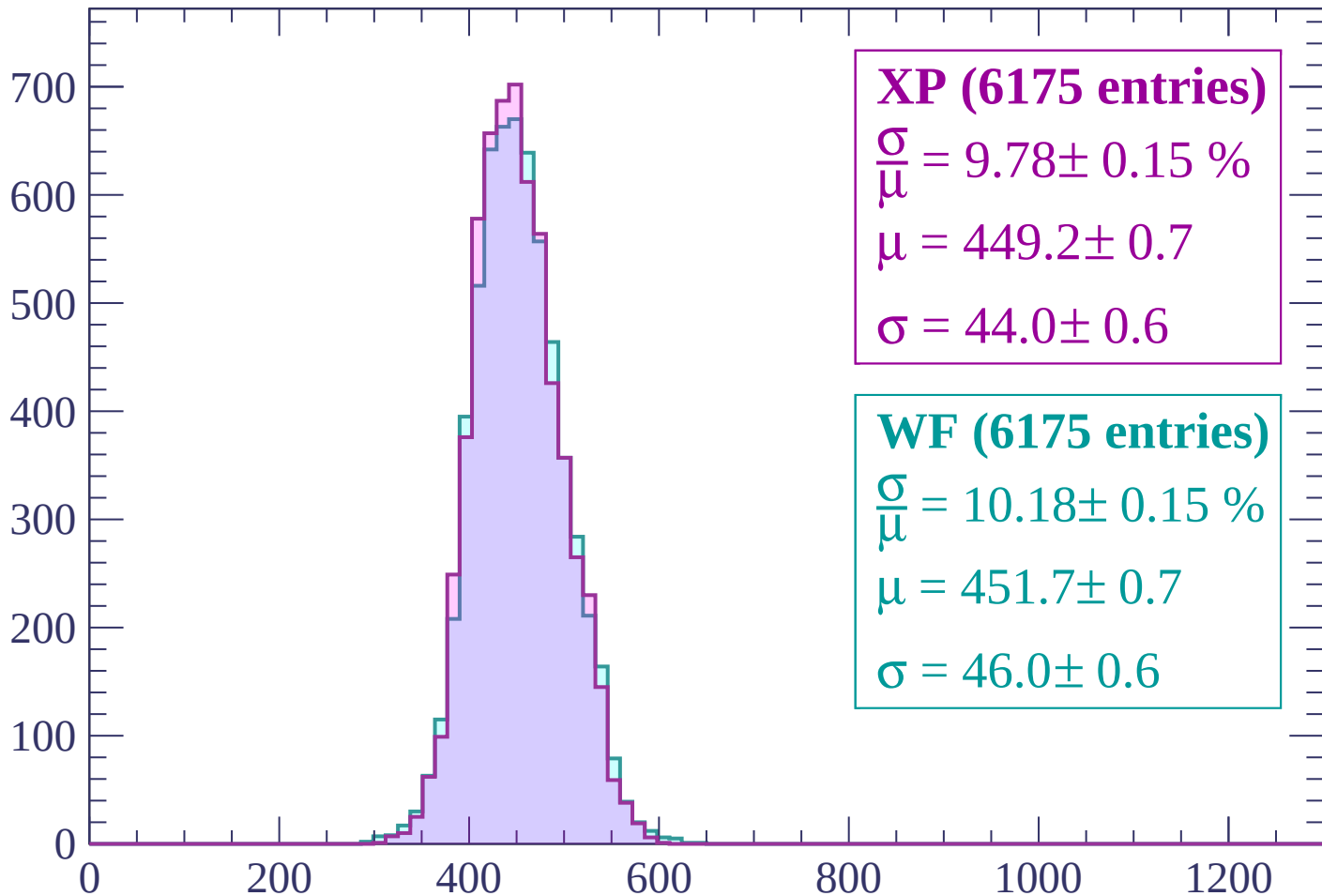
$$\mu = 455.0 \pm 0.8$$

$$\sigma = 48.6 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $-1020 < p < -960$

Count



XP (6175 entries)

$$\frac{\sigma}{\mu} = 9.78 \pm 0.15 \%$$

$$\mu = 449.2 \pm 0.7$$

$$\sigma = 44.0 \pm 0.6$$

WF (6175 entries)

$$\frac{\sigma}{\mu} = 10.18 \pm 0.15 \%$$

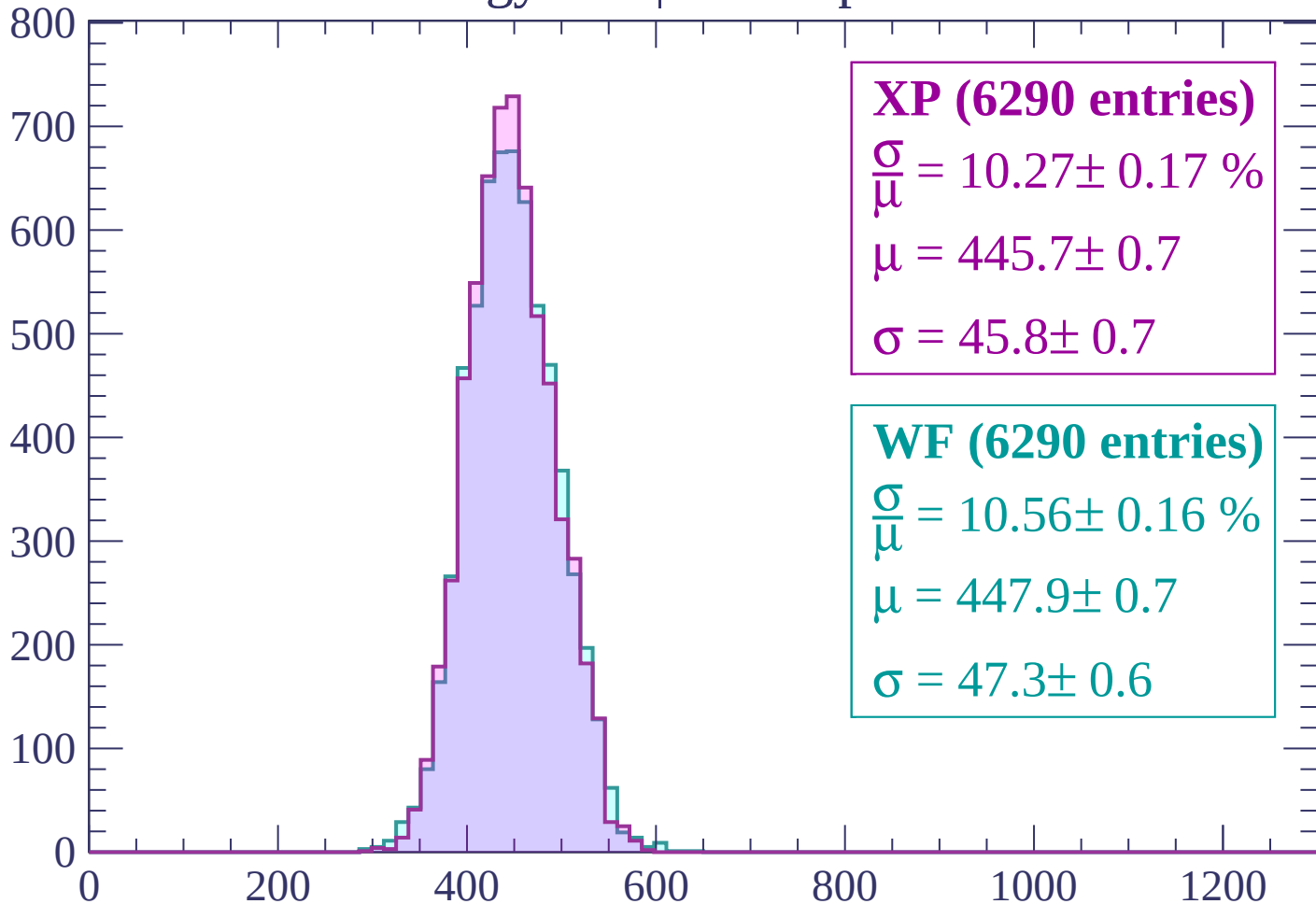
$$\mu = 451.7 \pm 0.7$$

$$\sigma = 46.0 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $-960 < p < -900$

Count



XP (6290 entries)

$\frac{\sigma}{\mu} = 10.27 \pm 0.17 \%$

$\mu = 445.7 \pm 0.7$

$\sigma = 45.8 \pm 0.7$

WF (6290 entries)

$\frac{\sigma}{\mu} = 10.56 \pm 0.16 \%$

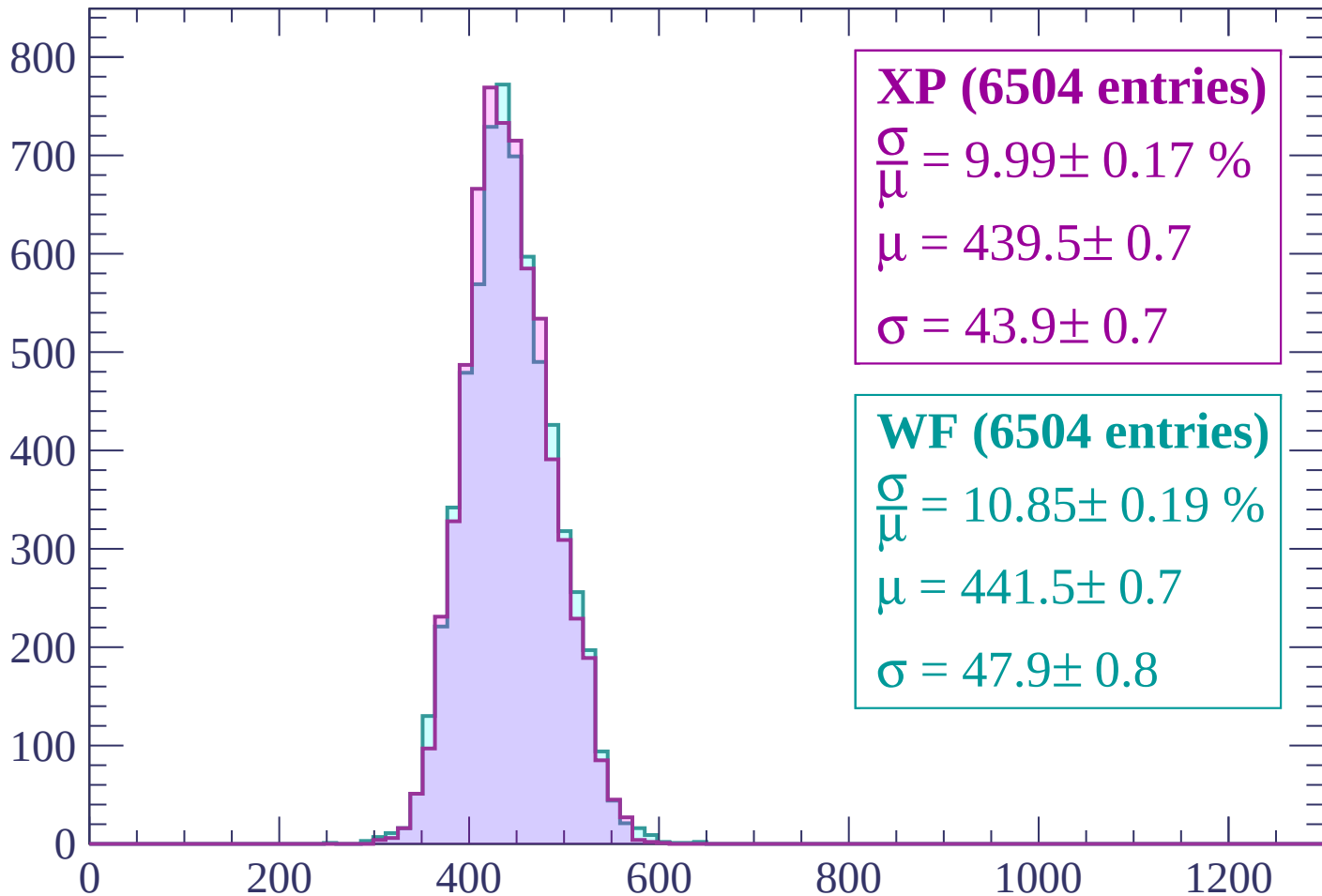
$\mu = 447.9 \pm 0.7$

$\sigma = 47.3 \pm 0.6$

dE/dx (ADC counts/cm)

Energy loss | $-900 < p < -840$

Count



XP (6504 entries)

$$\frac{\sigma}{\mu} = 9.99 \pm 0.17 \%$$

$$\mu = 439.5 \pm 0.7$$

$$\sigma = 43.9 \pm 0.7$$

WF (6504 entries)

$$\frac{\sigma}{\mu} = 10.85 \pm 0.19 \%$$

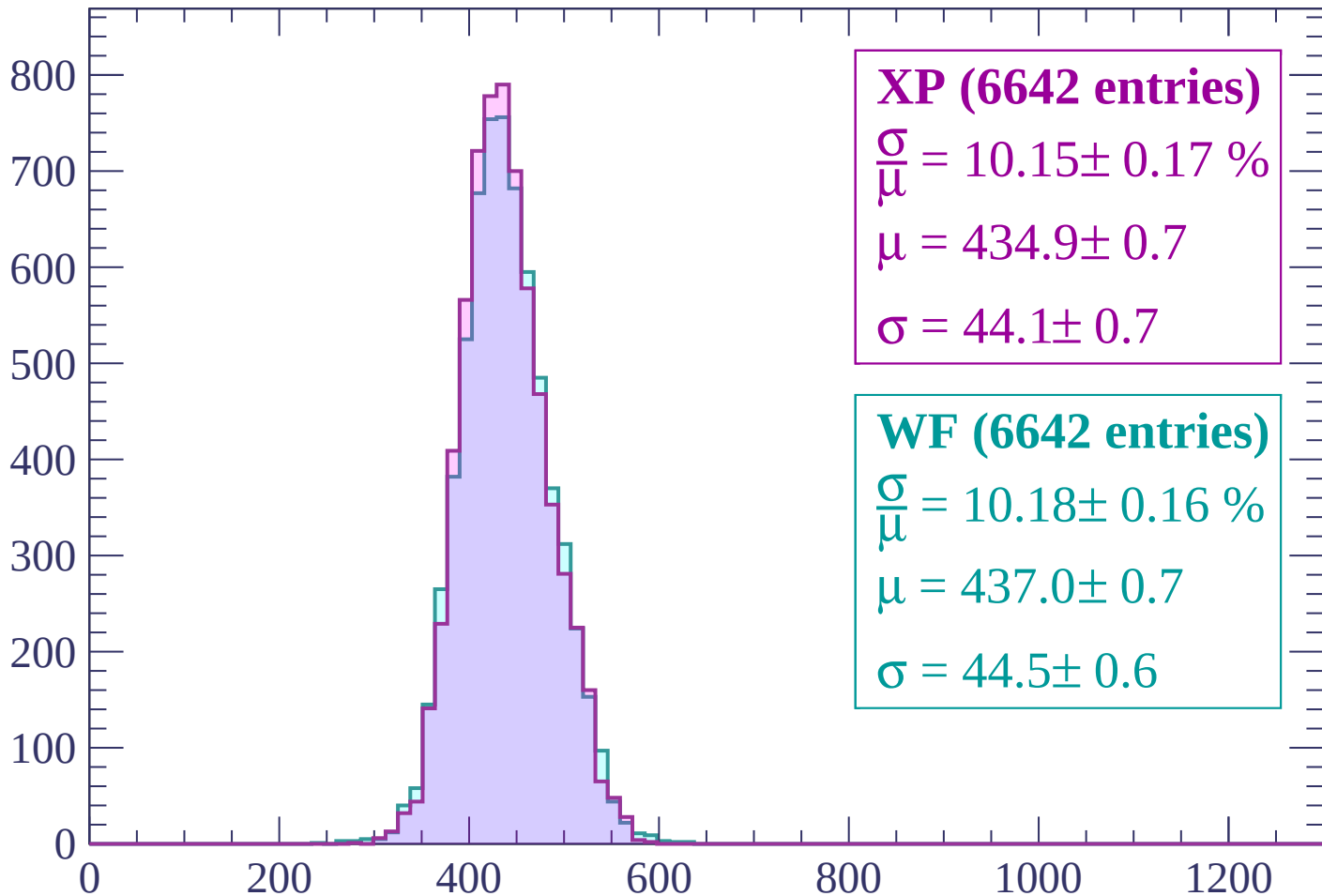
$$\mu = 441.5 \pm 0.7$$

$$\sigma = 47.9 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $-840 < p < -780$

Count



XP (6642 entries)

$\frac{\sigma}{\mu} = 10.15 \pm 0.17 \%$

$\mu = 434.9 \pm 0.7$

$\sigma = 44.1 \pm 0.7$

WF (6642 entries)

$\frac{\sigma}{\mu} = 10.18 \pm 0.16 \%$

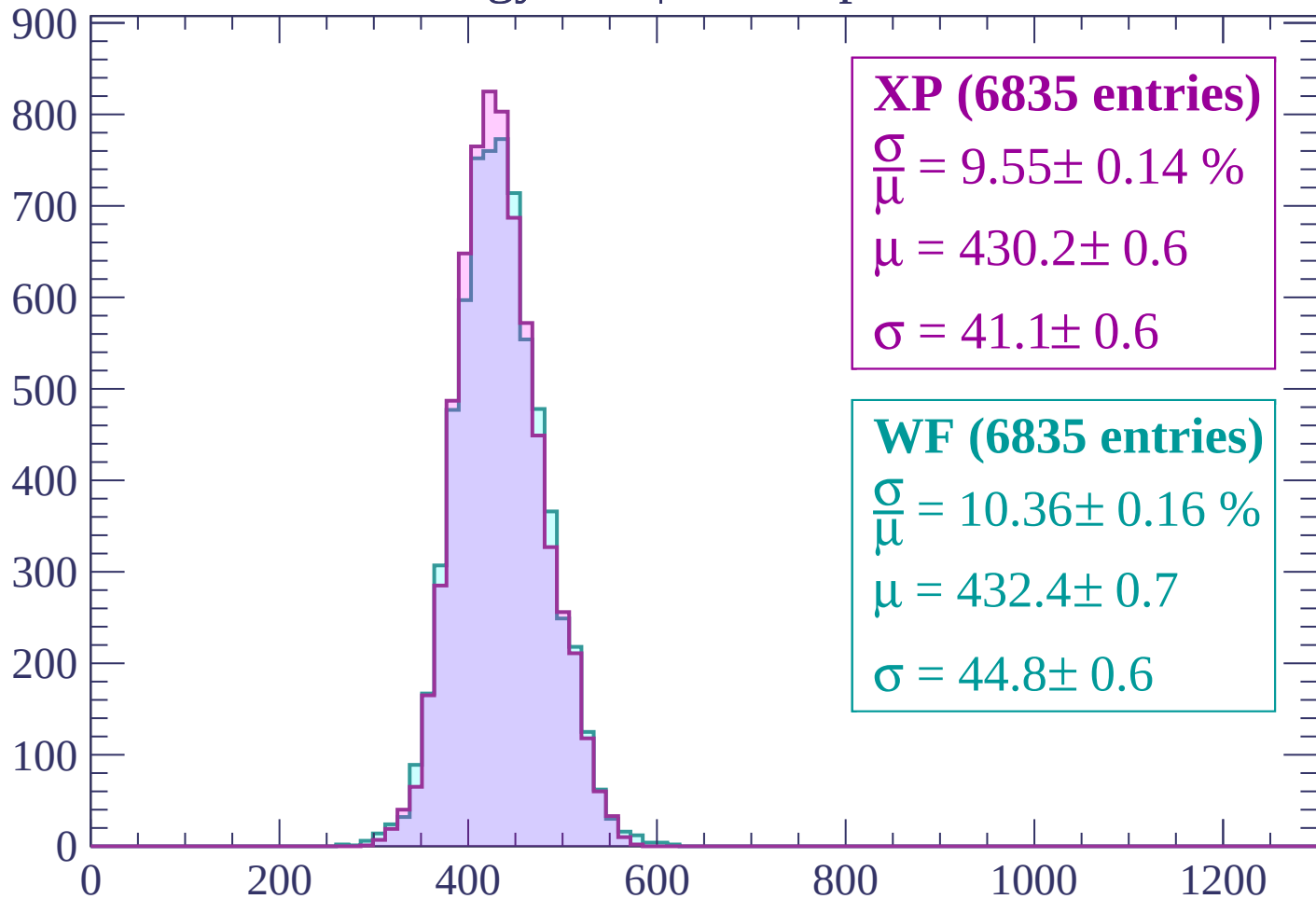
$\mu = 437.0 \pm 0.7$

$\sigma = 44.5 \pm 0.6$

dE/dx (ADC counts/cm)

Energy loss | $-780 < p < -720$

Count



XP (6835 entries)

$$\frac{\sigma}{\mu} = 9.55 \pm 0.14 \%$$

$$\mu = 430.2 \pm 0.6$$

$$\sigma = 41.1 \pm 0.6$$

WF (6835 entries)

$$\frac{\sigma}{\mu} = 10.36 \pm 0.16 \%$$

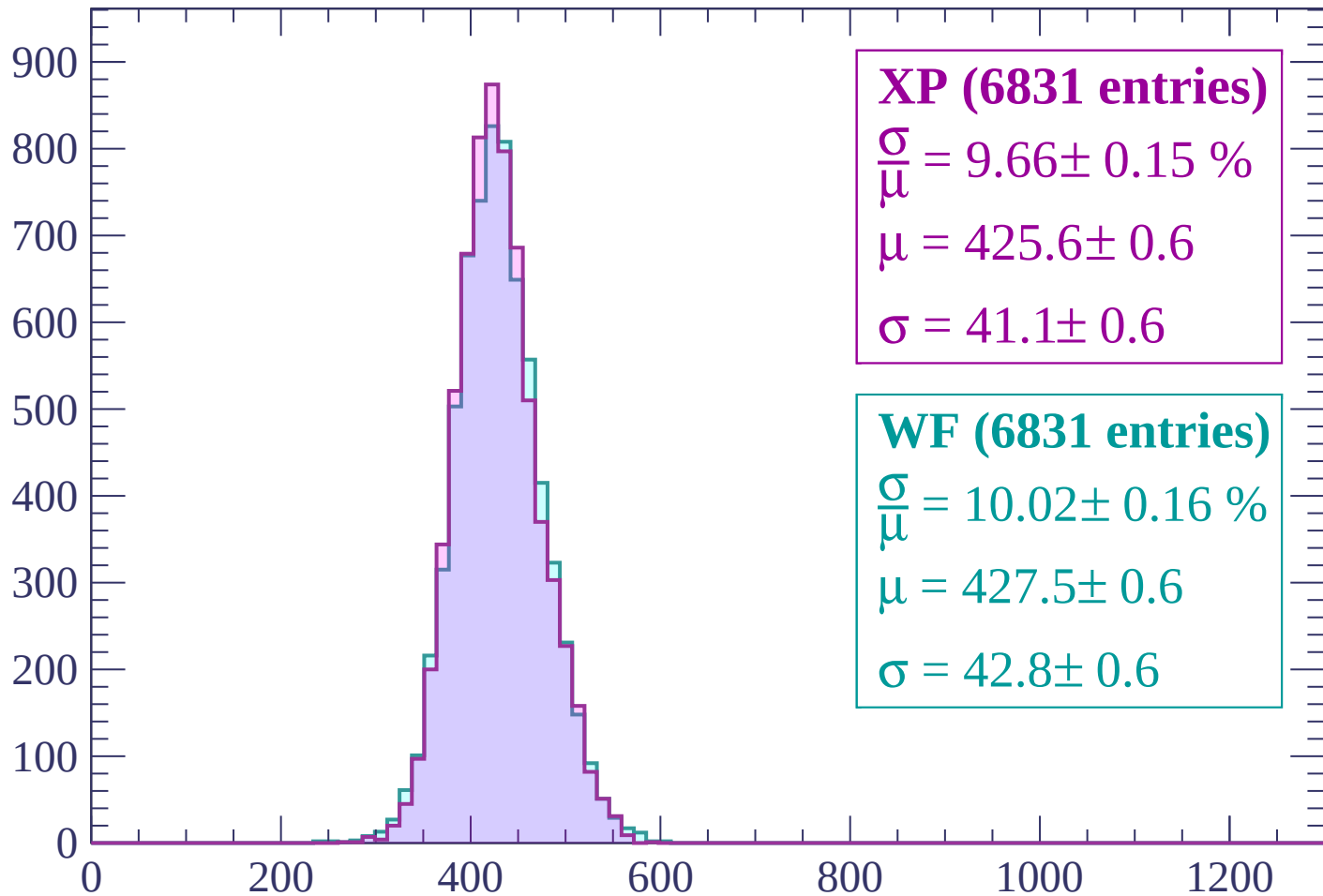
$$\mu = 432.4 \pm 0.7$$

$$\sigma = 44.8 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $-720 < p < -660$

Count



XP (6831 entries)

$$\frac{\sigma}{\mu} = 9.66 \pm 0.15 \%$$

$$\mu = 425.6 \pm 0.6$$

$$\sigma = 41.1 \pm 0.6$$

WF (6831 entries)

$$\frac{\sigma}{\mu} = 10.02 \pm 0.16 \%$$

$$\mu = 427.5 \pm 0.6$$

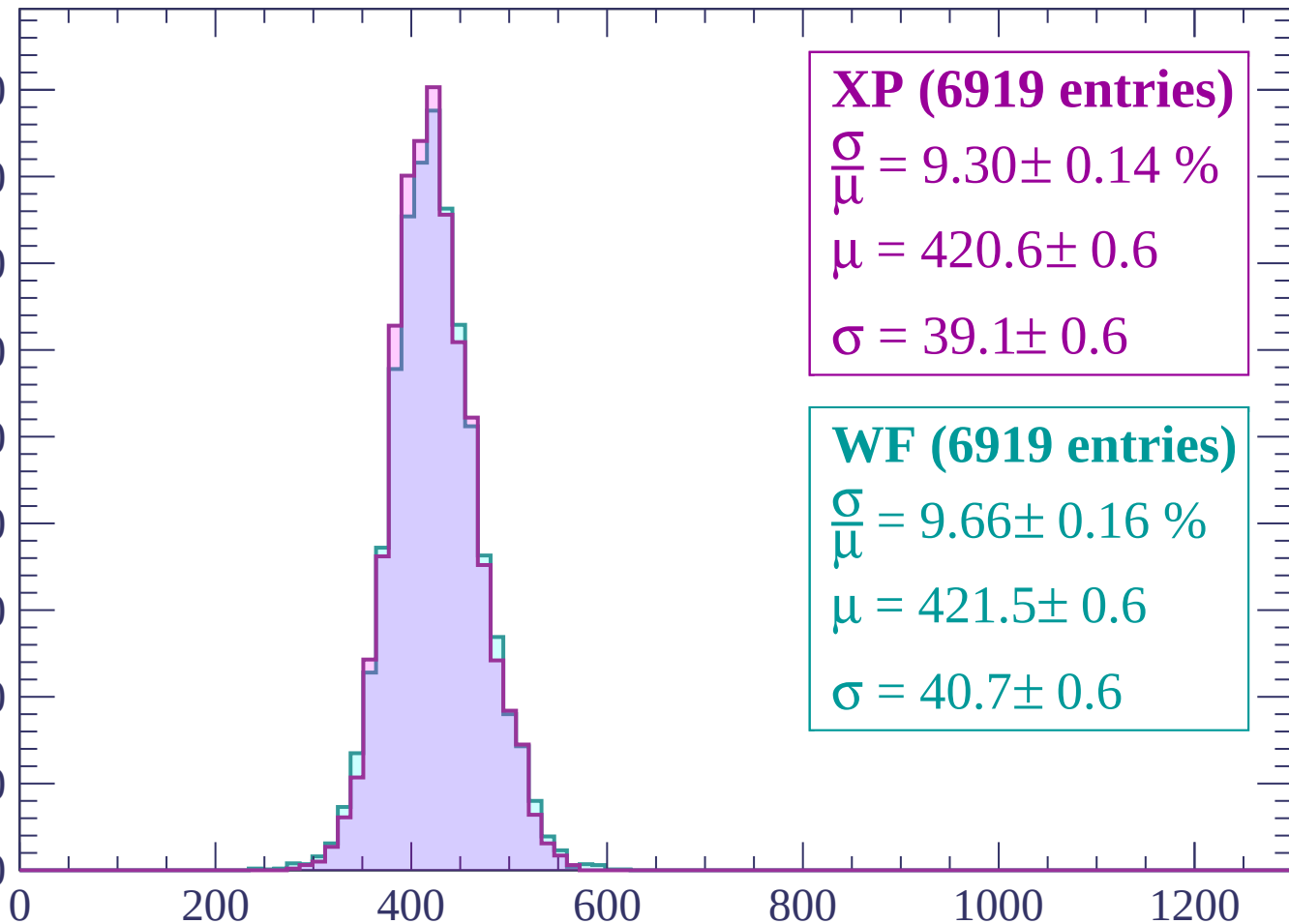
$$\sigma = 42.8 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $-660 < p < -600$

Count

900
800
700
600
500
400
300
200
100
0



XP (6919 entries)

$\frac{\sigma}{\mu} = 9.30 \pm 0.14 \%$

$\mu = 420.6 \pm 0.6$

$\sigma = 39.1 \pm 0.6$

WF (6919 entries)

$\frac{\sigma}{\mu} = 9.66 \pm 0.16 \%$

$\mu = 421.5 \pm 0.6$

$\sigma = 40.7 \pm 0.6$

dE/dx (ADC counts/cm)

Energy loss | $-600 < p < -540$

Count

1000

800

600

400

200

0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (6941 entries)

$\frac{\sigma}{\mu} = 9.22 \pm 0.14 \%$

$\mu = 415.9 \pm 0.6$

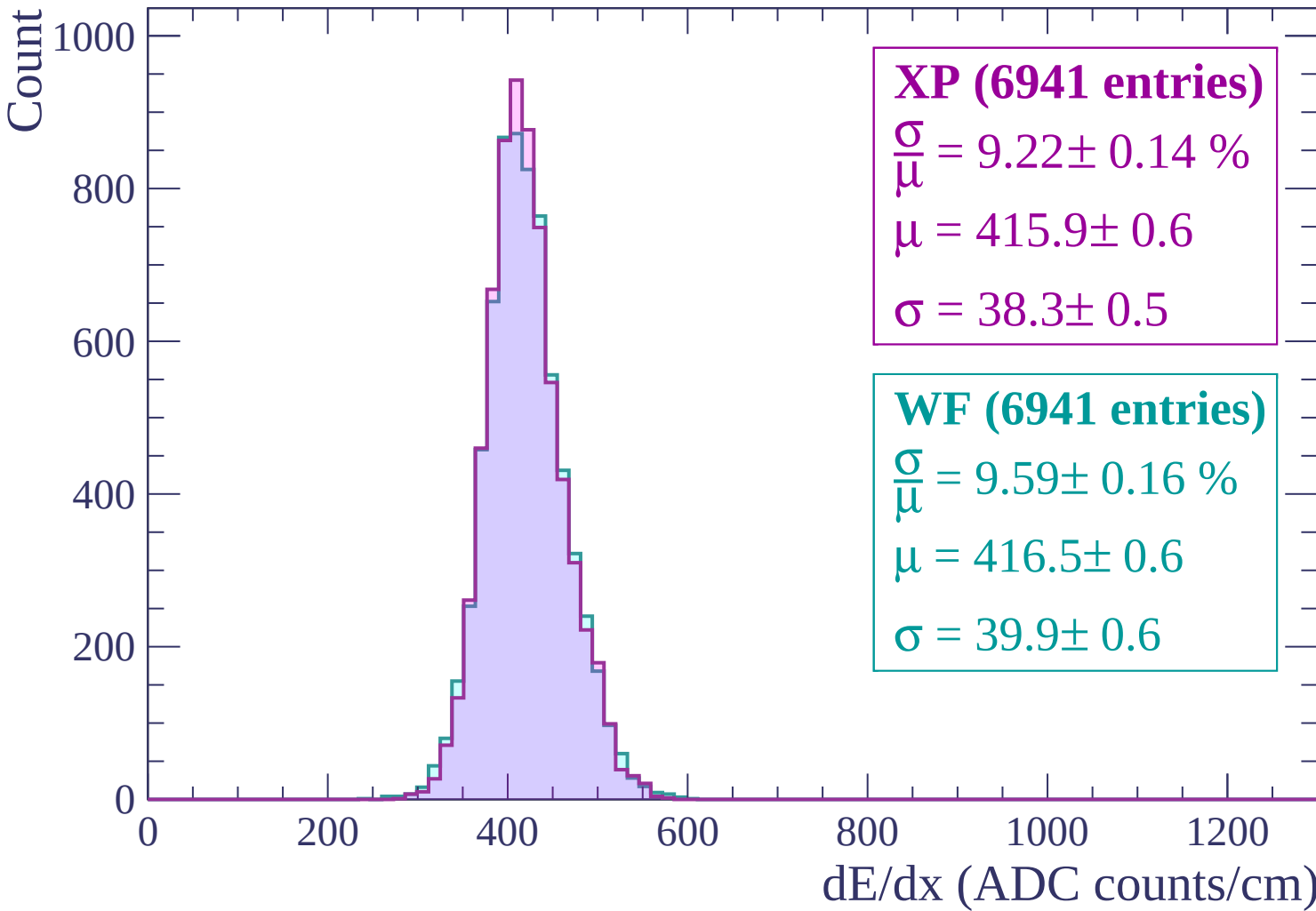
$\sigma = 38.3 \pm 0.5$

WF (6941 entries)

$\frac{\sigma}{\mu} = 9.59 \pm 0.16 \%$

$\mu = 416.5 \pm 0.6$

$\sigma = 39.9 \pm 0.6$



Energy loss | $-540 < p < -480$

Count

1000

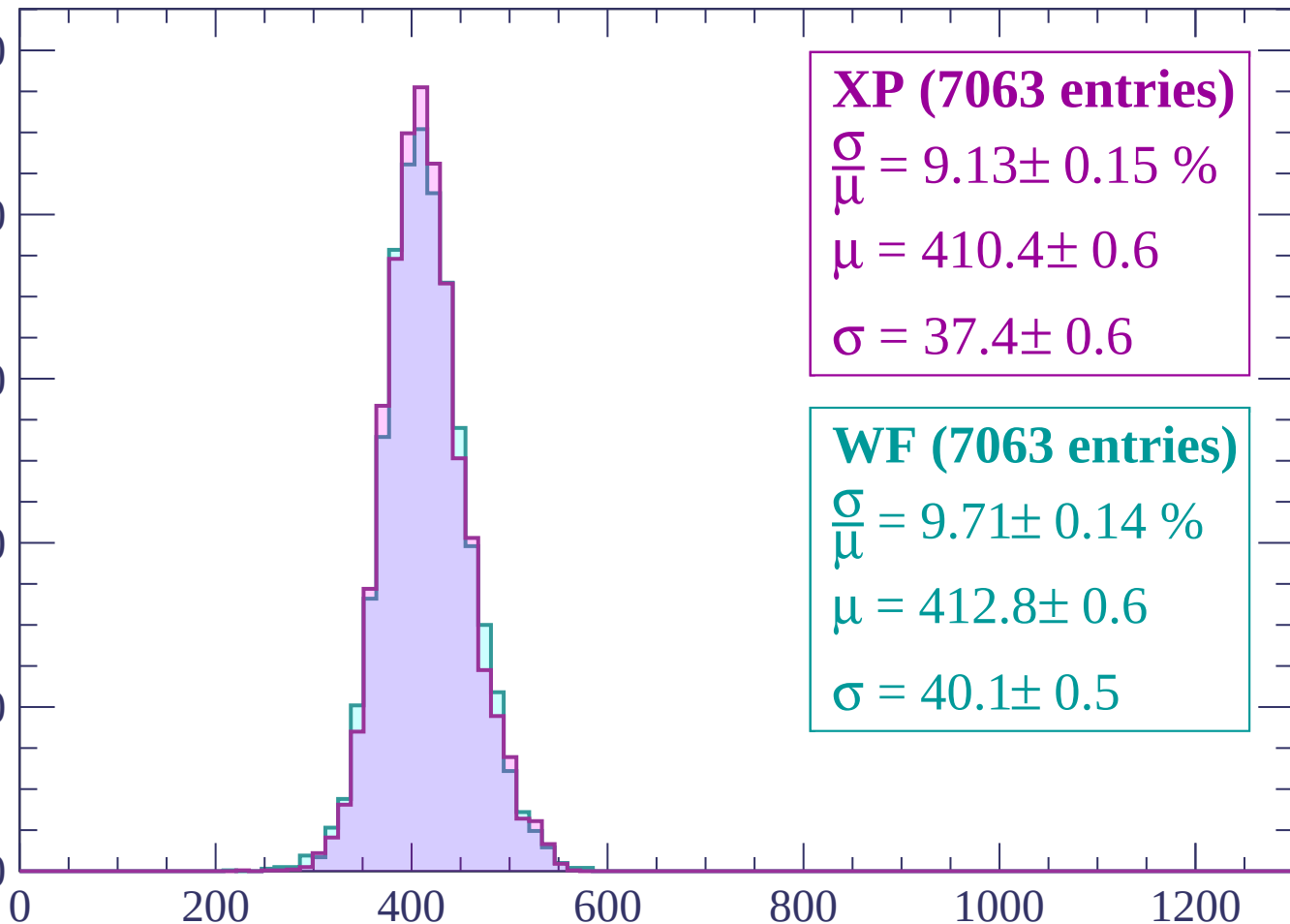
800

600

400

200

0



XP (7063 entries)

$\frac{\sigma}{\mu} = 9.13 \pm 0.15 \%$

$\mu = 410.4 \pm 0.6$

$\sigma = 37.4 \pm 0.6$

WF (7063 entries)

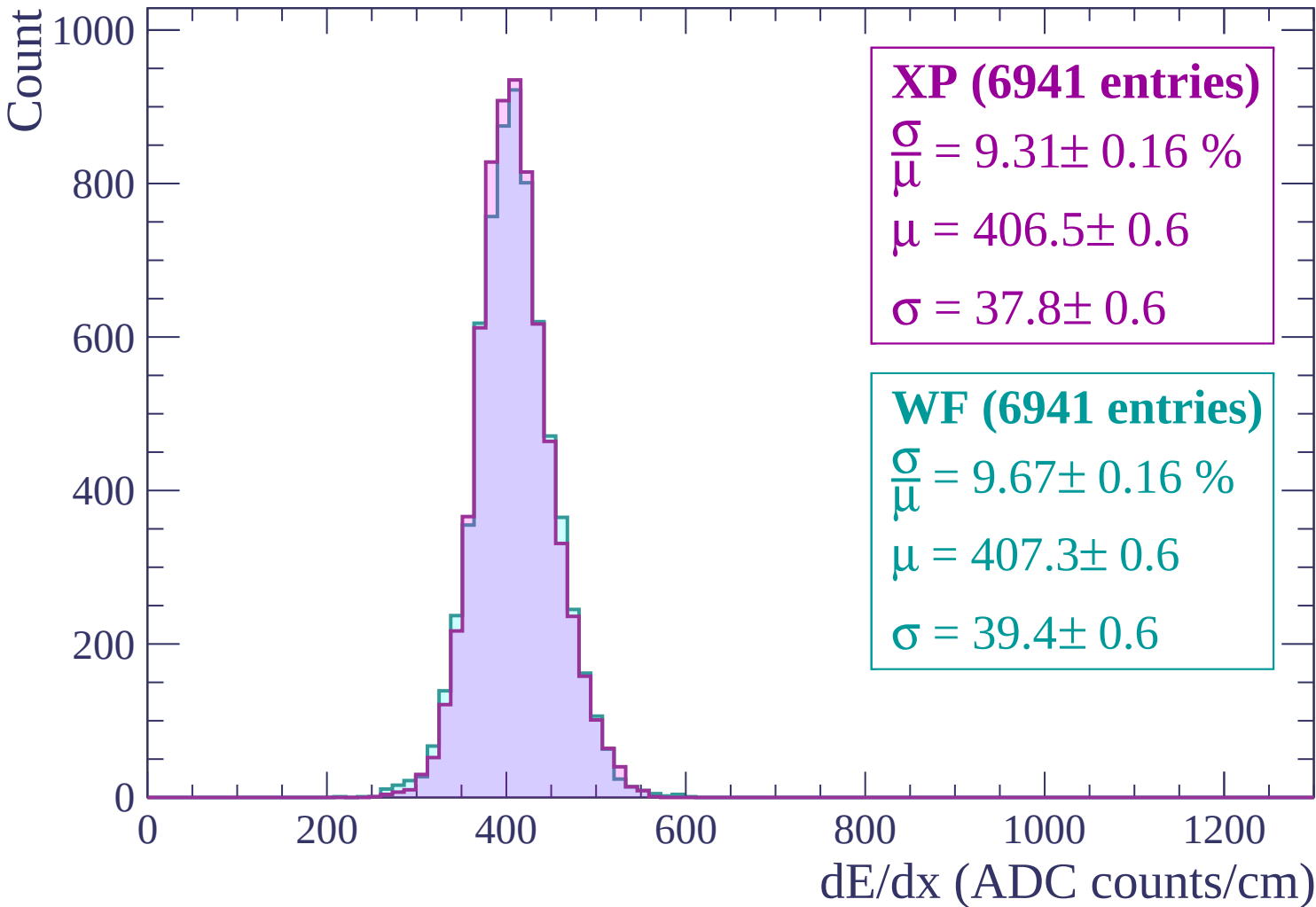
$\frac{\sigma}{\mu} = 9.71 \pm 0.14 \%$

$\mu = 412.8 \pm 0.6$

$\sigma = 40.1 \pm 0.5$

dE/dx (ADC counts/cm)

Energy loss | $-480 < p < -420$



Energy loss | $-420 < p < -360$

Count

1000

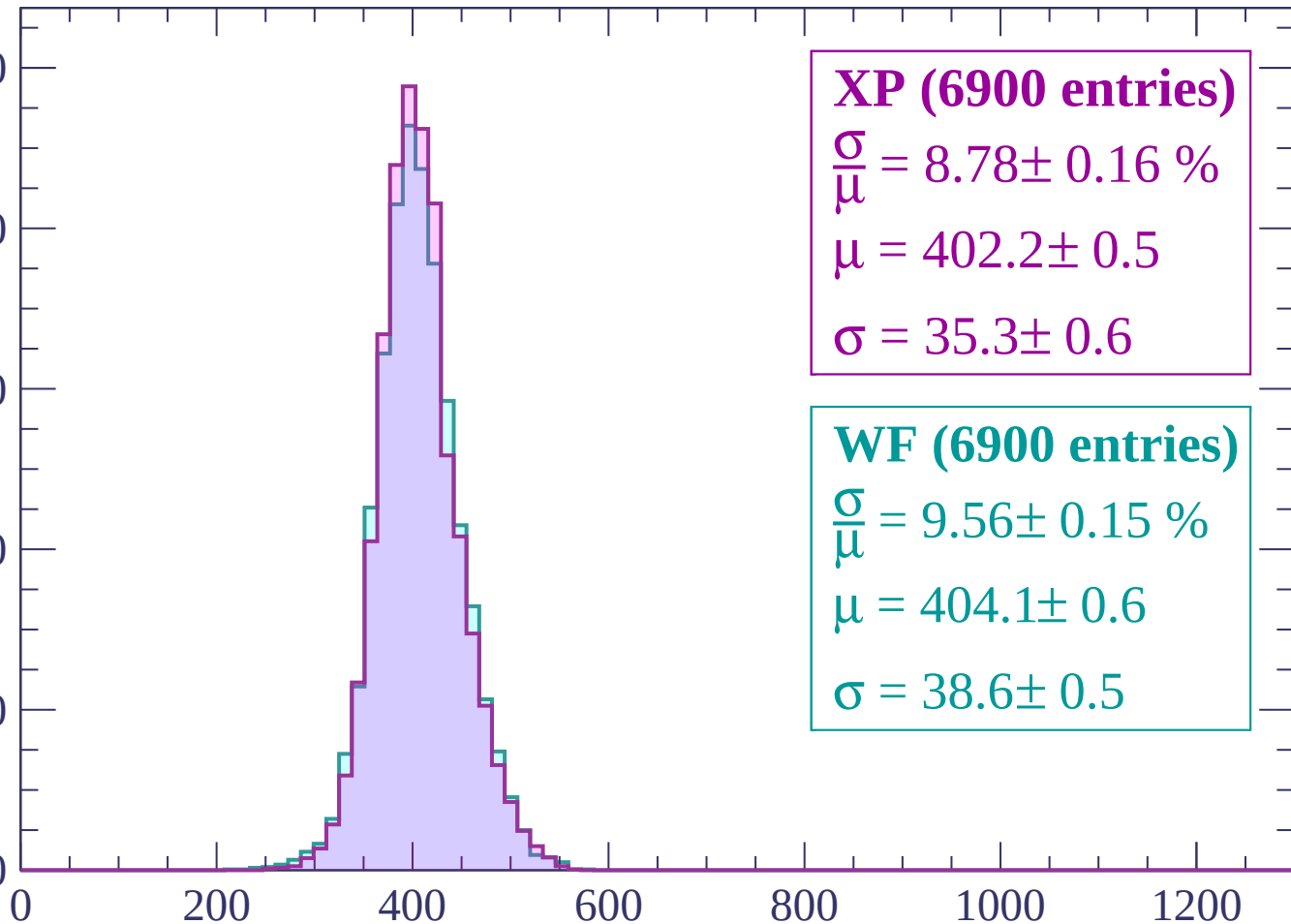
800

600

400

200

0



XP (6900 entries)

$\frac{\sigma}{\mu} = 8.78 \pm 0.16 \%$

$\mu = 402.2 \pm 0.5$

$\sigma = 35.3 \pm 0.6$

WF (6900 entries)

$\frac{\sigma}{\mu} = 9.56 \pm 0.15 \%$

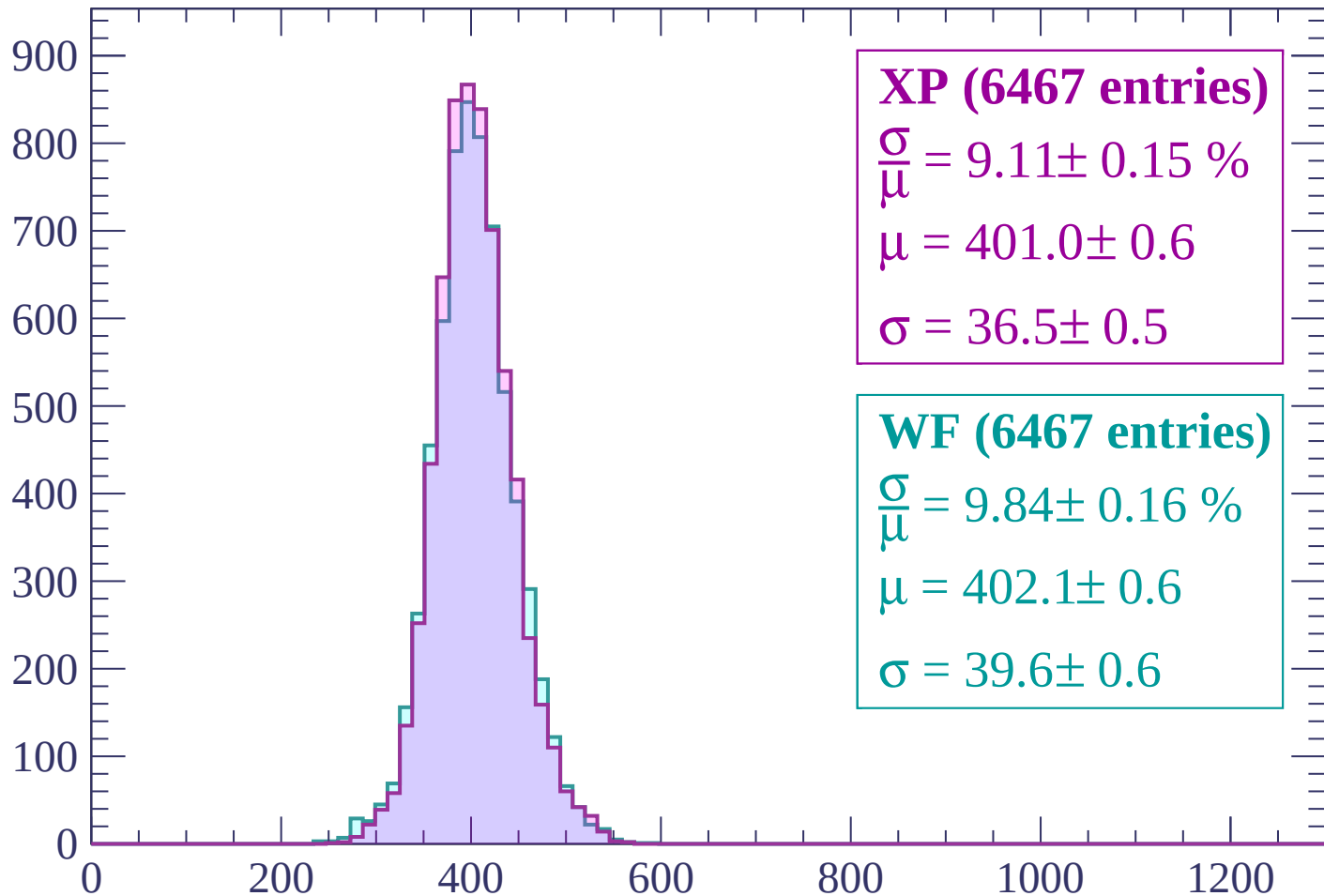
$\mu = 404.1 \pm 0.6$

$\sigma = 38.6 \pm 0.5$

dE/dx (ADC counts/cm)

Energy loss | $-360 < p < -300$

Count



XP (6467 entries)

$$\frac{\sigma}{\mu} = 9.11 \pm 0.15 \%$$

$$\mu = 401.0 \pm 0.6$$

$$\sigma = 36.5 \pm 0.5$$

WF (6467 entries)

$$\frac{\sigma}{\mu} = 9.84 \pm 0.16 \%$$

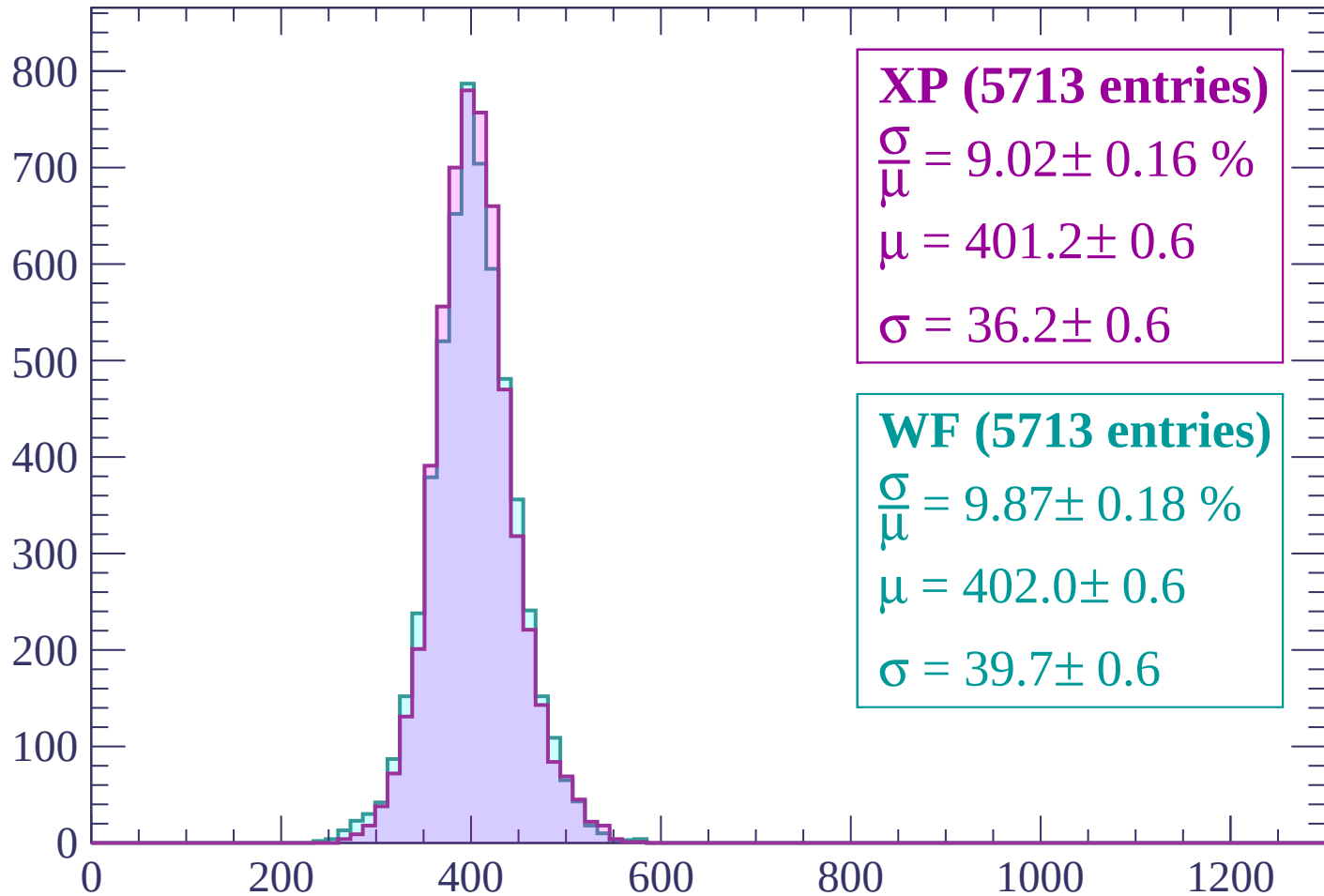
$$\mu = 402.1 \pm 0.6$$

$$\sigma = 39.6 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $-300 < p < -240$

Count



XP (5713 entries)

$\frac{\sigma}{\mu} = 9.02 \pm 0.16 \%$

$\mu = 401.2 \pm 0.6$

$\sigma = 36.2 \pm 0.6$

WF (5713 entries)

$\frac{\sigma}{\mu} = 9.87 \pm 0.18 \%$

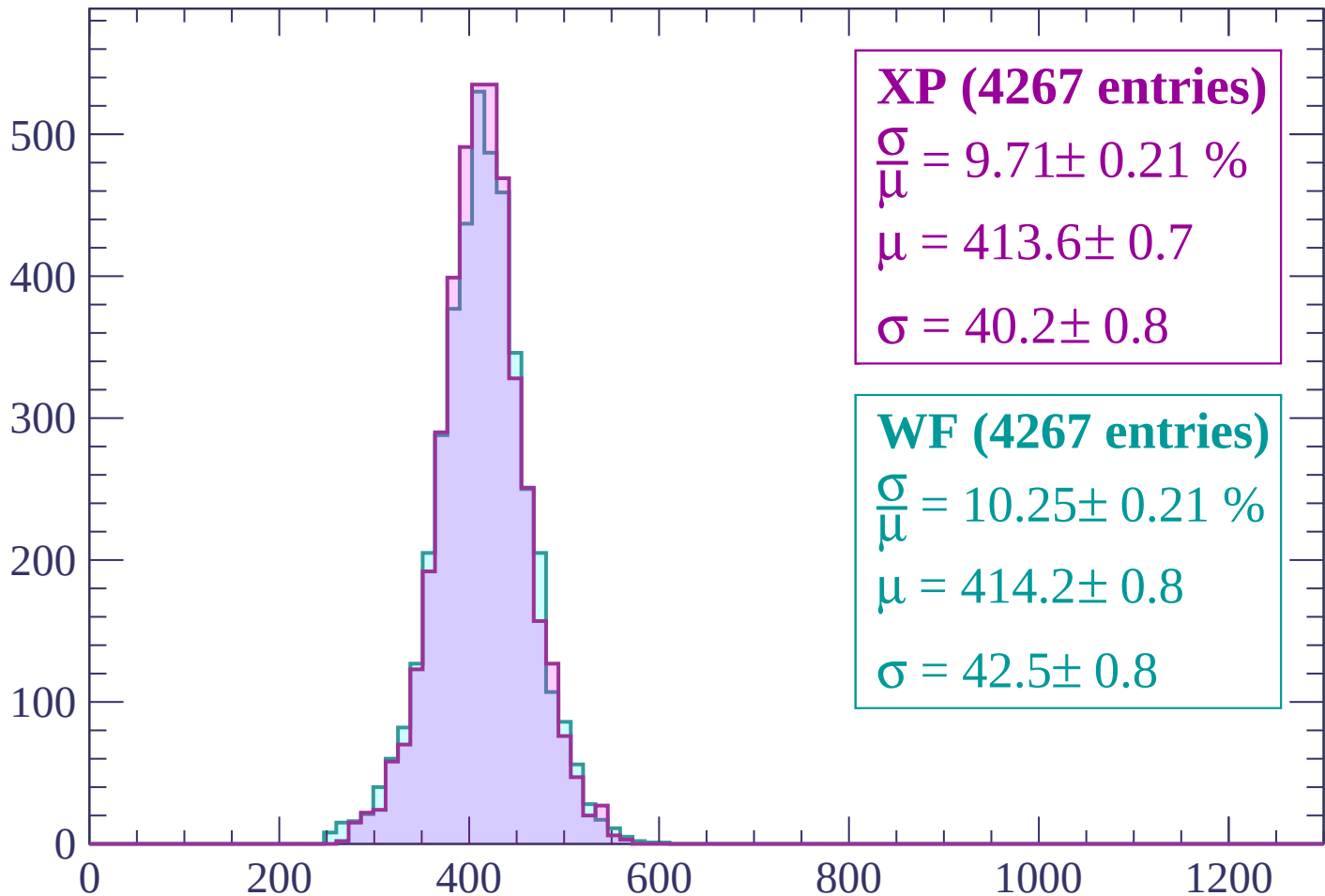
$\mu = 402.0 \pm 0.6$

$\sigma = 39.7 \pm 0.6$

dE/dx (ADC counts/cm)

Energy loss | $-240 < p < -180$

Count



XP (4267 entries)

$$\frac{\sigma}{\mu} = 9.71 \pm 0.21 \%$$

$$\mu = 413.6 \pm 0.7$$

$$\sigma = 40.2 \pm 0.8$$

WF (4267 entries)

$$\frac{\sigma}{\mu} = 10.25 \pm 0.21 \%$$

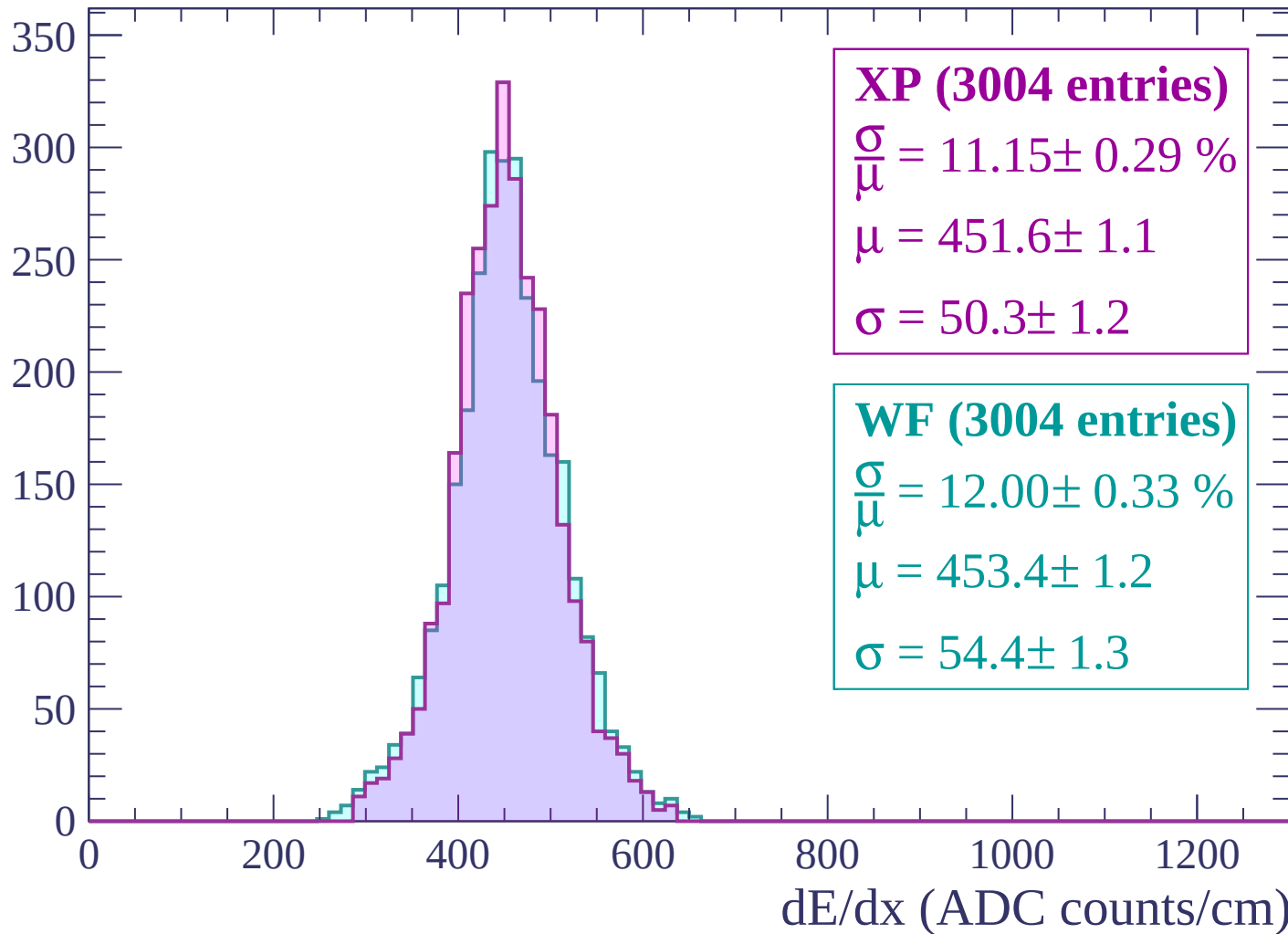
$$\mu = 414.2 \pm 0.8$$

$$\sigma = 42.5 \pm 0.8$$

dE/dx (ADC counts/cm)

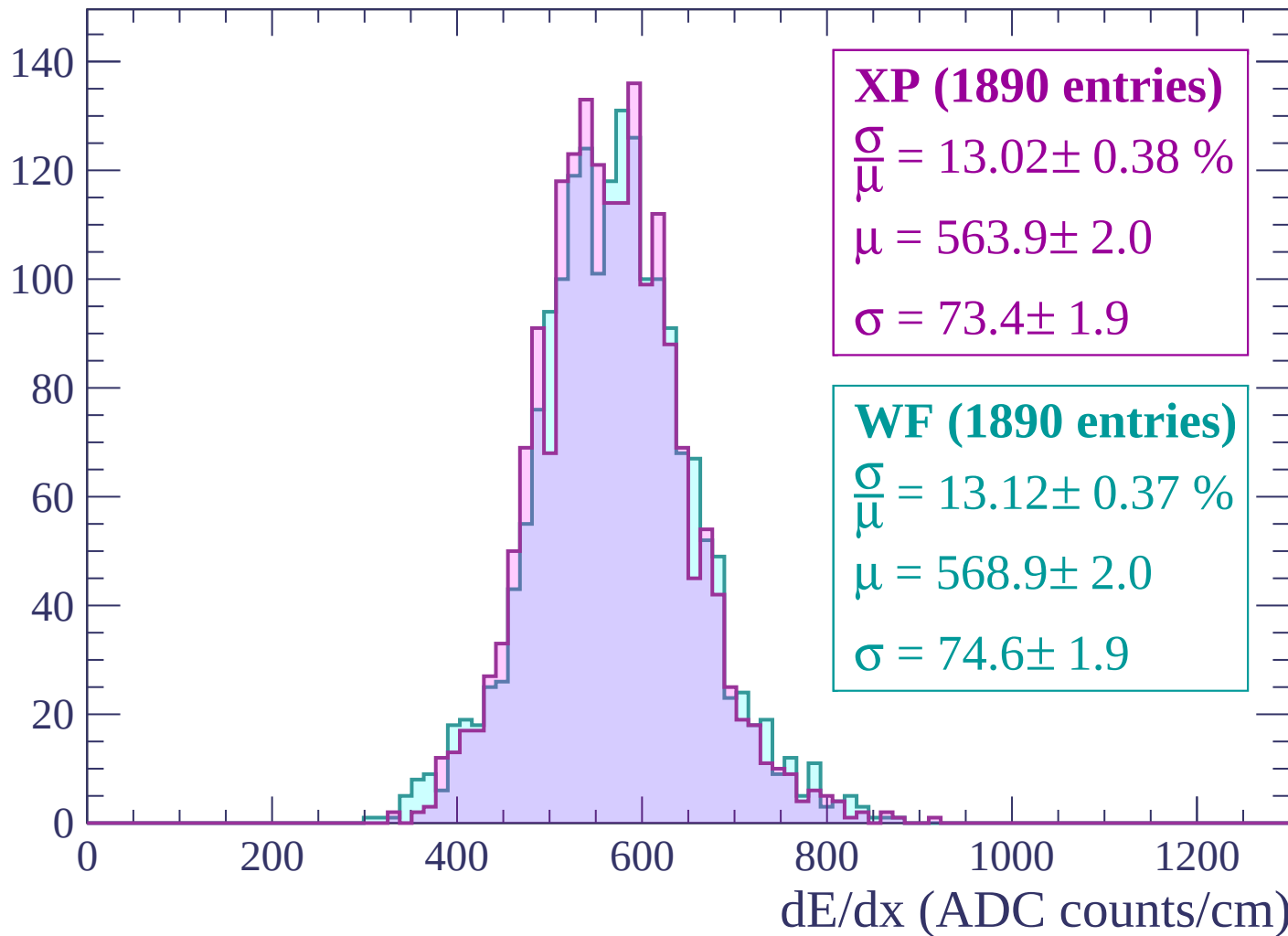
Energy loss | $-180 < p < -120$

Count



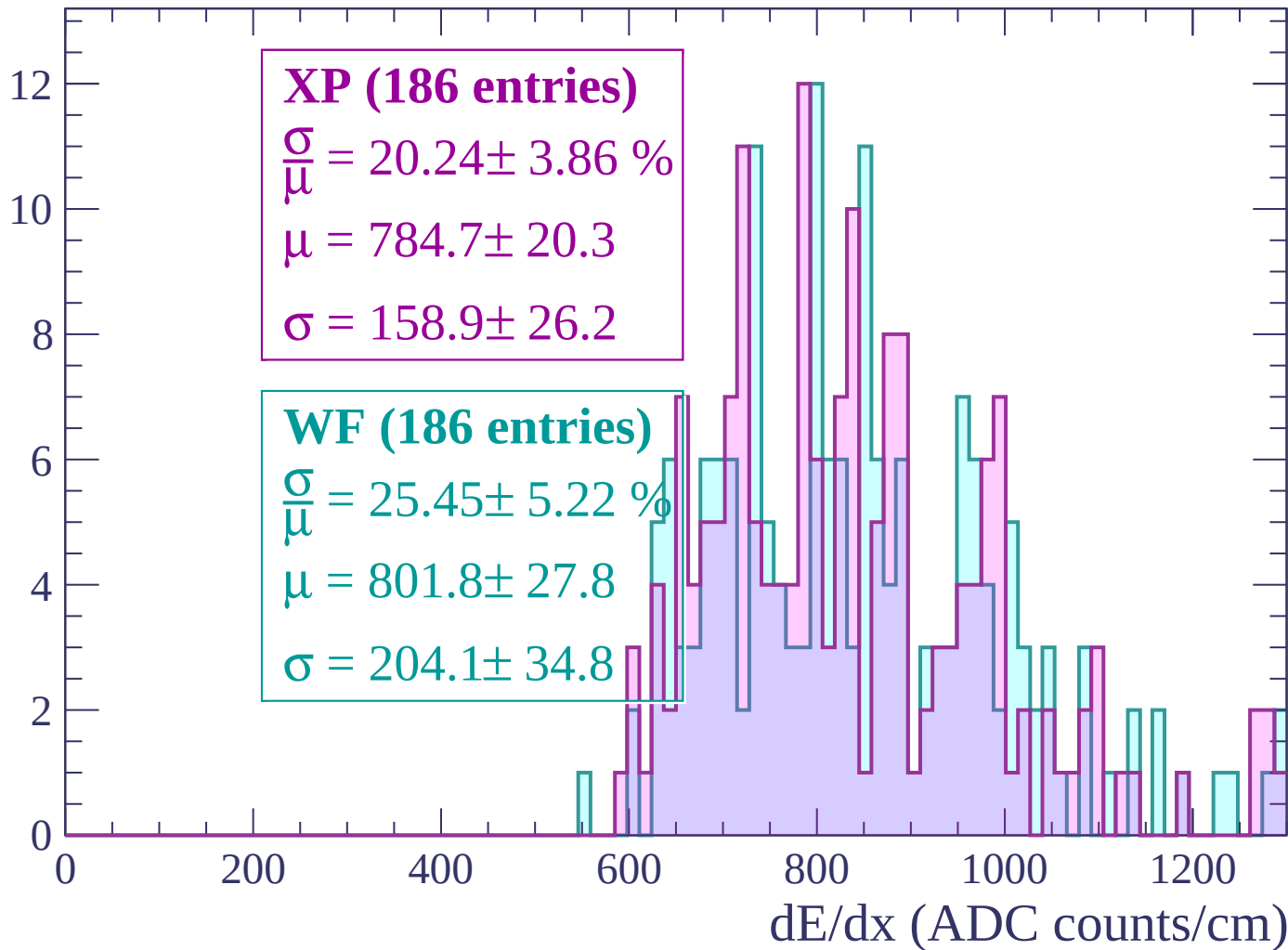
Energy loss | $-120 < p < -60$

Count



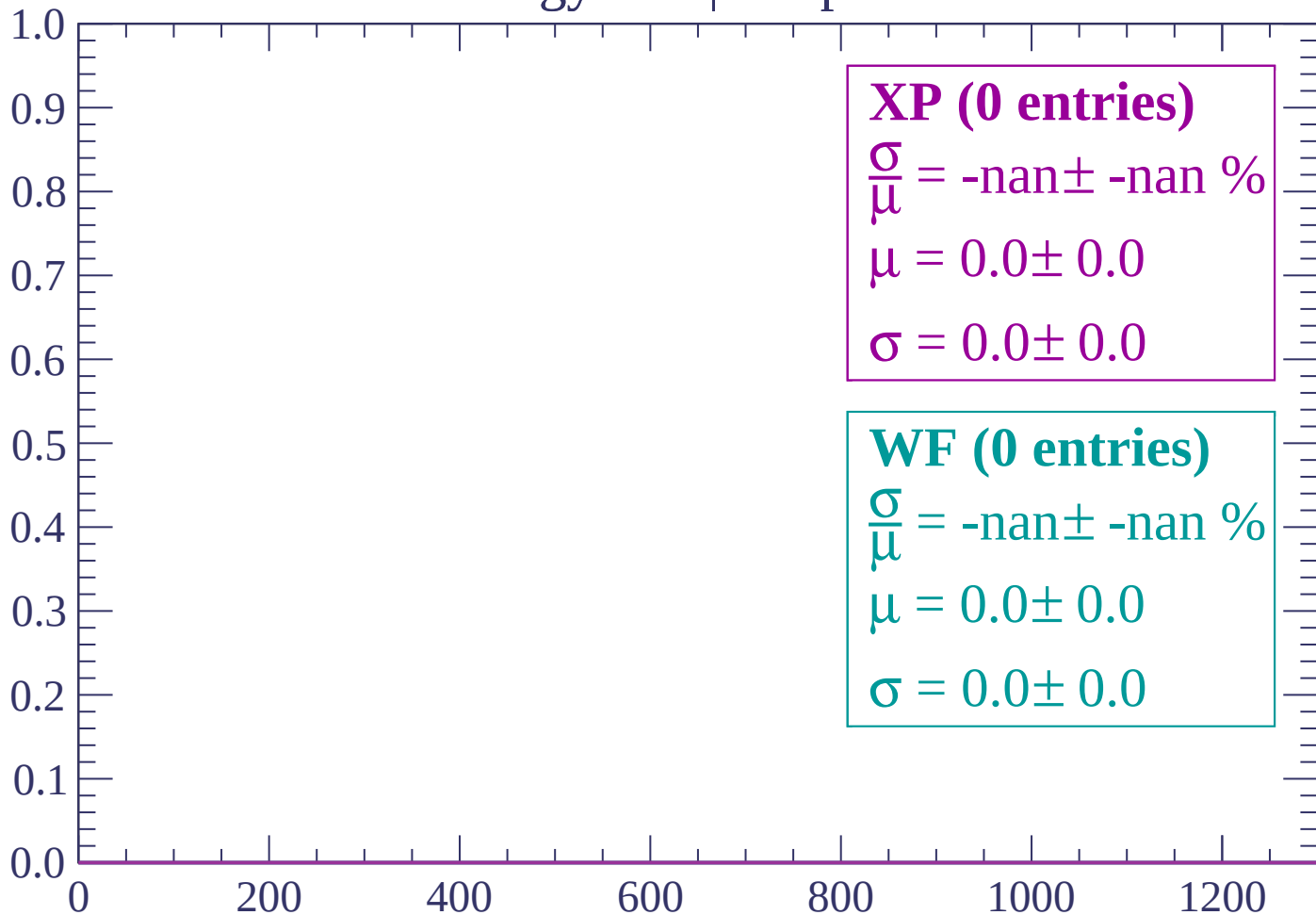
Energy loss | $-60 < p < 0$

Count



Energy loss | $0 < p < 60$

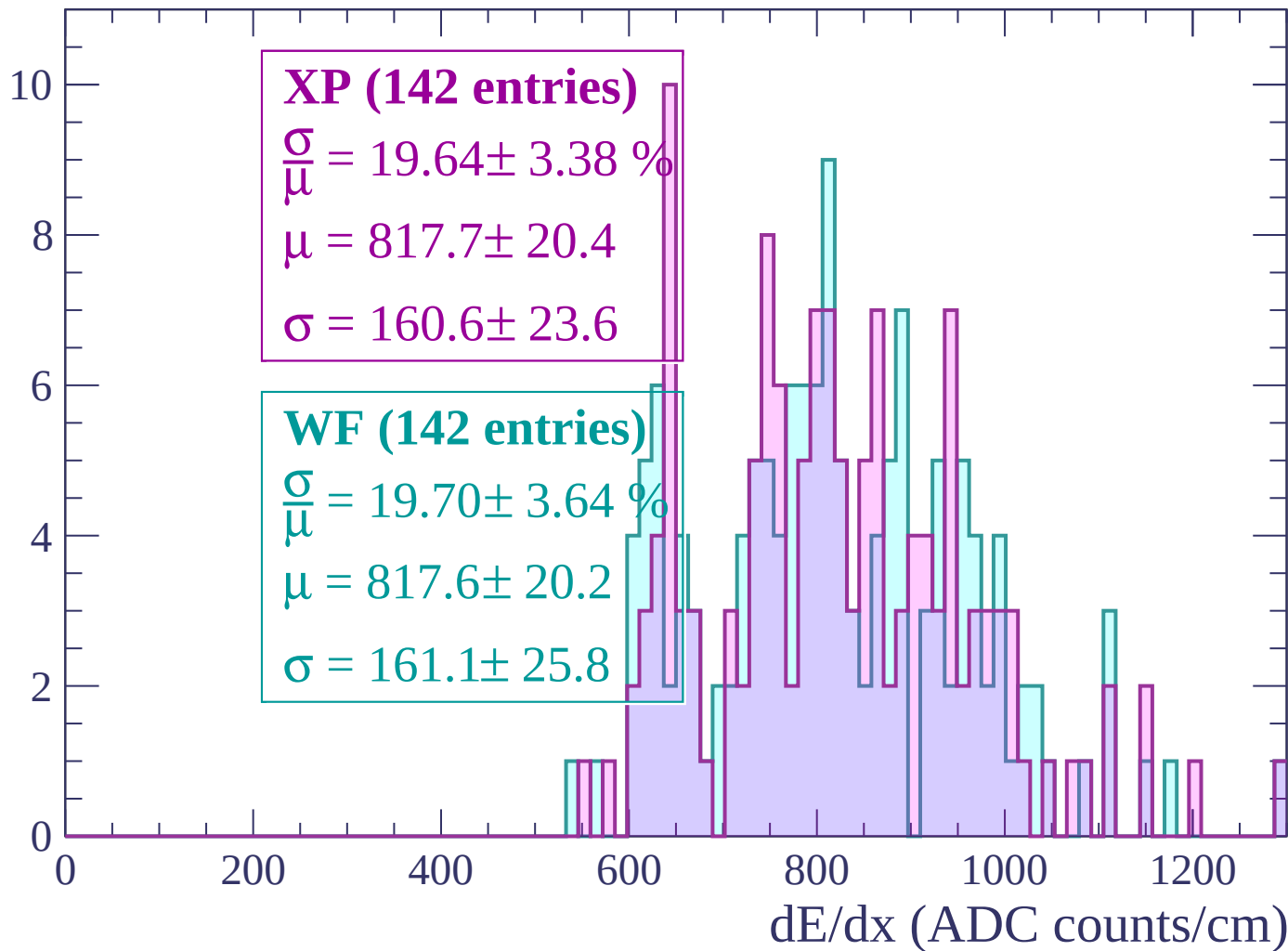
Count



dE/dx (ADC counts/cm)

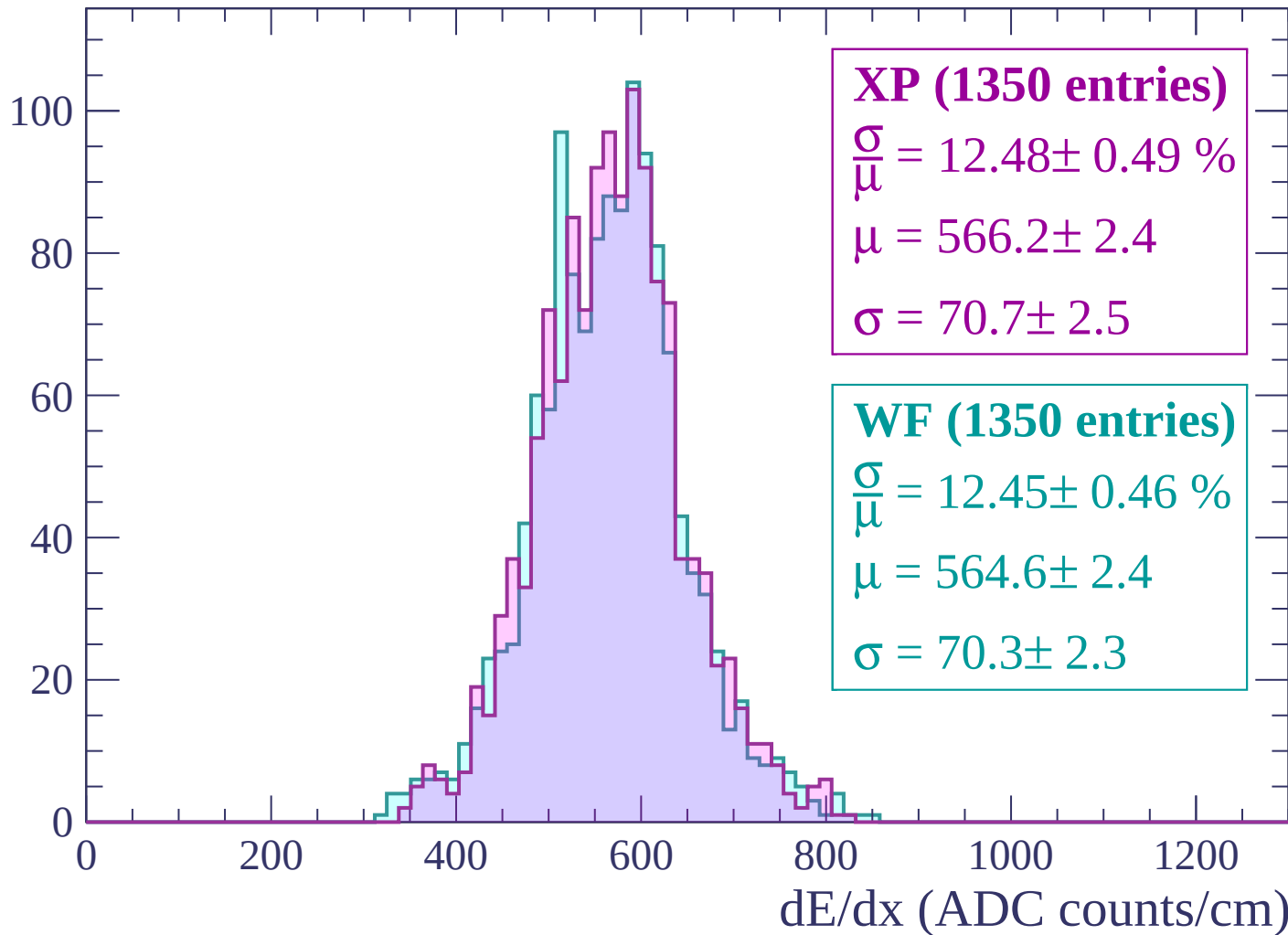
Energy loss | $60 < p < 120$

Count



Energy loss | $120 < p < 180$

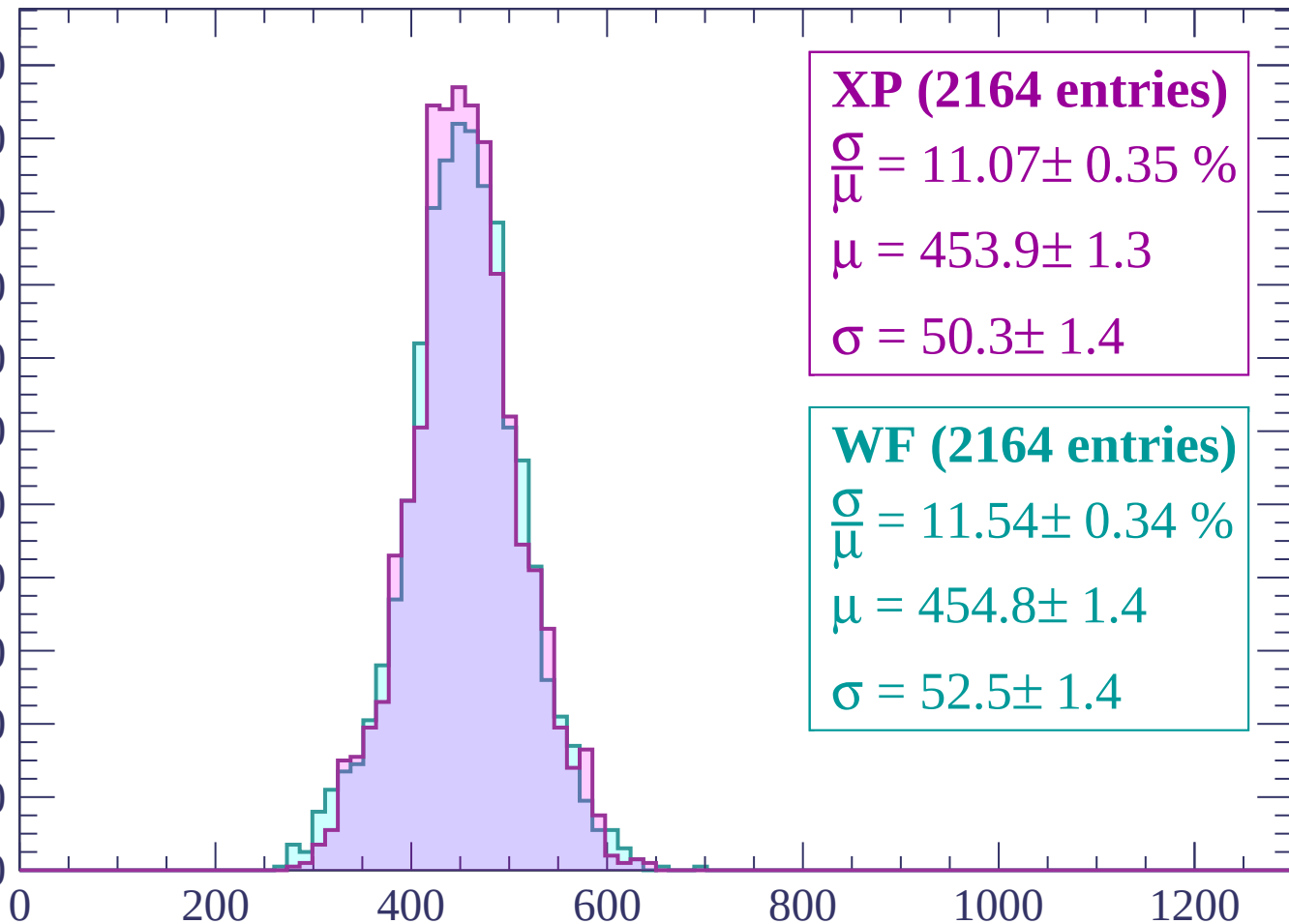
Count



Energy loss | $180 < p < 240$

Count

220
200
180
160
140
120
100
80
60
40
20
0



XP (2164 entries)

$\frac{\sigma}{\mu} = 11.07 \pm 0.35 \%$

$\mu = 453.9 \pm 1.3$

$\sigma = 50.3 \pm 1.4$

WF (2164 entries)

$\frac{\sigma}{\mu} = 11.54 \pm 0.34 \%$

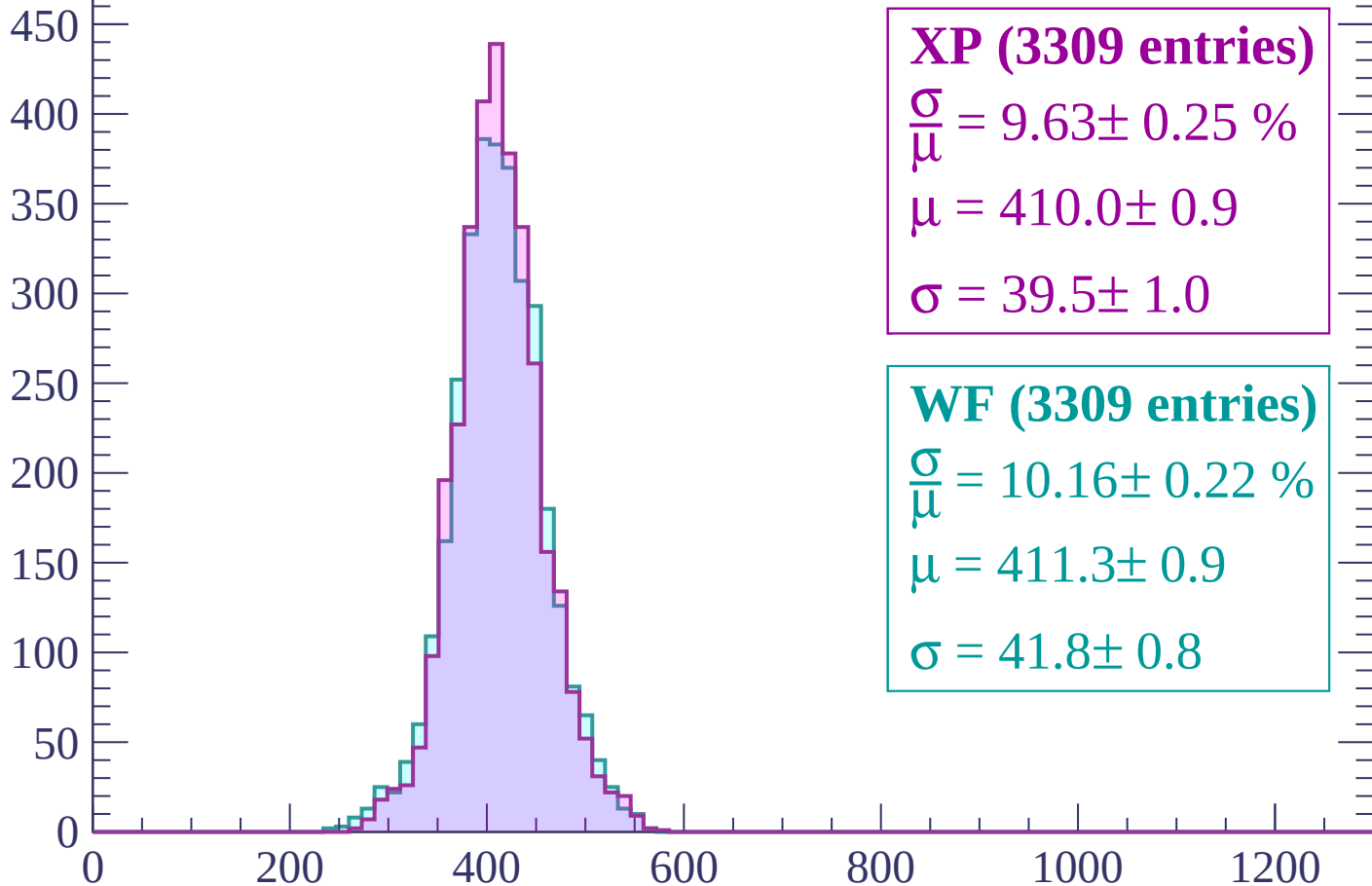
$\mu = 454.8 \pm 1.4$

$\sigma = 52.5 \pm 1.4$

dE/dx (ADC counts/cm)

Energy loss | $240 < p < 300$

Count



XP (3309 entries)

$$\frac{\sigma}{\mu} = 9.63 \pm 0.25 \%$$

$$\mu = 410.0 \pm 0.9$$

$$\sigma = 39.5 \pm 1.0$$

WF (3309 entries)

$$\frac{\sigma}{\mu} = 10.16 \pm 0.22 \%$$

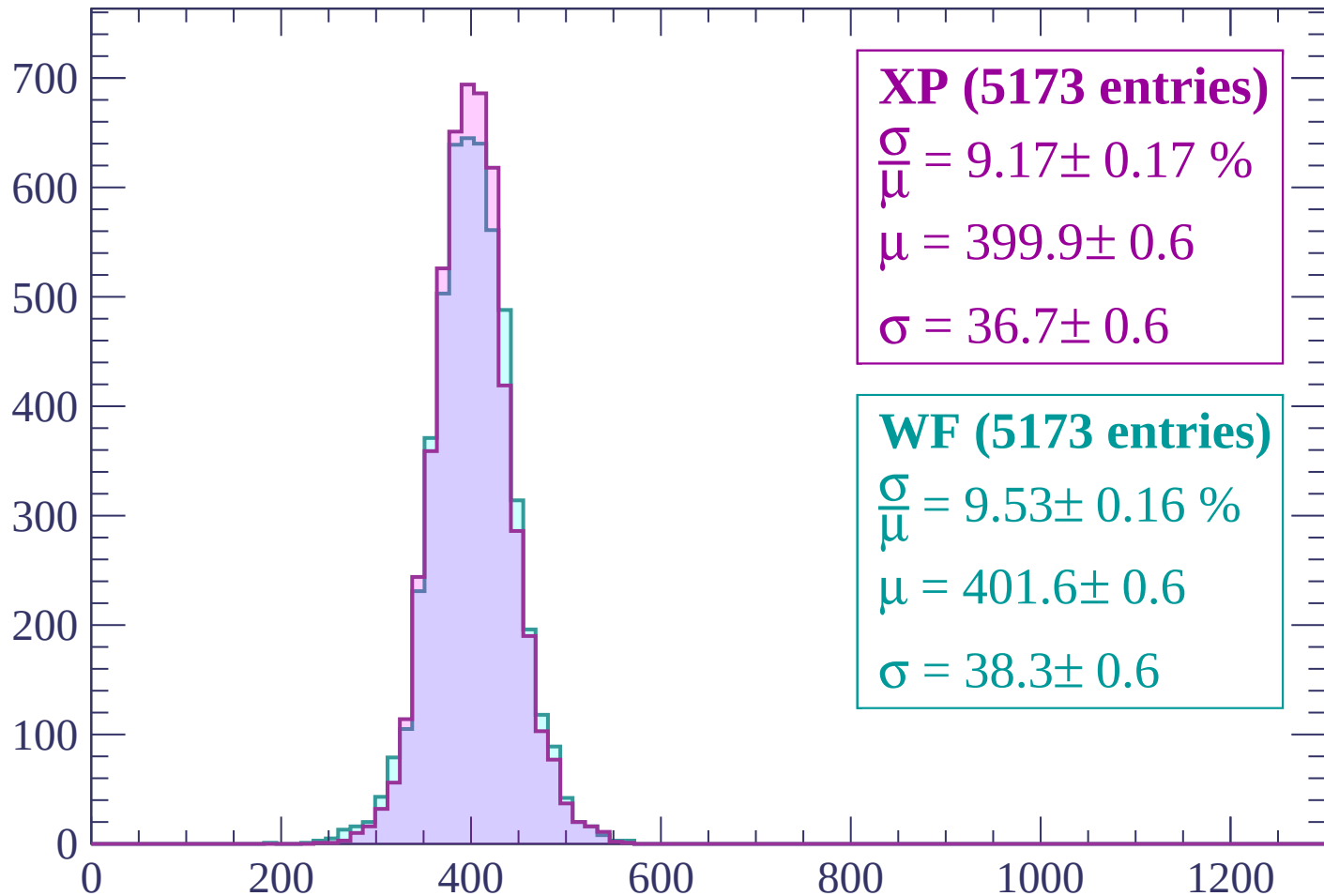
$$\mu = 411.3 \pm 0.9$$

$$\sigma = 41.8 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $300 < p < 360$

Count



XP (5173 entries)

$\frac{\sigma}{\mu} = 9.17 \pm 0.17 \%$

$\mu = 399.9 \pm 0.6$

$\sigma = 36.7 \pm 0.6$

WF (5173 entries)

$\frac{\sigma}{\mu} = 9.53 \pm 0.16 \%$

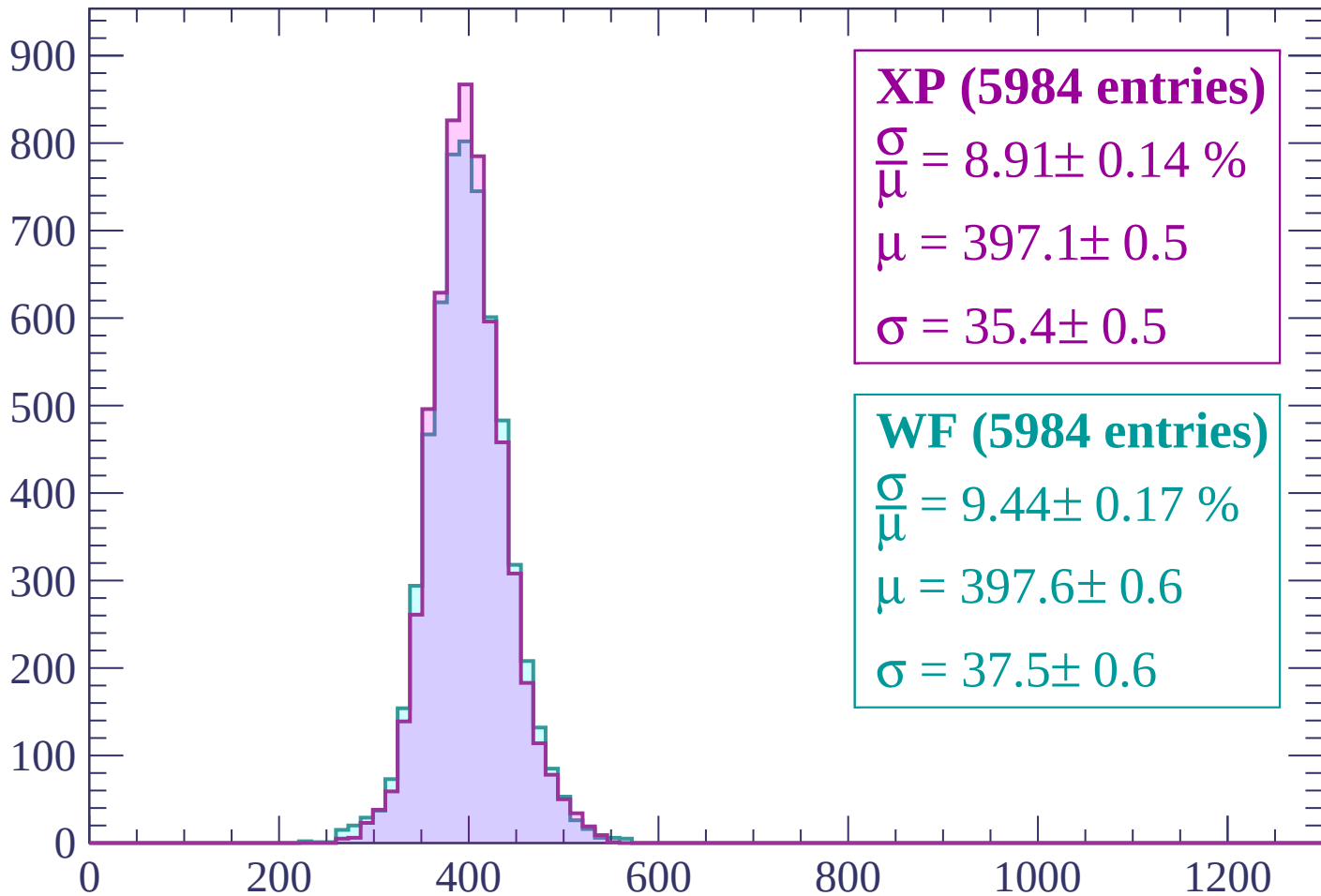
$\mu = 401.6 \pm 0.6$

$\sigma = 38.3 \pm 0.6$

dE/dx (ADC counts/cm)

Energy loss | $360 < p < 420$

Count



XP (5984 entries)

$\frac{\sigma}{\mu} = 8.91 \pm 0.14 \%$

$\mu = 397.1 \pm 0.5$

$\sigma = 35.4 \pm 0.5$

WF (5984 entries)

$\frac{\sigma}{\mu} = 9.44 \pm 0.17 \%$

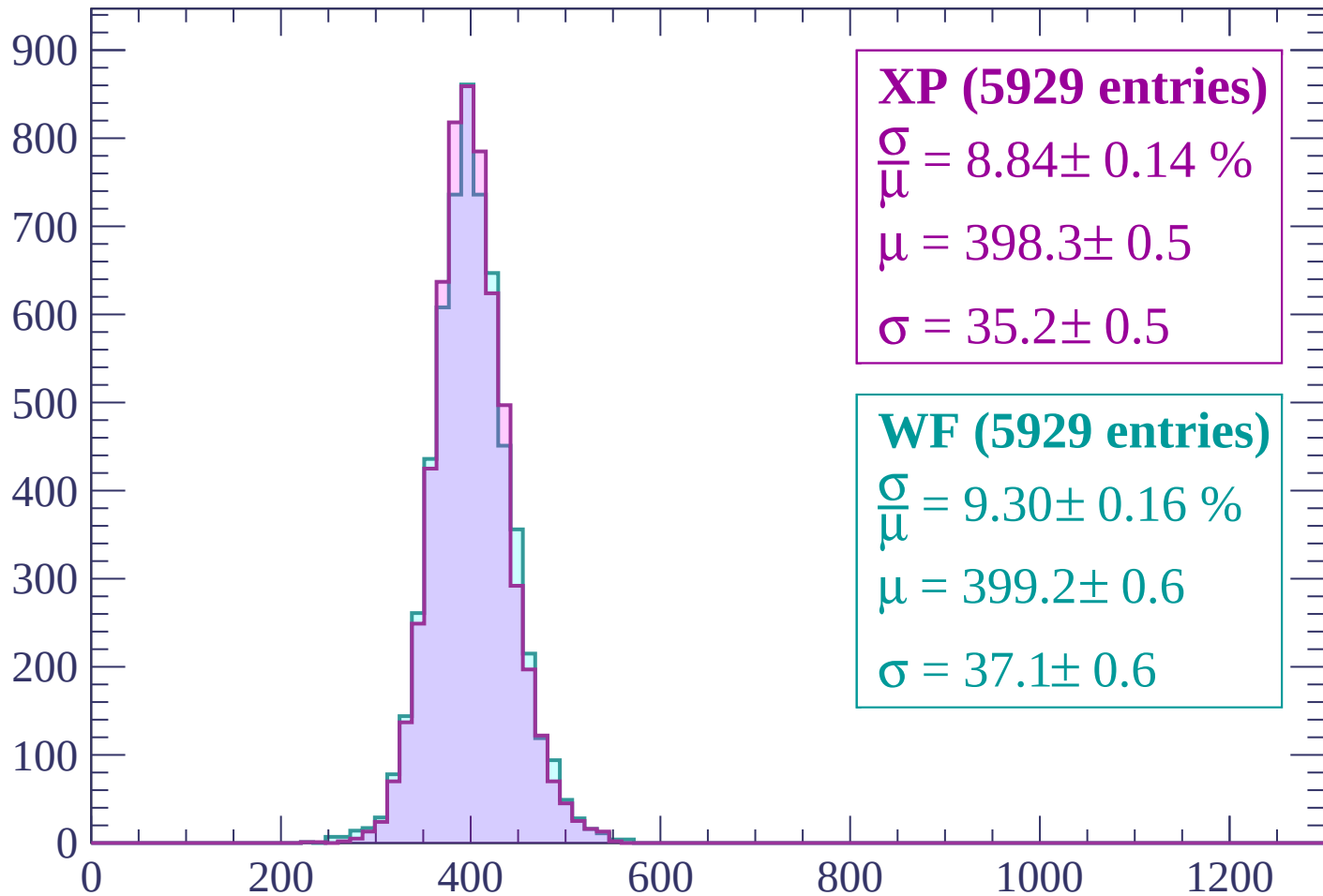
$\mu = 397.6 \pm 0.6$

$\sigma = 37.5 \pm 0.6$

dE/dx (ADC counts/cm)

Energy loss | $420 < p < 480$

Count



XP (5929 entries)

$$\frac{\sigma}{\mu} = 8.84 \pm 0.14 \%$$

$$\mu = 398.3 \pm 0.5$$

$$\sigma = 35.2 \pm 0.5$$

WF (5929 entries)

$$\frac{\sigma}{\mu} = 9.30 \pm 0.16 \%$$

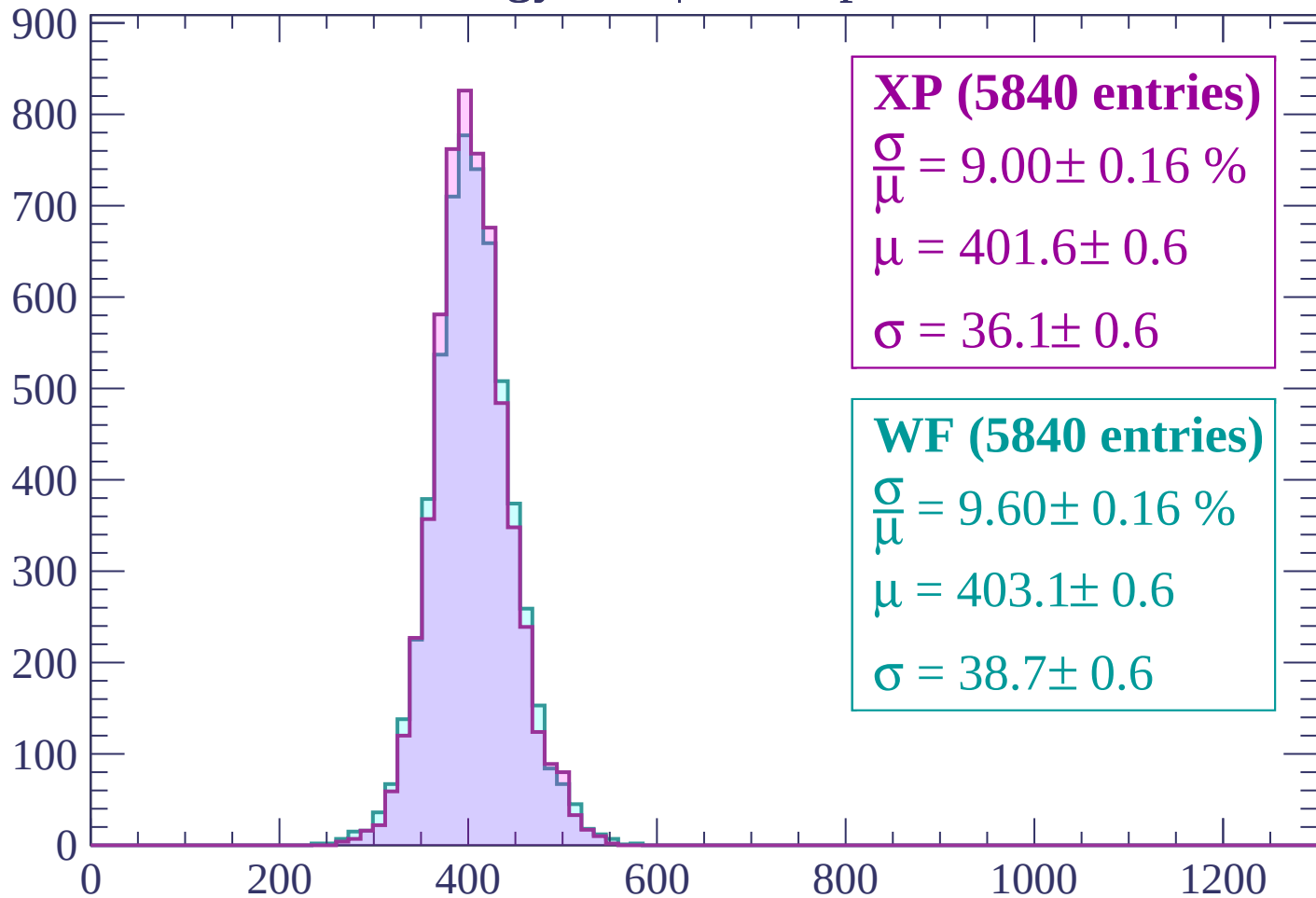
$$\mu = 399.2 \pm 0.6$$

$$\sigma = 37.1 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $480 < p < 540$

Count



XP (5840 entries)

$$\frac{\sigma}{\mu} = 9.00 \pm 0.16 \%$$

$$\mu = 401.6 \pm 0.6$$

$$\sigma = 36.1 \pm 0.6$$

WF (5840 entries)

$$\frac{\sigma}{\mu} = 9.60 \pm 0.16 \%$$

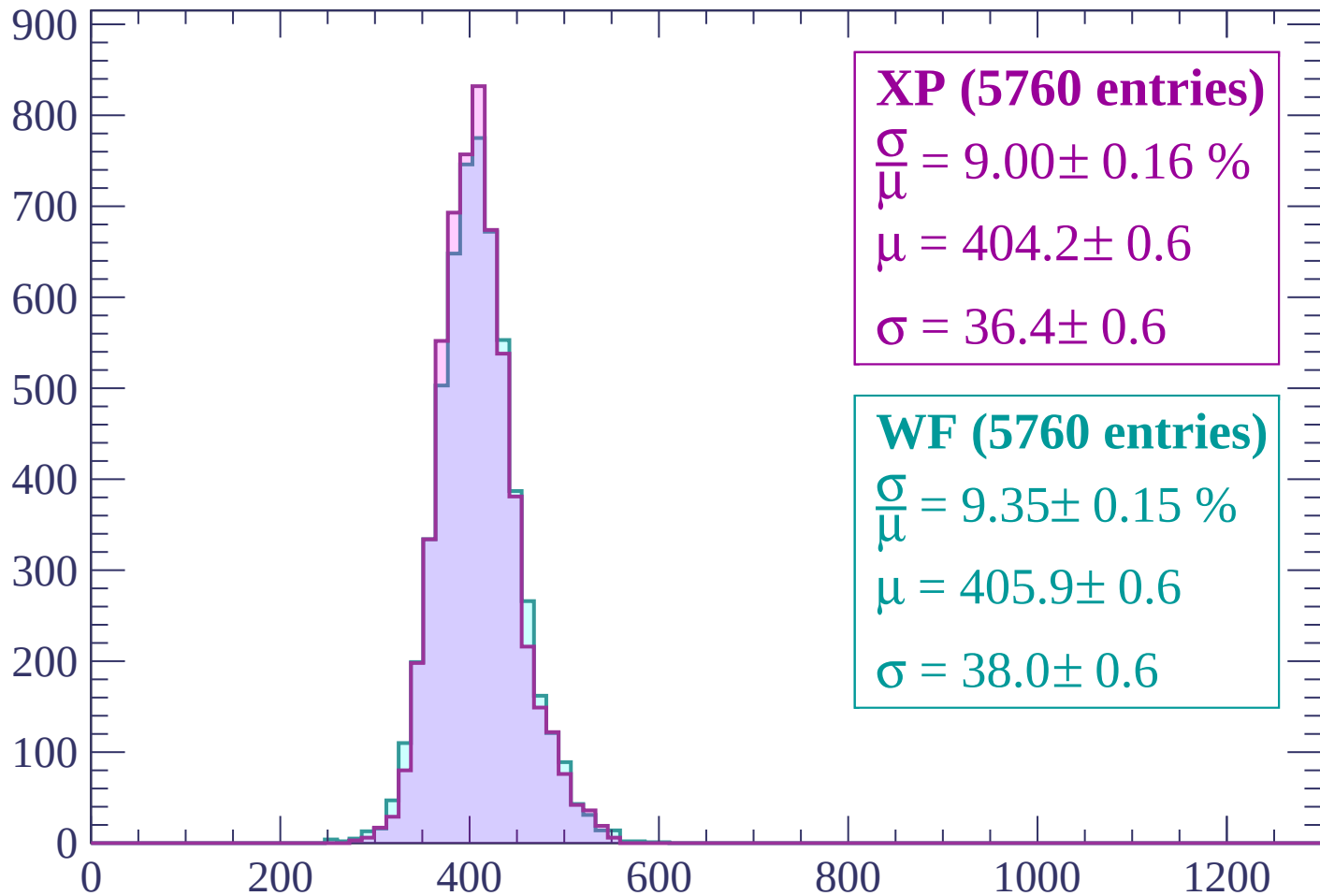
$$\mu = 403.1 \pm 0.6$$

$$\sigma = 38.7 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $540 < p < 600$

Count



XP (5760 entries)

$\frac{\sigma}{\mu} = 9.00 \pm 0.16 \%$

$\mu = 404.2 \pm 0.6$

$\sigma = 36.4 \pm 0.6$

WF (5760 entries)

$\frac{\sigma}{\mu} = 9.35 \pm 0.15 \%$

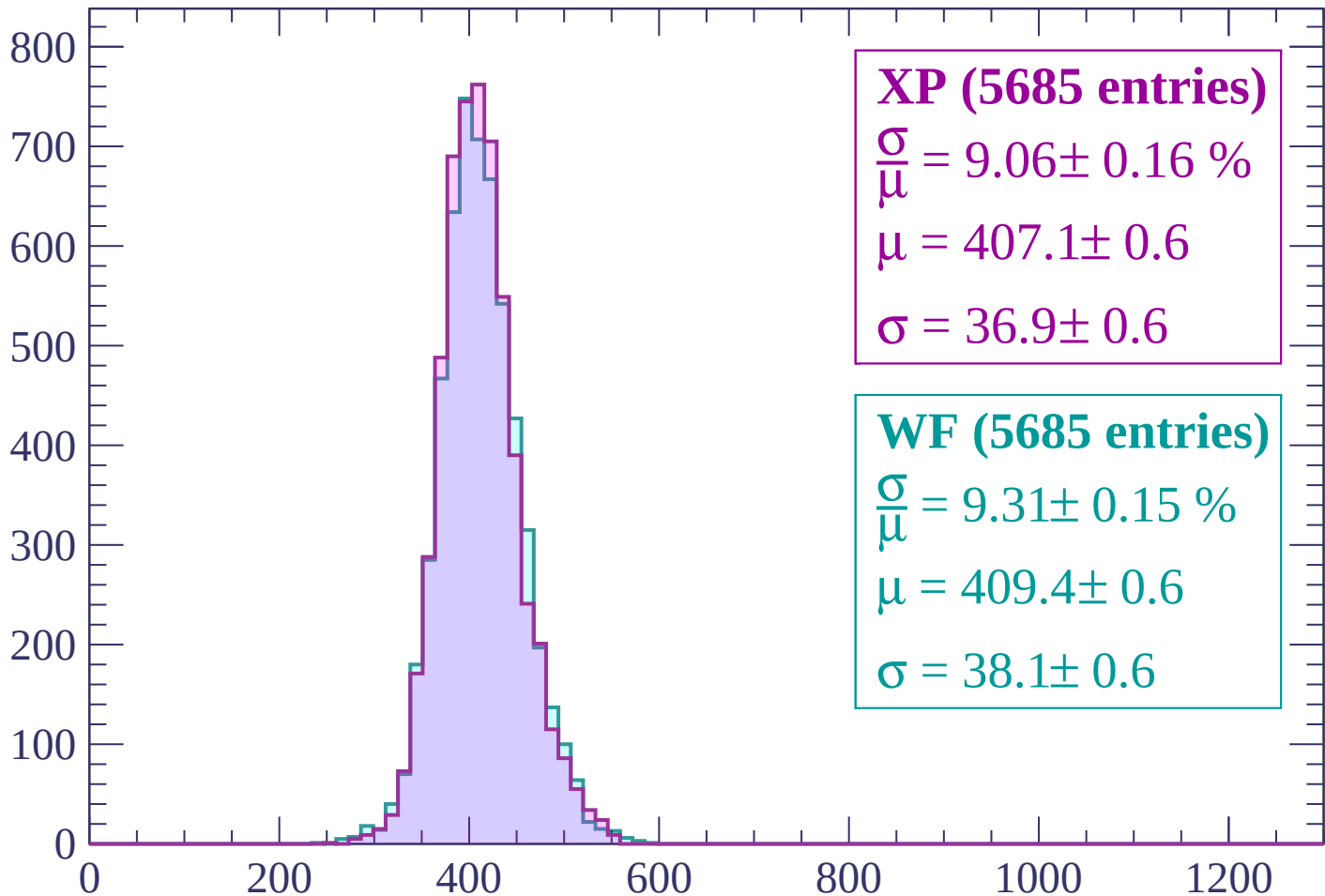
$\mu = 405.9 \pm 0.6$

$\sigma = 38.0 \pm 0.6$

dE/dx (ADC counts/cm)

Energy loss | $600 < p < 660$

Count



XP (5685 entries)

$$\frac{\sigma}{\mu} = 9.06 \pm 0.16 \%$$

$$\mu = 407.1 \pm 0.6$$

$$\sigma = 36.9 \pm 0.6$$

WF (5685 entries)

$$\frac{\sigma}{\mu} = 9.31 \pm 0.15 \%$$

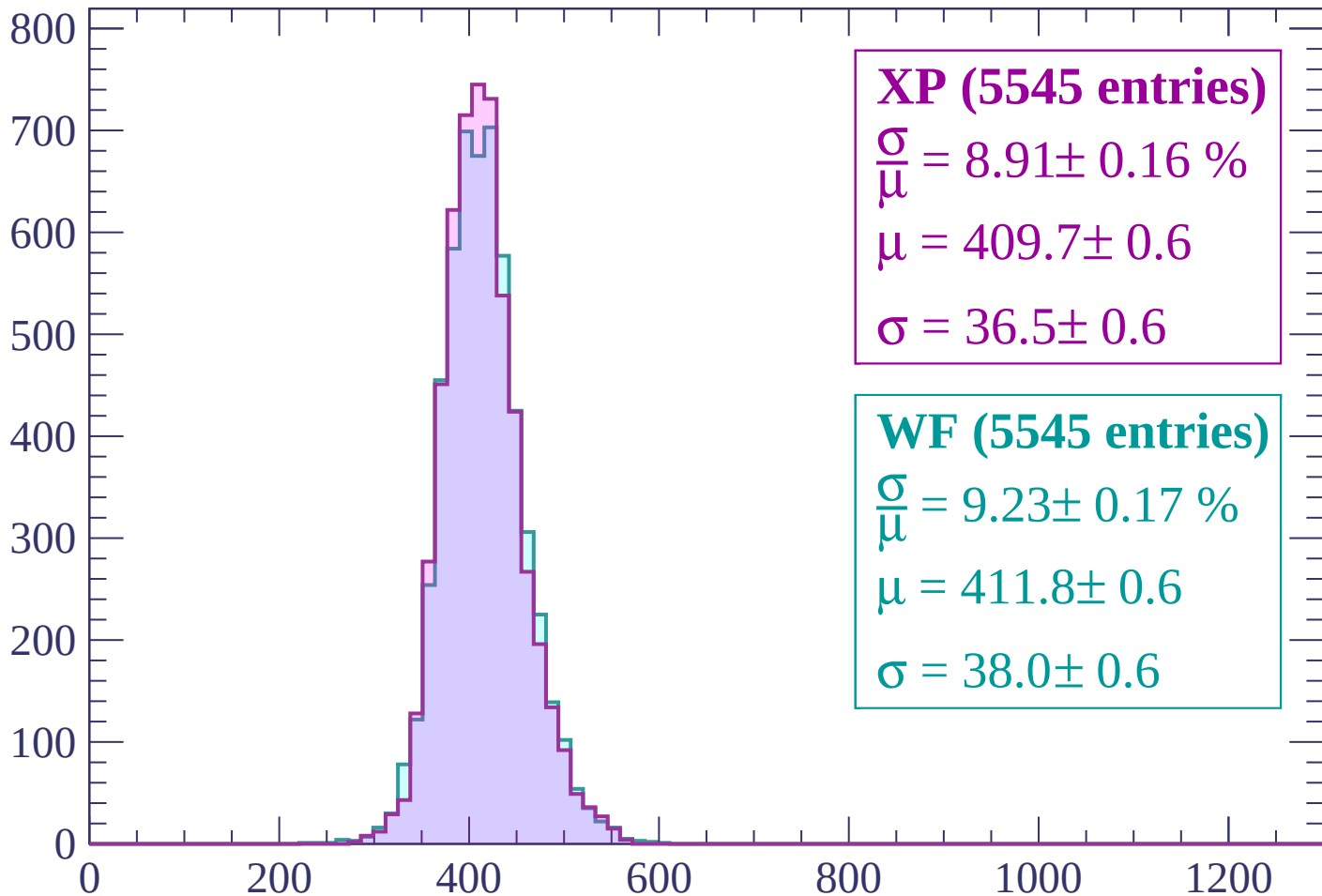
$$\mu = 409.4 \pm 0.6$$

$$\sigma = 38.1 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $660 < p < 720$

Count



XP (5545 entries)

$$\frac{\sigma}{\mu} = 8.91 \pm 0.16 \%$$

$$\mu = 409.7 \pm 0.6$$

$$\sigma = 36.5 \pm 0.6$$

WF (5545 entries)

$$\frac{\sigma}{\mu} = 9.23 \pm 0.17 \%$$

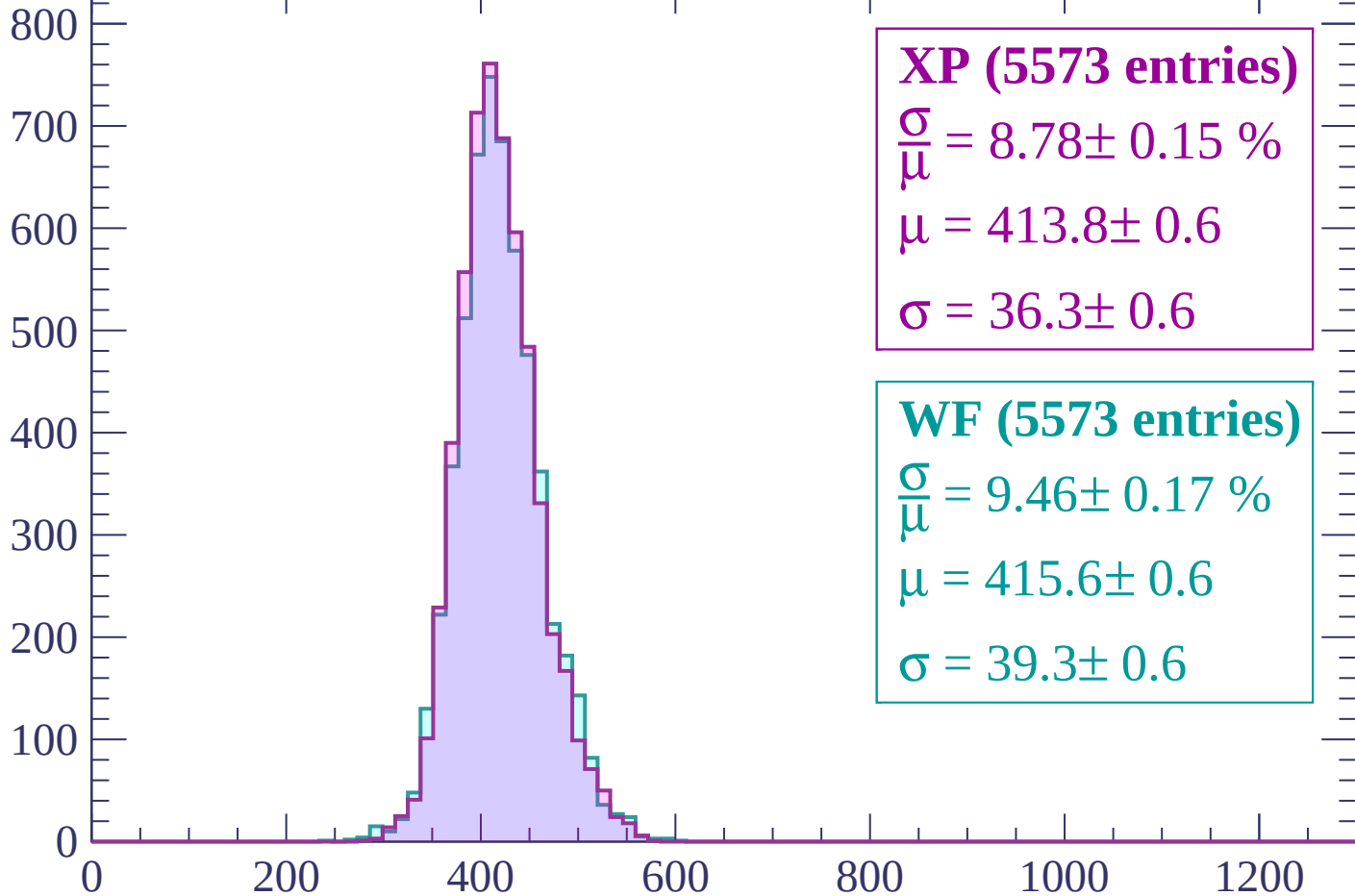
$$\mu = 411.8 \pm 0.6$$

$$\sigma = 38.0 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $720 < p < 780$

Count



XP (5573 entries)

$$\frac{\sigma}{\mu} = 8.78 \pm 0.15 \%$$

$$\mu = 413.8 \pm 0.6$$

$$\sigma = 36.3 \pm 0.6$$

WF (5573 entries)

$$\frac{\sigma}{\mu} = 9.46 \pm 0.17 \%$$

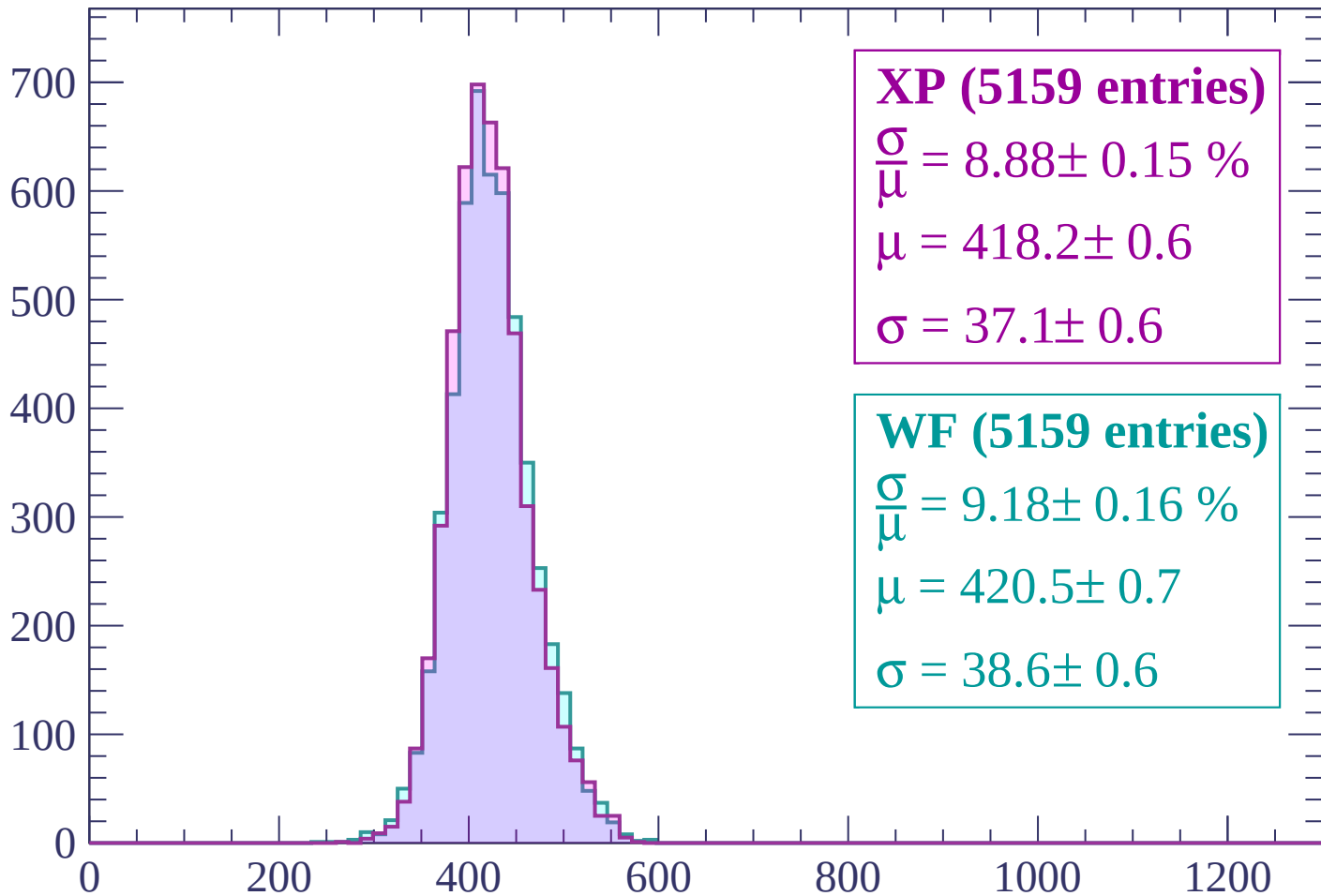
$$\mu = 415.6 \pm 0.6$$

$$\sigma = 39.3 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $780 < p < 840$

Count



XP (5159 entries)

$$\frac{\sigma}{\mu} = 8.88 \pm 0.15 \%$$

$$\mu = 418.2 \pm 0.6$$

$$\sigma = 37.1 \pm 0.6$$

WF (5159 entries)

$$\frac{\sigma}{\mu} = 9.18 \pm 0.16 \%$$

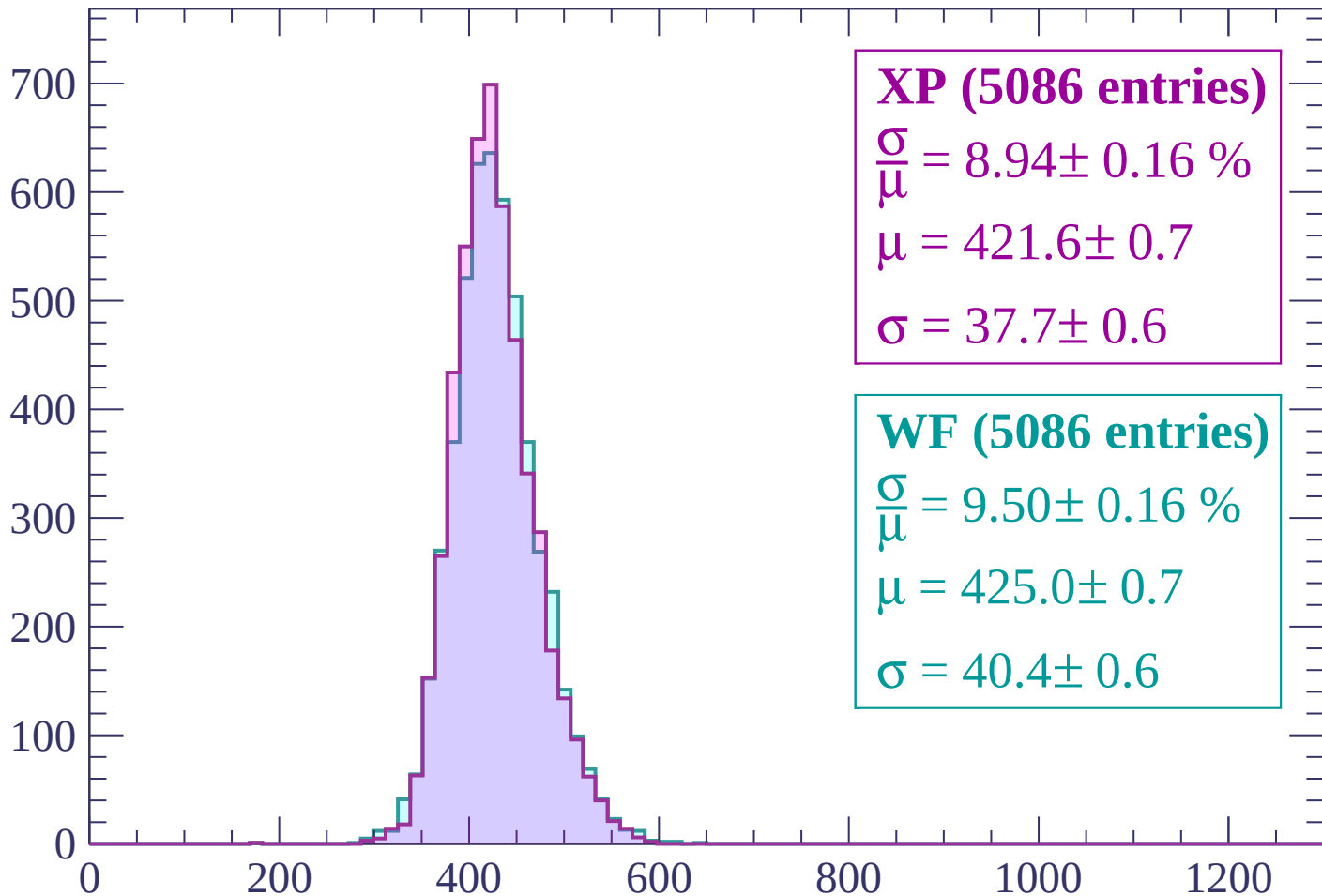
$$\mu = 420.5 \pm 0.7$$

$$\sigma = 38.6 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $840 < p < 900$

Count



XP (5086 entries)

$$\frac{\sigma}{\mu} = 8.94 \pm 0.16 \%$$

$$\mu = 421.6 \pm 0.7$$

$$\sigma = 37.7 \pm 0.6$$

WF (5086 entries)

$$\frac{\sigma}{\mu} = 9.50 \pm 0.16 \%$$

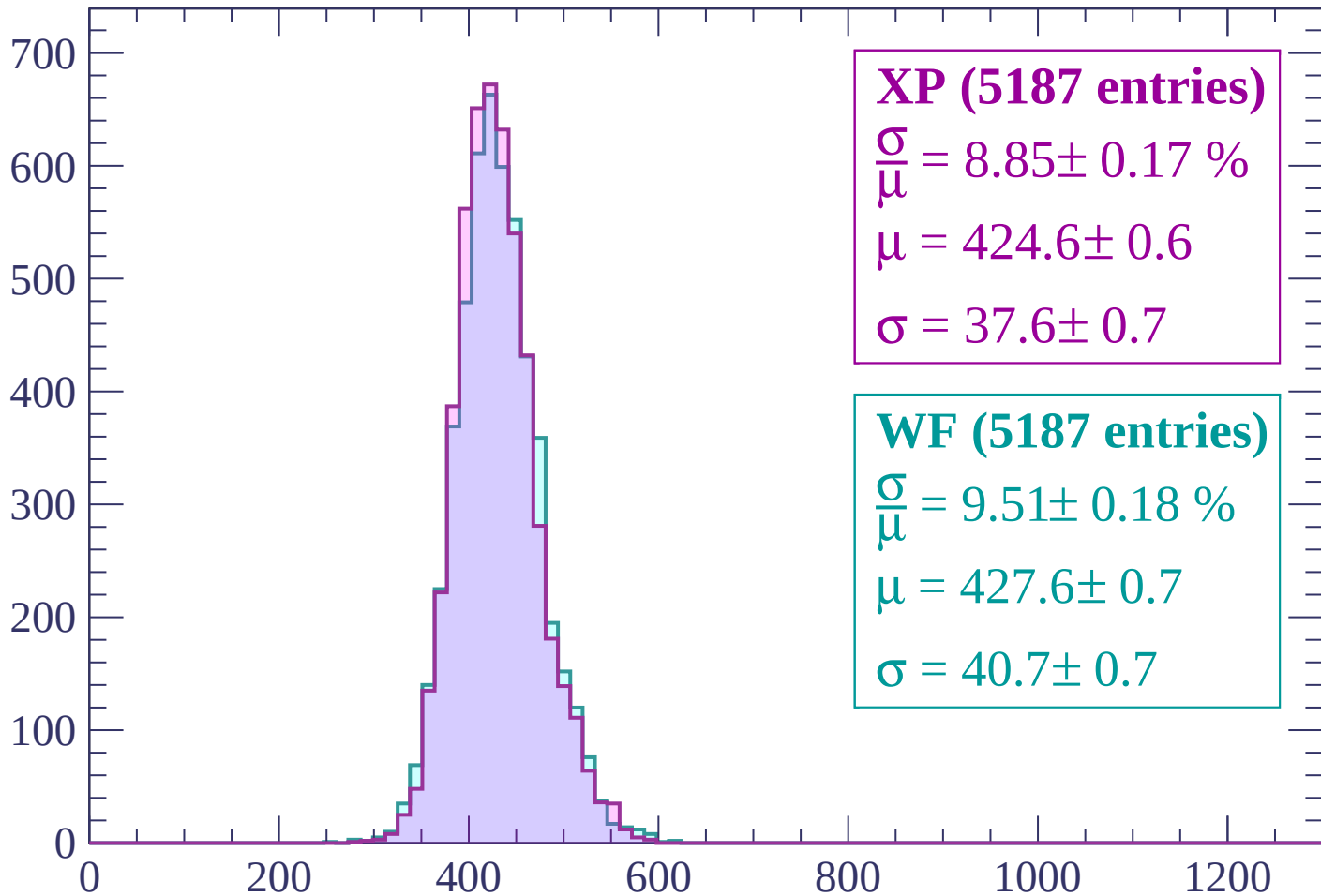
$$\mu = 425.0 \pm 0.7$$

$$\sigma = 40.4 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $900 < p < 960$

Count



XP (5187 entries)

$\frac{\sigma}{\mu} = 8.85 \pm 0.17 \%$

$\mu = 424.6 \pm 0.6$

$\sigma = 37.6 \pm 0.7$

WF (5187 entries)

$\frac{\sigma}{\mu} = 9.51 \pm 0.18 \%$

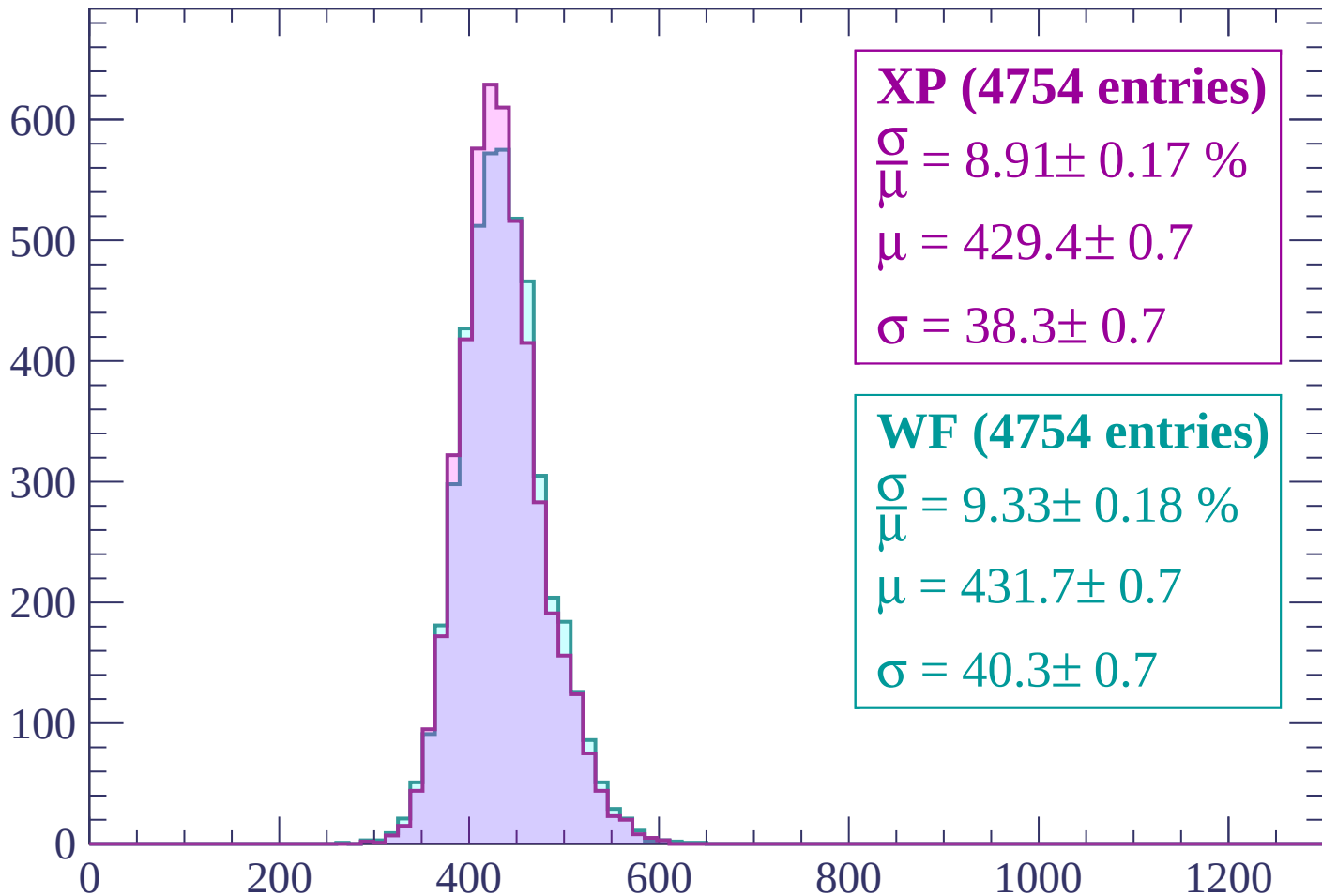
$\mu = 427.6 \pm 0.7$

$\sigma = 40.7 \pm 0.7$

dE/dx (ADC counts/cm)

Energy loss | $960 < p < 1020$

Count



XP (4754 entries)

$$\frac{\sigma}{\mu} = 8.91 \pm 0.17 \%$$

$$\mu = 429.4 \pm 0.7$$

$$\sigma = 38.3 \pm 0.7$$

WF (4754 entries)

$$\frac{\sigma}{\mu} = 9.33 \pm 0.18 \%$$

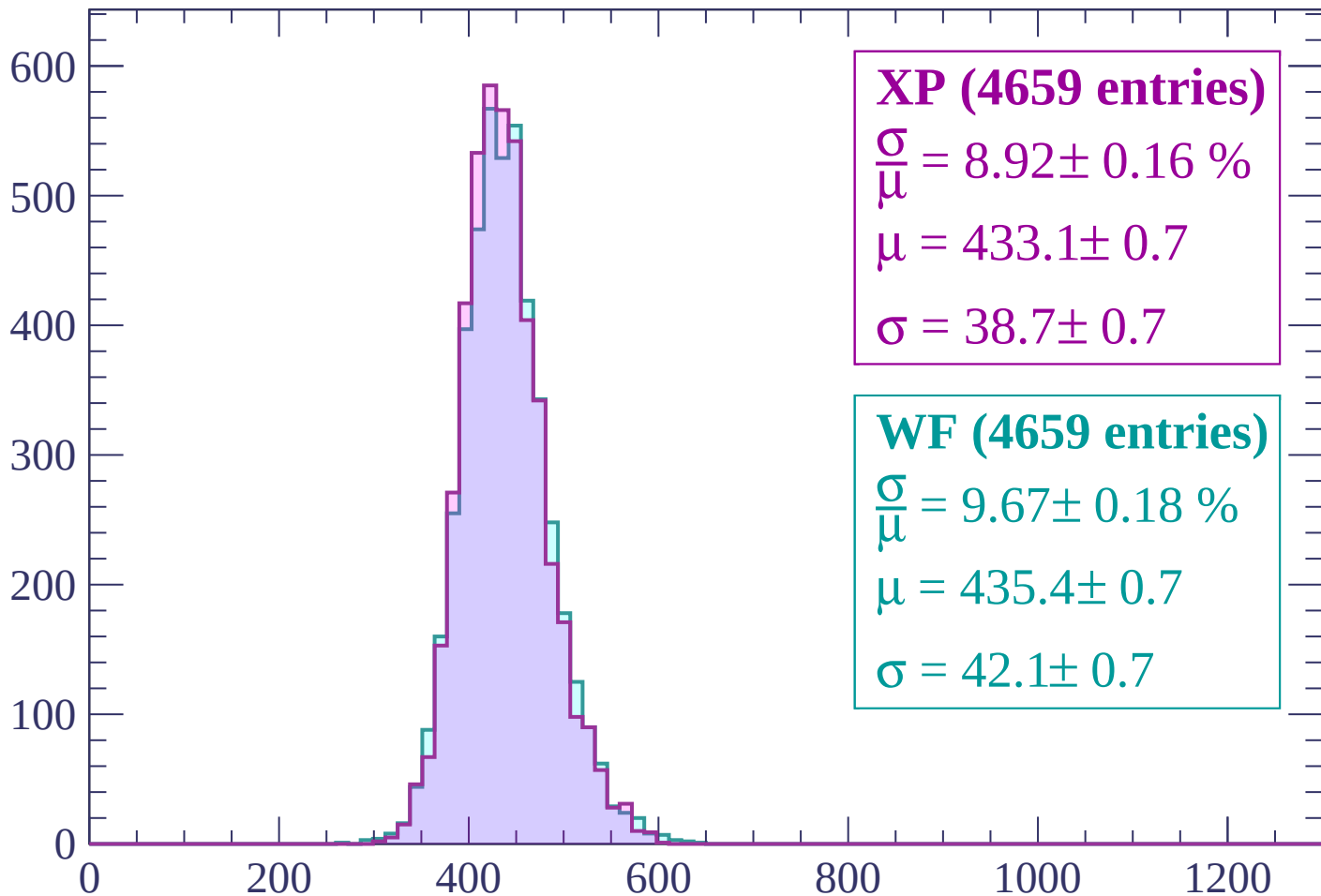
$$\mu = 431.7 \pm 0.7$$

$$\sigma = 40.3 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $1020 < p < 1080$

Count



XP (4659 entries)

$$\frac{\sigma}{\mu} = 8.92 \pm 0.16 \%$$

$$\mu = 433.1 \pm 0.7$$

$$\sigma = 38.7 \pm 0.7$$

WF (4659 entries)

$$\frac{\sigma}{\mu} = 9.67 \pm 0.18 \%$$

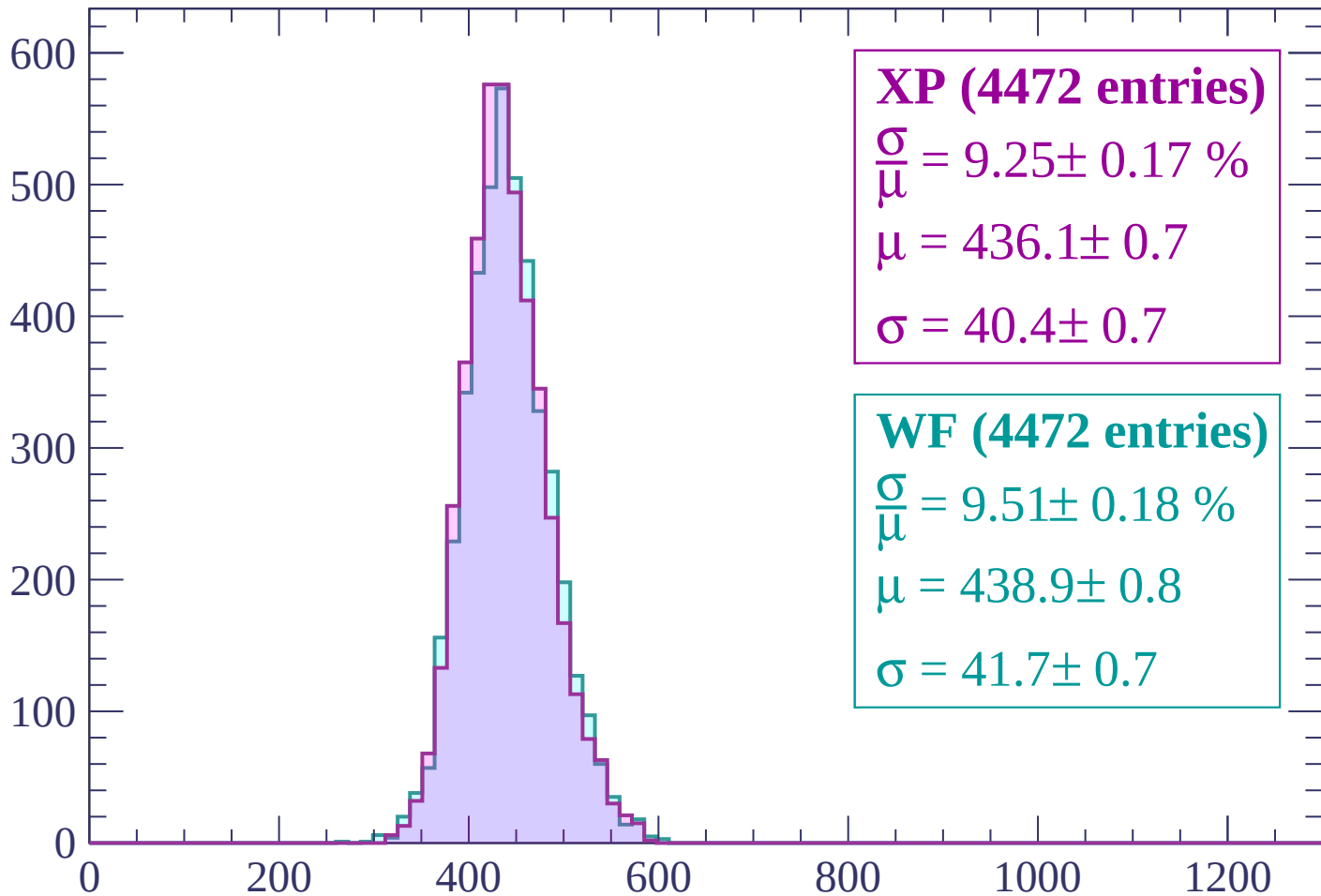
$$\mu = 435.4 \pm 0.7$$

$$\sigma = 42.1 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $1080 < p < 1140$

Count



XP (4472 entries)

$\frac{\sigma}{\mu} = 9.25 \pm 0.17 \%$

$\mu = 436.1 \pm 0.7$

$\sigma = 40.4 \pm 0.7$

WF (4472 entries)

$\frac{\sigma}{\mu} = 9.51 \pm 0.18 \%$

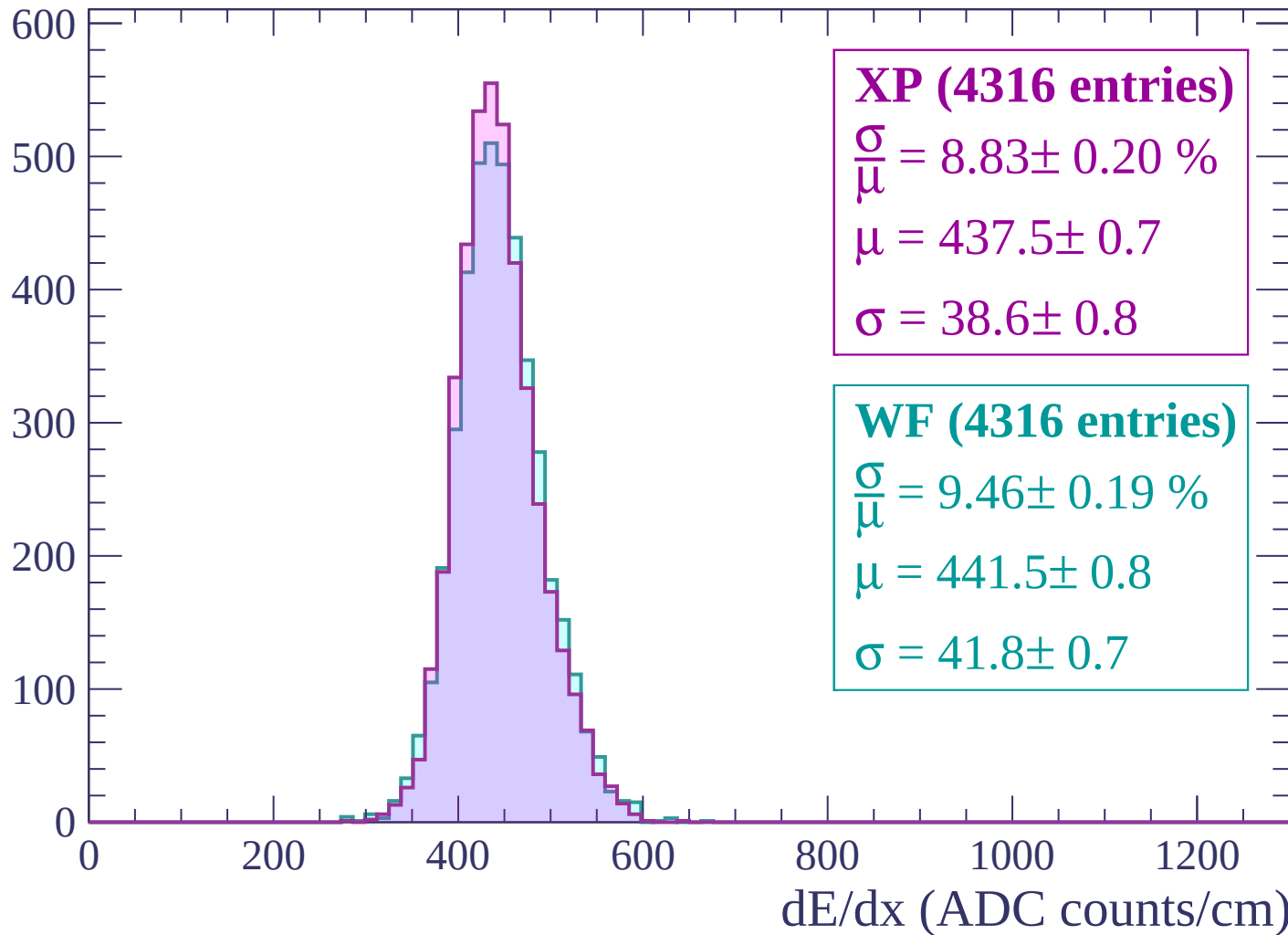
$\mu = 438.9 \pm 0.8$

$\sigma = 41.7 \pm 0.7$

dE/dx (ADC counts/cm)

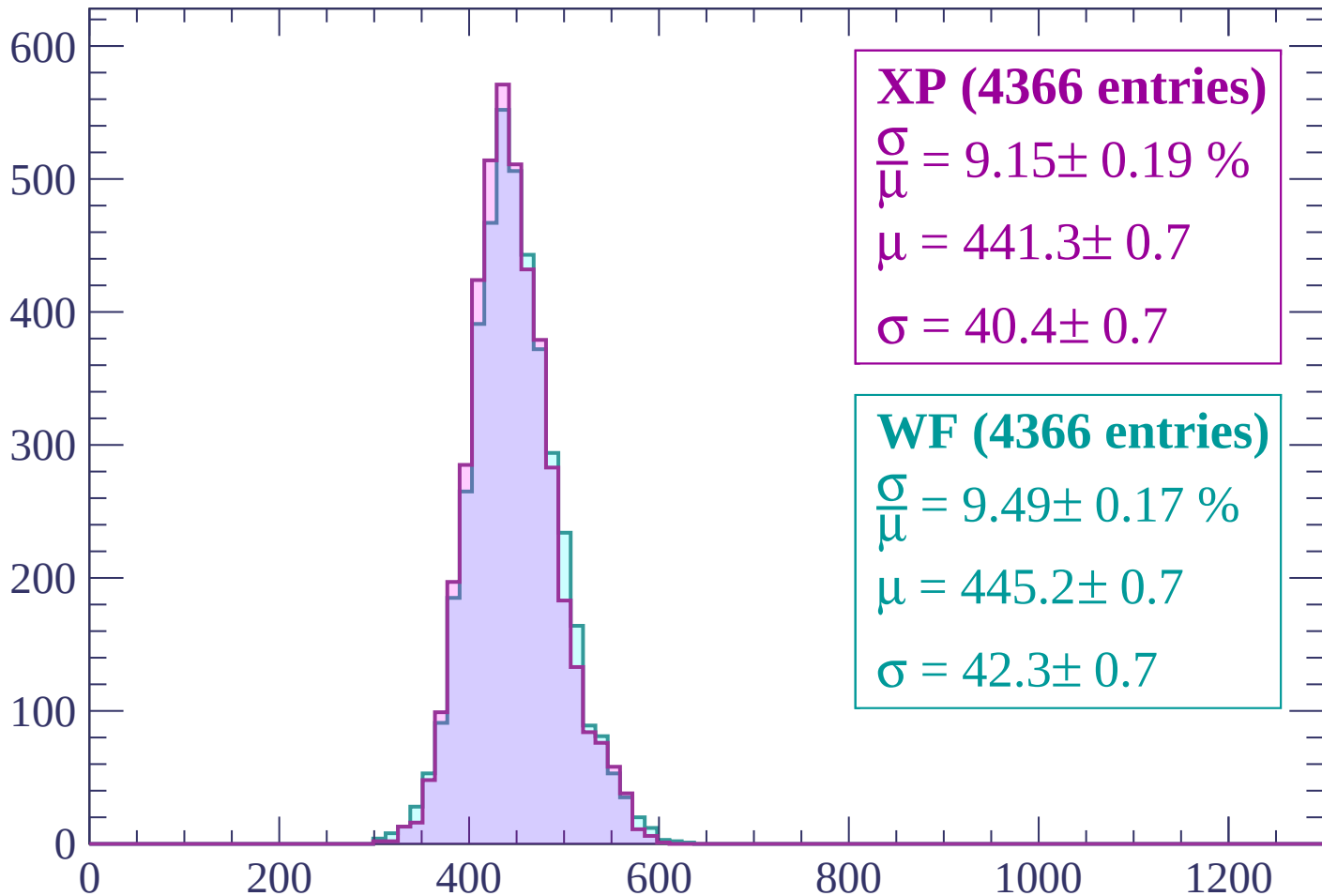
Energy loss | $1140 < p < 1200$

Count



Energy loss | $1200 < p < 1260$

Count



XP (4366 entries)

$\frac{\sigma}{\mu} = 9.15 \pm 0.19 \%$

$\mu = 441.3 \pm 0.7$

$\sigma = 40.4 \pm 0.7$

WF (4366 entries)

$\frac{\sigma}{\mu} = 9.49 \pm 0.17 \%$

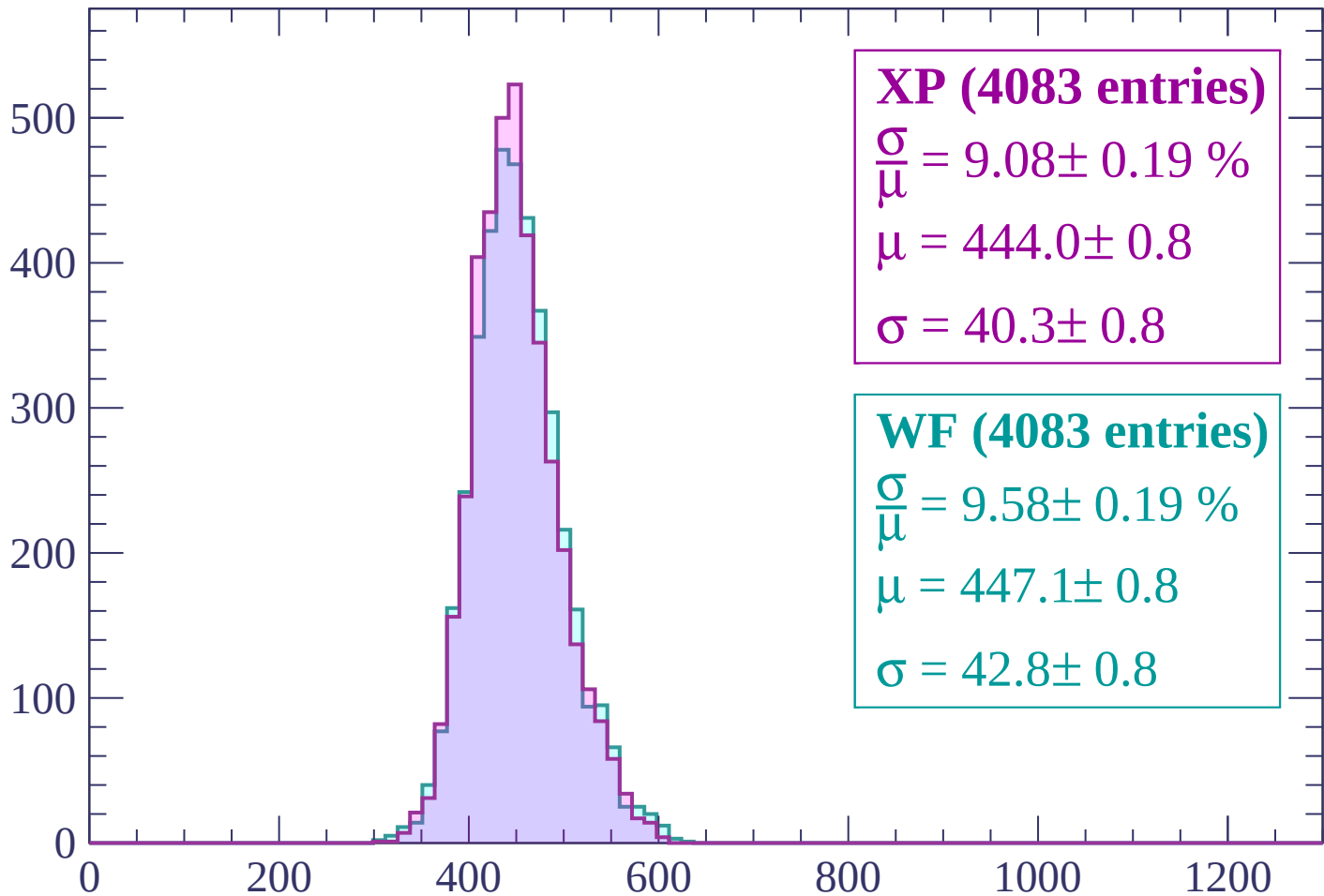
$\mu = 445.2 \pm 0.7$

$\sigma = 42.3 \pm 0.7$

dE/dx (ADC counts/cm)

Energy loss | $1260 < p < 1320$

Count



XP (4083 entries)

$$\frac{\sigma}{\mu} = 9.08 \pm 0.19 \%$$

$$\mu = 444.0 \pm 0.8$$

$$\sigma = 40.3 \pm 0.8$$

WF (4083 entries)

$$\frac{\sigma}{\mu} = 9.58 \pm 0.19 \%$$

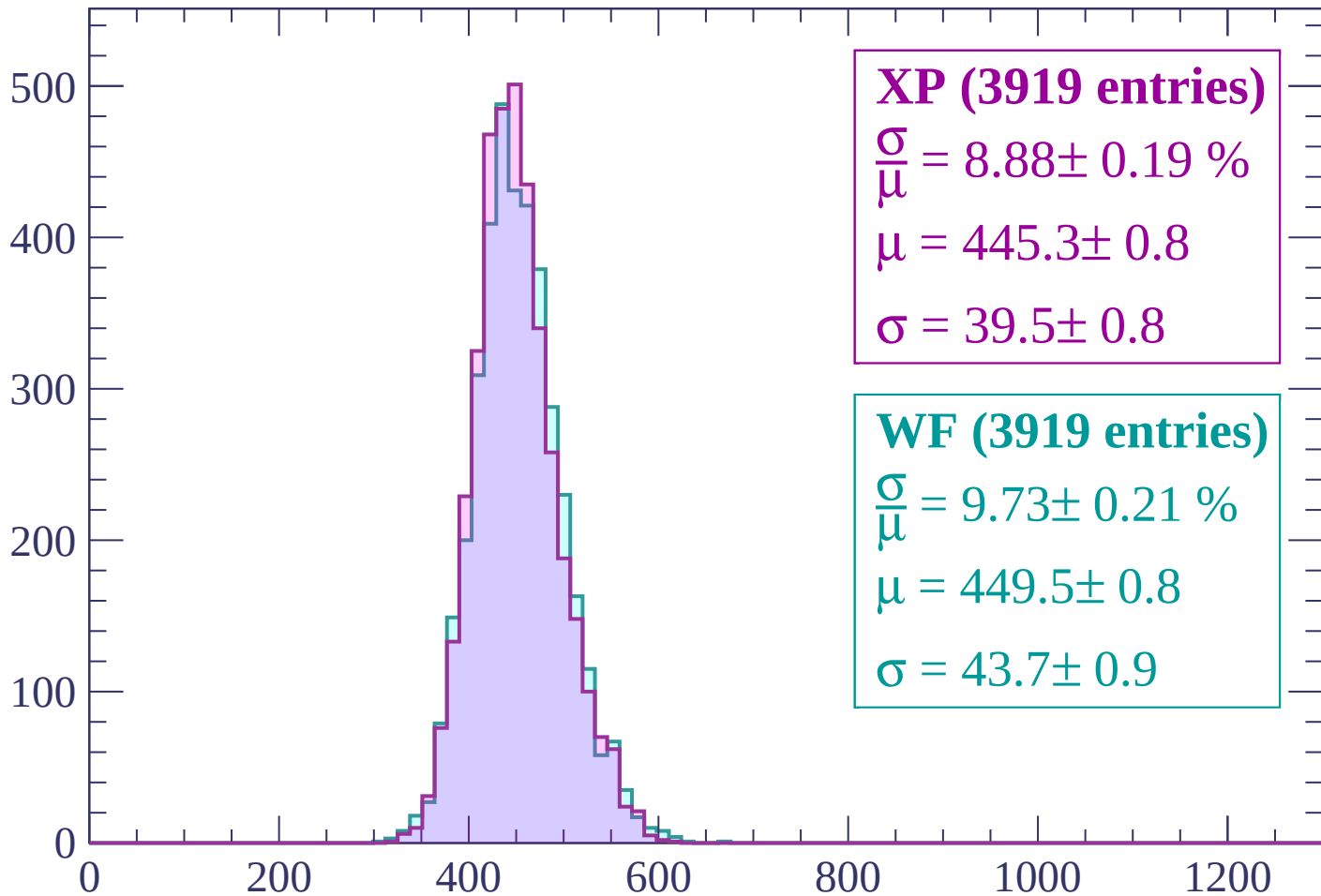
$$\mu = 447.1 \pm 0.8$$

$$\sigma = 42.8 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $1320 < p < 1380$

Count



XP (3919 entries)

$$\frac{\sigma}{\mu} = 8.88 \pm 0.19 \%$$

$$\mu = 445.3 \pm 0.8$$

$$\sigma = 39.5 \pm 0.8$$

WF (3919 entries)

$$\frac{\sigma}{\mu} = 9.73 \pm 0.21 \%$$

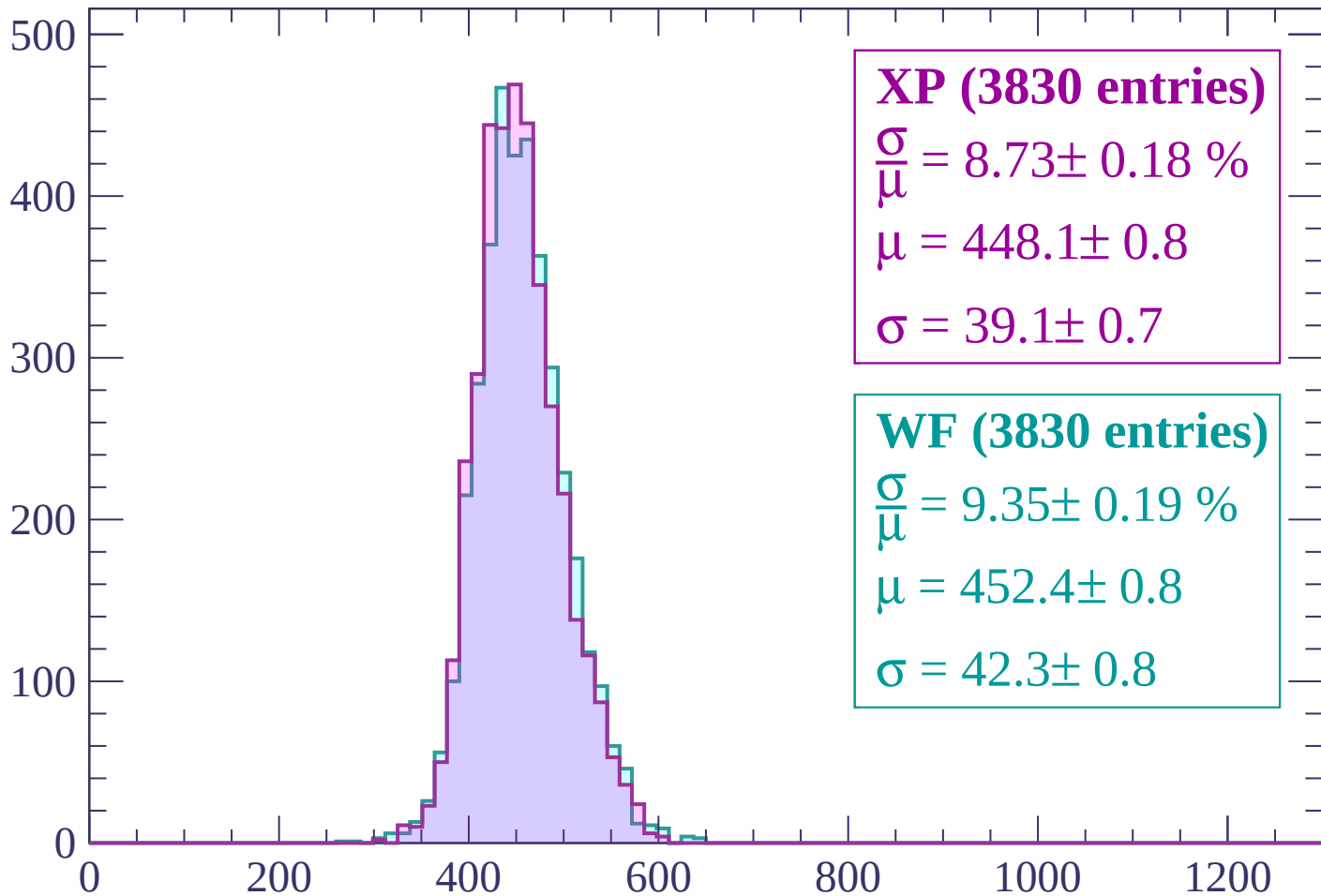
$$\mu = 449.5 \pm 0.8$$

$$\sigma = 43.7 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | $1380 < p < 1440$

Count



XP (3830 entries)

$$\frac{\sigma}{\mu} = 8.73 \pm 0.18 \%$$

$$\mu = 448.1 \pm 0.8$$

$$\sigma = 39.1 \pm 0.7$$

WF (3830 entries)

$$\frac{\sigma}{\mu} = 9.35 \pm 0.19 \%$$

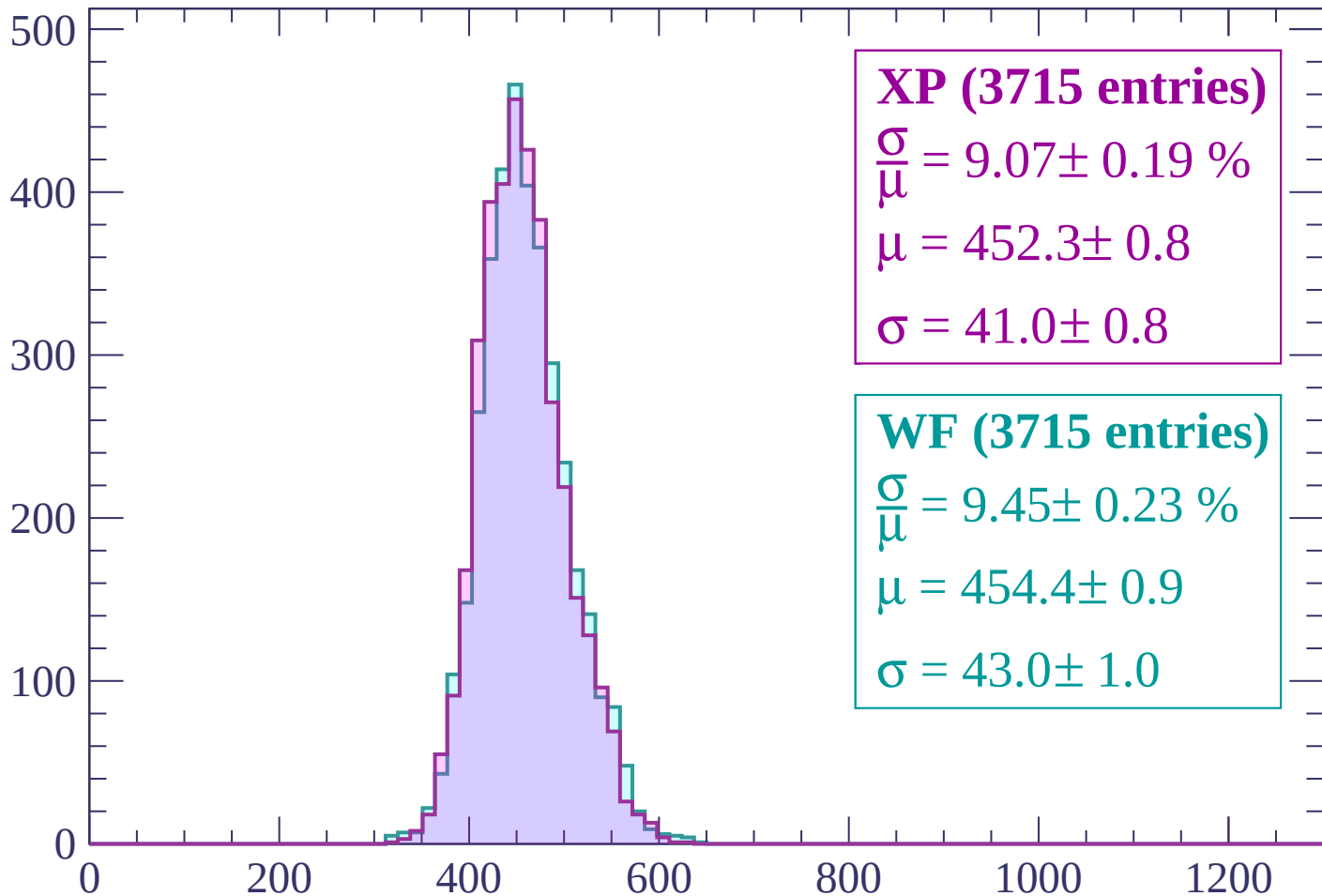
$$\mu = 452.4 \pm 0.8$$

$$\sigma = 42.3 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $1440 < p < 1500$

Count



XP (3715 entries)

$\frac{\sigma}{\mu} = 9.07 \pm 0.19 \%$

$\mu = 452.3 \pm 0.8$

$\sigma = 41.0 \pm 0.8$

WF (3715 entries)

$\frac{\sigma}{\mu} = 9.45 \pm 0.23 \%$

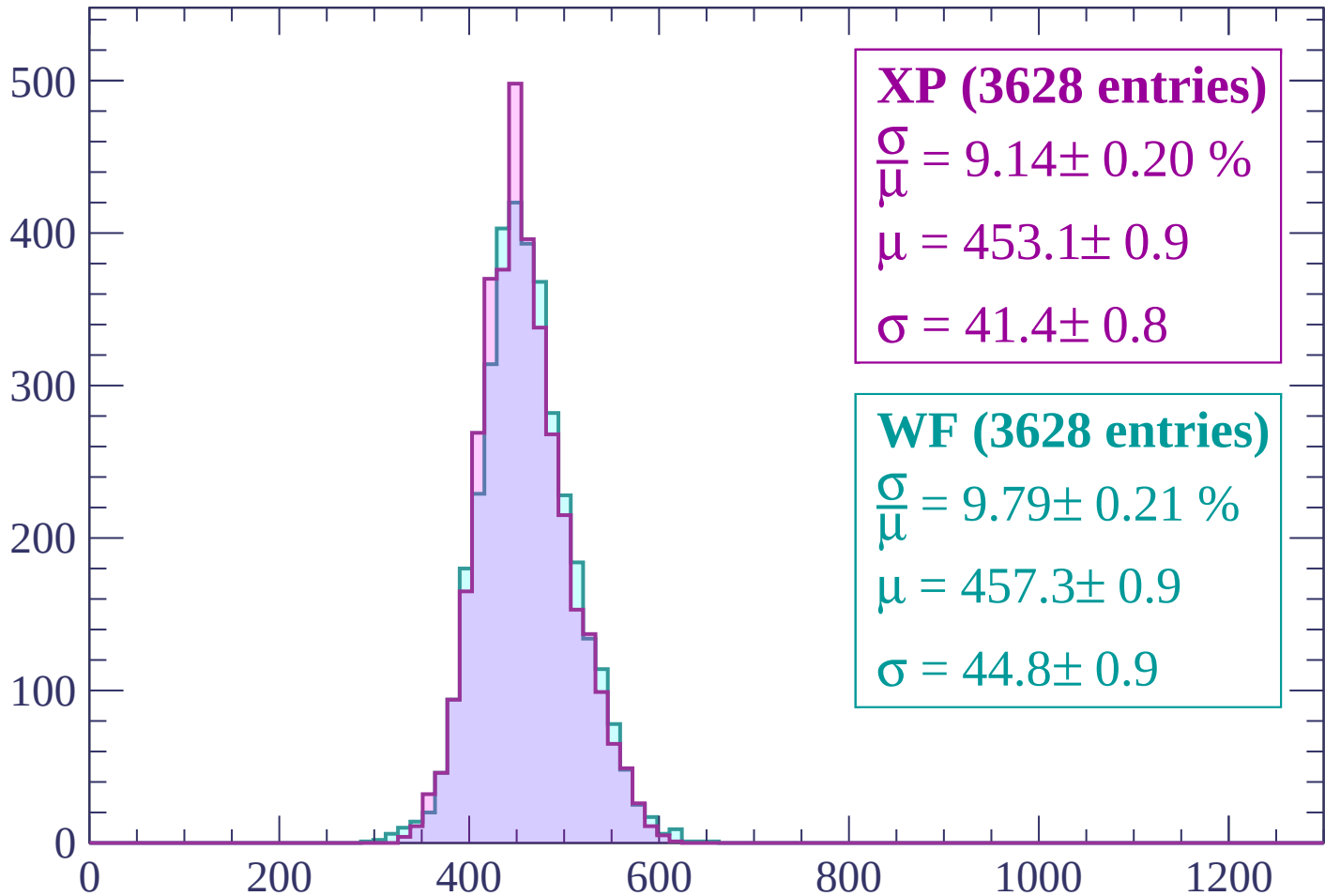
$\mu = 454.4 \pm 0.9$

$\sigma = 43.0 \pm 1.0$

dE/dx (ADC counts/cm)

Energy loss | $1500 < p < 1560$

Count



XP (3628 entries)

$\frac{\sigma}{\mu} = 9.14 \pm 0.20 \%$

$\mu = 453.1 \pm 0.9$

$\sigma = 41.4 \pm 0.8$

WF (3628 entries)

$\frac{\sigma}{\mu} = 9.79 \pm 0.21 \%$

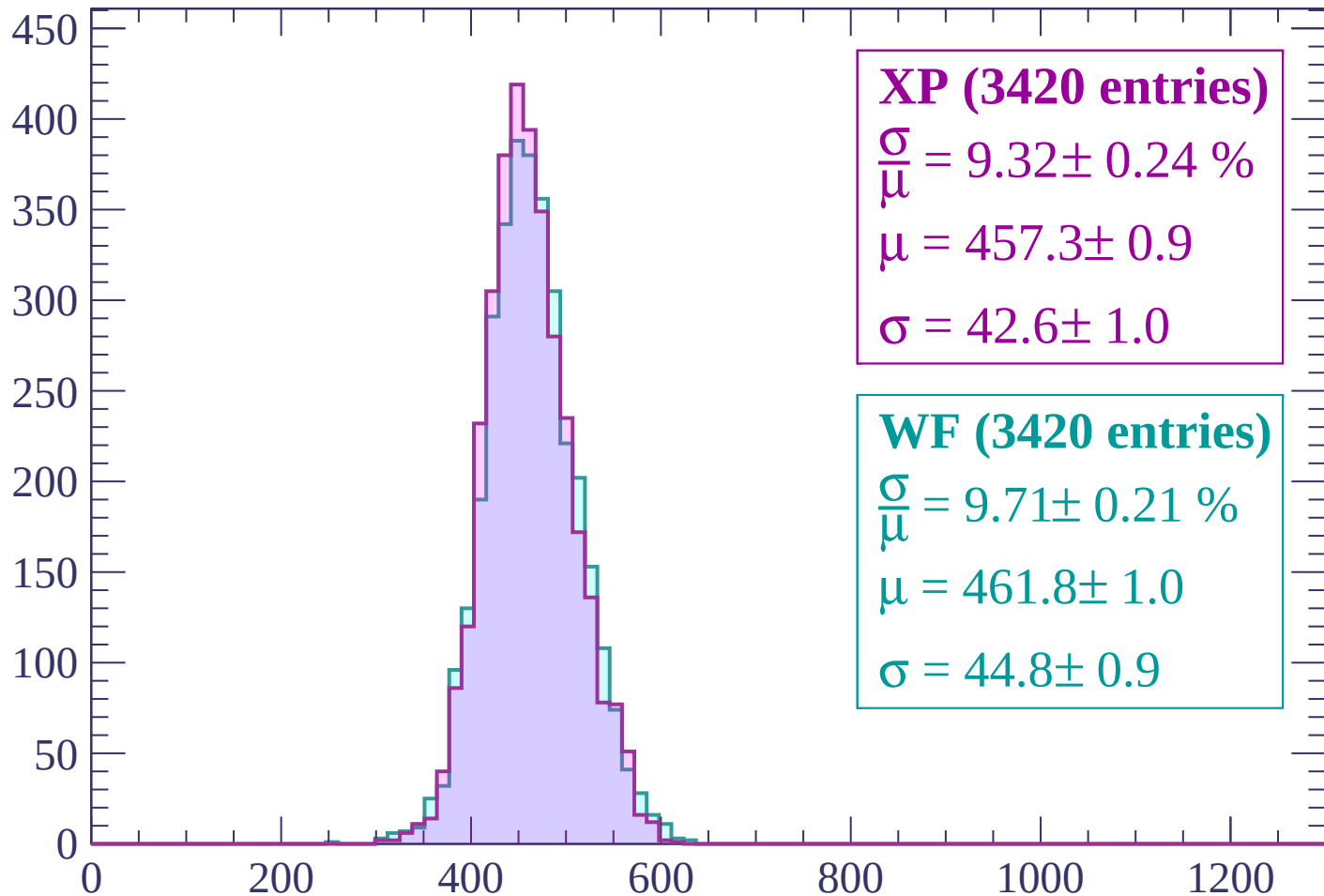
$\mu = 457.3 \pm 0.9$

$\sigma = 44.8 \pm 0.9$

dE/dx (ADC counts/cm)

Energy loss | $1560 < p < 1620$

Count



XP (3420 entries)

$\frac{\sigma}{\mu} = 9.32 \pm 0.24 \%$

$\mu = 457.3 \pm 0.9$

$\sigma = 42.6 \pm 1.0$

WF (3420 entries)

$\frac{\sigma}{\mu} = 9.71 \pm 0.21 \%$

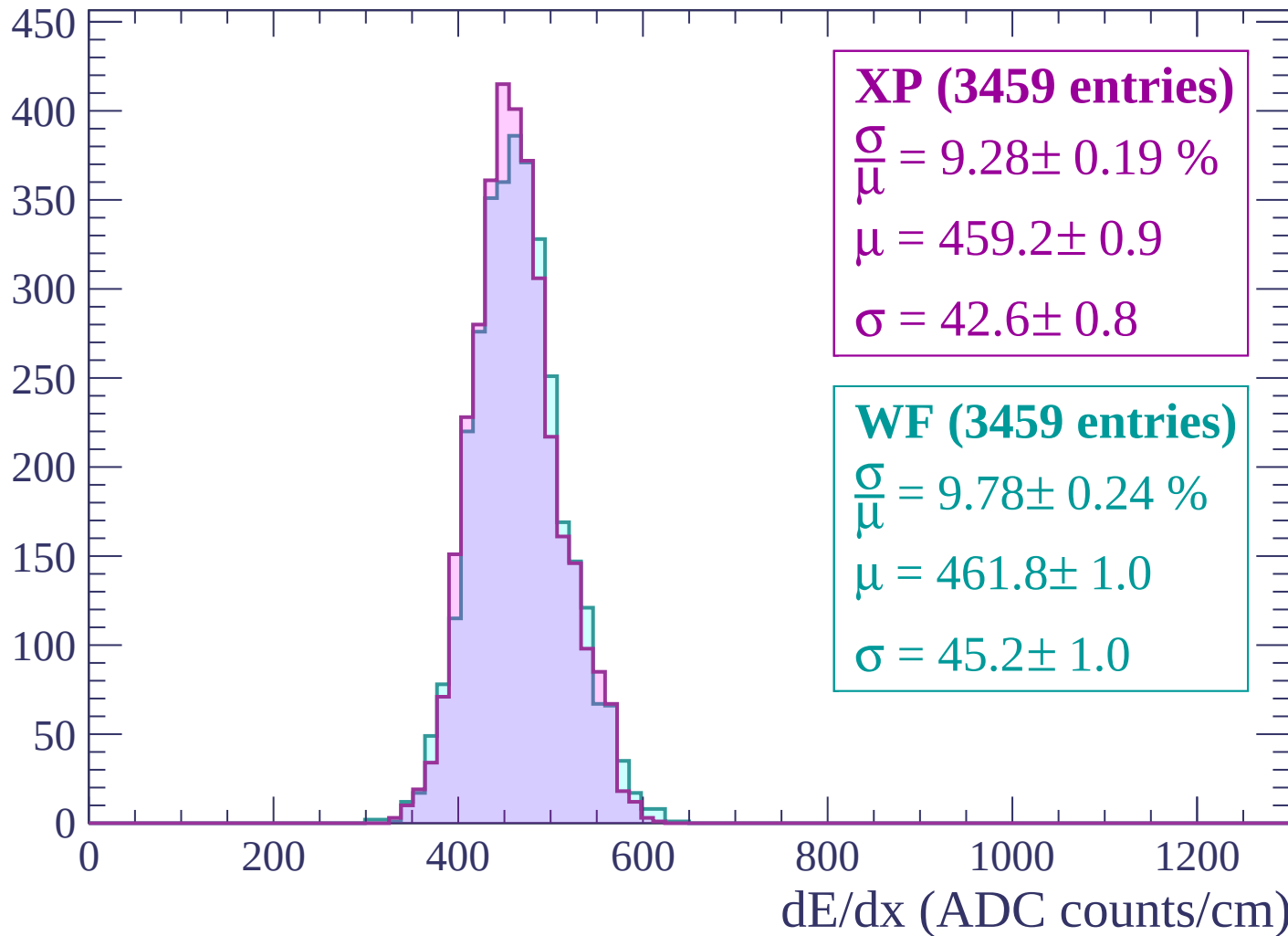
$\mu = 461.8 \pm 1.0$

$\sigma = 44.8 \pm 0.9$

dE/dx (ADC counts/cm)

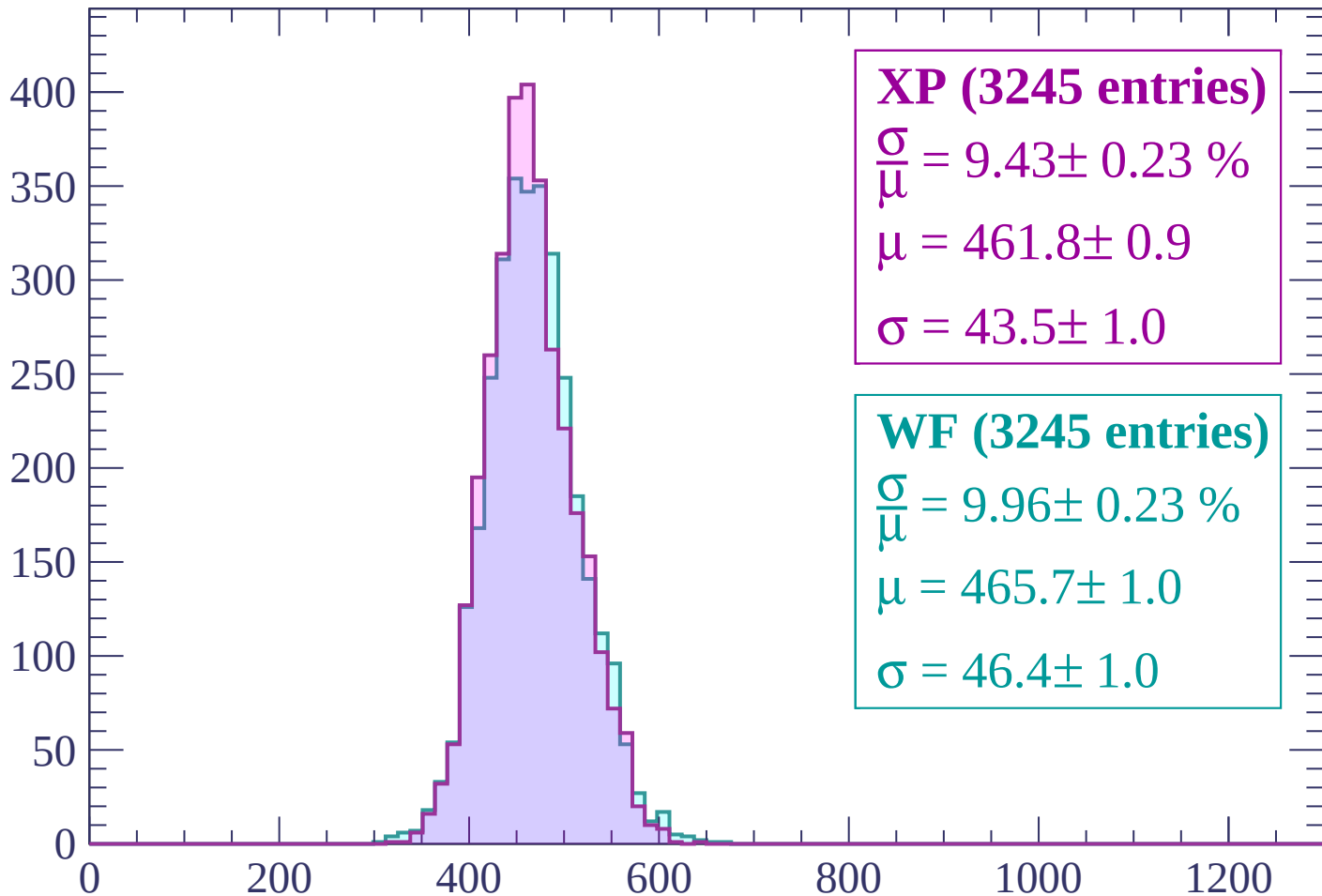
Energy loss | $1620 < p < 1680$

Count



Energy loss | $1680 < p < 1740$

Count



XP (3245 entries)

$\frac{\sigma}{\mu} = 9.43 \pm 0.23 \%$

$\mu = 461.8 \pm 0.9$

$\sigma = 43.5 \pm 1.0$

WF (3245 entries)

$\frac{\sigma}{\mu} = 9.96 \pm 0.23 \%$

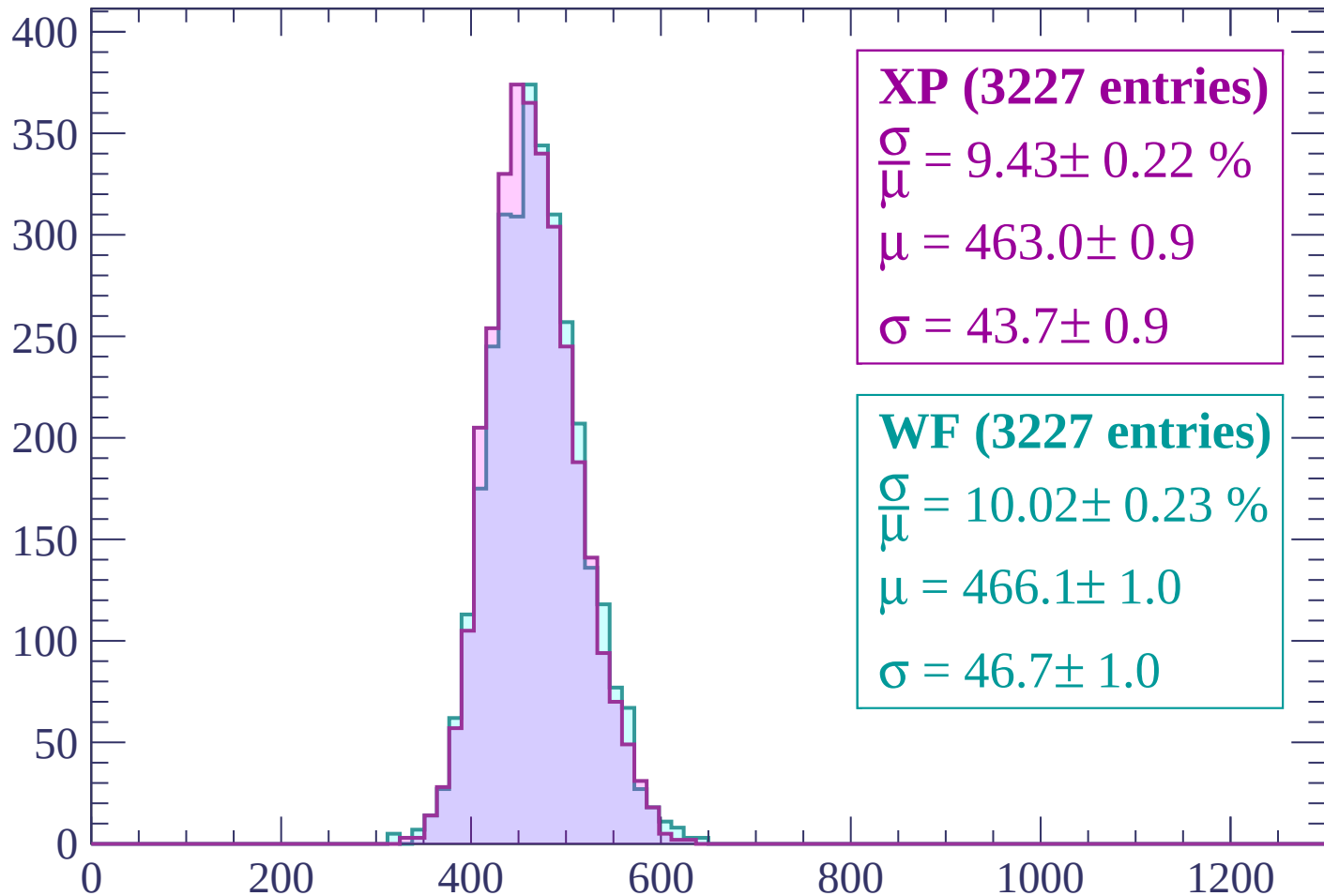
$\mu = 465.7 \pm 1.0$

$\sigma = 46.4 \pm 1.0$

dE/dx (ADC counts/cm)

Energy loss | $1740 < p < 1800$

Count



XP (3227 entries)

$$\frac{\sigma}{\mu} = 9.43 \pm 0.22 \%$$

$$\mu = 463.0 \pm 0.9$$

$$\sigma = 43.7 \pm 0.9$$

WF (3227 entries)

$$\frac{\sigma}{\mu} = 10.02 \pm 0.23 \%$$

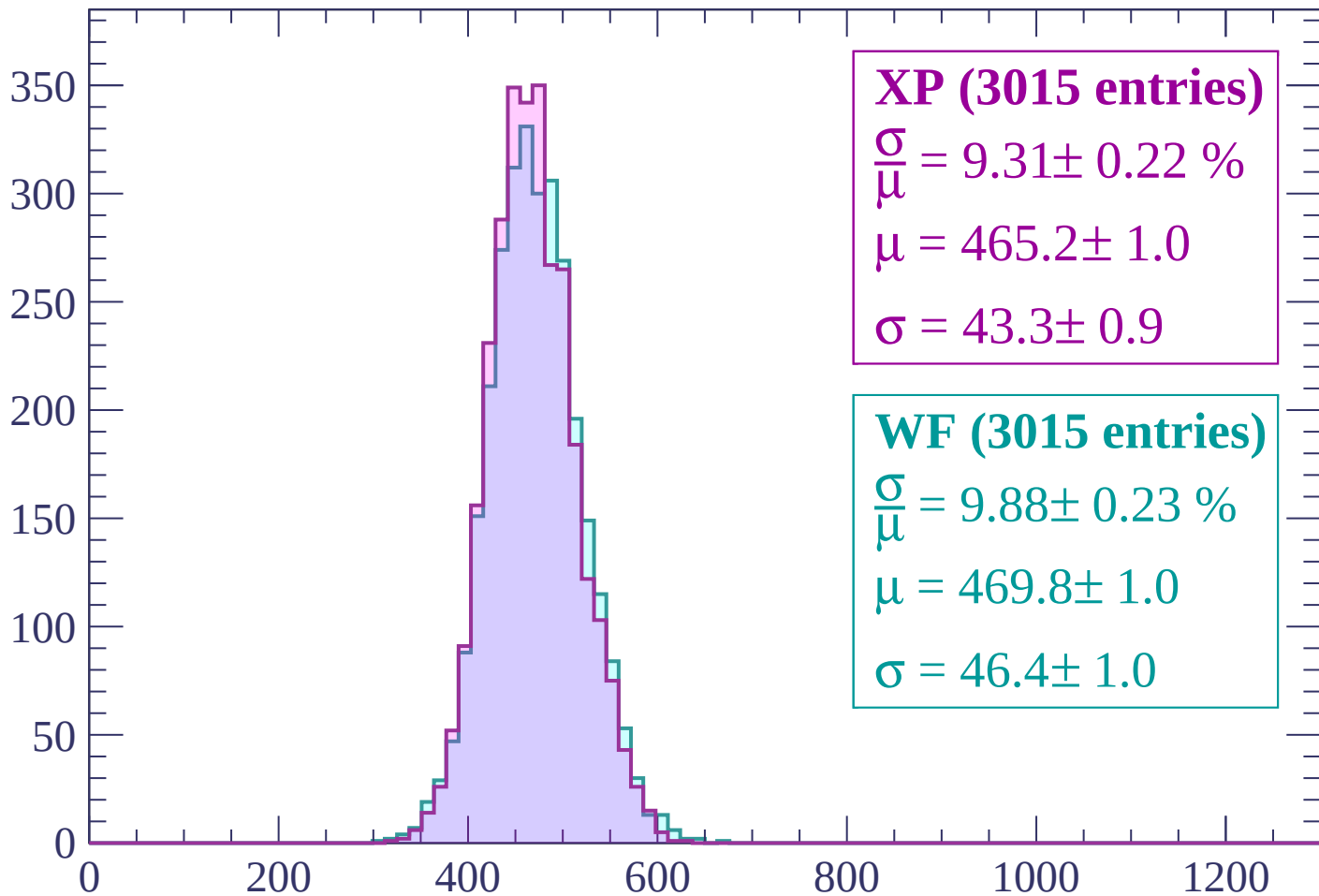
$$\mu = 466.1 \pm 1.0$$

$$\sigma = 46.7 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss | $1800 < p < 1860$

Count



XP (3015 entries)

$$\frac{\sigma}{\mu} = 9.31 \pm 0.22 \%$$

$$\mu = 465.2 \pm 1.0$$

$$\sigma = 43.3 \pm 0.9$$

WF (3015 entries)

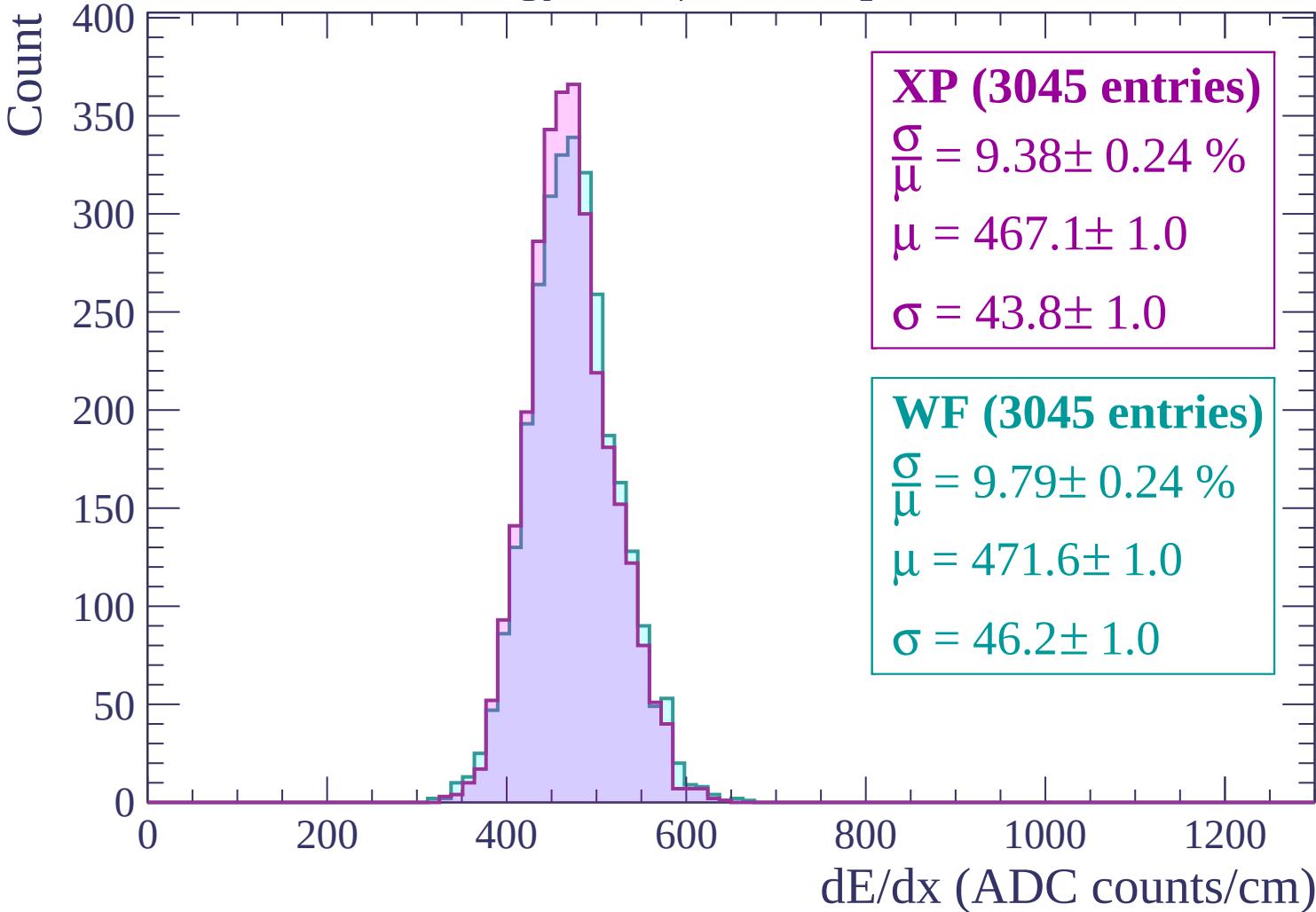
$$\frac{\sigma}{\mu} = 9.88 \pm 0.23 \%$$

$$\mu = 469.8 \pm 1.0$$

$$\sigma = 46.4 \pm 1.0$$

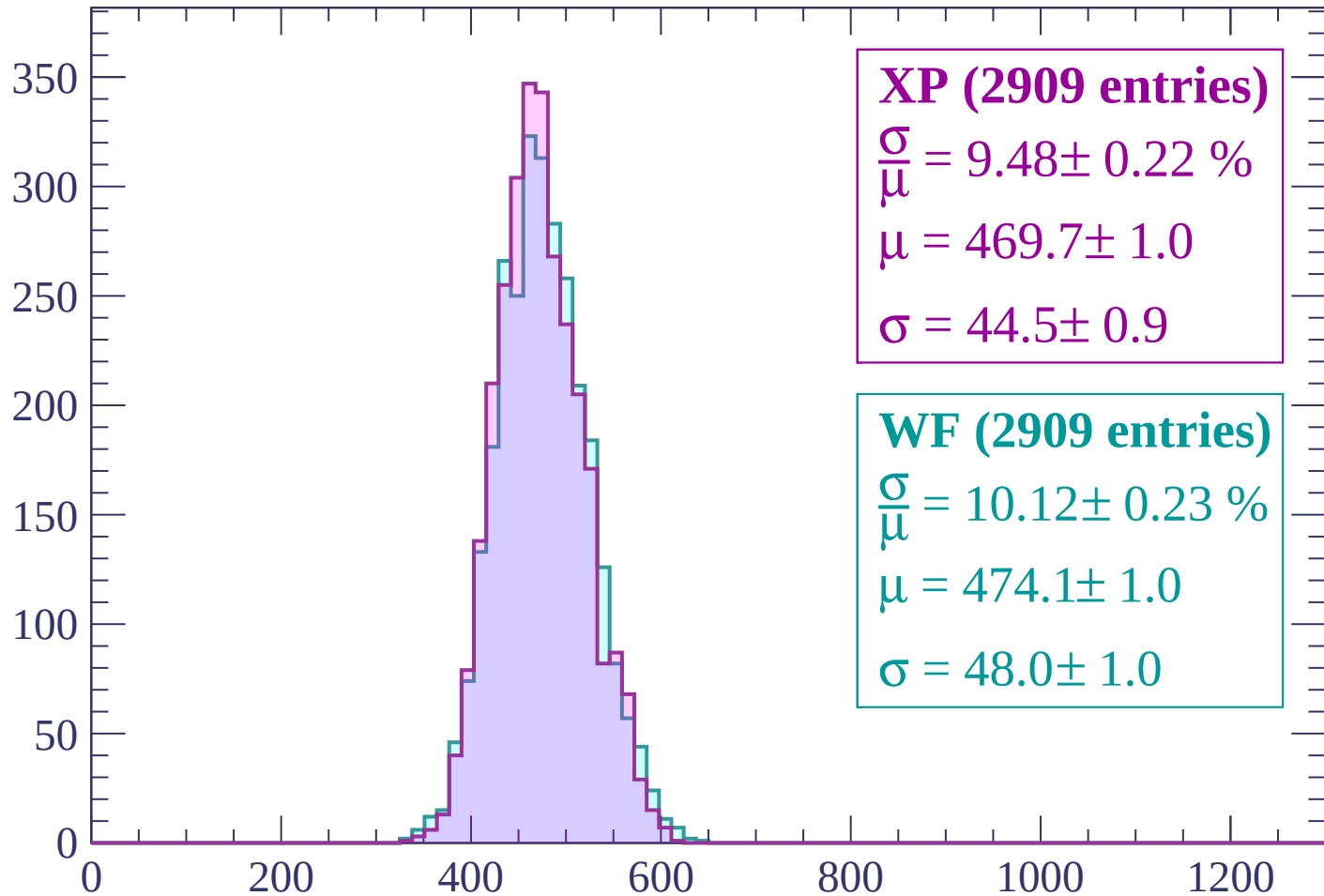
dE/dx (ADC counts/cm)

Energy loss | $1860 < p < 1920$



Energy loss | $1920 < p < 1980$

Count



XP (2909 entries)

$$\frac{\sigma}{\mu} = 9.48 \pm 0.22 \%$$

$$\mu = 469.7 \pm 1.0$$

$$\sigma = 44.5 \pm 0.9$$

WF (2909 entries)

$$\frac{\sigma}{\mu} = 10.12 \pm 0.23 \%$$

$$\mu = 474.1 \pm 1.0$$

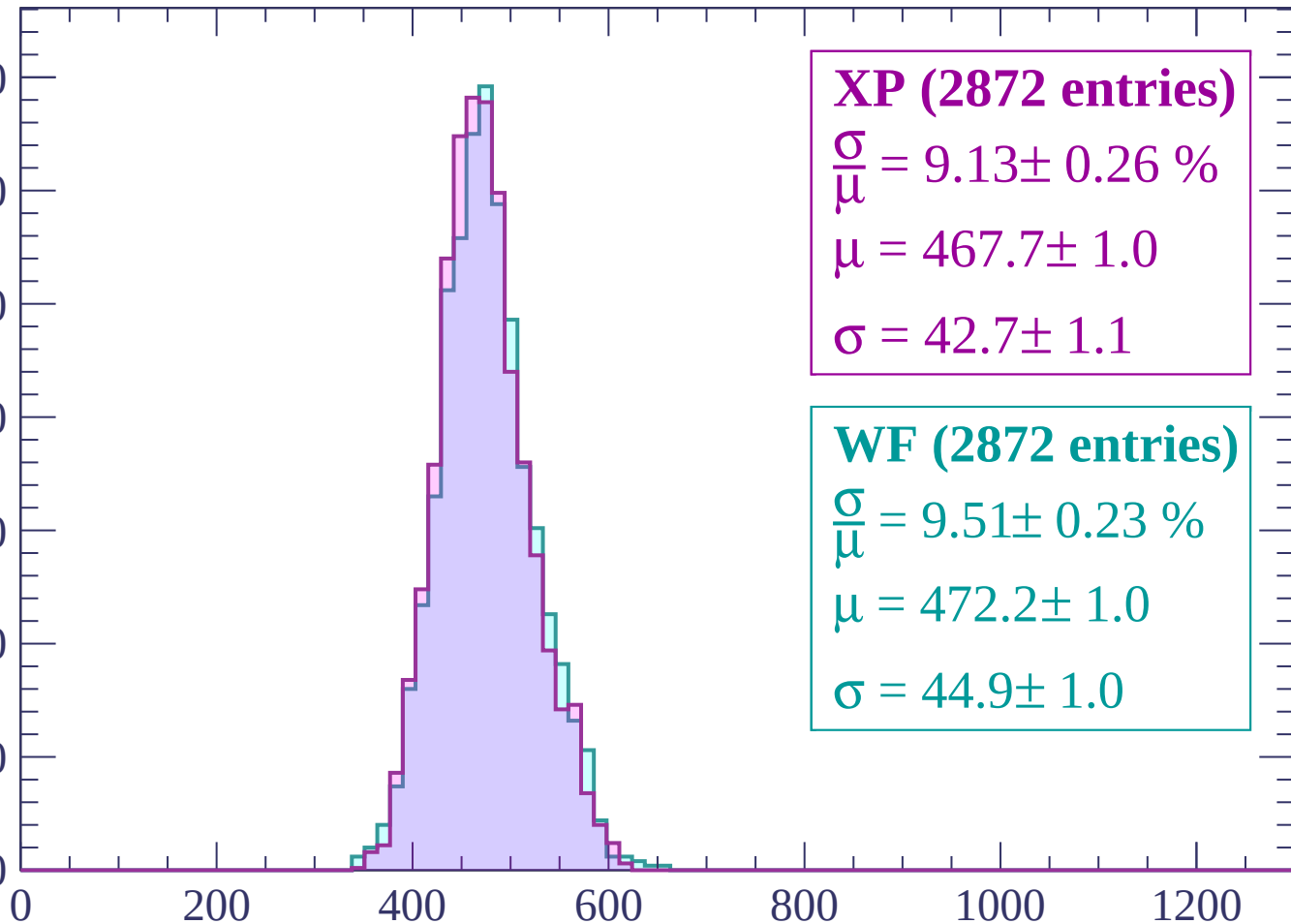
$$\sigma = 48.0 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss | $1980 < p < 2040$

Count

350
300
250
200
150
100
50
0



XP (2872 entries)

$\frac{\sigma}{\mu} = 9.13 \pm 0.26 \%$

$\mu = 467.7 \pm 1.0$

$\sigma = 42.7 \pm 1.1$

WF (2872 entries)

$\frac{\sigma}{\mu} = 9.51 \pm 0.23 \%$

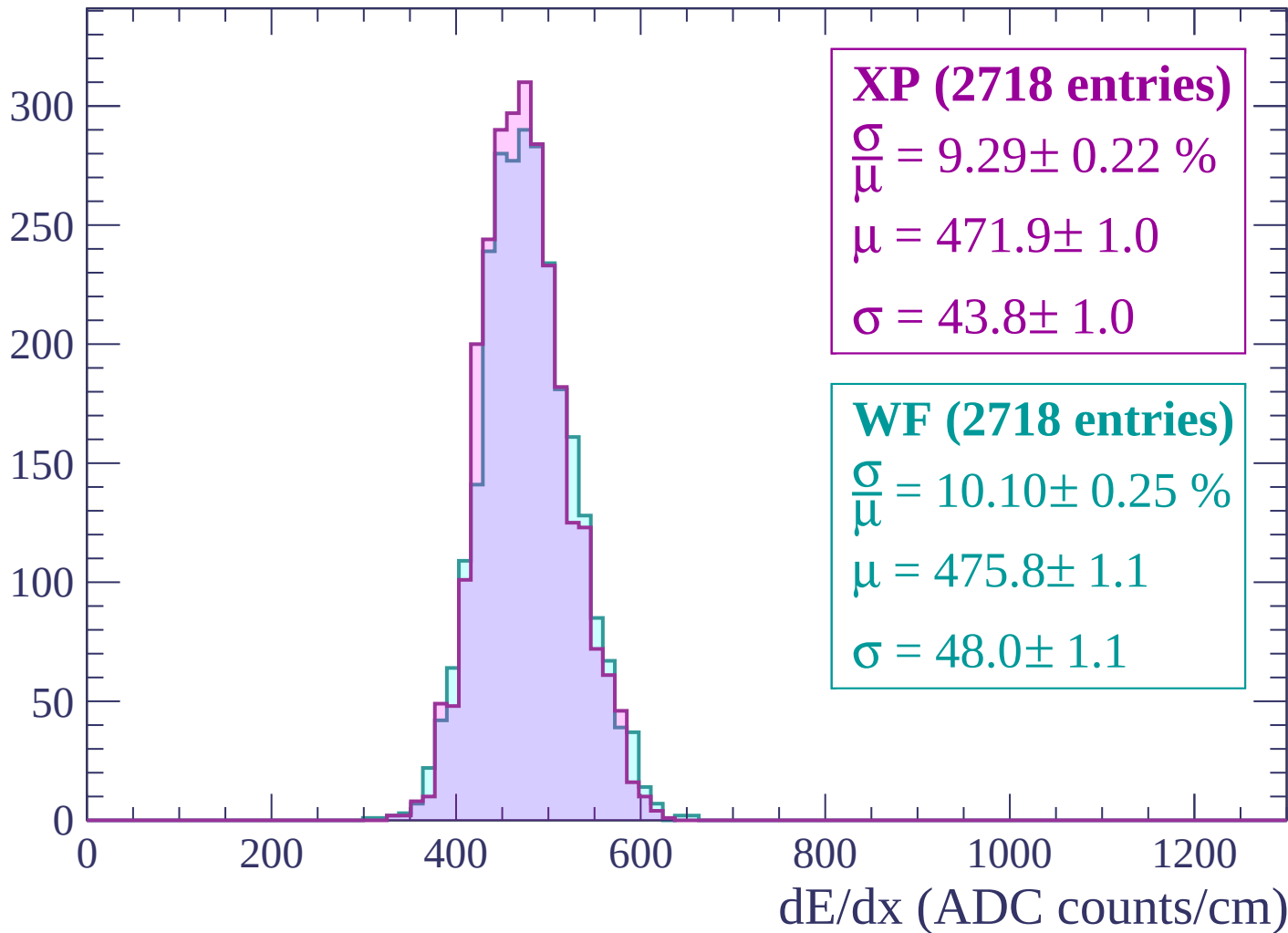
$\mu = 472.2 \pm 1.0$

$\sigma = 44.9 \pm 1.0$

dE/dx (ADC counts/cm)

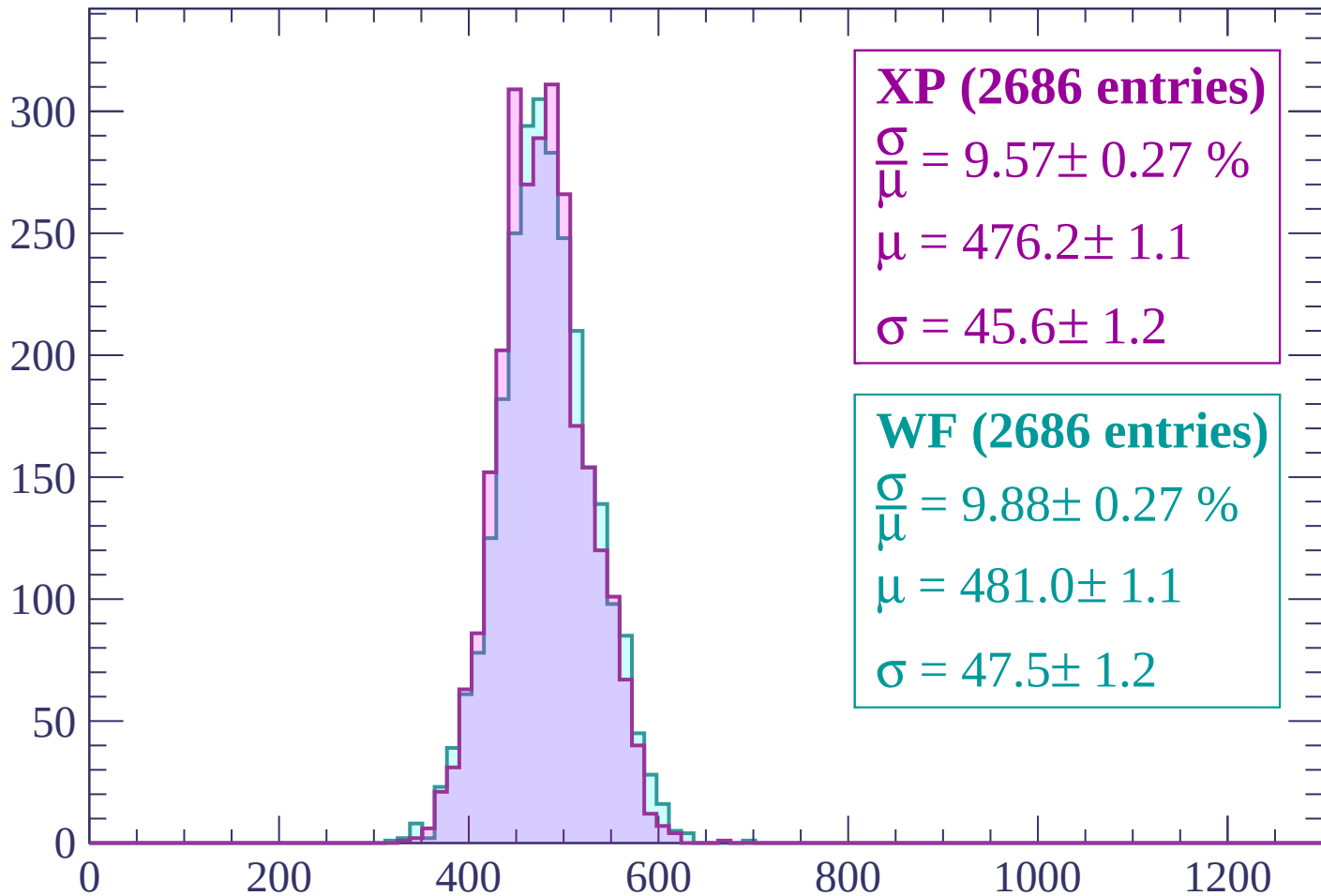
Energy loss | $2040 < p < 2100$

Count



Energy loss | $2100 < p < 2160$

Count



XP (2686 entries)

$$\frac{\sigma}{\mu} = 9.57 \pm 0.27 \%$$

$$\mu = 476.2 \pm 1.1$$

$$\sigma = 45.6 \pm 1.2$$

WF (2686 entries)

$$\frac{\sigma}{\mu} = 9.88 \pm 0.27 \%$$

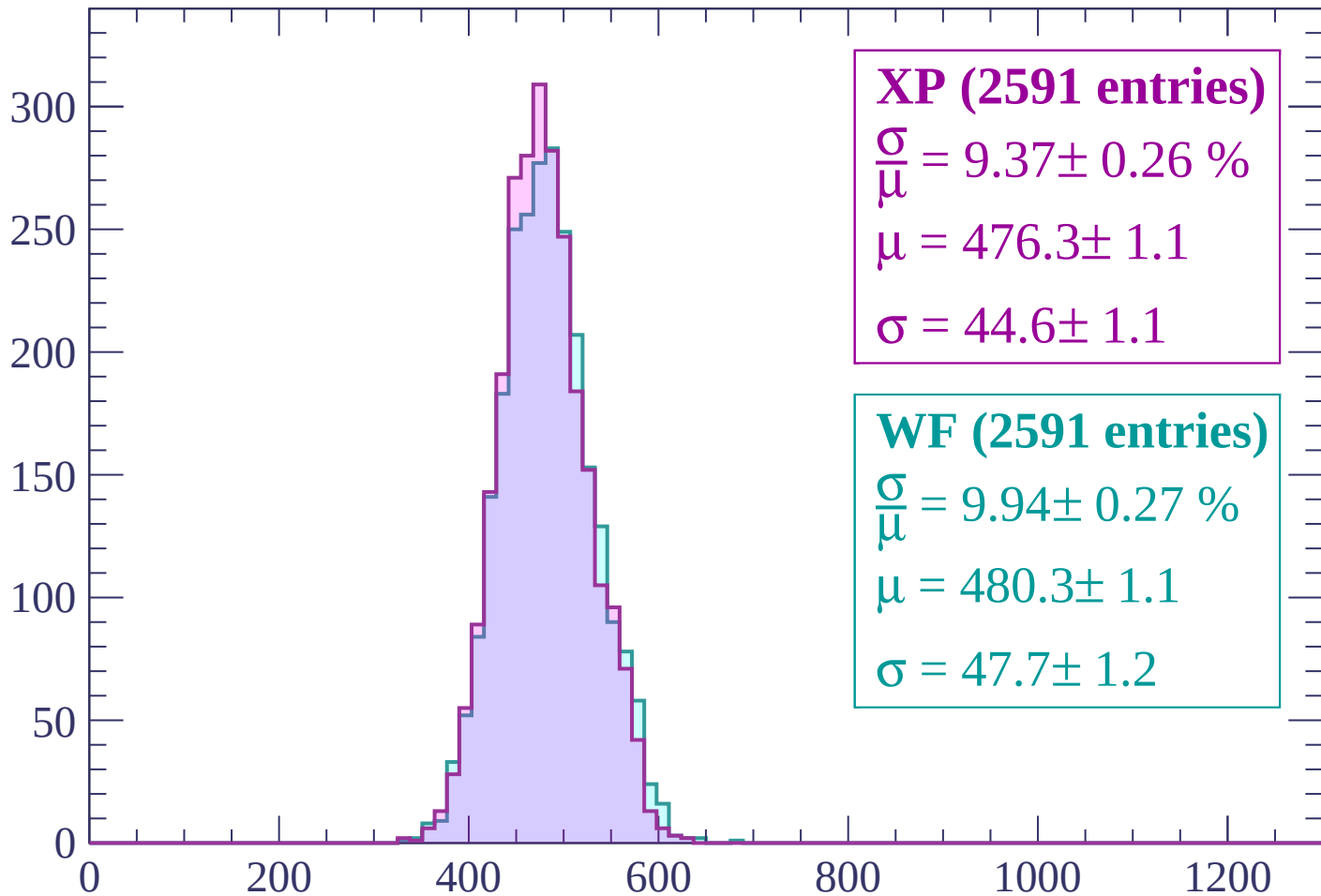
$$\mu = 481.0 \pm 1.1$$

$$\sigma = 47.5 \pm 1.2$$

dE/dx (ADC counts/cm)

Energy loss | $2160 < p < 2220$

Count



XP (2591 entries)

$$\frac{\sigma}{\mu} = 9.37 \pm 0.26 \%$$

$$\mu = 476.3 \pm 1.1$$

$$\sigma = 44.6 \pm 1.1$$

WF (2591 entries)

$$\frac{\sigma}{\mu} = 9.94 \pm 0.27 \%$$

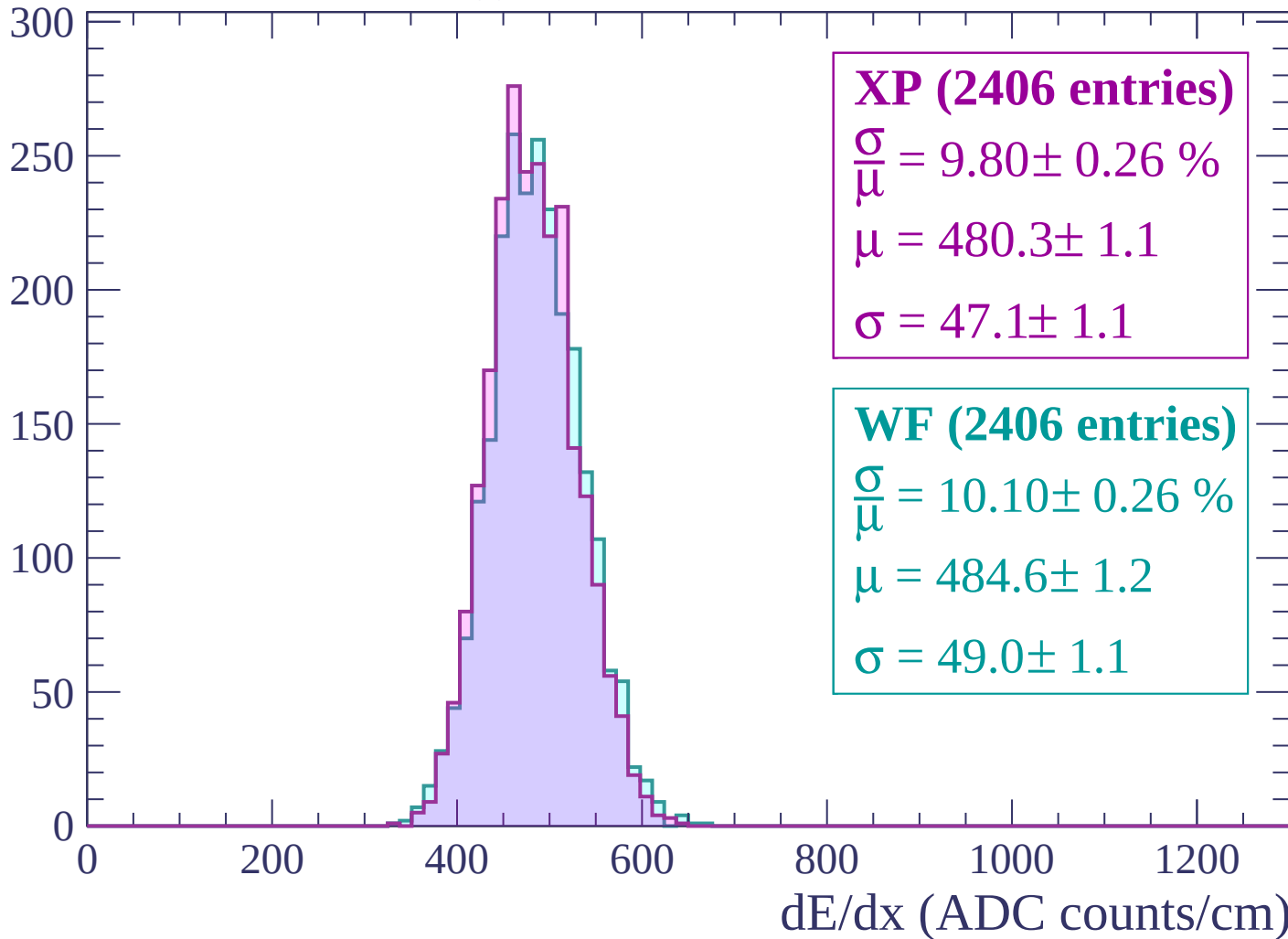
$$\mu = 480.3 \pm 1.1$$

$$\sigma = 47.7 \pm 1.2$$

dE/dx (ADC counts/cm)

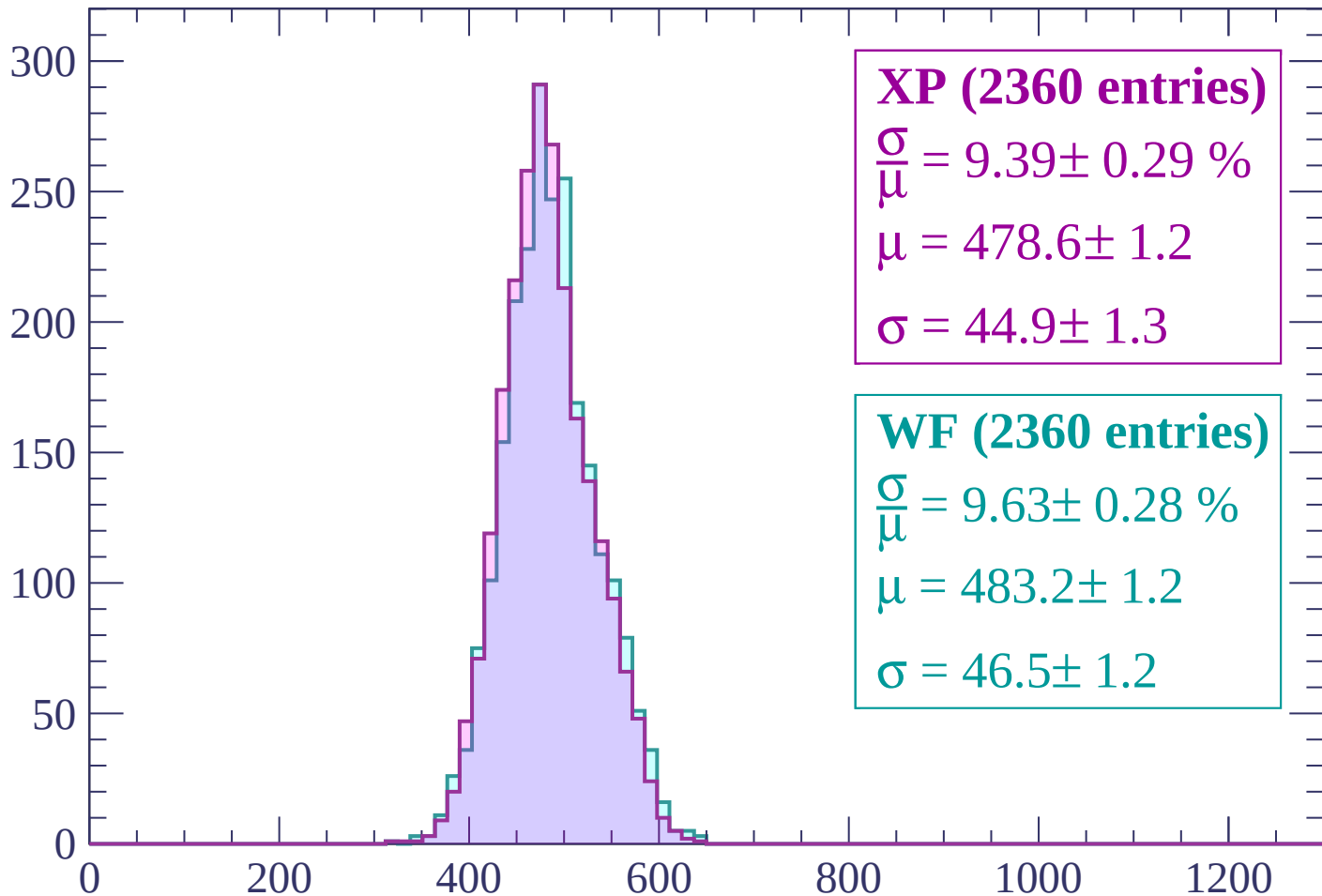
Energy loss | $2220 < p < 2280$

Count



Energy loss | $2280 < p < 2340$

Count



XP (2360 entries)

$$\frac{\sigma}{\mu} = 9.39 \pm 0.29 \%$$

$$\mu = 478.6 \pm 1.2$$

$$\sigma = 44.9 \pm 1.3$$

WF (2360 entries)

$$\frac{\sigma}{\mu} = 9.63 \pm 0.28 \%$$

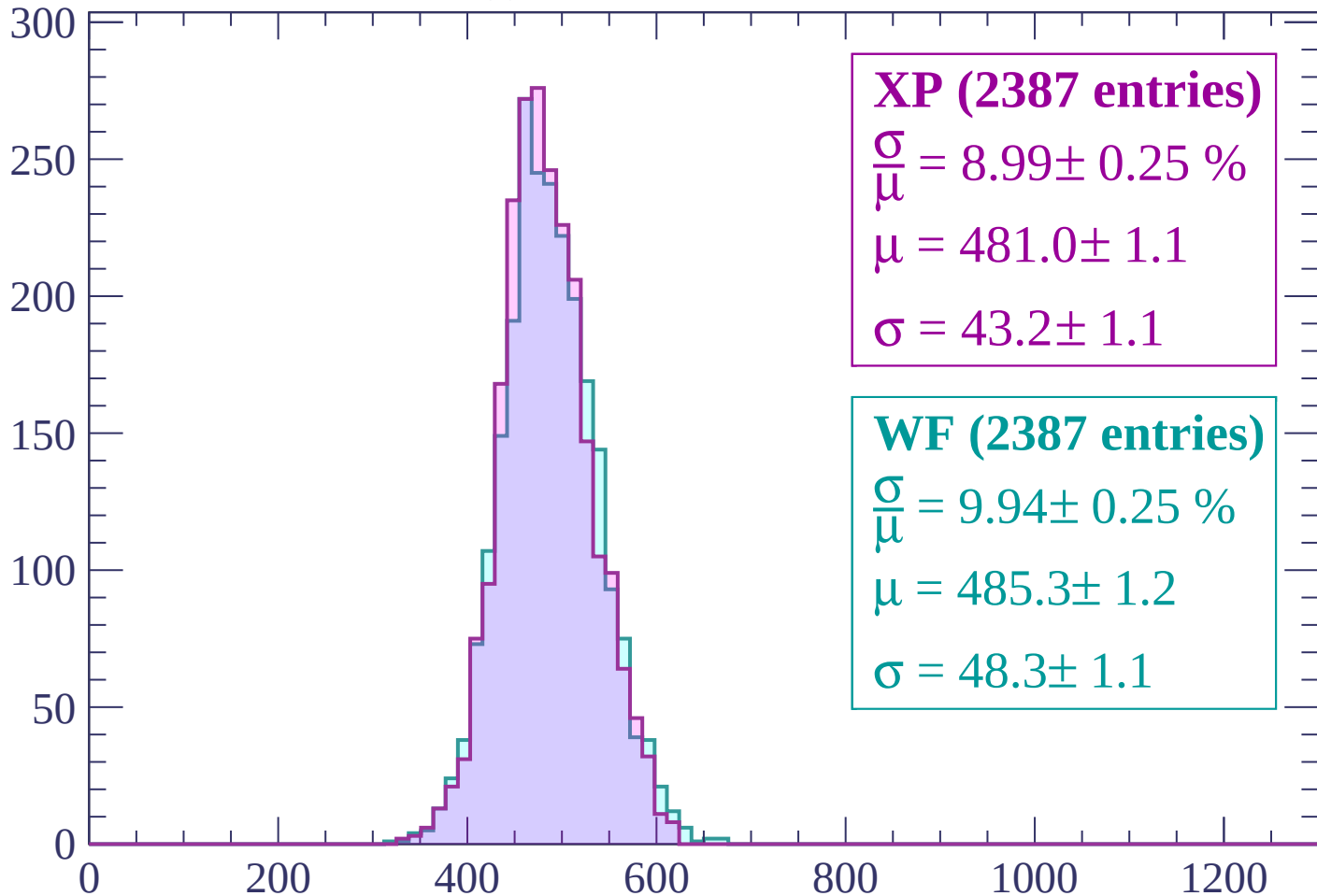
$$\mu = 483.2 \pm 1.2$$

$$\sigma = 46.5 \pm 1.2$$

dE/dx (ADC counts/cm)

Energy loss | $2340 < p < 2400$

Count



XP (2387 entries)

$$\frac{\sigma}{\mu} = 8.99 \pm 0.25 \%$$

$$\mu = 481.0 \pm 1.1$$

$$\sigma = 43.2 \pm 1.1$$

WF (2387 entries)

$$\frac{\sigma}{\mu} = 9.94 \pm 0.25 \%$$

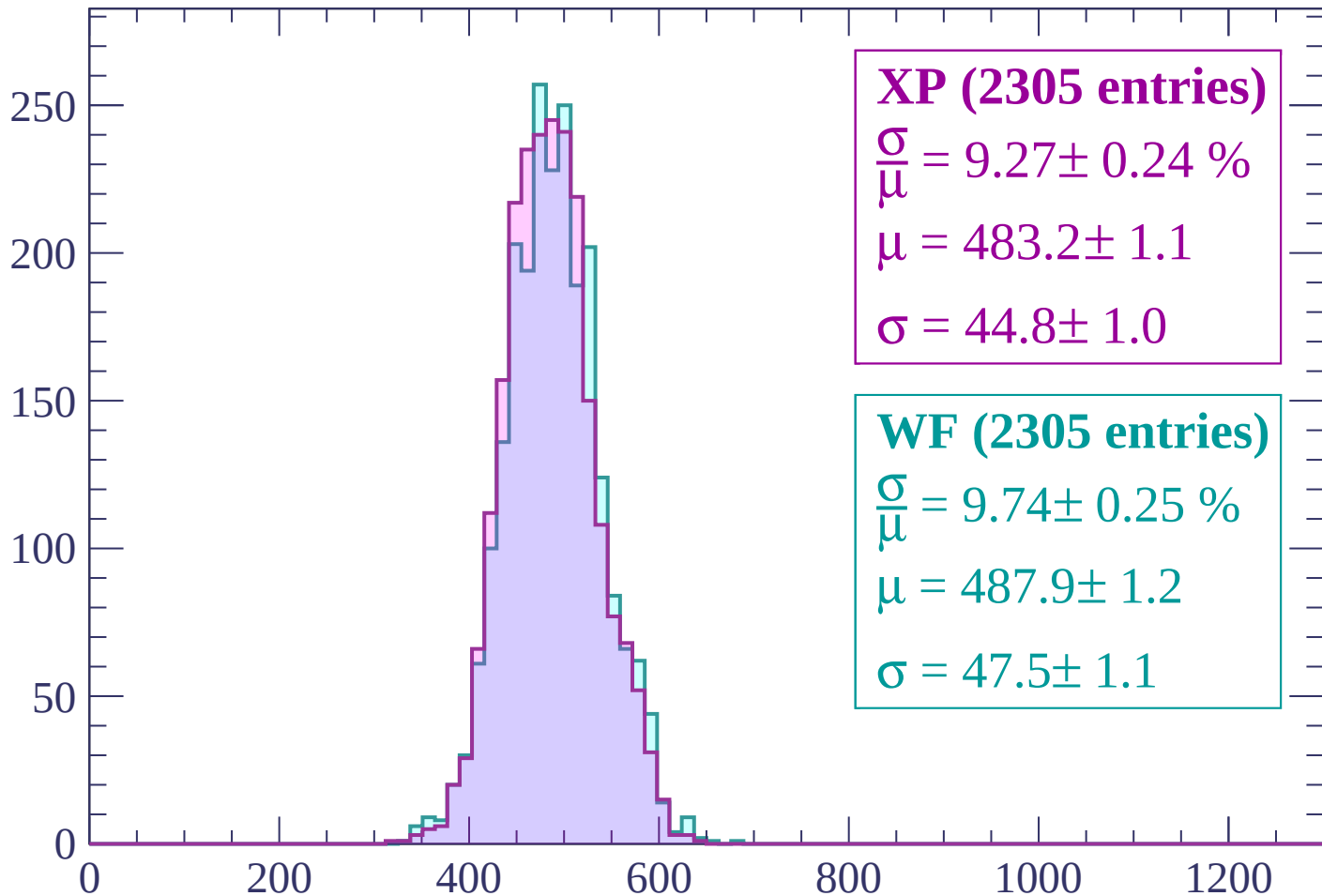
$$\mu = 485.3 \pm 1.2$$

$$\sigma = 48.3 \pm 1.1$$

dE/dx (ADC counts/cm)

Energy loss | $2400 < p < 2460$

Count



XP (2305 entries)

$$\frac{\sigma}{\mu} = 9.27 \pm 0.24 \%$$

$$\mu = 483.2 \pm 1.1$$

$$\sigma = 44.8 \pm 1.0$$

WF (2305 entries)

$$\frac{\sigma}{\mu} = 9.74 \pm 0.25 \%$$

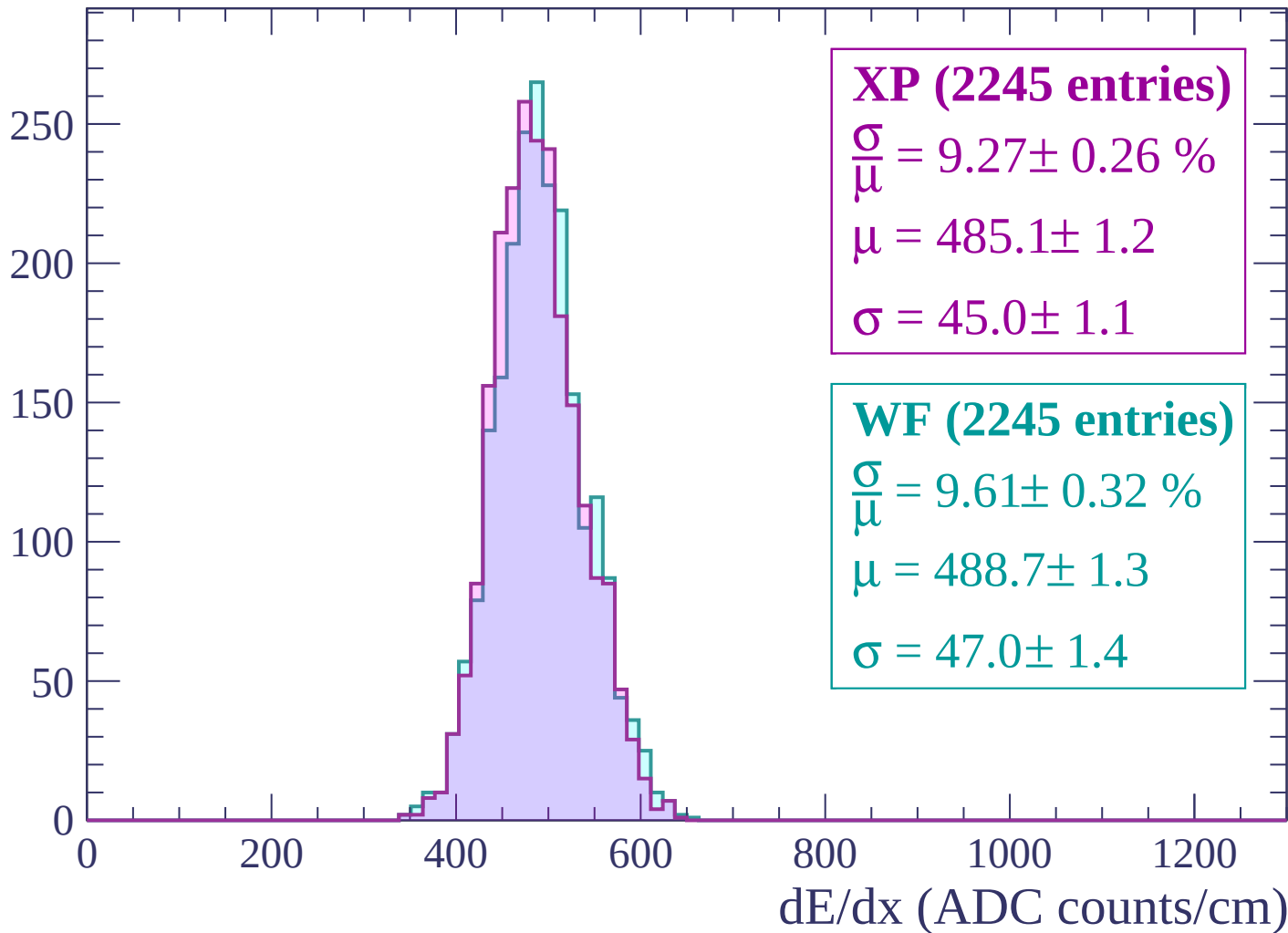
$$\mu = 487.9 \pm 1.2$$

$$\sigma = 47.5 \pm 1.1$$

dE/dx (ADC counts/cm)

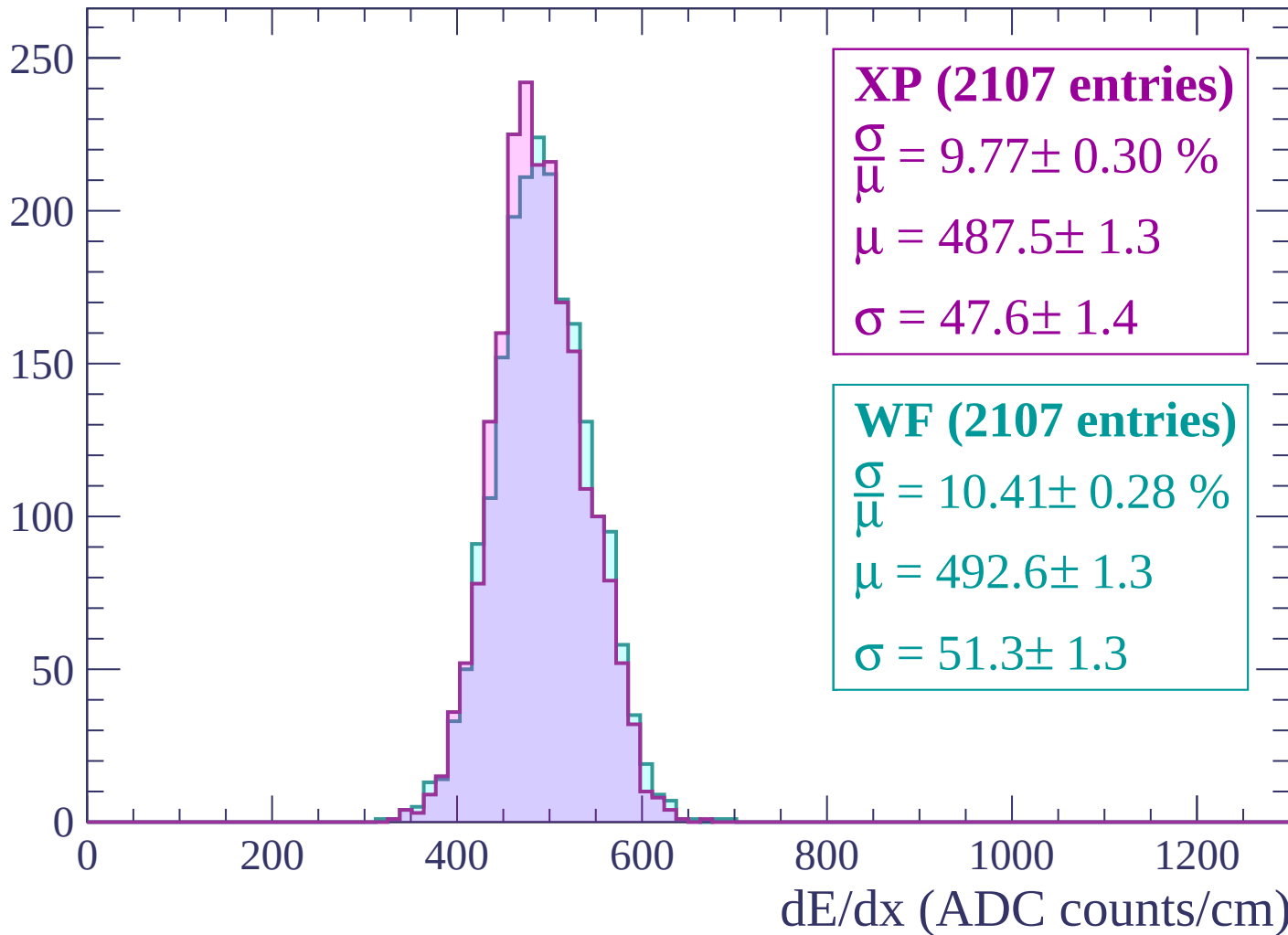
Energy loss | $2460 < p < 2520$

Count



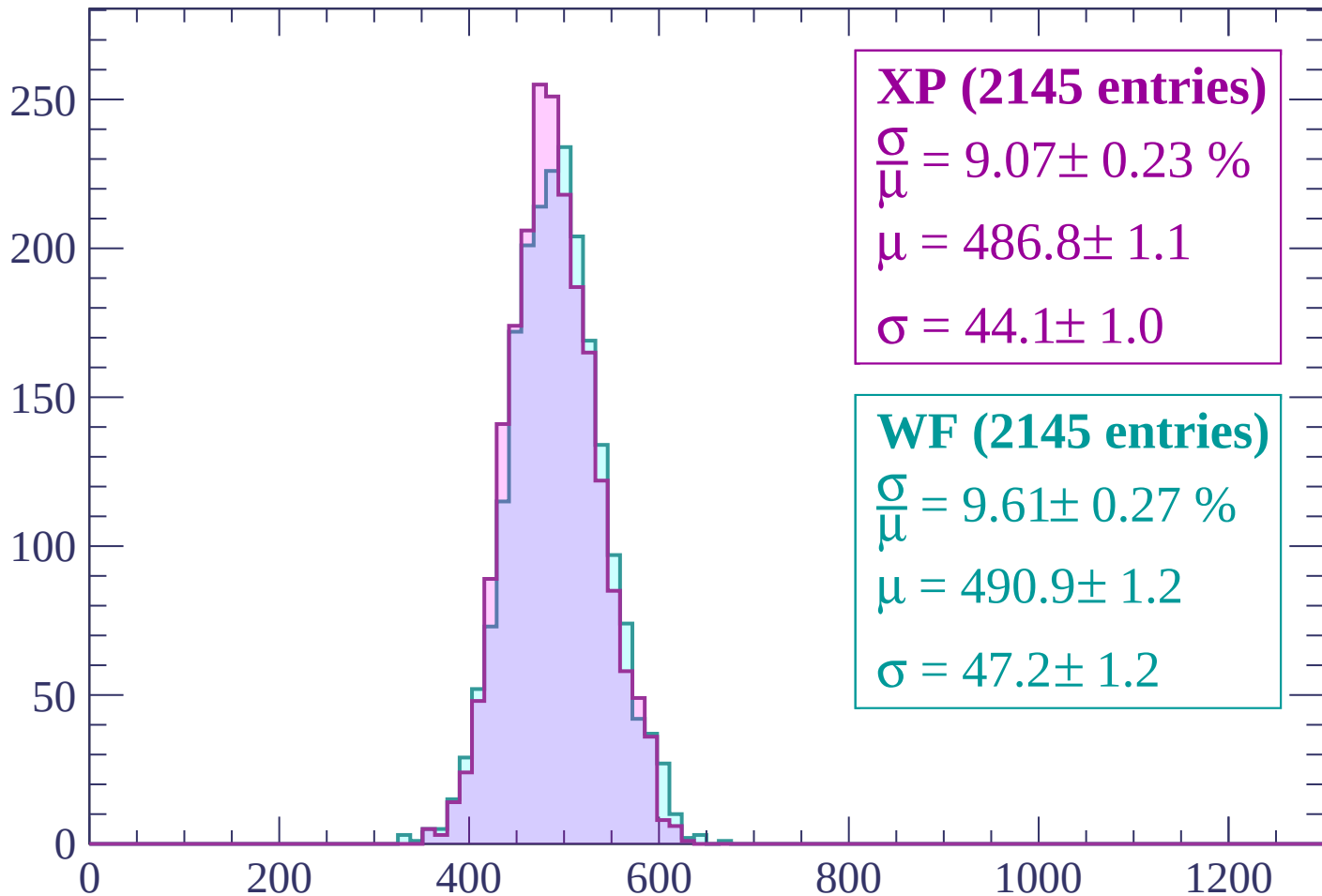
Energy loss | $2520 < p < 2580$

Count



Energy loss | $2580 < p < 2640$

Count



XP (2145 entries)

$$\frac{\sigma}{\mu} = 9.07 \pm 0.23 \%$$

$$\mu = 486.8 \pm 1.1$$

$$\sigma = 44.1 \pm 1.0$$

WF (2145 entries)

$$\frac{\sigma}{\mu} = 9.61 \pm 0.27 \%$$

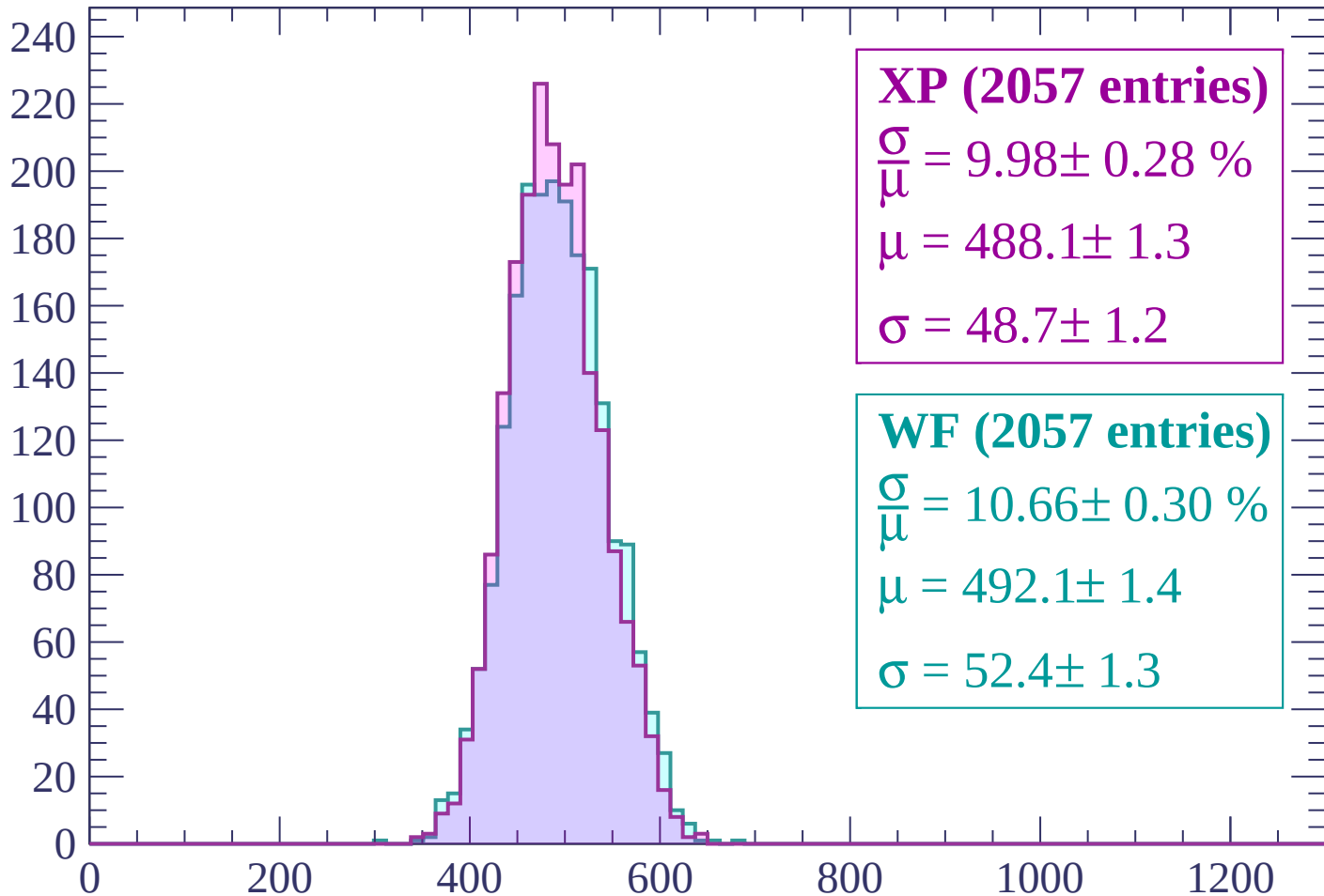
$$\mu = 490.9 \pm 1.2$$

$$\sigma = 47.2 \pm 1.2$$

dE/dx (ADC counts/cm)

Energy loss | $2640 < p < 2700$

Count



XP (2057 entries)

$$\frac{\sigma}{\mu} = 9.98 \pm 0.28 \%$$

$$\mu = 488.1 \pm 1.3$$

$$\sigma = 48.7 \pm 1.2$$

WF (2057 entries)

$$\frac{\sigma}{\mu} = 10.66 \pm 0.30 \%$$

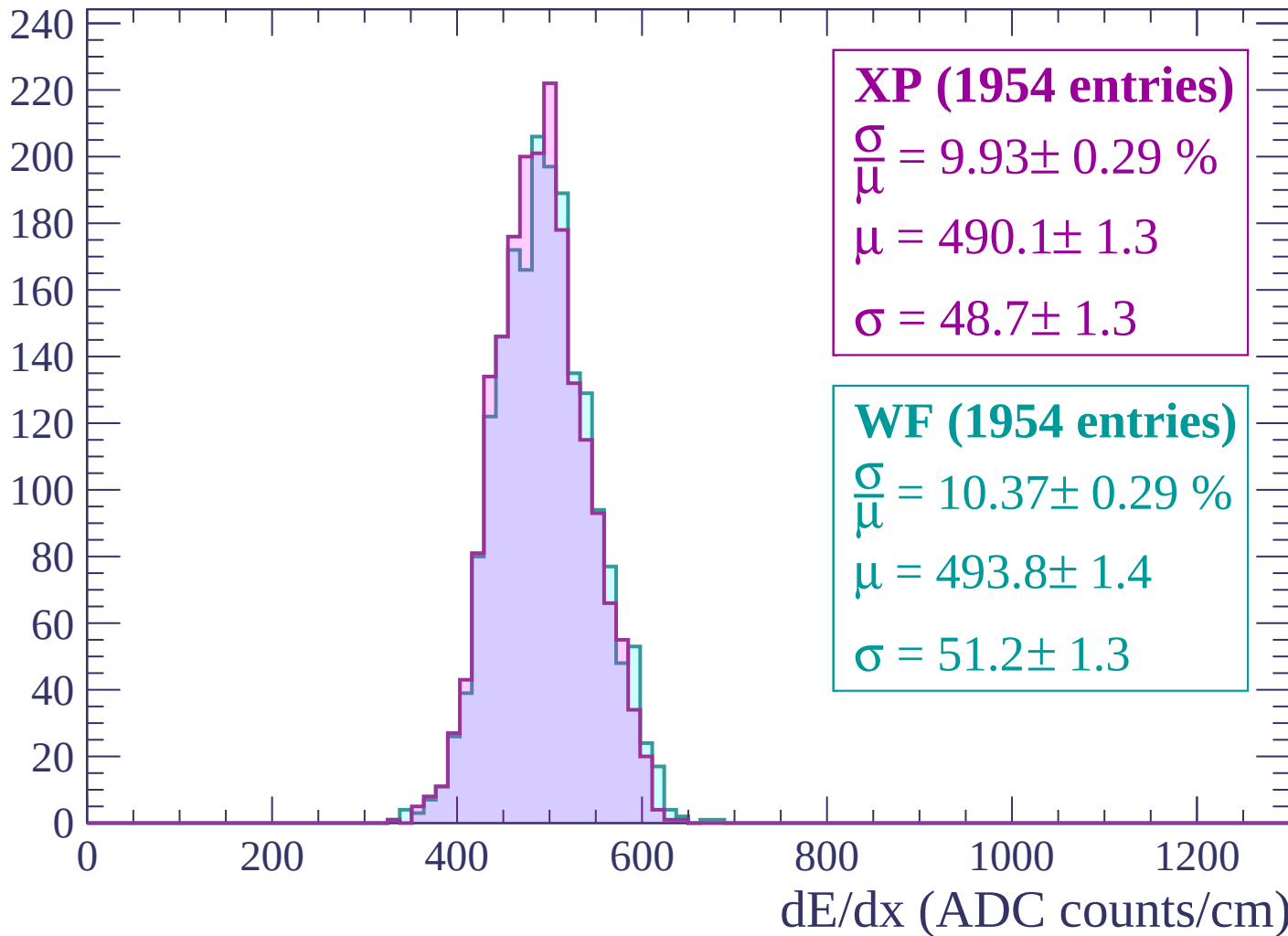
$$\mu = 492.1 \pm 1.4$$

$$\sigma = 52.4 \pm 1.3$$

dE/dx (ADC counts/cm)

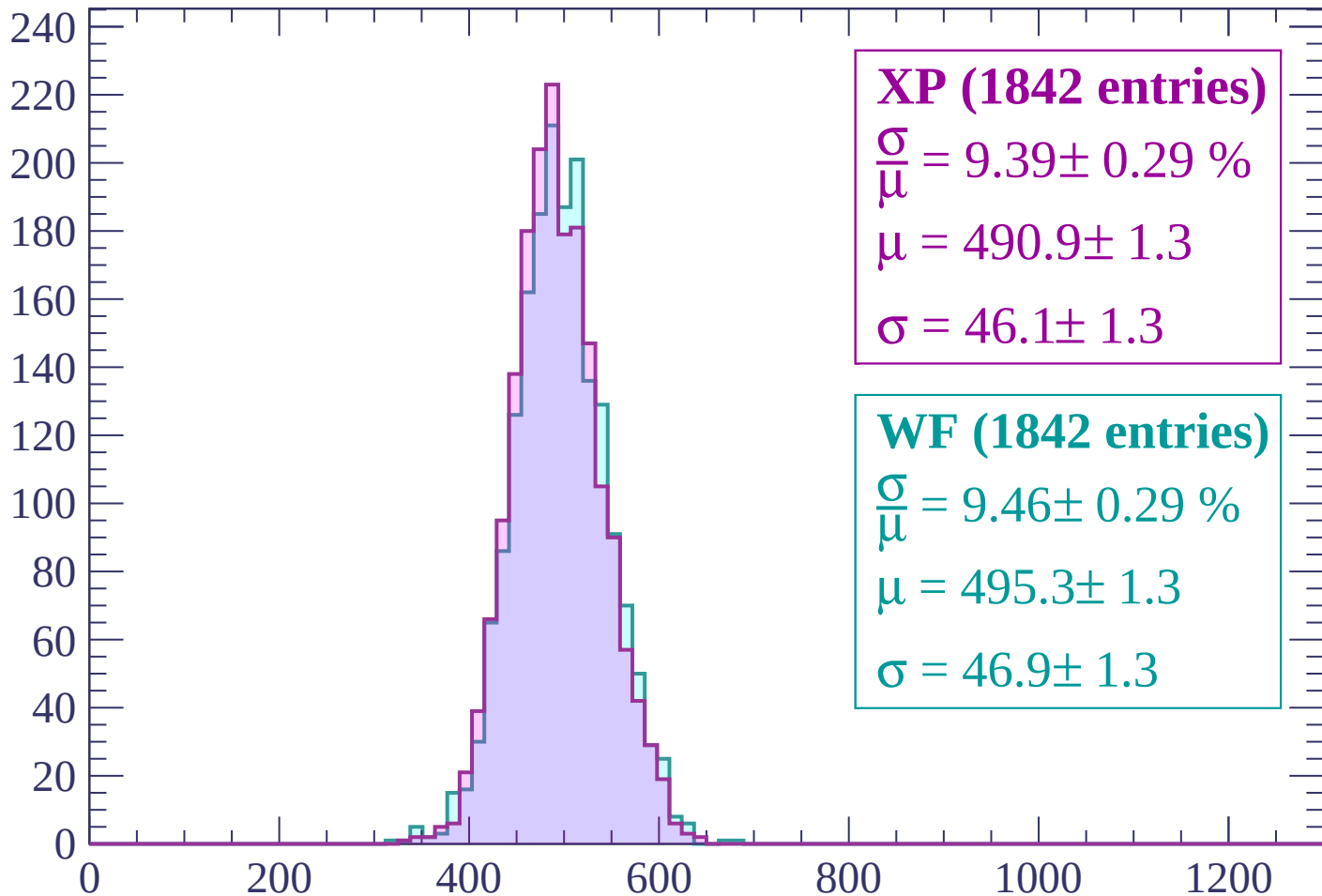
Energy loss | $2700 < p < 2760$

Count



Energy loss | $2760 < p < 2820$

Count



XP (1842 entries)

$$\frac{\sigma}{\mu} = 9.39 \pm 0.29 \%$$

$$\mu = 490.9 \pm 1.3$$

$$\sigma = 46.1 \pm 1.3$$

WF (1842 entries)

$$\frac{\sigma}{\mu} = 9.46 \pm 0.29 \%$$

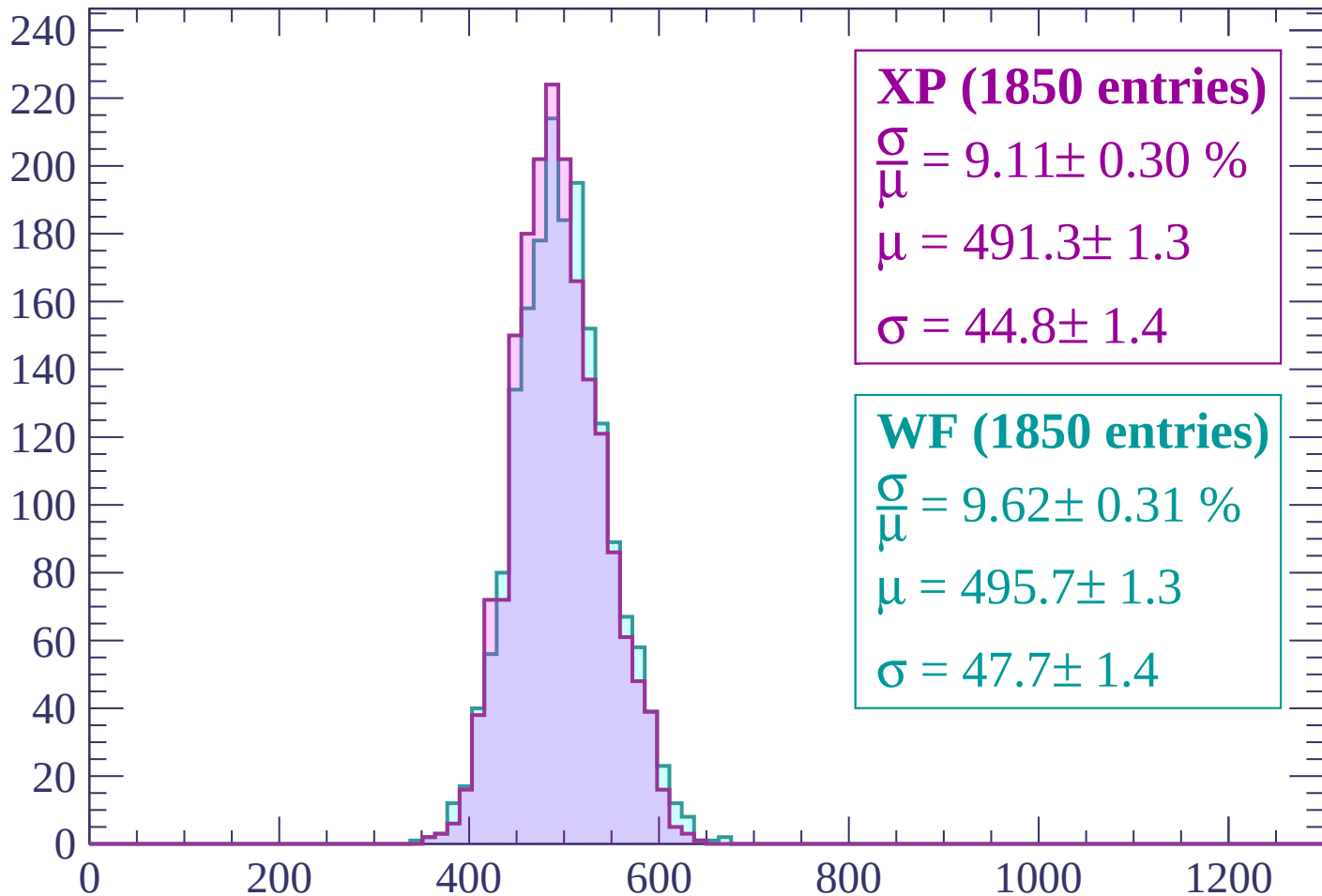
$$\mu = 495.3 \pm 1.3$$

$$\sigma = 46.9 \pm 1.3$$

dE/dx (ADC counts/cm)

Energy loss | $2820 < p < 2880$

Count



XP (1850 entries)

$$\frac{\sigma}{\mu} = 9.11 \pm 0.30 \%$$

$$\mu = 491.3 \pm 1.3$$

$$\sigma = 44.8 \pm 1.4$$

WF (1850 entries)

$$\frac{\sigma}{\mu} = 9.62 \pm 0.31 \%$$

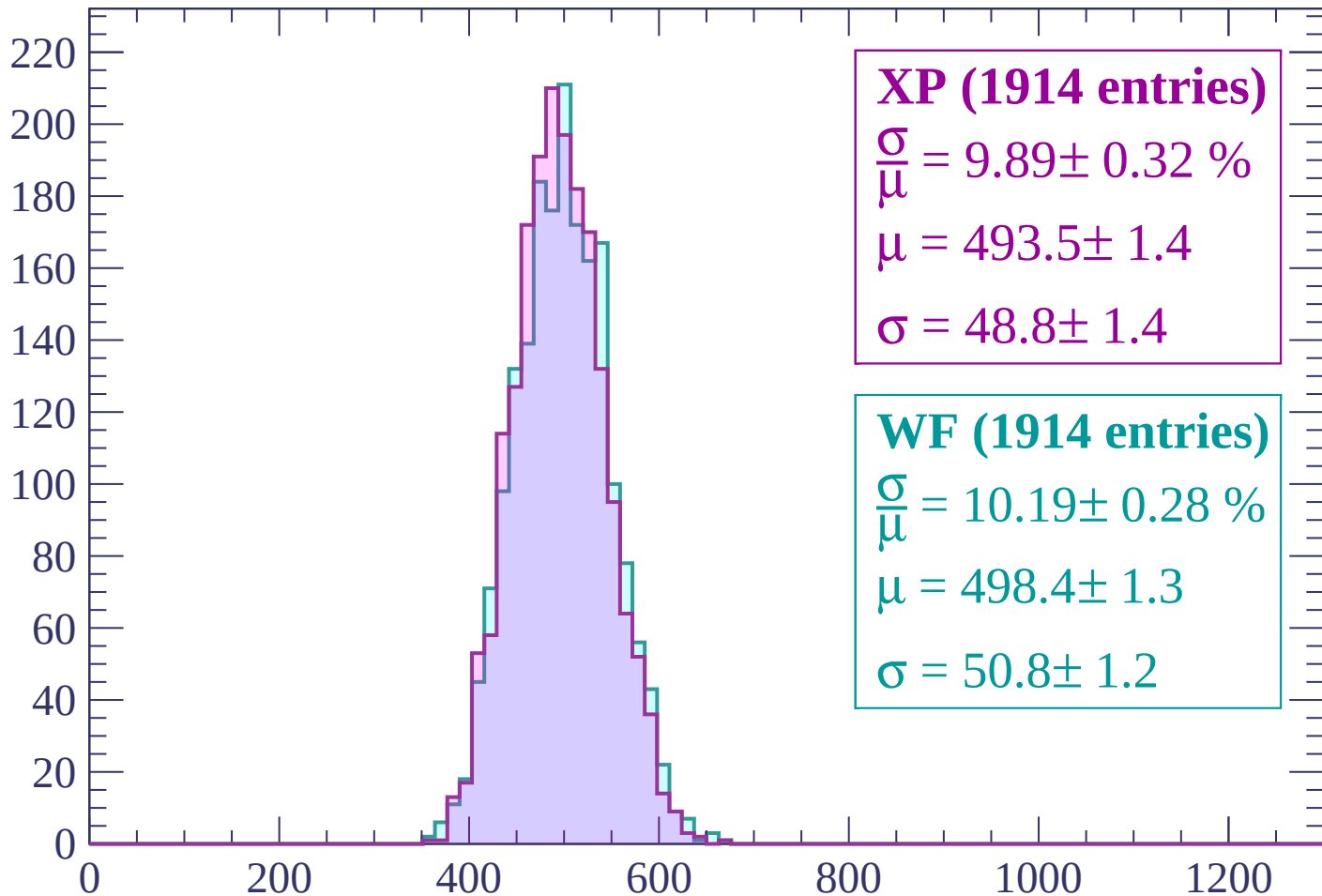
$$\mu = 495.7 \pm 1.3$$

$$\sigma = 47.7 \pm 1.4$$

dE/dx (ADC counts/cm)

Energy loss | $2880 < p < 2940$

Count



XP (1914 entries)

$$\frac{\sigma}{\mu} = 9.89 \pm 0.32 \%$$

$$\mu = 493.5 \pm 1.4$$

$$\sigma = 48.8 \pm 1.4$$

WF (1914 entries)

$$\frac{\sigma}{\mu} = 10.19 \pm 0.28 \%$$

$$\mu = 498.4 \pm 1.3$$

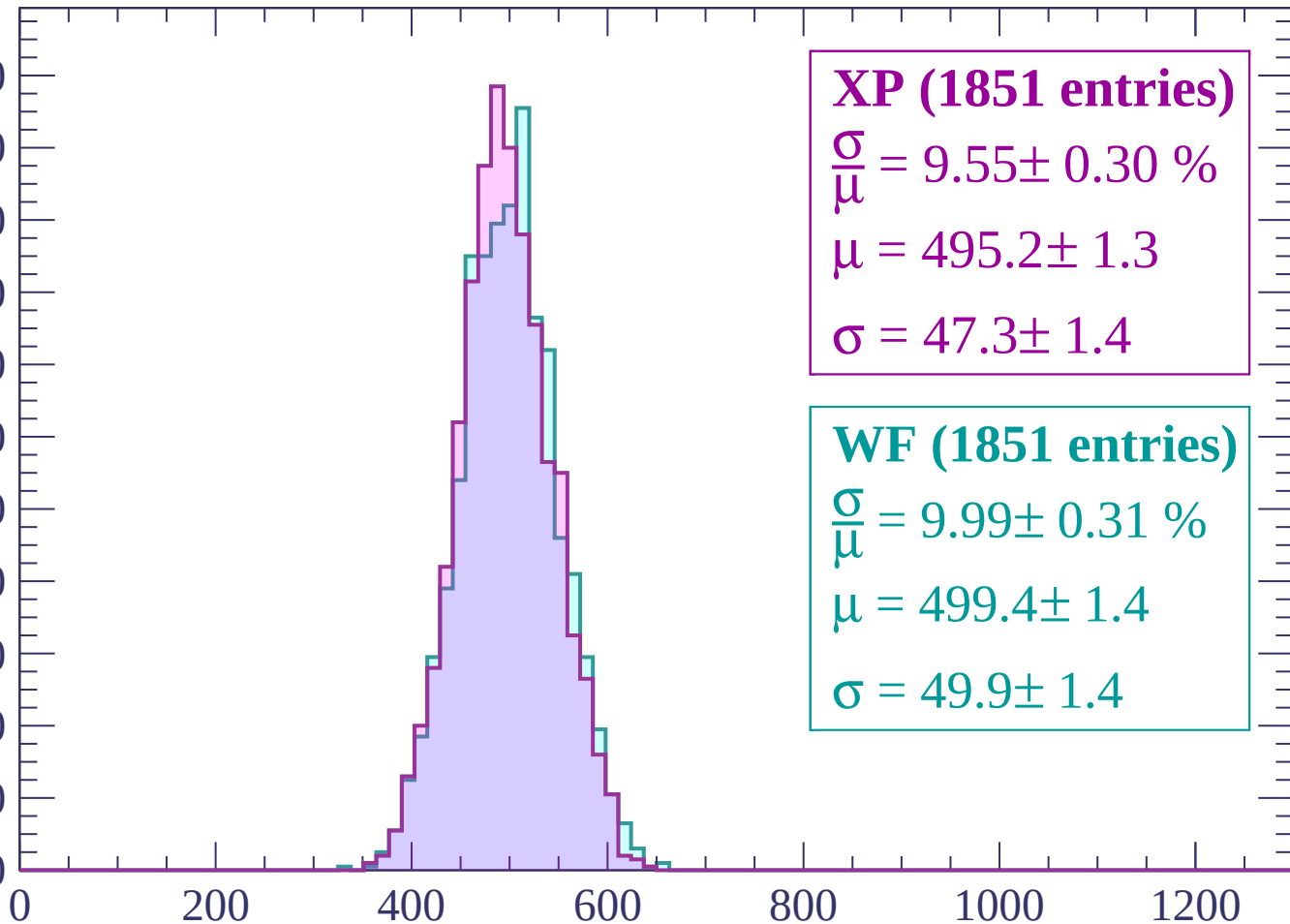
$$\sigma = 50.8 \pm 1.2$$

dE/dx (ADC counts/cm)

Energy loss | $2940 < p < 3000$

Count

220
200
180
160
140
120
100
80
60
40
20
0



XP (1851 entries)

$$\frac{\sigma}{\mu} = 9.55 \pm 0.30 \%$$

$$\mu = 495.2 \pm 1.3$$

$$\sigma = 47.3 \pm 1.4$$

WF (1851 entries)

$$\frac{\sigma}{\mu} = 9.99 \pm 0.31 \%$$

$$\mu = 499.4 \pm 1.4$$

$$\sigma = 49.9 \pm 1.4$$

dE/dx (ADC counts/cm)

Energy loss | $3000 < p < 3060$

Count

